



CSA B149.1:25

National Standard of Canada



Natural gas and propane installation code



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***Natural gas and propane
installation code***

IGAC

Interprovincial/Territorial Gas Advisory Council



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This edition of CSA B149.1 is dedicated to the life of our friend, colleague, and member: Lance Beaulieu.

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Preface

This is the seventeenth edition of CSA B149.1, *Natural gas and propane installation code*. It supersedes the previous editions, published in 2020, 2015, 2010, 2005, and 2000 by CSA Group (CSA) as CAN/CSA-B149.1, in 1995, 1991, 1986, 1980, 1978, 1976, and 1974 by the Canadian Gas Association (CGA), and in 1971, 1966, 1962, and 1958 by CSA Group.

The major changes from the previous edition include the following:

- a) updated Code Scope inclusions to clarify the threshold for propane systems between CSA B149.1 and CSA B149.2 [Clause [1.1](#) b)];
- b) corrected terminology to “vehicle-fuelling appliance” to align with other CSA standards [Clause [1.1](#) c)];
- c) clarified exclusion of “vehicle-refuelling stations as defined by CSA B108.1” [Clause [1.2](#) g)];
- d) added hydrogen-natural gas blends to the list of gases included in this Code (Clause [1.3](#)) and added reference to hydrogen-natural gas blends to applicable clauses (Clauses [4.5.4](#), [4.24.1](#), [5.5.1.4](#), [6.2.3.1](#), [6.2.3.2](#), [6.3.2](#) to [6.3.5](#), [6.3.8.1](#), and [6.9.6](#) and Tables [5.3](#) and [6.1](#));
- e) updated or added the following definitions:
 - i) updated definition of “high pressure regulator” to “industrial regulator” (Clause [3](#));
 - ii) updated definition of “line pressure regulator” (Clause [3](#));
 - iii) added definition for “manual shut-off valve” (Clause [3](#)); and
 - iv) added definition for “vent limiter” and sub-definition “limiting orifice type” (Clause [3](#));
- f) added reference to CSA B149.3 for field-approved appliances and equipment installation (Clause [4.1.4](#));
- g) added reference to CSA B149.6 for biogas use (Clause [4.2.4](#));
- h) updated rating plate requirements to show maximum percentage of hydrogen for blends with natural gas (Clause [4.5.4](#));
- i) updated bonding requirements to align with the *Canadian Electrical Code, Part I* (Clauses [4.7.3](#), [4.7.4](#), and [4.7.5](#));
- j) hydrogen-natural gas blends as a fuel gas to be odourized or monitored by a detection system (Clause [4.24.1](#));
- k) restructured Clause [5](#) as follows:
 - i) consolidated exemptions for outdoor gas vent terminations accounting for vent limiters, indoor ventilated spaces;
 - ii) clarified overpressure protection devices used with certified line pressure regulators must be factory pre-assembled and shipped as a unit with the regulator; and
 - iii) added the ability to use industrial regulators when flows/pressures are outside the scope of certified line pressure regulators, and must be interlocked with a high/low gas pressure device that serves the same appliance;
- l) updated clearances for gas vent terminations (Table [5.3](#)), now based on relief capacity and aligned with CSA Z662:23 (Annex [D](#));
- m) prohibited the use of hoses certified to CSA 8.3 for liquid propane systems (Clause [6.2.12](#));
- n) updated requirements to permit the use of press-connect fittings certified to CSA/ANSI LC 4/CSA 6.32 for piping of NPS 2-1/2 to 4 (Clause [6.9.2](#));
- o) prohibited the use of tape as a sealant for pipe sizes larger than NPS 1-1/2 for hydrogen-natural gas blends (Clause [6.9.6](#));
- p) added minimum separation requirements between the gas system and the electrical system sharing a common trench (Clause [6.15.15](#));

- q) added sleeve requirements for piping or tubing that passes through an exterior wall (Clauses [6.16.9](#) and [6.16.10](#));
- r) added requirements for a manual shutoff valve and signage for underground propane storage tanks (Clause [6.18.10](#));
- s) clarified leakage test requirements and allowable leakage rates (Clauses [6.22.2](#), [6.22.3](#), and [6.22.4](#) and Annex [K](#));
- t) added requirements for monitoring by a combustible gas indicator during purging (Clauses [6.23.4](#) and [6.23.7](#));
- u) stationary engine generator assemblies must be certified to ANSI/CAN UL/ULC 2200 (Clause [7.2.4.1](#));
- v) aligned venting requirements for engine or turbine fuel trains with other appliances (Clause [7.2.4.4](#));
- w) flexible foil noncombustible ducts on commercial-type clothes dryers must be certified to UL 2158A (Clause [7.4.3](#));
- x) permitted approved engineering designs for the installation of direct-fired makeup air heaters where food preparation areas are interconnected to areas frequented by the public (Clauses [7.20.14](#) and [7.21.13](#));
- y) added the ability to install recirculating direct gas-fired industrial air heaters and amended requirements accordingly (Clause [7.21](#));
- z) added method for safe interlocking of an appliance to exhaust fans (Clause [7.21.9](#));
- aa) added requirements for overpressure protection for engine venting systems (Clause [8.32](#));
- ab) clarified RFA and VFA requirements (Clause [10](#));
- ac) updated customer's meter and service regulator installations based on the latest edition of CSA Z662 (Annex [D](#));
- ad) removed informative annex for pressure regulators and overpressure protection devices;
- ae) added informative annex for allowable leakage of gas appliance valve train components (Annex [K](#)); and
- af) added informative annex for special considerations for industrial occupancy (Annex [O](#)).

CSA Group acknowledges that the development of this Code was made possible, in part, by the financial support of Natural Resources Canada.

This Code was prepared by the Technical Committee on Installation Code for Natural Gas and Propane Appliances, under the Fuels and Appliances Strategic Steering Committee, and has been formally approved by the Technical Committee and the Interprovincial/Territorial Gas Advisory Council.

This Code has been developed in compliance with Standards Council of Canada requirements for National Standards of Canada. It has been published as a National Standard of Canada by CSA Group.

Notes:

- 1) *Use of the singular does not exclude the plural (and vice versa) when the sense allows.*
- 2) *Although the intended primary application of this Code is stated in its Scope, it is important to note that it remains the responsibility of the users of the Code to judge its suitability for their particular purpose.*
- 3) *This Code was developed by consensus, which is defined by CSA Policy governing standardization — Code of good practice for standardization as “substantial agreement. Consensus implies much more than a simple majority, but not necessarily unanimity”. It is consistent with this definition that a member may be included in the Technical Committee list and yet not be in full agreement with all clauses of this Code.*
- 4) *A request for interpretation can be submitted to clarify the meaning of a provision in this Standard. To submit a request for interpretation, please refer to www.csagroup.org/request-for-interpretation. Committee interpretations are processed in accordance with the CSA Directives and guidelines governing standardization and are available on the Standards Activities page at standardsactivities.csagroup.org.*

- 5) *This Code is subject to review within five years from the date of publication. Suggestions for its improvement will be referred to the appropriate committee. To submit a proposal for change, please send the following information to inquiries@csagroup.org and include "Proposal for change" in the subject line:*
- a) *Code designation (number);*
 - b) *relevant clause, table, and/or figure number;*
 - c) *wording of the proposed change; and*
 - d) *rationale for the change.*

Sustainable Development Goals (SDG)

Foreword

CSA Group develops and maintains standards-based solutions across a broad range of topics, most of which support the United Nations Sustainable Development Goals (UN SDGs) towards shaping a sustainable and resilient future.

Through a robust mapping process, connections between CSA B149.1:25 and the following SDGs have been identified:

SDG		
	7.1	11

CSA B149.1:25 has notable linkages with the following SDGs:

- SDG 7: *Affordable and Clean Energy*
- SDG 11: *Sustainable Cities and Communities*

For further information on CSA Group's SDG Mapping initiative, please visit:

<https://www.csagroup.org/sdg/>

Note: *It is important to note that although some standards-based solutions explicitly support SDG targets, not all standards and documents link to the SDGs. Users of CSA documents should always take care and be specific when claiming their support of SDGs through the use of standards and other documents. The SDG mapping outcomes made available by CSA Group are intended to assist users in their evaluation of how the application of a standard or other document can support their work towards SDG achievement.*

CSA B149.1:25

Natural gas and propane installation code

0 Introduction

In 1958, CSA Group published the first edition of CSA B149, *Installation Code for Gas Burning Appliances and Equipment*. It was superseded by later editions in 1962, 1966, and 1971. Following the publication of the 1966 edition, the decision was made to split the Code into two parts: B149.1, dealing with the installation of appliances and equipment burning natural gas, and B149.2, dealing with the installation of appliances and equipment burning propane. As a first step, B149.2 was prepared and first published in 1969.

The Canadian Gas Association was accredited by the Standards Council of Canada as the standards development organization responsible for preparing standards for gas-burning appliances and equipment, and in this connection took over responsibility for the B149 Code in 1974. New editions of the Code were subsequently published in 1974, 1976, 1978, 1980, 1986, 1991, and 1995.

On June 30, 1997, CSA Group acquired International Approval Services (IAS), which was until then a joint venture of the American Gas Association (AGA) and the Canadian Gas Association. Under this agreement, CSA Group acquired the complete range of IAS standards administration, certification, and registration products and services for appliances and accessories fuelled by natural and liquefied petroleum gases. In 1998, the CSA B149 Installation Code Committee agreed to publish a Natural Gas and Propane Installation Code that would amalgamate the first seven sections of CAN/CGA-B149.1 and CAN/CGA-B149.2 to become CAN/CSA-B149.1-00. This amalgamation was in response to the trend among the authorities having jurisdiction of combining licensing and training for natural gas and propane. The remaining Clauses 8 to 14 of CAN/CGA-B149.2 became CAN/CSA-B149.2-00, *Propane Storage and Handling Code*.

1 Scope

1.1 Inclusions

This Code applies to the installation of

- a) appliances, equipment, components, and accessories where gas is to be used for fuel purposes;
- b) piping and tubing systems extending downstream from the termination of the utility installation or from any equipment extending downstream from the outlet of any propane container pressure regulator (commonly referred to as “first-stage regulator”);
- c) vehicle-fuelling appliances and associated equipment meeting the requirements of an appliance or the intended use of the appliance to fill a natural-gas-fuelled vehicle; and
- d) stationary gas engines and turbines.

1.2 Exclusions

This Code does not apply to

- a) marine or pipeline terminals;
- b) gas where used as a feedstock in petroleum refineries or chemical plants;
- c) utility pipeline distribution and transmission pipelines;
- d) storage and handling of liquefied natural gas or underground formations for natural gas;

- e) the installation of natural gas vehicle (NGV) fuel systems, containers, and refuelling stations;
- f) the storage and utilization of compressed natural gas on boats;
- g) the installation of vehicle-refuelling stations as defined by CSA B108.1;
- h) refrigerated storage or underground formations for propane;
- i) propane used on boats;
- j) propane used as a propellant in aerosol containers;
- k) butane fuel cylinders of 6.2 oz (175 g) capacity or less;
- l) the installation of containers and equipment to be used for propane in distribution locations and filling plants and on tank trucks, tank trailers, and cargo liners; and
- m) propane used as refrigerant.

1.3 Fuel

Where the term “gas” is used, the requirements of this Code include, and apply equally to, any of the following gases or mixtures of them: natural gas, manufactured gas, hydrogen-natural gas blends or mixtures of propane gas and air, propane, propylene, butanes (normal butane or isobutane), and butylenes.

1.4 No guarantee

This Code and any Standards referenced in it do not make or imply any assurance or guarantee with respect to the life expectancy, durability, or operating performance of equipment and materials referenced in the Code.

1.5 Units of measurement

The values given in US customary units are the units of record for the purposes of this Code. The values given in parentheses are for information and comparison only.

1.6 Terminology

In this Code, “shall” is used to express a requirement, i.e., a provision that the user is obliged to satisfy in order to comply with the Code; “should” is used to express a recommendation or that which is advised but not required; and “may” is used to express an option or that which is permissible within the limits of the Code.

Notes accompanying clauses do not include requirements or alternative requirements; the purpose of a note accompanying a clause is to separate from the text explanatory or informative material.

Notes to tables and figures are considered part of the table or figure and may be written as requirements.

Annexes are designated normative (mandatory) or informative (non-mandatory) to define their application.

2 Reference publications

This Code refers to the following publications, and where such reference is made, it shall be to the edition listed below, including all amendments published thereto.

CSA Group

Note: *CGA Standards, Recommended Practices, and Codes are now published by CSA Group.*

B51:24

Boiler, pressure vessel, and pressure piping code

B108:23

Natural gas refuelling stations installation code

B137 Series:23

Thermoplastic pressure piping standards package

B137.4:23

Polyethylene (PE) piping systems for gas services

B137.4.1:23

Electrofusion-type polyethylene (PE) fittings for gas services

B149.2:25

Propane storage and handling code

B149.3:25

Code for the field approval of fuel-burning appliances and equipment

CSA/ANSI B149.6:25

Code for biogas generation and utilization

B365-17 (R2022)

Installation code for solid-fuel-burning appliances and equipment

B366.1-11 (R2020)

Solid-fuel-fired central heating appliances

C22.1:24

Canadian Electrical Code, Part I

C282:24

Emergency Electrical Power Supply for Buildings

CGA Certification Laboratory Requirement LAB-009-1989 (revised 1994)

Flexible gas tubing for interior natural and propane gas piping systems

F300:22

Residential depressurization

CSA/ANSI LC 1:23/CSA 6.26:23

Fuel gas piping systems using corrugated stainless steel tubing

CSA/ANSI LC 4:22/CSA 6.32:22

Press-connect metallic fittings for use in fuel gas distribution systems

CSA/ANSI NGV 5.1:23

Residential fueling appliances

CSA/ANSI NGV 5.2:23

Vehicle fueling appliances (VFA)

W117.2:19 (R2023)

Safety in welding, cutting, and allied processes

CSA/ANSI Z21.13:22/CSA 4.9:22

Gas-fired low pressure steam and hot water boilers

CSA/ANSI Z21.15:21/CSA 9.1:21

Manually operated gas valves for appliances, appliance connector valves and hose-end valves

CSA/ANSI Z21.18:19 (R2024)/CSA 6.3:19 (R2024)

Gas appliance pressure regulators

CSA/ANSI Z21.21:23/CSA 6.5:23

Automatic valves for gas appliances

CSA/ANSI Z21.24:22/CSA 6.10:22

Connectors for gas appliances

CSA/ANSI Z21.41:23/CSA 6.9:23

Quick disconnect devices for use with gas fuel appliances

ANSI Z21.50:19 (R2024)/CSA 2.22:19 (R2024)

Vented decorative gas appliances

CSA/ANSI Z21.54:19 (R2024)/CSA 8.4:19 (R2024)

Gas hose connectors for portable outdoor gas-fired appliances

ANSI Z21.69-2015 (R2020)/CSA 6.16-2015 (R2020)

Connectors for moveable gas appliances

ANSI Z21.75-2016 (R2020)/CSA 6.27-2016 (R2021)

Connectors for outdoor gas appliances and manufactured homes

ANSI Z21.78-2010 (R2020)/CSA 6.20-2010 (R2020)

Combination Gas Controls For Gas Appliances

CSA/ANSI Z21.80:19 (R2024)/CSA 6.22:19 (R2024)

Line pressure regulators

CSA/ANSI Z21.90:19 (R2024)/CSA 6.24:19 (R2024)

Gas convenience outlets and optional enclosures

CSA/ANSI Z21.101:22/CSA 8.5:22

Gas hose connectors for portable and moveable gas appliances

ANSI Z83.4-2017 (R2022)/CSA 3.7-2017 (R2022)

Non-recirculating direct gas-fired heating and forced ventilation appliances for commercial and industrial application

ANSI Z83.11-2016 (R2021)/CSA 1.8-2016 (R2021)

Gas food service equipment

ANSI Z83.25-2017 (R2021)/CSA 3.19-2017 (R2021)

Direct gas-fired process air heaters

Z240 MH Series-16 (R2021)

Manufactured homes:

Z240.4.1-16 (R2021)

Installation requirements for gas-burning appliances in manufactured homes

Z240 RV Series:23

Recreational vehicles

Z240.4.2:23

Installation requirements for propane appliances and equipment in recreational vehicles

Z662:23

Oil and gas pipeline systems

CAN/CSA 1-3.7-1977 (withdrawn)

Direct gas-fired non-recirculating make-up air heaters

CSA 2.17:17 (R2022)

Gas-fired appliances for use at high altitudes

CAN1-3.12-78 (withdrawn)

Direct gas fired door air heaters

CSA 3.11:15 (R2020)

Lever operated pressure lubricated plug type gas shut-off valves

CSA 3.16:21

Lever operated pressure lubricated plug type gas shut-off valves

CSA 6.18-02 (R2022)

Service regulators for natural gas

CSA 8.1:15 (R2020)

Elastomeric composite hose and hose couplings for conducting propane and natural gas

CSA 8.3-2015 (R2020)

Thermoplastic hose and hose couplings for conducting propane and natural gas

CSA 12.6-04 (withdrawn)

Vehicle Refuelling Appliances

American Gas Association

Manual for Determination of Supercompressibility Factors for Natural Gas

ASME International

B1.20.1-2013 (R2018)

Pipe Threads, General Purpose (Inch)

B16 series of standards

B16.3-2021

Malleable Iron Threaded Fittings: Classes 150 and 300

B31.3-2024

Process Piping

B36.10-2022

*Welded and Seamless Wrought Steel Pipe***ASTM International**

A53/A53M-22

Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless

A106/A106M-19

Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service

A179/A179M-24

Standard Specification for Seamless Cold-Drawn Low-Carbon Steel Heat-Exchanger and Condenser Tubes

ASTM A312/A312M-24b

Standard Specification for Seamless, Welded, and Heavily Cold Worked Austenitic Stainless Steel Pipes

B88-22

Standard Specification for Seamless Copper Water Tube

B837-19

*Standard Specification for Seamless Copper Tube for Natural Gas and Liquefied Petroleum (LP) Gas Fuel Distribution Systems***CGA (Canadian Gas Association)**

3.7-1976 (withdrawn)

Direct-fired make-up air heaters (DFMAH)

CAN1-3.7-77 (R1996)

Direct-fired make-up air heaters (DFMAH)

CAN1-3.12-78 (R1996)

Direct Gas-Fired Door Air Heaters

12.6-M94 (R1999)

Vehicle Refuelling Appliances

CR96-001

Flexible Metallic Hose

CGSB (Canadian General Standards Board)

CAN/CGSB-3.13-M88 (withdrawn)

Liquefied Petroleum Gas (Butanes)

CAN/CGSB-3.14-2023

*Propane for Fuel Purposes***Gas Processors Association**

Engineering Data Book

Government of Canada*Transportation of Dangerous Goods Act*, S.C. 1992, c. 34, and the *Transportation of Dangerous Goods Regulations*, as amended from time to time*Weights and Measures Act*, R.S.C. 1985, c. W-6**NFPA (National Fire Protection Association)**

211-2024

*Standard for Chimneys, Fireplaces, Vents, and Solid Fuel-Burning Appliances***NRC (National Research Council Canada)***National Building Code of Canada*, 2020**Polyflo Computer Company**

B.C. Shebeko, Polyflo Flow Computer, 1974

ULC (Underwriters' Laboratories of Canada)

CAN/ULC-S110-13 (R2023)

Standard Methods of Test for Air Ducts

CAN/ULC-S114-2018 (R2024)

Standard Method of Test for Determination of Non-Combustibility in Building Materials

CAN/ULC-S508:2023

Standard for the Rating and Fire Testing of Fire Extinguishers

CAN/ULC-S604:2022

Standard for Factory-Built Type A Chimneys

CAN/ULC-S605:2022

Standard for Gas Vents

CAN/ULC-S609:2022

Standard for Low Temperature Vents Type L and PL

CAN/ULC-S629:2022

Standard for 650 °C Factory-Built Chimneys

CAN/ULC-S635:2016 (R2022)

Standard for Lining Systems for Existing Masonry or Factory-Built Chimneys and Vents

S636-08

Standard for Type BH Gas Venting Systems

CAN/ULC-S642:2021

Standard for Compounds and Tapes for Threaded Pipe Joints

ULSE Inc.

UL 144:2021

LP-Gas Regulators and Regulator Systems

ANSI/CAN/UL 536:2021

Flexible Metallic Hose

UL 959-2010 (R2019)

Medium Heat Appliance Factory-Built Chimneys

UL 2158A-2013 (R2023)

Clothes Dryer Transition Duct

ANSI/CAN UL/ULC 2200:2020 (R2022)

Stationary Engine Generator Assemblies

Other publication

Crocker, S. *Piping Handbook*, 4th ed., McGraw-Hill, 1945.

3 Definitions

The following definitions shall apply in this Code:

Accessory — a part capable of performing an independent function and contributing to the operation of the appliance or gas piping system that it serves.

Air supply (with respect to the installation of an appliance) — combustion air, flue gas dilution air, and ventilation air.

Combustion air — the air required for satisfactory combustion of gas, including excess air.

Excess air — that portion of the combustion air that is supplied to the combustion zone in excess of that which is theoretically required for complete combustion.

Flue gas dilution air — the ambient air that is admitted to a venting system at the draft hood, draft diverter, or draft regulator.

Ventilation air — air that is admitted to a space containing an appliance to replace air exhausted through a ventilation opening or by means of exfiltration.

Appliance — a device to convert gas into energy or compress gas for the purpose of fuelling; the term includes any component, control, wiring, piping, or tubing required to be part of the device.

Category I appliance — an appliance that operates with a nonpositive vent static pressure and with a flue loss not less than 17%.

Note: *This category consists of draft-hood-equipped appliances, appliances labelled as Category I, and fan-assisted appliances for venting into Type B vents.*

Category II appliance — an appliance that operates with a nonpositive vent static pressure and with a flue loss less than 17%.

Category III appliance — an appliance that operates with a positive vent static pressure and with a flue loss not less than 17%.

Category IV appliance — an appliance that operates with a positive vent static pressure and with a flue loss less than 17%.

Fuelling appliances — see **Residential fuelling appliances** and **Vehicle fuelling appliances**.

Approved — acceptable to the authority having jurisdiction.

Authority having jurisdiction — the governmental body responsible for the enforcement of any part of this Code, or the official or agency designated by that body to exercise such a function.

Automatic vent damper device — a device intended for installation at the outlet or downstream of an individual appliance draft hood and designed to automatically open the venting system before or shortly after the main burner is activated and to automatically close the venting system after the main burner is deactivated.

Electrically operated vent damper device — an automatic vent damper device that is controlled by electrical energy.

Thermally actuated vent damper device — an automatic vent damper device dependent for operation exclusively upon the direct conversion of the thermal energy of the vent gases into mechanical energy.

Baffle — a stationary device used to divert the flow of fluid (air, water, or steam) and flue gases (in the case of a flue baffle) or used to shield parts of an appliance from the effects of flame (in the case of a flame baffle) or heat (in the case of a heat baffle).

Bathroom — a room used primarily for bathing and containing a bathtub and/or shower.

Bedroom — a room furnished with a bed and used primarily for sleeping.

Bed-sitting room — a one-room apartment serving as both a bedroom and a sitting room.

Bleed vent — a vent where the expiration or inspiration of air or gas occurs from or to one side of a diaphragm of any accessory, component, or equipment such as a valve, pressure regulator, or switch.

Boiler — an appliance intended to supply hot liquid or vapour for space-heating, processing, or power purposes (does not include appliances certified as water heaters).

Branch line — the part of a piping or tubing system that conveys gas from the main piping or tubing or header to an appliance(s).

Building — a structure or part thereof used or intended for supporting or sheltering persons, animals, or property and classified by its occupancy in accordance with the applicable building code of the authority having jurisdiction or, in the absence of such a code, in accordance with the *National Building Code of Canada*.

Aircraft hangar — a building or other structure in any part of which aircraft are housed or stored and in which aircraft can undergo servicing, repairs, or alterations.

Assembly building — any building or part of a building used by a gathering of persons for civic, political, travel, religious, social, educational, recreational, or like purposes, or for the consumption of food or drink.

Care or detention occupancy building — any building or part thereof used or occupied by persons who require special care or treatment because of cognitive or physical limitations, or by persons who are incapable of self-preservation and who are restrained from self-destruction by security measures not under their control.

Commercial building — any building used in connection with direct trade with, or service of, the public.

Industrial building — any building used in connection with production or process work or with storage or warehousing.

Residential building — any building or part thereof used by persons for whom sleeping accommodation is provided but who are not harboured or detained to receive medical care or treatment or are not involuntarily detained.

Building opening — a planned aperture that is intended to permit air infiltration but does not include an exhaust vent or a non-openable window.

Burner — a device, or group of devices, that forms an integral unit for the introduction of gas, with or without air or oxygen, into the combustion zone for ignition.

Fan-assisted burner — a burner in which the combustion air is supplied by a mechanical device such as a fan or blower at sufficient pressure to overcome the resistance of the burner only.

Forced-draft burner — a burner in which the combustion air is supplied by a mechanical device such as a fan or blower at sufficient pressure to overcome the resistance of the burner and the appliance.

Natural-draft burner — a burner that is not equipped with a mechanical device for supplying combustion air.

Certified (with respect to any appliance, accessory, component, equipment, or manufacturer's installation instructions) — investigated and identified by a designated testing organization as conforming to recognized standards, requirements, or accepted test reports.

Chimney — a primarily vertical shaft that encloses at least one flue for conducting flue gases outdoors.

Factory-built chimney — a chimney that consists entirely of factory-made parts, each designed to be assembled with the other without requiring fabrication on site.

Masonry or concrete chimney — a chimney of brick, stone, concrete, or approved masonry units constructed on site.

Metal chimney (smokestack) — a single-wall chimney of metal constructed on site.

Chimney draft — see **Draft**.

Combustible — material that fails to conform to CAN/ULC-S114 requirements for noncombustibility.

Combustion air — see **Air supply**.

Combustion products — constituents that result from the combustion of gas with the oxygen of the air and include inert gases but exclude excess air.

Combustion safety control (flame safeguard) — a primary safety control that senses the presence of flame and causes gas to be shut off in the event of flame or ignition failure.

Commercial- and industrial-type appliance or equipment — an appliance or equipment other than a residential or recreational type.

Commercial building — see **Building**.

Commercial cooking appliance — an appliance that complies with the applicable CSA Standard covering hotel and restaurant ranges and unit broilers, hotel and restaurant deep fat fryers, commercial baking and roasting ovens, counter appliances, kettles, steam cookers, or steam generators.

Component — an essential part of an appliance or equipment.

Concealed piping or tubing — piping or tubing that, when in place in a wall, floor, or ceiling of a finished building, is hidden from view and can only be exposed by use of a tool. The term does not apply to piping or tubing that passes directly through a wall or partition.

Concrete chimney — see **Chimney (masonry or concrete chimney)**.

Condensate (condensation) — a liquid separated from a gas (including flue gas) due to a reduction in temperature or an increase in pressure.

Container (with respect to NGV/propane storage) — either a cylinder or a tank.

Continuous pilot — see **Pilot**.

Cylinder (with respect to NGV/propane storage) — a container designed and fabricated in accordance with the specifications of Transport Canada or the US Department of Transportation for the storage and transportation of gas.

Damper — a plate or valve for regulating the flow of air or flue gas.

Barometric damper — see **Draft regulator**.

Delivery pressure — the outlet gas pressure from the service regulator for natural gas or a second stage propane regulator for propane.

Depressurization — the maximum appliance input rating of a Category I appliance equipped with a draft hood that could be attached to the vent when the appliance is located in a structure that can experience sustained depressurization of up to 0.02 in w.c. (5 Pa), e.g., a tight structure with a mechanically ventilated structure. See Annex [C](#).

Design pressure — the maximum inlet pressure a gas piping system or valve train is capable and intended to continuously sustain, contain, or control under normal conditions.

Direct-fired appliance — an appliance in which the combustion products or flue gases are intermixed with the medium being heated.

Direct-vent appliance — an appliance constructed so that all the combustion air is supplied directly from, and the products of combustion are vented directly to, the outdoors by independent enclosed passageways connected directly to the appliance.

Dirt pocket (dust pocket) — a pocket in a piping system designed for the collection of dirt and from which the dirt can be removed.

Draft — the flow of air or combustion products, or both, through an appliance and its venting system.

Chimney draft — the available natural draft of the chimney measured at or near the base of the chimney.

Mechanical draft — a draft produced by a mechanical device, such as a fan, blower, or aspirator, that can supplement natural draft.

Forced draft — a mechanical draft produced by a device upstream from the combustion zone of an appliance.

Induced draft — a mechanical draft produced by a device downstream from the combustion zone of an appliance.

Natural draft — a draft other than a mechanical draft.

Draft-control device — either a draft hood or a draft regulator.

Draft hood — a draft-control device having neither movable nor adjustable parts. A draft hood may be built into an appliance, attached to an appliance, or made part of a vent connector. It is designed to

- a) ensure the ready escape of flue gases from the combustion chamber in the event of either no draft or stoppage downstream from the draft hood;
- b) prevent a backdraft from entering the combustion chamber of the appliance; and
- c) neutralize the effect of stack action of either a chimney or a vent upon the operation of the appliance.

Draft regulator (barometric damper) — a draft-control device intended to stabilize the natural draft in an appliance by admitting room air to the venting system. A double-acting draft regulator is one whose balancing damper is free to move in either direction.

Drip pocket (drip) — a pocket in a piping system designed for the collection of condensate and from which the condensate can be removed.

Dust pocket — see **Dirt pocket**.

Dwelling unit — a housekeeping unit used or intended to be used as a domicile by one or more persons, and usually containing cooking, eating, living, sleeping, and sanitary facilities.

Emergency generators (generators) — engines that operate to provide power to critical operational support such as protection of property, firefighting activities, and building evacuation.

Enclosure — a secondary structure (room) within or attached to a structure (building) in which an appliance is installed.

Engine — a device that performs mechanical work that is used to operate other machinery and equipment.

Reciprocating engine (also known as a piston engine) — an engine that utilizes one or more pistons in order to convert pressure into a rotating motion.

Turbine engine — a rotary engine that extracts energy from a flow of combustion gas. It has an upstream compressor coupled to a downstream turbine and a combustion chamber in between. (“Gas turbine” can also refer to the turbine element.)

Equipment — a device, other than an appliance, accessory, or component, that is connected to a piping or tubing system.

Excess air — see **Air supply**.

Expanding pilot — see **Pilot**.

Factory-built chimney — see **Chimney**.

False ceiling space — ceiling space that is enclosed with tiles or panels that are removable without the use of a tool. A typical type is the T-bar-constructed suspended ceiling.

Fan-assisted burner — see **Burner**.

Fan-assisted combustion system — an appliance equipped with an integral mechanical means to either draw or force products of combustion through the combustion chamber and/or heat exchanger. See Annex [C](#).

FAN Max — the maximum appliance input rating of a Category I appliance with a fan-assisted combustion system that could be attached to the vent. See Annex [C](#).

FAN Min — the minimum appliance input rating of a Category I appliance with a fan-assisted combustion system that could be attached to the vent. See Annex [C](#).

FAN+FAN — the maximum combined input rating of two or more fan-assisted appliances attached to the common vent. See Annex [C](#).

FAN+NAT — the maximum combined input rating of one or more fan-assisted appliances and one or more draft-hood-equipped appliances attached to the common vent. See Annex [C](#).

Fast-closing valve — see **Valve**.

Fireplace — a device for burning solid fuel that has the major portion of one or more essentially vertical sides open or openable for refuelling and for the visual effects of the burning fuel.

Fitting — an item in a piping or tubing system that is used as a means of connection, such as an elbow, return bend, tee, union, bushing, coupling, or cross, but does not include such functioning items as a valve or pressure regulator.

Fixed-liquid-level gauge (when referring to the liquid level in containers) — a type of liquid-level gauge that uses a small bleed valve and is designed to indicate when the liquid level in a container being filled reaches the point at which the gauge or its connecting tube communicates with the interior of the container.

Flame baffle — see **Baffle**.

Flame safeguard — see **Combustion safety control**.

Flammable liquid — a liquid that has a flashpoint below 100 °F (38 °C) and that has a vapour pressure not exceeding 40 psia (276 kPa absolute) at 100 °F (38 °C).

Flashpoint — the minimum temperature at which a liquid within a container gives off vapour in sufficient concentration to form an ignitable mixture with air near the surface of the liquid.

Floor furnace — a furnace that is suspended from the floor of the space being heated and that supplies warm air to such space through integral floor or wall grilles without the use of ducts.

Flue — an enclosed passageway for conveying flue gases.

Flue backflow preventer — a system or device used in common venting of positive pressure appliances to prevent the exhaust flue from active appliances sharing the system from flowing back into the vent of dormant or idling appliance(s).

Flue baffle — see **Baffle**.

Flue collar — that portion of an appliance designed for the attachment of a draft hood, vent connector, or venting system. See Annex C.

Flue damper — a movable plate for regulating the flow of flue gases and intended for installation either in the flue outlet of any gas appliance or in the vent connector from an individual appliance that is not equipped with a draft-control device.

Automatic flue damper — a flue damper that is designed to fully open the venting system automatically before ignition of the main burner and is either interlocked to automatically close off the supply of gas to the appliance or provided with means to ensure that the damper will fall fully open upon loss of the driving medium.

Manually operated flue damper — a flue damper that is adjustable and manually set and locked in the desired position, and is designed or constructed to provide a fixed minimum opening.

Flue gas dilution air — see **Air supply**.

Flue gases — combustion products and excess air.

Forced draft — see **Draft (mechanical draft)**.

Forced-draft burner — see **Burner**.

Furnace — an indirect-fired, flue-connected, space-heating appliance that uses warm air as the heating medium and usually has provision for the attachment of ducts.

Gallon (except where specifically indicated otherwise) — the Canadian standard gallon as defined under the Canadian *Weights and Measures Act*.

Note: For the purpose of this Code, 1 imperial gallon is considered equal to 1 Canadian standard gallon.

Garage —

Private garage — a building, designed for the parking, storage, or repair of vehicles used to transport propane, that is approved by the authority having jurisdiction for its purpose and is located on property owned or leased by the owner of the vehicles.

Repair garage — a building or part of a building where facilities are provided for the repair or servicing of a motor vehicle.

Storage garage — a building or part of a building that is used or intended for either the storage or parking of motor vehicles and that contains no provision for either the repair or servicing of such vehicles.

Gas connector — a factory-fabricated assembly consisting of gas conduit and related fittings designed to convey gaseous fuel from a gas supply piping to the gas inlet of an appliance. A gas connector is not intended for vibration isolation, nor expansion or contraction control. For the purposes of this Code, solid gas connector includes

- a) gas connectors for gas appliances that are certified to CSA/ANSI Z21.24/CSA 6.10 and are not designed for repeated movement after being connected nor for repeated disconnecting and connecting and are not intended to be used with quick-disconnect devices or convenience outlets;
- b) gas connectors for movable gas appliances that are certified to ANSI Z21.69/CSA 6.16 and are designed for repeated movement on a regular basis and could be used with quick-disconnect devices or gas convenience outlets;
- c) gas connectors for outdoor gas appliances and manufactured homes that are certified to ANSI Z21.75/CSA 6.27 and are designed for supplying gas to an appliance for outdoor installation that is not frequently moved after installation and to the gas inlet on a mobile home;
- d) gas hose connectors for movable appliances that are certified to CSA/ANSI Z21.101/CSA 8.5 and are designed for repeated movement on a regular basis and must be used with gas convenience outlets; and
- e) gas hose connectors for outdoor portable gas-fired appliances that are certified to CSA/ANSI Z21.54/CSA 8.4 and are designed for use in unconcealed outdoor locations where the gas pressure does not exceed 1/2 psig (3.45 kPa).

Gas convenience outlet — a permanently mounted hand-operated device that is certified to CSA/ANSI Z21.90/CSA 6.24 providing a means for connecting and disconnecting an appliance to the gas supply piping by way of a gas hose or gas connector. The device includes an integral, manually operated gas valve with a non-displaceable valve member so that disconnection can be accomplished only when the manually operated gas valve is in the closed position, and is equipped with

- a) a thermal shut-off that stops the flow of gas within a specified ambient temperature range; and
- b) an optional enclosure.

Gas hose — a factory-fabricated flexible hose assembly and related fittings designed to convey gaseous fuel from a gas supply piping to the gas inlet to an appliance.

Gas piping system — all components that convey gas or liquids, such as piping, tubing, valves, hoses, and fittings, from the point of delivery to the inlet of the appliance.

Heat baffle — see **Baffle**.

Heat reclaimer — a device installed either externally or internally to a venting system to extract heat from flue gases.

Heater —

Catalytic heater — a heater that employs a porous matrix on or in which is distributed a catalytic agent such as platinum; the role of the matrix is to support the catalyst and to provide a surface on which the combustion of the hydrocarbon fuel vapour takes place.

Construction heater — a portable heater intended only for temporary use in heating buildings under construction, alteration, or repair.

Direct-fired make-up air heater (DFMAH) — a self-contained direct-fired air heater used only to heat outside air to replace inside air that is exhausted; it is not intended for building heating.

Direct gas-fired process air heater (DFPAH) — a direct gas-fired air heater that can be capable of either or both of two operating modes:

- a) the ventilation mode, suitable for use when people are present, in which all the products of combustion generated by the gas-burning device are released into the airstream being heated, and whose purpose is to offset building heat loss by heating only incoming outside air; and
- b) the process mode, for use when people are not present, in which the heater can operate as a direct gas-fired heater intended for the drying, baking, or curing of product.

Infrared heater — a heater that transfers heat from the source to the heated objects without heating the intervening air.

Non-recirculating direct gas-fired industrial air heater (DFIAH) — a heater in which all the products of combustion generated by the gas-burning device are released into the airstream being heated and whose purpose is to offset building heat loss by heating only incoming outside air.

Radiant heater — a heater that radiates heat to the surrounding air.

Ignition — the establishment of a flame.

Intermittent ignition — a source of ignition that continues to function during the entire period that the flame is present.

Interrupted ignition — a source of ignition that ceases to function after the trial-for-ignition period.

Indirect-fired appliance — an appliance in which the combustion products or flue gases are not mixed within the appliance with the medium that is being heated.

Induced draft — see **Draft (Mechanical draft)**.

Industrial appliance — see **Commercial- and industrial-type appliance or equipment**.

Industrial building — see **Building**.

Installer — any individual, firm, corporation, or company that either directly or through a representative is engaged in the installation, replacement, repair, or servicing of gas piping systems, venting systems, appliances, components, accessories, or equipment, and whose representative is either experienced or trained, or both, in such work and has complied with the requirements of the authority having jurisdiction.

Insulating millboard — a factory-fabricated board formed with noncombustible materials, normally fibres, and having a thermal conductivity not exceeding

$$1 \frac{\text{Btu} \cdot \text{in}}{\text{h} \cdot \text{ft}^2 \cdot \text{F}} = (0.144 \frac{\text{W}}{\text{m} \cdot \text{K}})$$

Lock-up (positive shut-off) — a feature of a pressure regulator that is capable of maintaining a reduced outlet pressure when the fuel flow condition is static.

Lubricated-plug-type valve — see **Valve**.

Manual shut-off valve — see **Valve**.

Masonry chimney — see **Chimney (masonry or concrete chimney)**.

Maximum operating pressure — the maximum pressure to which any component or portion of the fuel system can be subjected.

Mechanical air intake — a means to mechanically provide ventilation and/or combustion and flue gas dilution air requirements to a building.

Mechanical draft — see **Draft**.

Mobile housing — a term that includes all of the following structures:

Mobile home — a dwelling that consists of a vehicular portable structure built on a chassis and designed to be used with or without a permanent foundation and to be connected to indicated utilities.

Mobile industrial or commercial structure — a structure that is not intended as a dwelling unit, is towable on its own chassis, and is designed for use without a permanent foundation. Such a structure is built specifically for commercial or industrial use, such as a construction office, bunkhouse, wash house, kitchen and dining unit, library, television unit, industrial display unit, laboratory unit, or medical clinic.

Multiple-section mobile home — a single structure composed of separate mobile units, each towable on its own chassis; when the units are towed to the site, they are coupled together mechanically and electrically to form the single structure. These structures are sometimes referred to as “double-wide mobile homes” when only two units are joined together.

Swing-out and expandable room-section mobile home — a mobile home that can be telescoped when towed and expanded later for additional capacity.

Mobile outdoor food service unit — a unit used outdoors for preparation and dispensing of food or beverages and that contains appliances or equipment operated by propane in the vapour state. The unit can be equipped with wheels and its own motive power.

N/A — not applicable due to physical or geometric constraints. See Annex [C](#).

NAT Max — the maximum appliance input rating of a Category I appliance equipped with a draft hood that could be attached to the vent. There are no minimum appliance input ratings for draft-hood-equipped appliances. See Annex [C](#).

NAT+NAT — the maximum combined input rating of two or more draft-hood-equipped appliances attached to the common vent. See Annex [C](#).

Natural draft — see **Draft**.

Natural-draft burner — see **Burner**.

NGV — natural gas stored in a gaseous state to be used as engine fuel for a highway vehicle.

Noncombustible — material that conforms to CAN/ULC-S114 requirements for noncombustibility.

NR — not recommended due to potential for condensate formation and/or pressurization of the venting system. See Annex [C](#).

Operating control — a control used to regulate or control the normal operation of an appliance.

Overpressure protection device — a device that under abnormal conditions will act to reduce, restrict, or shut off the supply of gas flowing into a system to prevent gas pressure in that system from exceeding the rated pressure of the system components.

Monitoring regulator — an overpressure protection device that functions as a second gas pressure regulator in series with the primary gas pressure regulator.

Overpressure relief device — an overpressure protection device that functions by discharging gas from the downstream system.

Overpressure shut-off device — an overpressure protection device that functions by completely shutting off the flow of gas into the downstream system.

Pilot — a flame that is used to ignite a gas/air or propane/air mixture at the main burner(s).

Continuous pilot — a pilot that burns without turndown throughout the entire time the burner is in service, whether the main burner is firing or not.

Expanding pilot — a pilot that burns at low turndown throughout the entire time the burner is in service, whether or not the main burner is firing, except that upon a call for heat, the fuel flow to the pilot is automatically increased to produce a flame that will reliably ignite the main burner fuel.

Intermittent pilot — a pilot that is automatically lighted each time there is a call for heat and that burns during the entire period that the main burner is firing.

Interrupted pilot — a pilot that is automatically lighted each time there is a call for heat and that is cut off automatically at the end of the trial-for-ignition period of the main burner.

Proved pilot — a pilot flame supervised by a primary safety control that senses the presence of the pilot prior to gas being admitted to the main burner.

Pipe wrap tape — adhesive tape made of PVC or polyethylene material with minimum thickness of 10 mil (0.3 mm), with an adhesive resistant to water.

Piping — components of a gas piping system that include fittings, rigid pipe, and flanges.

Point of transfer — the dispensing hose inlet connection.

Power venter — a field-installed device that provides mechanical draft and, when installed, is located between the appliance flue outlet or draft-control device and the vent or chimney termination, or at the vent or chimney termination.

Pressure regulator — a device, either adjustable or nonadjustable, for controlling, maintaining within acceptable limits, a uniform outlet pressure.

Appliance regulator — a pressure regulator located in the valve train of an appliance.

Industrial regulator — a pressure regulator installed downstream of the service regulator and upstream of utilization equipment or another pressure regulator and that is either not certified, or certified but used in an application different than the scope of its certification.

Note: *An example of an industrial pressure regulator is one that is not certified to UL 144, CSA 6.3, or CSA 6.22. Another example is a pressure regulator that is certified to UL 144, CSA 6.3, or CSA 6.22, but it is installed in an application that is outside of the scope of its certification.*

Line pressure regulator — a pressure regulator installed downstream of the service regulator or second stage propane service regulator and upstream of gas utilization equipment, and certified to CSA/ANSI Z21.80/CSA 6.22 or UL 144.

Lock-up (positive shut-off) regulator — a regulator that is capable of maintaining a reduced outlet pressure when the fuel flow condition is static.

Service regulator — a pressure regulator installed on a service line to control the pressure of the gas delivered to the customer.

Second stage propane regulator — in a propane application, a pressure regulator at the service entrance intended to reduce first-stage pressure to control the gas pressure inside a building. This regulator is also sometimes known as a “2 psi liquefied propane regulator”.

Products of combustion — see **Combustion products**.

Purge — to replace the existing fluid (gaseous or liquid) in piping, tubing, equipment, a container, or an appliance with a desired fluid.

Purge burner — a burner equipped with a constant ignition source and a flame arrestor, intended to burn the escaping (discharged) gas during purging operations.

Quick-disconnect device — a hand-operated device that is used for connecting and disconnecting an appliance or a hose or certain types of gas connectors to a gas supply, that is equipped with an automatic means to shut off the gas supply when the device is disconnected, and that is certified to CSA/ANSI Z21.41/CSA 6.9.

Rated pressure — the maximum pressure that the materials, gas piping, devices, valve train components, operating controls, or safety controls are designed to contain or control.

Readily accessible — capable of being reached quickly for operation, renewal, servicing, or inspection, without requiring climbing over, or the removal of, an obstacle or the use of a portable ladder.

Recreational vehicle — a portable structure intended as temporary accommodation for travel, vacation, or recreational use. Such a structure includes a chassis-mounted camper, motorized home, slide-in camper, tent trailer, or travel trailer.

Chassis-mounted camper — an accommodation body that attaches onto a truck chassis and is not intended for removal.

Motorized home — a vehicular portable structure of self-propelled design.

Slide-in camper — an accommodation body that fits into a standard vehicle and is designed to be easily removable.

Tent trailer — a vehicular portable structure built on its own chassis and having a rigid or canvas top and side walls that can be folded or otherwise condensed for transit.

Travel trailer — a vehicular portable structure that is intended to be towed by a motor vehicle and that does not fold up or reduce in size for transit.

Relief device — a device designed to open to prevent a rise of gas pressure in excess of a specified value due to an emergency or abnormal conditions.

Residential appliance — an appliance commonly used in, but not restricted to use in, a dwelling unit.

Residential building — see **Building**.

Residential fuelling appliance (RFA) — an appliance that dispenses natural gas for vehicles directly into the vehicle natural gas fuel storage system.

Safe location (for venting of gas) — a location that allows for the destruction or dispersal of vented gas so that it can reasonably be expected that the following will be prevented:

- a) blocking of the vent termination by snow, ice, water, or any other object or thing;
- b) gas accumulating in or under a building or enclosure;
- c) gas accumulating near a source of ignition, hot surface, electrical equipment, or operating control;
- d) gas accumulating in an area where a person would likely have difficulty in quickly leaving the area;
or
- e) in an industrial application, gas discharging directly toward a person, walkway, staircase, or ladder.

Note: *In industrial applications, due consideration should be given to the pressure and constituents of the gas in order to ensure the vent location does not create a hazard.*

Safety limit control — a safety control intended to prevent an unsafe condition of temperature, pressure, or liquid level.

Safety shut-off valve — see **Valve**.

Spray area — any fully enclosed, partly enclosed, or unenclosed area within a building in which spray processes are performed that could result in dangerous quantities of flammable or combustible vapours, mists, residues, dust, or deposits. Spray areas can include any area in the direct path of a spray operation process, a spray booth, a spray room, or a spray workstation.

Structure — the entire building in which an appliance is installed.

Supply pressure — the gas pressure at the manual shut-off valve of an appliance or equipment.

Tank (with respect to NGV/propane storage) — the class of container for the storage and transportation of gas, designed and fabricated in accordance with CSA B51.

Tubing — components of a gas piping system that include fittings and flexible pipe, tube, corrugated stainless steel tubing (CSST), and feedback lines.

Two-stage regulation — a propane gas vapour delivery system that utilizes a first-stage regulator and a second-stage regulator(s) or utilizes an integral two-stage regulator or automatic changeover regulator.

Valve — a device by which the flow of a fluid can be started, stopped, or regulated by a movable part that opens or obstructs passage.

Back check valve — a valve that is normally closed and allows flow in only one direction.

Cylinder valve — a valve fitted to a cylinder.

Emergency shut-off valve — a valve that is part of a system that is designed to limit and shut down the flow of propane in the event that a vehicle moves away from a transfer point with the transfer hose or swivel-type piping connected to it.

Excess-flow valve — a valve designed to close when the liquid or vapour passing through it exceeds a prescribed flow rate as determined by a pressure drop across the valve.

Fast-closing valve — an automatic valve that has a closing time of less than 5 s upon being de-energized.

Hydrostatic relief valve — a pressure relief valve installed in a liquid propane line.

Internal excess-flow valve — an excess-flow valve that remains functional within the tank when any portion of the valve external to the tank's perimeter is sheared off or otherwise damaged. An internal excess-flow valve can be integral to another valve.

Note: *The perimeter of the tank is taken to mean not only the surface of the tank's shell and ends but also the outline of any boss, spigot, or nozzle welded to the tank so as to project outwards from it. The outline of welded brackets, mountings, guards, subcompartments, and the like are not considered part of the perimeter for the purpose of this definition.*

Internal relief valve — an overpressure relief device that is built into the body of the diaphragm assembly of a pressure regulator.

Internal valve — a valve designed and installed so that its seat is within a tank and the arrangement of the parts of the valve are such that damage to the parts outside the tank will not prevent effective seating of the valve.

Line relief valve — an overpressure relief device installed in the piping or tubing system downstream of a pressure regulator.

Lubricated-plug-type valve — a manually operated valve of the plug and barrel type that is

- a) provided with means for maintaining a lubricant between its bearing surfaces;
- b) so designed that the lapped bearing surfaces can be lubricated and the lubricant level maintained without removing the valve from service;
- c) so constructed that the lubricant can be stored in a reservoir that enables the lubricant to be distributed evenly over the entire lapped bearing surfaces of the valve when the plug is rotated; and
- d) equipped with built-in stops to limit the rotation of the plug to one quarter turn when fully opening or fully closing the valve.

Manual shut-off valve — a manually operated valve in a gas piping system or a valve train for shutting off the fuel for maintenance, testing, and safety purposes.

Safety shut-off valve — a valve that automatically shuts off the supply of gas when de-energized by a combustion safety control, safety limit control, or loss of actuating medium.

Stop-fill valve — a device in a container that is intended to automatically shut off the flow of liquid into the container when a predetermined fixed level is achieved.

Valve train — all gas-confining valves, controls, piping, fittings, hoses, and tubing of an appliance downstream of the manual shut-off valve to the inlet of the burner.

Vapourizer — an appliance for converting liquid propane to vapour by means other than atmospheric heat transfer through the surface of the container.

Direct-fired vapourizer — a vapourizer in which heat furnished by a flame is directly applied to a heat-exchange surface in contact with the liquid propane to be vaporized.

Indirect vapourizer — a vapourizer in which heat furnished by steam, hot water, or another heating medium is applied to a vaporizing chamber's tubing, pipe coils, or other heat-exchange surface

containing the liquid propane to be vapourized. The heating of the medium being used occurs at a point remote from the vapourizer.

Vehicle fuelling appliance (VFA) — an appliance that compresses natural gas for vehicles and dispenses directly into onboard vehicle storage or delivers to external natural gas storage systems.

Vent — that portion of a venting system designed to convey flue gases directly to the outdoors from either a vent connector or an appliance when a vent connector is not used.

Type B vent — a vent complying with CAN/ULC-S605 consisting entirely of factory-made parts, each designed to be assembled with the others without requiring field fabrication, and intended for venting gas appliances.

Type BH vent — a vent complying with ULC S636 consisting entirely of factory-made parts, each designed to be assembled with the others without requiring field fabrication, and intended for venting gas appliances.

Type BW vent — a vent complying with CAN/ULC-S605 consisting entirely of factory-made parts, each designed to be assembled with the others without requiring field fabrication, and intended for venting only wall furnaces for use with this type of vent.

Type L vent — a vent complying with CAN/ULC-S609 consisting of factory-made parts, each designed to be assembled with the others without requiring field fabrication.

Vent connector — that part of a venting system that conducts the flue gases from the flue collar of an appliance to a chimney or vent, and that may include a draft-control device.

Vent limiter — a means that limits the flow of gas from a pressure regulator's atmospheric diaphragm chamber to the atmosphere in the event of a diaphragm rupture. This may be either a limiting orifice or a limiting device.

Limiting orifice type — a vent limiter where the flow through the limiter is the same in both directions

Ventilated space — a space where there is an air change by means of natural ventilation or mechanical means, or where the space communicates with the rest of the structure by means of permanent openings.

Ventilation (with respect to the space in which an appliance is installed) — the removal of inside air, leaked or spilled products of combustion, or flue gases from the space in which an appliance is installed to outside the space, and the replacement of same by air from outside the space.

Ventilation air — see **Air supply**.

Venting system — a system for the removal of flue gases to the outdoors by means of a chimney, vent connector, vent, or a natural or mechanical exhaust system.

Special venting system — a venting system certified with the appliance and either supplied or specified by the appliance manufacturer.

Wash-mobile — a mobile outdoor unit that uses propane-heated water, or a solution, for the purpose of cleaning.

Zero governor — a regulating device that is adjusted to deliver gas at atmospheric pressure within its flow rating.

4 General

4.1 Application

4.1.1 Requirements

For the purpose of this Code, the requirements contained in CAN/CGSB-3.14 for propane or CAN/CGSB-3.13 for butane shall apply.

4.1.2 Code conflicts

When a specification or document referenced in Clause [2](#) contains a requirement that conflicts with a requirement in this Code, the requirement in this Code shall govern.

4.1.3 Installation instructions

Except as required by Clause [4.1.4](#), an appliance, accessory, component, equipment, or any other item shall be installed in accordance with the manufacturer's certified instructions and this Code.

4.1.4 Field-approved appliance installation

A field-approved appliance or equipment shall be installed in accordance with the Clause 10.1 of CSA B149.3 and this Code.

4.1.5 Conflicts

Where a conflict exists between the manufacturer's instructions and this Code, the requirements of this Code shall prevail unless otherwise approved.

4.2 Approval of appliances, accessories, components, equipment, and material

4.2.1 Installation

An appliance, accessory, component, equipment, or material used in an installation shall be of a type and rating approved for the specific purpose for which it is employed.

4.2.2 Deviation

When deviation from or postponement of these requirements is necessary, permission in writing shall be obtained from the authority having jurisdiction before the work proceeds, and this permission shall apply only to the particular installation for which it is given.

4.2.3 Assembly or construction approval

The approval of the assembly or construction of an appliance is subject to the authority having jurisdiction. (CSA B149.3 contains provisions for the assembly and construction of appliances.)

4.2.4 Biogas

When using biogas, this Code shall be used in conjunction with CSA/ANSI B149.6.

4.3 Responsibilities of the installer

4.3.1 Initial installation

Before leaving installations, installers shall ensure that the appliance, accessory, component, equipment, or piping and tubing they installed complies with the Code requirements, and the person initially activating the appliance shall ensure that the appliance is in safe working order.

4.3.2 User instructions

Installers shall instruct the user in the safe and correct operation of all appliances or equipment that they install.

4.3.3 Manufacturer instructions

The installer shall ensure that the manufacturer's instructions supplied with the appliance are left with the user.

4.3.4 Replacement parts

Before installing any replacement part of an appliance, the installer shall ensure that the replacement part provides operational characteristics at least equivalent to those of the original part.

4.3.5 Appliance conversion

When the installation or conversion of an appliance constitutes a conversion from another form of energy, the installer shall advise the user of the appliance, at the time of installation or conversion, to have the former form of energy either removed or left safe and secure from accidental activation. For example, the user shall be advised

- a) in the case of a fuel oil supply tank
 - i) to remove the fill pipe, and cap or plug the exposed fill pipe opening to an inside tank;
 - ii) to shut off the tank outlet valve, remove the filter, and plug or cap the valve outlet; and
 - iii) where the tank is located outdoors, to disconnect all exposed piping or tubing, and cap or plug the piping or tubing as close as practicable to the tank;
- b) in the case of a fuel oil central distributing system
 - i) to shut off the fuel oil supply line valve located within the building; and
 - ii) to disconnect the fuel oil supply line immediately downstream of the meter, and cap or plug the outlet of the meter;
- c) in the case of a propane system
 - i) to shut off the cylinder or tank valve; and
 - ii) to disconnect and cap or plug the propane gas piping system outdoors; and
- d) in the case of an electrical appliance
 - i) to shut off the power supply to the electrical appliance at the switch; and
 - ii) to ensure that the overcurrent protection, fuse, or circuit breaker has been removed or put in the off position.

4.3.6 User procedures

The installer installing the installation or conversion, as specified in Clause [4.3.5](#), shall advise the user of the appliance in writing of the procedures to be followed in discontinuing the supply of the former form of energy.

4.3.7 Piping pressure test

It shall be the responsibility of the installer of a gas piping system to perform pressure tests in accordance with Clause [6.22.2](#) and to ensure that the piping or tubing system is gas-tight at the completion of the tests.

4.3.8 Gas tight test

It shall be the responsibility of the installer of an appliance to perform tests in accordance with Clause [6.22.3](#) and to ensure that the system is gas-tight at the completion of the tests.

4.4 Training and quality of labour

4.4.1 Work quality

All work shall be done in a skillful, thorough manner. Careful attention shall be paid not only to the mechanical execution of the work but also to the arrangement of the installation.

4.4.2 Training

Personnel performing installation, operation, and maintenance work shall be properly trained in such functions.

4.5 Suitability of use

4.5.1 Type of gas and pressure

An appliance shall not be installed unless it is designed for use with the type of gas to which it is to be connected and is suitable for the pressure supplied.

Note: When considering fuels that contain hydrogen-natural gas blends, caution should be taken to ensure that all appliances, accessories, components, equipment, and material are suitable for the fuel application and that the impacts from the introduction of hydrogen into the fuel, such as hydrogen embrittlement, are taken into consideration.

4.5.2 Hazards

The use of an appliance, accessory, component, equipment, or material shall be prohibited where a hazard is created.

4.5.3 Appliance conversion

When an appliance is converted from the gas or fuel specified on the rating plate, the conversion shall be in accordance with the manufacturer's certified instructions. If there are no manufacturer's instructions for conversion of the appliance, the converted appliance shall be approved.

4.5.4 Rating plate conversion

If an appliance is converted from one gas to another, the gas to which it is converted shall be marked on the appliance rating plate by the fitter making the conversion. For hydrogen-natural gas blends, the maximum percentage of hydrogen shall be marked.

4.5.5 Damaged appliances and piping

Appliances, accessories, components, equipment, and gas piping systems that have been exposed to fire, explosion, flood, or other damage shall not be offered for sale, installed, reactivated, or reconnected to the supply until the appliance, accessory, component, equipment, gas piping system has

been inspected by a person acceptable to the authority having jurisdiction. The inspection of a gas piping system exposed to flood shall include a check for water in the gas piping system.

4.5.6 Used appliances

A used appliance shall be inspected and determined to be safe for continued use by the installer before reconnection to the gas piping system.

4.6 Meter and service regulator installations

4.6.1 Installations

Meter and service regulator installations shall be in accordance with CSA Z662 (see Annex [D](#)).

4.6.2 Supplier distributor systems

No person other than an employee or person authorized by the supplier or distributor shall perform any alterations, repairs, tests, services, removals, changes, installations, connections, or any other type of work on the supplier's or distributor's system.

4.7 Electrical connections and components

4.7.1 Code requirements

Electrical connections between an appliance and building wiring shall comply with the local electrical code or, in the absence of such, with the *Canadian Electrical Code, Part I*.

4.7.2 Electrical circuit

An electrical circuit employed for operating an automatic main control valve, automatic pilot, room-temperature thermostat, safety limit control, or another electrical device used with an appliance shall be in accordance with the appliance wiring diagram.

4.7.3 Equipotential bonding

All building metal gas piping systems connected to one or more gas-fired appliances, shall be made electrically continuous and shall be equipotentially bonded as follows:

- a) using a conductor not smaller than a No. 6 AWG copper or No. 4 AWG aluminum, with a connection that is accessible after installation;
- b) meeting the requirements of the local electrical code or, in the absence of such, the *Canadian Electrical Code, Part I*; or
- c) as permitted in Clause [4.7.4](#).

4.7.4 Bonding through appliance

Gas piping systems shall be considered to be bonded to the electrical system when the metal gas piping system or CSST certified with an arc-resistant jacket or coating system in compliance with CSA/ANSI LC 1/CSA 6.26 used in a gas piping system is connected to one or more appliances permanently connected to the bonding conductor of the circuit supplying the appliance.

4.7.5 Bonding non arc-resistance CSST

In addition to the requirements of Clause [4.7.3](#), CSST that does not have an arc-resistant jacket or coating system in compliance with CSA/ANSI LC 1/CSA 6.26 used in a gas piping system shall be bonded according to the CSST manufacturer's installation instructions.

4.8 Mobile homes and recreational vehicles

4.8.1 Appliance installation

The installation of gas-burning appliances and the gas piping system in mobile homes shall be in accordance with CSA Z240.4.1. In a recreational vehicle, it shall comply with the requirements of CSA Z240.4.2.

4.8.2 Placement

When a vehicle ceases to be used as a mobile home or recreational vehicle and is placed at a location in a permanent fixed manner, the system shall comply with all applicable requirements of this Code.

4.8.3 Appliance application

An appliance in the application described in Clause [4.8.2](#) shall not be required to be certified specifically for use within a mobile home.

4.9 Hazardous and corrosive locations

4.9.1 Installation

An appliance shall not be installed in a location that has an environment corrosive to an appliance or venting system.

4.9.2 Approved installation

An appliance, unless certified or approved for installation in a hazardous location, shall not be installed in any location where a flammable vapour, combustible dust or fibres, or an explosive mixture is present.

4.10 Smoking

Smoking or providing any other source of ignition shall not be permitted in the area where work is being done on piping, tubing, or equipment that either contains or has contained gas, unless the gas piping system or equipment has been purged of all gas as outlined in Clause [6.23](#).

4.11 Isolation of safety devices

Isolating or rendering inoperative a safety shut-off valve, safety limit control, or relief valve shall be prohibited.

4.12 Leak detection

4.12.1 Sources of ignition

A match, candle, flame, or other source of ignition shall not be used to check for a gas leak.

4.12.2 Light and/or flashlight use

A light, including a flashlight, used in connection with a search for gas leakage shall be of the Class I, Group IIA type.

4.12.3 Electric switches

An electric switch either in or adjacent to an area of gas leakage shall not be operated unless it is a Class I, Group IIA type.

4.13 Appliance clearances to combustible material

4.13.1 Clearances

The clearances required in Clause [7](#) between an appliance and combustible material shall be considered the minimum without protection and shall be measured from the appliance, disregarding either the burner or any other projecting component.

4.13.2 Clearance reduction

The clearances to combustible material specified in Clause [7](#) shall not be reduced unless

- a) such reduced clearance is certified as safe by a nationally recognized certification organization acceptable to the authority having jurisdiction and so marked on the appliance nameplate; or
- b) protection is provided for the combustible material, and such protection and such reduced clearance are in accordance with Table [4.1](#).

4.13.3 Combustible floor material

An appliance with an input up to and including 400 000 Btu/h (120 kW) and certified for installation on noncombustible flooring may be installed on a floor constructed of combustible material, provided that

- a) the floor is protected with at least two continuous courses of 4 in (100 mm) thick hollow masonry units covered with sheet metal at least 0.0195 in (0.56 mm) thick;
- b) the masonry units are arranged so that the hollow cores will permit air circulation through them; and
- c) the base as specified in Items a) and b) extends not less than 6 in (150 mm) beyond the sides of the appliance.

4.13.4 Aircraft storage installations

A heater located in an aircraft storage or servicing area shall be installed so that no portion of an aircraft that can occupy the area is within the clearance to combustible material, as marked on the appliance rating plate. Clearances specified in Table [4.1](#) shall not apply.

Table 4.1
Appliance clearance reductions with specified forms of protection
 (See Clauses [4.13.2](#), [4.13.4](#), [7.4.7](#), and [7.27.3](#).)

a) Imperial

Type of protection (applied to the combustible material unless otherwise specified, and covering all surfaces within the distance specified as the required clearance with no protection)	Required clearance (without protection)											
	36 in and greater		18 to 35 in		12 to 17 in		9 to 11 in		6 to 8 in		3 to 5 in	
	Clear- ance may be reduced by											
Minimum thicknesses	Above	Sides and rear	Above	Sides and rear	Above	Sides and rear	Above	Sides and re- ar	Above	Sides and re- ar	Above	Sides and rear
1/4 in insulating millboard spaced out 1 in	6	18	3	9	3	6	3	5	3	4	2	2
28 gauge sheet metal on 1/4 in insulating millboard	12	18	6	9	3	6	3	5	3	4	2	2
28 gauge sheet metal spaced out 1 in	18	24	9	12	6	8	5	6	4	4	2	2
28 gauge sheet metal on 1/8 in insulating millboard spaced out 1 in	18	24	9	12	6	8	5	6	4	4	2	2
1/4 in insulating millboard on 1 in mineral wool batts reinforced with wire mesh or equivalent	18	24	12	12	8	8	6	6	4	4	2	2
22 gauge sheet metal on 1 in mineral wool batts reinforced with wire or equivalent	18	24	12	15	10	10	7	7	4	4	2	2

(Continued)

Table 4.1 (Continued)**a) Imperial**

Type of protection (applied to the combustible material unless otherwise specified, and covering all surfaces within the distance specified as the required clearance with no protection)	Required clearance (without protection)												
	36 in and greater		18 to 35 in		12 to 17 in		9 to 11 in		6 to 8 in		3 to 5 in		
	Clear- ance may be reduced by												
	Minimum thicknesses		Above	Sides and rear	Above	Sides and rear	Above	Sides and rear	Above	Sides and re- ar	Above	Sides and re- ar	Above
1/4 in insulating millboard	0	0	0	0	0	0	0	0	0	2	2	1	1

Note: All clearances shall be measured from the outer surface of the equipment to the combustible material, disregarding any intervening protection applied to the combustible material.

(Continued)

Table 4.1 (Continued)

b) Metric												
Type of protection (applied to the combustible material unless otherwise specified, and covering all surfaces within the distance specified as the required clearance with no protection)	Required clearance (without protection)											
	900 mm and greater		450 to 900 mm		300 to 430 mm		225 to 327 mm		150 to 200 mm		75 to 125 mm	
	Clearance may be reduced by											
Minimum thicknesses	Above	Sides and rear	Above	Sides and rear	Above	Sides and rear	Above	Sides and rear	Above	Sid- es and re- ar	Above	Sides and rear
6 mm insulating millboard spaced out 25 mm	150	450	75	225	75	150	75	125	75	100	50	50
0.3 mm sheet metal on 6 mm insulating millboard	300	450	150	225	75	150	75	125	75	100	50	50
0.3 mm sheet metal spaced out 25 mm	450	600	225	300	150	200	125	150	100	100	50	50
0.3 mm sheet metal on 3 mm insulating millboard spaced out 25 mm	450	600	225	300	150	200	125	150	100	100	50	50
6 mm insulating millboard on 25 mm mineral wool batts reinforced with wire mesh or equivalent	450	600	300	300	200	200	150	150	100	100	50	50
0.68 mm sheet metal on 25 mm mineral wool batts reinforced with wire or equivalent	450	600	300	375	250	250	175	175	100	100	50	50
6 mm insulating millboard	0	0	0	0	0	0	0	0	50	50	25	25

Note: All clearances shall be measured from the outer surface of the equipment to the combustible material, disregarding any intervening protection applied to the combustible material.

4.14 Accessibility

4.14.1 Installation

An appliance shall be installed so that it is accessible for servicing.

4.14.2 Service clearances

An appliance shall be installed with a minimum service clearance of 24 in (610 mm) to any side, top, or bottom where service could be necessary, except where

- a) a greater distance is indicated on the appliance rating plate; or
- b) the distance is not sufficient for the removal, replacement, or repair of a component, an accessory, or any equipment either forming an integral part of the appliance or connected to the appliance, in which case minimum service clearance that is adequate to effect such removal, replacement, or repair shall be provided.

All measurements shall be taken between the outermost projection of the appliance and surrounding structure or property line after the appliance has been installed.

4.14.3 Access dimensions

An access opening with minimum dimensions of 24 × 30 in (610 × 760 mm) shall be provided to the space in which an appliance is located.

4.14.4 Appliance passageways

A clear and unobstructed passageway at least 36 in (900 mm) high and 36 in (900 mm) wide shall be provided to each appliance.

4.14.5 Roof access

An appliance shall not be installed on a roof

- a) exceeding 13 ft (4 m) in height from grade to roof elevation unless fixed access to the roof is provided; and
- b) exceeding 26 ft (8 m) in height from grade to roof elevation unless permanent fixed access to the roof by means of either a stairway or a stairway leading to a ladder not exceeding 13 ft (4 m) in height is provided.

4.14.6 Roof installation clearances

When an appliance is installed on a roof

- a) the appliance shall be installed on a well-drained surface. When water stands on the roof, either at the appliance or in the passageways to the appliance, or when the roof is sloped or has a water seal, a suitable anti-skid walkway shall be provided. Such a walkway shall be located adjacent to the appliance and control panels, and when the appliance is located on a sloped roof, the walkway shall extend from the appliance to the point of access and be equipped with guardrails so that the appliance can be safely accessed and serviced;
- b) the clearance between the appliance and the edge of the roof or other hazard shall be at least 6 ft (2 m); and
- c) if the appliance is enclosed, such an enclosure shall permit easy entry and movement, be of reasonable height, and have at least 2 ft (600 mm) clearance on either side of the service access panel of each appliance in the enclosure.

4.14.7 Means of service

An appliance installed at a distance of 10 ft (3 m) or more from either the floor or finished grade level, as measured from the lowest point of the appliance, shall be provided with either

- a) a permanent accessible service platform that permits access to all parts of the appliance requiring service; or
- b) other approved means of service access.

4.15 Outdoor installations

4.15.1 Appliance approval

An appliance installed outdoors shall be approved for outdoor use.

4.15.2 Flue gases

An appliance installed outdoors shall be located to prevent circulation of flue gases into the combustion air inlet or circulating airstream of an adjacent appliance.

4.15.3 Space heater or pool heater installation

A space-heating or pool-heating appliance installed outdoors at grade level shall be placed on a base consisting of poured-in-place concrete or a reinforced concrete slab of the preformed type, extending at least

- a) 6 in (150 mm) beyond all sides of the appliance; and
- b) 2 in (50 mm) above grade level.

The ground shall first be prepared and provided with gravel for drainage.

4.16 Appliances in garages

4.16.1 Protection

An appliance in a garage shall be protected against damage.

4.16.2 Flammable vapours

In a storage garage, except for an appliance certified as flammable vapours ignition resistant (FVIR), an appliance shall be installed so that a component capable of igniting a flammable vapour is located not less than 18 in (450 mm) above the floor.

4.16.3 Flammable vapour clearance

In a repair garage, an appliance shall be installed so that a component capable of igniting a flammable vapour is located not less than 4.5 ft (1400 mm) above the floor.

4.16.4 Building Code requirements

The installation of a forced-air appliance in a garage shall be in accordance with local building codes or, in the absence of local codes, the *National Building Code of Canada*.

4.17 Appliance ductwork connections

4.17.1 Approved use

Ductwork shall not be connected to an appliance unless the appliance is approved for use with ductwork.

4.17.2 Return air ducts

Return air ducts installed in an enclosure shall be in accordance with the provisions set out for duct systems in the *National Building Code of Canada*.

4.18 Combined heating systems

4.18.1 Solid fuel heated air

Air heated by a solid-fuel-fired appliance or fireplace shall not be introduced into any part of the ductwork system of a gas-fired appliance, except where either

- a) a certified combination gas and solid-fuel-fired appliance is installed; or
- b) a certified solid-fuel-fired furnace downstream series add-on to a gas-fired furnace is installed.

Note: A solid-fuel-fired furnace downstream series add-on is an appliance that shares the same heated air duct system and is installed downstream of a gas-fired furnace as per CSA B366.1.

4.18.2 Building code requirements

The installation of a solid-fuel-fired portion of the appliance referred to in Clause [4.18.1 a\)](#) and the installation of a solid-fuel-fired downstream series add-on furnace referred to in Clause [4.18.1 b\)](#) shall conform to the applicable provincial or territorial building code or, in its absence, to CAN/CSA-B365.

4.19 Appliances protected by automatic fire-extinguishing systems

4.19.1 Interlock system

When an exhaust system protected by an automatic fire-extinguishing system is installed over an appliance not provided with a flame safeguard, the operation of the fire-extinguishing system shall be interlocked with the gas supply to the appliance so as to automatically shut off the gas, including the pilot, to the appliance to be protected by the system and also to any other appliance that can be affected by the extinguishing system.

4.19.2 Interlock valve

The valve used to shut off the gas supply referred to in Clause [4.19.1](#) shall be located outside the protected area, be identified as to its function, and have permanent legible relighting instructions posted adjacent to it. The valve shall be either

- a) an approved mechanical non-electric fast-closing valve of the manual-reset type; or
- b) an automatic electrically operated fast-closing valve
 - i) of the manual-reset type; or
 - ii) provided with a remote manual-reset function to open.

4.19.3 Manual shutoff valve

A manual shut-off valve shall be installed immediately upstream of the valve referred to in Clause [4.19.2](#).

4.20 Control of appliances with self-energizing pilots

When two or more appliances with self-energizing pilots are installed, each shall be independently controlled by a separate actuating device such as a thermostat.

4.21 Defective heat exchangers

4.21.1 Replacement

Where the heat exchanger of a furnace installed in a dwelling unit is found to be defective, it shall be replaced.

4.21.2 Temporary repair

The heat exchanger referred to in Clause [4.21.1](#) may be temporarily repaired if necessary, and the repair shall be in accordance with procedures acceptable to the authority having jurisdiction.

4.21.3 Commercial or industrial repair

Where the heat exchanger of a commercial or industrial appliance is found to be defective, it may be repaired, and the repair shall be in accordance with procedures acceptable to the authority having jurisdiction.

4.22 High-altitude installations

4.22.1 Certification and adjustment

For high-altitude installations, appliances shall be certified in compliance with CSA 2.17 and shall be adjusted to the high-altitude rating shown on the nameplate following the manufacturer's instructions when installed at elevations between 2000 and 4500 ft (600 and 1350 m) above sea level.

4.22.2 Elevation input rating

When an appliance is installed at elevations above the maximum elevation option provided by the manufacturer, the input rate shall be reduced following the manufacturer's certified instructions. If no instructions are given for configuring the appliance for higher altitudes, the input rate shall be reduced at the rate of 4% for each additional 1000 ft (300 m). The input rate shall be reduced from the rate indicated for the highest altitude option provided.

4.23 Protection of appliances from physical damage

Where an appliance is installed in an area where physical damage can be incurred, the appliance shall be protected from such damage.

4.24 Odourization

4.24.1 Natural gas odourization

Natural gas or hydrogen-natural gas blends used for fuel purposes supplying an occupied building shall be odourized in accordance with CSA Z662 or be otherwise readily detectable, or the building shall be equipped with an approved means of gas detection.

4.24.2 Propane odourization

Propane distributed for fuel purposes shall be odourized in accordance with CAN/CGSB-3.14.

4.24.3 Propane odourization responsibility

Odourization of the propane shall be the responsibility of the producer or processor, who shall indicate on the shipping document its compliance with CAN/CGSB-3.14 as referenced in Clause [4.24.2](#) of this Code.

4.25 Mobile homes and recreational vehicles

4.25.1 Appliance type and separation

Every heating appliance, water heater, or refrigerator installed in a mobile home or a vehicle, other than a canvas-top tent trailer, shall be of the direct-vent appliance type or equivalent, and shall be installed to provide complete separation of the combustion system from the atmosphere of the space provided for living.

4.25.2 Air inlet or flue gas outlet clearances

A combustion air inlet or flue gas outlet of an appliance or any other vehicle opening shall be located at least 3 ft (0.9 m) from any container filler spout or fixed-liquid-level gauge of a vehicle if the intake, outlet, or opening is located above or at the same level. If any portion of such inlet, outlet, or opening is located below the spout or fixed-liquid-level gauge, the clearance shall be the sum of the vertical distance below the spout or fixed-liquid-level gauge plus 3 ft (0.9 m).

4.25.3 Propane vapour supply pressure

Propane vapour, at a pressure not in excess of 13 in w.c. (3.2 kPa), shall be supplied into the piping or tubing supplying any appliance.

4.25.4 Appliance certification

An appliance installed in a propane-equipped mobile home or recreational vehicle shall be certified for use with propane.

4.25.5 Combustion air

Provision shall be made to ensure a supply of combustion air for an appliance, other than an appliance of the direct-vent type, as described in Clause [8](#).

4.25.6 Alternate means of combustion or ventilation air

An open door may be used as an alternative means of providing combustion or ventilation air to a wash-mobile or food service vehicle, provided that the door is interlocked to the propane supply to ensure that the door remains open during appliance operation.

4.25.7 Cargo heater

A cargo heater shall be installed in a readily accessible location.

4.25.8 Cargo heater installation

A cargo heater shall be protected to prevent damage or impaired operation resulting from the shifting or handling of cargo.

4.25.9 Propane warning labels

A durable label in both English and French made of a material that is not adversely affected by water, employing an adhesive that is not water soluble, and measuring not less than 4.5 × 5.75 in

(100 × 125 mm) shall be provided. This label shall be located on the vehicle, adjacent to the propane container, and shall be worded as follows:

a) in English:

WARNING

This system is designed for use with PROPANE only.

DO NOT CONNECT NATURAL GAS TO THIS SYSTEM.

Before turning on propane:

- Be certain appliances are certified for propane and are equipped with correct burner orifices.
- Make certain all propane connections have been made tight, all appliance valves are turned off, and any unconnected outlets are capped.

After turning on propane:

- Light all pilots.
- Each connection, including those at appliances, regulators, and cylinders, shall be leak-tested periodically with soapy water by the occupant. Never use a lighted match or other flame when checking for leaks.
- Do not leave system turned on or containers connected until system has been proven to be propane-tight.
- Cooking appliances must not be used for space heating.
- When the containers are disconnected, the propane supply line must be capped or plugged.

b) in French:

AVERTISSEMENT

Cette installation est conçue pour fonctionner au PROPANE seulement.

NE PAS ALIMENTER CETTE INSTALLATION AVEC DU GAZ NATUREL.

Avant d'admettre le propane :

- S'assurer que l'appareil est certifié pour fonctionner au propane et qu'il est muni des orifices de brûleur appropriés.
- S'assurer que tous les raccordements de propane sont étanches, que tous les robinets d'appareils sont fermés et que toutes les sorties non raccordées sont bouchées.

Après avoir ouvert l'admission de propane :

- Allumer toutes les veilleuses.
- L'occupant doit s'assurer périodiquement, à l'aide d'eau savonneuse, qu'il n'y a aucune fuite aux points de raccordement des appareils, des régulateurs et des bouteilles. Ne jamais utiliser une allumette allumée ou toute autre flamme pour détecter une fuite.
- Ne pas laisser le propane ni les récipients branchés avant de s'être assuré que l'installation ne présente aucune fuite de propane.
- Ne pas utiliser des appareils de cuisson pour réchauffer une pièce.
- Lorsque les récipients sont débranchés, la tuyauterie d'alimentation en propane est obligée d'être bouchée.

The words “WARNING” and “PROPANE” and the phrase “DO NOT CONNECT NATURAL GAS TO THIS SYSTEM” (in French: “AVERTISSEMENT”, “PROPANE”, and “NE PAS ALIMENTER CETTE INSTALLATION AVEC DU GAZ NATUREL”) shall be a minimum of 1/4 in (6.4 mm) in height, and the remainder of the wording shall be a minimum of 1/8 in (3.2 mm) in height.

4.25.10 Appliance warning labels

On all vehicles equipped with appliances, a durable label, made of material not adversely affected by water and employing an adhesive that is not water soluble, shall be provided in both English and French. This label shall be attached adjacent to any fuel-filling locations (gasoline, diesel, or NGV) and propane cylinders. The label shall be worded as follows:

a) in English:

WARNING

Engine ignition and all appliance pilot lights shall be turned off before and during refuelling of motor fuel tanks or any mounted propane container.

b) in French:

AVERTISSEMENT

Le moteur et toutes les veilleuses des appareils doivent être éteintes avant et pendant le remplissage des réservoirs de combustible et de tout récipient de propane installé.

The words “WARNING” and “AVERTISSEMENT” shall be a minimum of 1/4 in (6.4 mm) in height, and the remainder of the wording shall be a minimum of 1/8 in (3.2 mm) in height.

5 Pressure controls

5.1 Delivery pressure

5.1.1 Gas piping system pressures

Subject to Clause [5.1.2](#), the gas pressure in a gas piping system, extending from the termination of the utility/distributor’s installation inside a building, shall not be higher than that shown in Table [5.1](#) in normal operation.

Table 5.1
Pressure inside buildings
 (See Clause [5.1.1](#) and [7.2.2.3.](#))

Type of building	Maximum pressure, psig (kPa)	
	Other than central boiler or mechanical room	Central boiler or mechanical room
One- and two-family dwellings and row housing	2 (14)	—
Hotels and motels	5 (35)	20 (140)
Residential, other than one- and two-family dwellings and row housing	5 (35)	20 (140)
Care or detention and assembly buildings	5 (35)	20 (140)
Business and commercial buildings	20 (140)	20 (140)*†
Industrial buildings	66 (455) Natural gas 20 (140) Propane	66 (455) Natural gas 20 (140) Propane
Central-heating plants	—	66 (455) Natural gas 20 (140) Propane
Building under construction using propane or natural gas (construction heater application)	66 (455) Natural gas 25 (175) Propane	66 (455) Natural gas 25 (175) Propane

* 20 psig (140 kPa) may be used in boiler or mechanical rooms located on the roof of commercial buildings for propane.

† 66 psig (455 kPa) may be used in boiler or mechanical rooms located on the roof of commercial buildings for natural gas.

5.1.2 Piping to central boilers or mechanical rooms

Piping to central boiler or mechanical rooms at gas pressures that are greater than allowed for other building locations shall not pass anywhere inside the building other than the central boiler or mechanical room.

5.1.3 Gas piping system design

A gas piping system shall be designed so that the gas pressure shall not exceed the rated pressure of any accessory, equipment, or appliance, under normal operation and in the event of a failure of an upstream pressure regulator.

5.1.4 Delivery pressure and overpressure protection set-points

The delivery pressure and overpressure protection set-points, as supplied by the gas utility or fuel distributor, shall be considered in the design and installation of a gas piping system, including any modifications to an existing gas piping system.

5.1.5 Residential applications

In a residential application using propane, propane vapour pressure in a gas piping system between the first-stage pressure regulator and second-stage propane regulator shall not be higher than 10 psig (70 kPa). In other applications using propane, means shall be provided to prevent liquefaction of propane.

5.1.6 Piping propane in the liquid phase

Propane shall not be piped into or within any building in the liquid phase, except when

- a) the building is used exclusively to house appliances or equipment for vapourization (including grain dryers), pressure reduction, propane/air mixing, or distribution;
- b) the building is a container-filling building;
- c) the fire-separated portion of the building is used exclusively for housing an internal combustion engine or industrial process; or
- d) the fire-separated portion of the building is occupied exclusively by research and experimental laboratories.

5.1.7 Line pressure regulator location

For applications using propane, a second stage propane regulator installed within a one- or two-family dwelling or row housing shall not be located more than 3 ft (0.9 m) from the point where the propane supply enters the dwelling.

5.2 Pressure regulators

5.2.1 General

5.2.1.1 Gas supply pressures

Gas shall be supplied to an appliance, equipment, or accessory at a normal operating pressure that is within the pressure range specified on the appliance's, equipment's, or accessory's rating plate, or as indicated by the manufacturer's instructions.

Note: *This requirement applies at maximum and minimum load demand on the gas piping system.*

5.2.1.2 Pressure regulators and overprotection devices

To meet Clause [5.2.1.1](#), when the delivery pressure is greater than the maximum rated pressure of the downstream valve train, appliance, or equipment, one or more or a combination of the following shall be installed:

- a) a line pressure regulator; or
- b) an industrial pressure regulator.

5.2.1.3 Pressure regulator features

Every pressure regulator shall be

- a) suitable for the gas;
- b) of sufficient size to provide the required flow of gas;
- c) factory set or field-adjusted to provide, under normal operating conditions, an outlet pressure required for the gas piping system at the extremes of inlet pressures to which the regulator can be exposed;
- d) capable of supplying the gas pressure as required by Clause [5.2.1.1](#);
- e) installed in accordance with the manufacturer's instructions and ratings; and

- f) be constructed so that the outlet pressure does not exceed 150% of the normal outlet operating pressure under no flow conditions when the downstream appliance or equipment is shut down.

5.2.1.4 Minimum clearances

The minimum clearance specified in Clauses [7.4.4](#) and [7.5.2](#) between a pressure regulator and the moisture-exhaust duct shall be maintained.

5.2.1.5 Bypassing pressure regulators

A pressure regulator shall not be bypassed, and a safety limit or a safety relief device shall not be isolated, bypassed, or in any way made ineffective by a valve or other device.

5.2.1.6 Lock-up type pressure regulators

A pressure regulator shall be of the lock-up (positive shut-off) type.

5.2.1.7 Manual shut-off valves and overpressure protection devices

A pressure regulator shall have

- a) a manual shut-off valve installed upstream of the pressure regulator; and
- b) an overpressure protection device in accordance with Clause [5.3](#).

5.2.1.8 Pressure regulators

A pressure regulator provided in a gas supply line shall not be installed

- a) where it is inaccessible for repair, replacement, servicing, or inspection;
- b) in a concealed location; or
- c) where it could be reasonably expected to be subject to physical or chemical damage.

5.2.1.9 Outdoor installations and unheated areas

A pressure regulator installed outdoors or in an unheated area shall be positioned so that the bonnet vent opening discharges vertically downward, except that

- a) when installed within a container dome, the vent opening shall be positioned downward not less than 15° from the horizontal; and
- b) a single-stage pressure regulator of a capacity not exceeding 150 000 Btu/h (45 kW) may be installed in a horizontal position, provided that it is protected from inclement weather.

5.2.1.10 Pressure regulators installed on vehicles

A pressure regulator shall be installed on a vehicle in such a manner that its safe operation will not be impeded by weather conditions, and it shall be protected by a substantial metal or plastic hood of the enclosed style.

5.2.1.11 Mounting cylinders on A-frames

When provision is made for mounting a cylinder on the A-frame of a vehicle, a rigidly mounted support bracket for mounting the pressure regulator shall be provided. The pressure regulator shall be protected in accordance with Clause [5.2.1.10](#).

5.2.2 Additional requirements for line pressure regulators

5.2.2.1 Line pressure regulator

A line pressure regulator, where installed shall be certified to either

- a) CSA/ANSI Z21.80/CSA 6.22; or
- b) UL 144.

Note: CSA/ANSI Z21.80/CSA 6.22 applies to both natural gas and propane, while UL 144 applies only to propane.

5.2.2.2 Two-stage regulation

Not less than two-stage regulation shall be utilized on all permanent propane installations.

5.2.3 Additional requirements for industrial pressure regulators

5.2.3.1 Industrial pressure regulator

An industrial pressure regulator, where installed, shall directly serve any of the following in the downstream gas piping system:

- a) a line pressure regulator that complies with Clause [5.2.2](#) and mounted upstream of the appliance, equipment, or valve train;
- b) another industrial pressure regulator; or
- c) a high and a low gas pressure device installed on the valve train of the appliance or in the downstream piping system upstream of the valve train of the appliance.

5.2.3.2 High and low gas pressure devices

Where high and low gas pressure devices are required in Clause [5.2.3.1](#), they shall be interlocked into the downstream appliance(s) to cause safety shut down when either device detects a fault condition.

5.2.3.3 Pressure regulator(s) downstream of another pressure regulator

Where a pressure regulator(s) is installed downstream of another pressure regulator, the downstream pressure regulator shall be protected by an overpressure protection device complying with Clause [5.3](#), if failure of the upstream pressure regulator could result in exposing the downstream pressure regulator to inlet pressure greater than its rated pressure.

5.3 Overpressure protection devices

5.3.1 Overpressure protection

Except as permitted in Clause [5.3.2](#), a line pressure regulator or an industrial pressure regulator shall be provided with an overpressure protection device, and based on its intended outlet pressure, set to not exceed the maximum allowable downstream pressures shown in Table [5.2](#).

5.3.2 Overpressure protection devices

An overpressure protection device shall not be required for a Class I line pressure regulator certified to CSA/ANSI Z21.80/CSA 6.22 and rated for inlet pressure of 2 psig (14 kPa), provided every appliance it directly serves is equipped with either an appliance regulator certified to CSA/ANSI Z21.18/CSA 6.3, an automatic gas valve certified to CSA/ANSI Z21.21/CSA 6.5, or a combination control valve certified to CSA/ANSI Z21.78/CSA 6.20.

5.3.3 Class I line pressure regulator

A Class I line pressure regulator certified to CSA ANSI Z21.80/CSA 6.22 with a rated inlet pressure of more than 2 psig (14 kPa) shall be installed only with the factory pre-assembled or supplied overpressure protection device with which the regulator is certified.

Note: Class I line pressure regulators certified to CSA/ANSI Z21.80/CSA 6.22, (i.e., capable of being adjusted to deliver an outlet pressure of ½ psig (3.5 kPa) or less), and rated for inlet pressures of either 5 psig (35 kPa) or 10 psig (70 kPa), must be supplied with an independent means to limit the downstream pressure to 2 psig (14 kPa) in the event of failure of the regulating mechanism, which is acceptable for certified appliances with a maximum inlet gas pressure of 14 in w.c. Line pressure regulators with separate overpressure protection devices must be factory pre-assembled, and supplied to the field as a unit.

5.3.4 Internal relief valve or line relief valve

If an internal relief valve or line relief valve is used as the overpressure protection device, it shall be sized to fully relieve the rated capacity of the line pressure regulator when in the wide-open position.

Table 5.2
Overpressure protection device setpoint requirements
(See Clause [5.3.1](#) and Figure [8.2](#).)

Appliance or equipment maximum rated inlet gas pressure	Maximum allowable downstream pressure
14 in w.c. (3.5 kPa) or less	2 psig (14 kPa) (see Clause 5.3.2 for limitation; use manufacturer's specification otherwise)
Greater than 14 in w.c. (3.5 kPa) up and including 2 psig (14 kPa)	5 psig (35 kPa)
Greater than 2 psig (14 kPa) up to and including 10 psig (70 kPa)	5 psig (35 kPa) or 2 times maximum rated inlet pressure, whichever is greater
Greater than 10 psig (70 kPa)	10 psig (70 kPa) over maximum rated inlet pressure

Note: The installer might need to consult with pressure regulator manufacturers to determine how best to remain within the maximum allowable downstream pressures.

5.4 Hydrostatic relief devices for propane applications

5.4.1 Hydrostatic relief valves

A hydrostatic relief valve (to relieve at a safe location outdoors) shall be installed between each pair of shut-off valves on a propane liquid piping system. The start-to-discharge pressure setting of such a hydrostatic relief valve shall be neither less than 375 psig (2500 kPa) nor more than 500 psig (3500 kPa).

5.4.2 Discharge lines from hydrostatic relief devices

Discharge lines from two or more hydrostatic relief devices may run into a common discharge header, provided that the cross-sectional area of the header is at least equal to the sum of the cross-sectional areas of the individual discharge lines and that the header is not connected to any vent or bleed line.

5.5 Venting of pressure control devices

5.5.1 Construction of vent piping for pressure control devices other than overpressure relief devices

5.5.1.1 Inclusions

The requirements in Clause [5.5.1](#) apply to bleed vents from pressure control devices including, but not limited to,

- a) automatic valves;
- b) diaphragm valves; and
- c) combination controls.

5.5.1.2 Exceptions

The requirements in Clause [5.5.1](#) do not apply to

- a) pressure regulators (including monitoring regulators) equipped with internal relief valves; and
- b) line overpressure relief valves.

Note: For the venting requirement of overpressure protection devices, including line relief valves and internal relief valves on pressure regulators, refer to Clause [5.5.2](#).

5.5.1.3 Venting

Except as permitted in Clauses [5.5.1.4](#) and [5.5.1.5](#) for natural gas and hydrogen-natural gas blend applications and Clauses [5.5.1.5](#) and [5.5.3.1](#) for propane, the bleed vent from a pressure control device shall

- a) be constructed of steel pipe, copper, seamless aluminum, or steel tubing that complies with Clause [6.2](#);
- b) be of a size at least equal to the nominal pipe size of the vent outlet of the valve, combination control, pressure regulator, or control device, and in no case shall the inside diameter be less than 0.25 in (6 mm).
- c) connect to any other bleed vent only as permitted in Clause [5.5.1.5](#); and
- d) terminate in accordance with Clause [5.6](#).

5.5.1.4 Natural gas and hydrogen-natural gas blend applications

For natural gas and hydrogen-natural gas blend applications, the bleed vent line from a diaphragm valve or combination control that is installed on an appliance that has rated inlet pressure not in excess of 0.5 psig (3.5 kPa) may terminate in the appliance combustion chamber adjacent to a continuous pilot, provided that the terminus of the bleed vent is in a burner tip having a melting point in excess of 1450 °F (790 °C) and that is securely held in a fixed position relative to the pilot flame and that will not adversely affect the operation of the thermal element.

5.5.1.5 Bleed vents

The bleed vents of any two or more pressure control devices of any type, or any combination thereof, may be connected into a single, common bleed vent, provided that

- a) there is compliance with
 - i) Clause [5.5.1.3](#) for inlet pressure not in excess of 0.5 psig (3.5 kPa); or
 - ii) Clause [5.5.2.5](#) a) and b) for inlet pressure in excess of 0.5 psig (3.5 kPa); and
- b) the single, common bleed vent line has an area of not less than twice the total area of the connected bleed vents upstream of the connection.

5.5.2 Construction of vent lines for overpressure relief devices

5.5.2.1 Inclusions

The requirements in Clause [5.5.2](#) apply to vent lines originating from

- a) pressure regulators (including monitoring regulators) equipped with internal relief valves; and
- b) line overpressure relief valves.

5.5.2.2 Exceptions

The requirements in Clause [5.5.2](#) do not apply to bleed vents from pressure control devices, including but not limited to

- a) automatic valves;
- b) diaphragm valves; and
- c) combination controls.

Note: For the venting requirement of pressure control devices, refer to Clause [5.5.1](#).

5.5.2.3 Vent line

A vent line shall be of sufficient size and configuration to prevent impedance upon a regulator.

5.5.2.4 Outdoor vent line termination

The outdoor vent line termination of a pressure regulator or a line relief device shall be equipped with a means to prevent the entry of water, insects, or foreign material.

5.5.2.5 Vent line material and sizing

Except as specified in Clause [5.5.2.6](#), when a pressure regulator with internal relief valve or a line relief valve is installed, it shall be vented separately and terminate in accordance with Clause [5.6](#) by a vent line

- a) of steel pipe, or of seamless steel tubing or copper tubing or CSST that complies with Clause [6.2](#); and
- b) of a size
 - i) at least equal to the nominal pipe size of the vent outlet of the valve or regulator increased as specified by the manufacturer's instructions and for CSST increased by one pipe size diameter; or
 - ii) in the absence of manufacturer's instructions, increased by one pipe size diameter for every 50 ft (15 m) or part thereof that the vent line extends beyond the initial 50 ft (15 m). This increase shall be made at the connection on the device.

5.5.2.6 Common vent lines

Except as required by Clause [5.5.2.7](#), when two or more gas overpressure relief valves are installed, they may be connected into a single vent line, provided that

- a) there is compliance with Clause [5.5.2.5](#);
- b) the single vent line has an area equal to the largest relief valve opening plus 50% of the total area of the other relief valve openings;
- c) the highest inlet pressure of any one line relief valve does not exceed 1.1 times the lowest inlet pressure of any other line relief valve, based on manufacturer's product literature; and
- d) the highest start-to-discharge pressure of one line relief valve does not exceed 1.1 times the lowest start-to-discharge pressure of any of the other line relief valves, based on manufacturer's product literature.

5.5.2.7 Vent line for internal relief valve

A vent line from a pressure regulator with an internal relief valve shall be piped independently to the outdoors and shall not be manifolded with any other vent line.

5.5.3 Additional requirements for venting of propane pressure control devices

Δ 5.5.3.1 Vent lines

For propane applications, except as specified in Clause [5.5.4](#), a pressure regulator, line relief device, or hydrostatic relief device on an appliance using gas heavier than air shall be equipped with a vent line

- a) in accordance with Clause [5.5.2.5](#) a) and b); and
- b) terminating outdoors in accordance with Clause [5.6](#).

5.5.3.2 Propane applications

For propane applications, a regulator vent, line relief device, or hydrostatic relief valve discharging vertically upward shall be provided with a loose-fitting rain cap. When discharging downward, it shall be provided with a protective screen.

5.5.4 Venting exemptions for pressure regulators

5.5.4.1 Exemptions

The following shall be exempt from Clause [5.6](#), provided the conditions in Clause [5.5.4.2](#) are also met:

- a) a line pressure regulator certified to CSA/ANSI Z21.80/CSA 6.22 that incorporates a vent limiter;
- b) an appliance pressure regulator certified to ANSI Z21.18/CSA 6.3 that incorporates a vent limiter, and with a rated pressure no higher than 2 psig (14 kPa); or
- c) an industrial pressure regulator equipped with a safety diaphragm if installed on a valve train of a CSA B149.3 approved appliance having a capacity greater than 400 000 btu/h (120 kW).

5.5.4.2 Exemption conditions

The conditions for exemption from Clause [5.6](#) shall be

- a) the vent limiter limits the escape of gas to
 - i) not more than 2.5 ft³/h (0.0706 m³/h) for natural gas; and
 - ii) not more than 1 ft³/h (0.0283 m³/h) for LP gases and subject to the limitation in Clause [5.5.4.3](#);
- b) the pressure regulator is installed in a ventilated space and to where the accumulation of fuel gas from the vent limiter does not exceed 25% of the gas's lower explosive limit; and
- c) the clearance from the vent limiter to any source of ignition is met, as listed in Table [5.3](#).

Note: Examples of unventilated spaces include, but are not limited to,

- a) a closet;
- b) the space between a suspended ceiling and the underside of above flooring; and
- c) a room having no openings (e.g., a hole in a door or wall) to an adjacent, ventilated room or to the outdoors.

5.5.4.3 Line pressure regulator rated for 5 psig (35 kPa) or 10 psig (70 kPa)

The exemption permitted in Clause [5.5.4.1](#) shall not apply to a line pressure regulator rated for 5 psig (35 kPa) or 10 psig (70 kPa) when the fuel is propane.

Note: This clause imposes the limitation to propane systems because it describes installation of regulators certified to CSA/ANSI Z21.80/CSA 6.22 on piping systems operating at either 5 psig (35 kPa) or 10 psig (70 kPa), and, by definition, a line pressure regulator certified to ANSI Z21.80/CSA 6.22 on propane systems is intended to be used between the propane 2 psig (14 kPa) service regulator and the gas utilization equipment.

5.6 Termination of vent lines

Except where permitted by Clauses [5.5.1.4](#) and [5.5.4](#), the vent line from overpressure protection devices, relief devices, and internal relief valves, and the termination of any other vent line not eligible to be vented into a ventilated space shall terminate outdoors with the clearances specified in Table [5.3](#).

Δ

Table 5.3
Clearance from vent line and bleed vent terminations, ft (m)
(See Clauses [5.5.4.2](#), [5.6](#), [8.14.8](#), and [10.1.8](#).)

	Natural gas and hydrogen-natural gas blend vents (relief or bleed) ≤ 50 ft ³ /h (1.5 m ³ /h)	Natural gas and hydrogen-natural gas blend relief vents (relief or bleed) > 50 ft ³ /h (1.5 m ³ /h) and ≤ 1900 ft ³ /h (55 m ³ /h)	Natural gas and hydrogen-natural gas blend relief vents (relief or bleed) > 1900 ft ³ /h (55 m ³ /h)	Propane relief vents and bleed vents
Building opening*	1 (0.3)	3 (0.9)	10 (3)	3 (0.9)
Appliance vent outlet†	Dimension A	Dimension B	Dimension B	Dimension B
Moisture exhaust duct‡	3 (0.9)	3 (0.9)	3 (0.9)	3 (0.9)
Mechanical air intake	3 (0.9)	10 (3)	10 (3)	10 (3)
Appliance air intake	1 (0.3)	3 (0.9)	10 (3)	10 (3)
Source of ignition	1 (0.3)	3 (0.9)	3 (0.9)	5 (1.5)

Notes: Dimension A = 1 ft (0.3 m) in any direction from the vent termination of a gas appliance, and additionally 3 ft (0.9 m) horizontally when within a vertical distance of 15 ft (4.6 m) between the appliance vent termination and the gas vent termination.

Dimension B = 3 ft (0.9 m) in any direction from the vent termination of a gas appliance, and additionally 3 ft (0.9 m) horizontally when within a vertical distance of 15 ft (4.6 m) between the appliance vent termination and the gas vent termination.

* Outdoor air intakes that are less than 8 in (200 mm) in diameter or equivalent area shall be considered a building opening in using this table.

† See also Clause [8.14.8](#).

‡ Applies to gas or electric dryer termination.

Note: The outdoor air intake referred to in this table is the ducting that goes from the outside of the structure and terminates into the return air plenum before the appliance, sometimes referred to as a “fresh-air intake”.

Note: CSA Z662 specifies the necessary minimum clearances from the vent of a regulator or an associated overpressure relief device that belongs to a utility or propane supplier is required to building features similar to those shown in this table. Annex [D](#) contains a reprinting of the mandated clearances specified in CSA Z662. Piping and appliance installers and repairers and other construction trades are reminded to maintain these clearances when planning and executing work.

Note: CSA B149.2 specifies the minimum clearances from the propane containers to a source of ignition.

5.7 Pressure regulators on appliances using propane

5.7.1 Excess propane pressures

An appliance that can be subjected, through supply pressure, design, creepage, or fluctuation, to propane pressure in excess of that for which it is rated shall be equipped with an appliance pressure regulator.

5.7.2 Pilots

When a pressure regulator is required by Clause [5.7.1](#), the propane supply to the pilot or group of pilots shall be regulated by an approved pressure regulator independent of the main burner propane supply.

6 Gas piping systems

6.1 General

6.1.1 Piping

Gas piping or tubing shall be of steel, copper, or plastic.

6.1.2 Removal

If removed from a gas installation, piping, tubing, and fittings shall not be reused unless thoroughly cleaned, inspected, and ascertained to be equivalent to new material. Piping, tubing, or fittings previously used with other gases may be reused with gas, provided that

- a) it is ascertained that the piping, tubing, or fittings to be used are equivalent to new material; and
- b) the piping, tubing, or fittings to be used are cleaned, inspected, and tested.

6.2 Material

6.2.1 Piping standard specification

Piping shall comply with ASTM A53/A53M or ASTM A106/A106M.

6.2.2 Fittings

A fitting used with steel pipe shall be

- a) either malleable iron or steel and shall comply with the material selection requirements of CSA Z662 or the applicable ASME B16 series of standards; or
- b) certified to CSA/ANSI LC 4/CSA 6.32.

When Schedule 80 pipe is required, the minimum Class of fitting used with the pipe shall be Class 300.

6.2.3 Piping schedule

6.2.3.1 Operating pressure equal to or less than 125 psig (860 kPa)

A gas piping system using natural gas, hydrogen-natural gas blends, or propane vapour phase with

operating pressures up to and including 125 psig (860 kPa) shall comply with the following as applicable:

- a) Piping shall be at least Schedule 10 for NPS 1/2 to 2. When using Schedule 10 to less than Schedule 40, piping shall be located indoors, and joints shall use fittings certified to CSA/ANSI LC 4/CSA 6.32.
- b) Piping shall be at least Schedule 40 for NPS 2-1/2 to 10.
- c) Pipe larger than NPS 10 shall be at least standard weight.

6.2.3.2 Operating pressure exceeding 125 psig (860 kPa)

A gas piping system using natural gas, hydrogen-natural gas blends, or propane vapour phase with operating pressures exceeding 125 psig (860 kPa) and all liquid piping systems shall comply with either of the following:

- a) For pipe sizes up to and including NPS 10
 - i) piping shall be at least Schedule 40 when using welded or flanged joints; or
 - ii) piping shall be at least Schedule 80 when using threaded joints. Threaded joints shall be threaded or threaded and back welded.
- b) Pipe larger than NPS 10 shall be at least standard weight.

6.2.4 Copper tubing for gas

Copper tubing used for gas shall be Type G, K, or L, and shall meet the requirements of one of the following Standards, as applicable:

- a) Type G tube shall meet ASTM B837; or
- b) Types K and L tube shall meet ASTM B88.

6.2.5 Copper tubing for propane

Copper tubing Types K and L specified in Clause [6.2.4](#) b) may be used for liquid propane or propane in the vapour phase.

6.2.6 Flare nuts

Flare nuts shall be forged from UNS C37700 brass and shall not be externally machined.

6.2.7 Tubing fittings rated pressure

Tubing fittings shall be rated for a working pressure of not less than 125 psig (860 kPa) for operating pressures of 125 psig (860 kPa) or less. For higher operating pressures, tubing and fittings shall be rated for a minimum of 250 psig (1725 kPa).

6.2.8 Copper tubing underground

Copper tubing for underground use shall be either Type L or G, externally coated with extruded polyethylene or PVC resin at the time of manufacture, or Type K, and any portion of the copper tubing that extends above ground shall be protected against physical damage.

6.2.9 Tubing

Tubing shall be one of the following:

- a) CSST;
- b) seamless copper; or
- c) seamless steel.

6.2.10 Hoses and hose fittings

Except as required in Clauses [6.2.11](#) and [6.2.12](#), every non-metallic hose and hose fitting shall have a minimum working pressure of 350 psig (2400 kPa) and shall comply with CSA 8.1 or CSA 8.3.

Note: Refer to Clause [6.20.5](#) for metallic hoses.

6.2.11 Hoses for cutting and welding

Every hose and hose connection used in cutting or welding systems shall comply with CAN/CSA-W117.2.

6.2.12 Liquid propane hose prohibition

Hoses certified to CSA 8.3 shall not be used in a liquid propane system.

6.2.13 Seamless steel tubing standard specification

Seamless steel tubing shall comply with ASTM A179/A179M.

6.2.14 Alternative material

Materials not specified in Clause [6.2](#) may be used if they conform to a nationally recognized standard or to a test report of a nationally recognized certification organization.

6.2.15 Plastic piping and tubing standard specification

Plastic piping and tubing shall comply with CSA B137.4.

6.2.16 Plastic piping and tubing fittings

Fittings for plastic piping and tubing systems shall comply with CSA B137.4 or CSA B137.4.1.

6.2.17 Plastic pipe prohibition

Plastic pipe shall not be used in a liquid propane system.

6.2.18 Plastic piping or tubing use

Except as mentioned in Clause [6.2.19](#), plastic piping or tubing shall only be used for outdoor underground service.

6.2.19 Plastic pipe propane prohibition

Plastic pipe shall not be used for, as a minimum, the first 10 ft (3 m) of piping on the downstream side of a vapourizer in a propane application and shall not exceed the temperatures specified in Clause [6.2.20](#).

6.2.20 Plastic piping or tubing temperature

Plastic piping or tubing shall not be used at ambient temperatures exceeding 122 °F (50 °C) or where the steady-state operating temperature of the materials will exceed 86 °F (30 °C).

6.2.21 Plastic piping or tubing termination

Plastic piping or tubing may terminate above ground and outside a building, provided that

- a) the aboveground portions are completely encased with a certified metallic sheathing or anodeless riser that extends a minimum of 6 in (15 cm) below grade; and
- b) the plastic piping or tubing is not subject to external loading stresses created by other piping, appliances, or equipment.

6.2.22 CSST standard specification

CSST and associated fittings shall comply with CSA/ANSI LC 1/CSA 6.26 or CSA publication CGA Certification Laboratory Requirement LAB-009.

6.2.23 CSST prohibition

CSST shall not be used as a gas connector.

6.3 Size

6.3.1 General

Piping, tubing, and hose shall be of sufficient size to provide a supply of gas to meet the requirements of volume and pressure at the point of use.

6.3.2 Pressure equal to or less than 14 in w.c. (3.5 kPa)

A gas piping system supplied at pressures up to and including 14 in w.c. (3.5 kPa) shall be designed to prevent the loss in pressure between the appliance and either the termination of the utility installation or the last-stage regulator from exceeding the maximum allowable pressure drop specified in Table 6.1. The minimum size of piping, tubing, and fittings shall be determined in accordance with good engineering practice, such as

- by the use of Tables A.1 and A.8 of Annex A for natural gas or hydrogen-natural gas blends, which include allowance for a reasonable number of fittings, when the maximum allowable pressure drop is 0.5 in w.c. (0.125 kPa);
- by the use of Tables A.2 and A.9 in Annex A for natural gas or hydrogen-natural gas blends or Tables B.1 and B.6 in Annex B for propane, which include allowance for a reasonable number of fittings, when the maximum allowable pressure drop is 1 in w.c. (0.25 kPa);
- by the method of calculation outlined in Annex A for natural gas or hydrogen-natural gas blends or Annex B for propane; or
- for CSST, by the use of sizing methods and tables supplied by the manufacturer.

Table 6.1
Maximum allowable pressure drop
(See Clause 6.3.2.)

	Piping and tubing system supply pressure	Maximum allowable pressure drop
Natural gas/hydrogen- natural gas blends	< 7 in w.c. (1.75 kPa)	0.5 in w.c. (0.125 kPa)
Natural gas/hydrogen- natural gas blends/ propane	7 in w.c. (1.75 kPa) up to 14 in w.c. (3.5 kPa)	1 in w.c. (0.25 kPa)

6.3.3 Pressure exceeding 14 in w.c. (3.5 kPa)

A gas piping system operating at a pressure exceeding 14 in w.c. (3.5 kPa) shall be designed to ensure an adequate supply of gas to each appliance served at the respective designated pressure rating, and to ensure that the appliance will not be overpressured under conditions of no flow. The minimum size of

piping, tubing, and fittings shall be determined in accordance with Clause [6.3.4](#) for [2](#) psig (14 kPa) systems or good engineering practice, such as

- a) by the use of the applicable tables in Annex [A](#) for natural gas or hydrogen-natural gas blends or Annex [B](#) for propane, making allowance for fittings as necessary;
- b) by the method of calculation outlined in Annex [A](#) for natural gas or hydrogen-natural gas blends or Annex [B](#) for propane; or
- c) for CSST, by the use of sizing methods and tables supplied by the manufacturer.

6.3.4 Pressure regulator

The gas piping system shall be designed to provide adequate gas pressure to the 2 psig (14 kPa) pressure regulator to match downstream load requirements. Pressure regulator sizing shall be subject to the minimum available inlet supply pressure. See Tables [A.2](#), [A.3](#), [A.4](#), [A.10](#), and [A.11](#) in Annex [A](#) for natural gas or hydrogen-natural gas blends or Tables [B.2](#) and [B.7](#) in Annex [B](#) for propane, which include allowance for a reasonable number of fittings.

6.3.5 Plastic piping

Plastic piping shall be sized

- a) by the use of Table [A.7](#) in Annex [A](#) for natural gas or hydrogen-natural gas blends or Tables [B.1](#) to [B.5](#) in Annex [B](#) for propane; or
- b) by the method of calculation outlined in Annex [A](#) for natural gas or Annex [B](#) for propane.

Note: See Table [A.17](#) for natural gas and Table [B.12](#) for propane.

6.3.6 Relative density

Annex [A](#) is based on natural gas of 0.60 relative density. For gas having a relative density other than 0.60, the multipliers given in Table [A.15](#) in Annex [A](#) shall be applied to the capacities listed. Annex [B](#) is based on propane of 1.52 relative density.

6.3.7 Equivalent length

Annexes [A](#) and [B](#) give the resistance of bends, fittings, and valves as an equivalent length of straight pipe in ft (m) to be added to the actual length to obtain the total equivalent length on which pressure loss calculations shall be based. See Tables [A.16](#) and [B.11](#).

6.3.8 Size less than NPS 1/2

6.3.8.1 Natural gas and hydrogen-natural gas blends threaded pipe and fittings

For natural gas and hydrogen-natural gas blends, threaded pipe and fittings less than NPS 1/2 used in a piping system shall be Schedule 80.

6.3.8.2 Propane piping location

For propane, piping less than NPS 1/2 shall not be used indoors, except that NPS 3/8 piping may be used as a branch line not exceeding 25 ft (7.5 m) in length.

6.3.9 Piping concealed location

Piping less than NPS 1/2 shall not be used in a concealed location.

6.4 Volume of gas to be used for sizing gas piping systems

6.4.1 Total volume

The total volume of gas required shall

- a) be determined as the total volume for all appliances supplied, except as permitted in Clause [6.4.3](#); and
- b) include an allowance for known future extensions.

6.4.2 Volume determination

The volume of gas required for each appliance shall be determined from either

- a) the appliance rating plate; or
- b) the appliance manufacturer when the rating is not shown on the appliance.

6.4.3 Diversity load

When a diversity of load has been established to the satisfaction of the authority having jurisdiction, a percentage of the total volume may be used.

6.5 Allowable pressure and pressure drop

6.5.1 Pressure determination

The gas pressure required for each appliance shall be determined from either

- a) the appliance rating plate; or
- b) the appliance manufacturer when the required pressure is not shown on the appliance.

6.5.2 Pressure regulator requirement

When the pressure is in excess of that determined by Clause [6.5.1](#) or when either excessive fluctuation or creepage exists, a pressure regulator shall be provided in the gas piping system to the appliance.

6.5.3 Pressure drop determination

In determining the pressure drop in the gas piping system, the resistance offered by bends, fittings, and valves shall be included in the calculations, together with the resistance offered by the length of piping or tubing.

6.6 Extensions

6.6.1 Adequate capacity

When an existing gas piping system is of adequate capacity for an additional appliance, any required extension that also contains the existing system shall be sized to meet the requirements of Clause [6.3](#).

6.6.2 Insufficient capacity

When an existing gas piping system is of insufficient capacity for an additional appliance, the supply piping or tubing to any extension shall be sized to meet the requirements of Clause [6.3](#) to within 24 in (610 mm) of the point of the supplier termination or the pressure regulator, at which point it may be reduced if necessary.

6.7 Location

6.7.1 Concealed piping or tubing

Concealed piping or tubing that contains fittings or joints shall not be run where the fittings or joints cannot be inspected and tested in accordance with Clause [6.22](#) in their final position prior to being concealed.

6.7.2 Piping or tubing installation prohibition

Piping or tubing shall not be installed

- a) in a stairwell other than a stairwell within a dwelling unit unless it is totally enclosed by a chase consisting of material that has the same fire-resistance rating as that required for the stairwell;
- b) in a chimney, flue, elevator, dumbwaiter or material lift shaft, or any chute including for linen or refuse use;
- c) in a heating or ventilating plenum, duct, or shaft, except as permitted in Clause [6.7.6](#); or
- d) in contact with either cinders, ashes, or other corrosive materials.

6.7.3 Corrosive chemical

Piping or tubing shall not be concealed in a location where a corrosive chemical is used.

6.7.4 Solid flooring

Piping or tubing in solid flooring, such as concrete, shall be laid in channels and suitably covered to permit access to the piping or tubing. Alternatively, the piping or tubing shall be encased in ducts so that there is free air space around the pipe or tube. Such a duct shall be ventilated.

6.7.5 Vertical piping

Each vertical piping chase shall have an opening at the top and bottom, and the opening shall have a minimum area equivalent to a round opening of 1 in (25 mm) in diameter.

6.7.6 False ceiling space

Piping or tubing may be installed in a false ceiling space, including one used as a return-air plenum of a central warm-air or air-conditioning system.

6.8 Piping practices

6.8.1 General

Piping and fittings shall be clear and free from cutting burrs, threading burrs, scale, and defects.

6.8.2 Piping ends

The ends of all piping shall be reamed.

6.8.3 Piping support

Piping or tubing shall not be supported by other piping or tubing, and shall be installed with individual supports of sufficient strength and quality. Supports shall be spaced in accordance with Table [6.2](#).

Table 6.2
Spacing of supports for piping and tubing*
 (See Clauses [6.8.3](#), [6.25.1](#), and [6.25.2](#).)

NPS	Maximum spacing of supports ft (m)
1/2 or less — horizontal	6 (2)
3/4–1 — horizontal	8 (2.5)
1-1/4–2-1/2 — horizontal	10 (3)
3–4 — horizontal	15 (5)
5–8 — horizontal	20 (6)
10 or larger — horizontal	25 (8)
All sizes — vertical	Every floor but not more than 125% of horizontal spacings
Tubing — all sizes — vertical and horizontal	6 (2)

* For piping and tubing installed on a rooftop, see Clause [6.25](#).

6.8.4 Structural integrity

A girder, beam, joist, or other structural member shall not be cut for the installation of piping in such a manner as to reduce its strength below that required for the purpose for which it is intended.

6.8.5 Back flow prevention

When air, oxygen, or other gas under pressure is used in connection with the gas supply, either a back-check valve or other suitable equipment shall be provided as close as practicable to the point of interconnection to prevent any interchange of such in the piping or tubing system.

6.8.6 Filter requirements

When a filter is used in a piping system, a low-pressure indicator shall be installed directly downstream of the filter.

6.8.7 Manufacturer's fabricated fittings

Manufacturer's fabricated fittings shall be used in welded gas piping systems except as permitted in Clause [6.8.8](#).

6.8.8 Supply header exceeds NPS 2-1/2

When a gas supply header is NPS 2-1/2 or larger, the branch line connection may be a job-fabricated welded fitting, provided that the branch line does not exceed 50% of the pipe diameter of the gas supply header, and in no case shall the branch connection be less than NPS 3/4.

6.8.9 Support material

Where piping and tubing supports are used, they shall be metallic and shall be installed to avoid galvanic action between the piping or tubing and the supports.

6.8.10 Test ports

A test port shall be installed immediately downstream of a line pressure regulator or an industrial pressure regulator except where the pressure regulator can be adjusted while measuring and observing the supply pressure at any appliance being served by the pressure regulator. Where opening the test port could create an uncontrolled release of gas, the test port shall be equipped with a manual shut-off valve that is either capped or plugged.

6.9 Joints and connections

6.9.1 Steel piping

Joints in steel piping shall be threaded, flanged, press-connected, or welded, and shall be as specified in Clause [6.15.2](#). When mating flanges, they shall be of the same face type and rating.

6.9.2 Type

Piping of NPS 2-1/2 to 4 shall have either press-connect fittings certified to CSA/ANSI LC 4/CSA 6.32 or welded pipe joints.

Piping greater than NPS 4 shall have welded pipe joints.

6.9.3 Welding

Welding of gas piping systems shall be performed in accordance with a procedure and by an operator registered under the applicable provincial or territorial legislation.

6.9.4 Welding acceptance criteria

The acceptance criteria for any welds shall

- a) for design pressures greater than 250 psig (1720 kPa), comply with the visual and radiographic inspection requirements of Clause [7](#) of CSA Z662 or other approved methods;
- b) for design pressures between 100 psig (700 kPa) and up to 250 psig (1720 kPa), be in accordance with the visual and radiographic inspection requirements of Chapter VI of ASME B31.3; or
- c) for design pressures of 100 psig (700 kPa) and less, be by visual inspection of the external weld surface, as well as the internal weld surface where accessible without the use of special tools.

Acceptance criteria of the weld shall be in accordance with the requirements of Annex [J](#).

Notes:

- 1) For clarification, the use of the word “inspection” in this Code means the same as “examination” in ASME B31.3.
- 2) Hand mirrors and flashlights are not considered to be special tools.

6.9.5 Piping or fitting thread

A piping or fitting thread shall be tapered and shall comply with ANSI/ASME B1.20.1.

6.9.6 Jointing sealant

When a jointing sealant is used, it shall be certified to CAN/ULC-S642 and shall be applied to the male threads of a metal pipe. Tape shall be stretched and applied in a clockwise direction, with a 50% overlap leaving the first two starter threads bare. Tape shall not be used for pipe sizes larger than 1-1/2 in (38 mm) nominal pipe size for hydrogen-natural gas blends.

6.9.7 Gasket material

Gasket materials shall be of either neoprene or other similar material resistant to any action of gas. Natural rubber shall not be used.

6.9.8 Lubricant use

A lubricant used in either a valve or a control shall be of a type approved for gas use. It shall be capable of withstanding the service conditions to which it can be subjected when used in accordance with the manufacturer's recommendations.

6.9.9 Seamless copper, brass, or steel tubing

A joint in seamless copper, brass, or steel tubing shall be made by

- a) a flare joint;
- b) an approved fitting, other than a metallic ball sleeve compression-type fitting;
- c) brazing with a material that has a melting point exceeding 1000 °F (525 °C); or
- d) a press-connect fitting.

6.9.10 Bushing

A bushing shall be made of

- a) steel for a change of one pipe size; or
- b) steel or malleable iron for a change of two or more pipe sizes.

6.9.11 Plastic pipe and fittings

Plastic pipe and fittings shall be joined by heat fusion, electrofusion, or mechanical methods. Such joining methods shall be compatible with the materials being joined, in accordance with CSA Z662 and the manufacturer's instructions.

6.10 Branch piping outlets

If a branch piping outlet is installed in the main gas piping system before it is known what size of piping is required to be connected to it, the outlet shall be of the same size as the piping which supplies it.

6.11 Appliance connections

6.11.1 General

The following requirements apply to appliance connections:

- a) Where approved, rigid piping may be used to connect an appliance to the building gas piping system.
- b) A vented appliance shall be connected directly to the gas supply with piping or tubing or by means specified in Clause [6.21](#).
- c) CSST and copper tubing shall only be connected to an appliance secured in place (nonmovable) or to rigid piping.

6.11.2 Gas hose or connector requirements

An appliance connection shall satisfy one of the following requirements:

- a) An appliance permitted by Clauses [6.20](#) and [6.21](#) for use with a gas hose may be connected to the building piping by means of a gas convenience outlet or by means of a quick-disconnect device, and where a quick-disconnect device is used, a readily accessible manual shut-off valve shall be installed upstream of, and as close as practicable to, the quick-disconnect device.

- b) An appliance permitted by Clauses 6.20 and 6.21 for use with a gas connector that is certified to CSA/ANSI Z21.24/CSA 6.10 shall be directly connected to the building gas piping system secure outlet.
- c) An appliance permitted by Clauses 6.20 and 6.21 for use with a gas connector that is certified to ANSI Z21.69/CSA 6.16 may be connected directly to the building gas piping system secure outlet or connected by means of a gas convenience outlet or a quick-disconnect device, and where a quick-disconnect device is used, a readily accessible manual shut-off valve shall be installed upstream of, and as close as practicable to, the quick-disconnect device.
- d) An appliance permitted by Clauses 6.20 and 6.21 for use with a gas connector that is certified to CSA/ANSI Z21.101/CSA 8.5 shall be connected to the building gas piping system by means of a gas convenience outlet.

6.11.3 Quick-disconnect location

A quick-disconnect device shall not be used to connect appliances indoors in a residential building.

6.11.4 Gas convenience outlet certification

A gas convenience outlet shall be certified to CSA/ANSI Z21.90/CSA 6.24.

6.11.5 Quick-disconnect certification

A quick-disconnect device shall be certified to CSA/ANSI Z21.41/CSA 6.9.

6.11.6 Support

An appliance shall be adequately supported and connected to the piping so that there is no undue strain on the connection.

6.12 Piping outlets

6.12.1 General

During the period when an appliance is not connected to an outlet of a piping system, the outlet shall be made tight by means of

- a) a plugged valve; or
- b) either a cap or plug made of material compatible with the material of the piping or tubing system.

6.12.2 Unthreaded piping

An unthreaded portion of a piping outlet shall extend at least 1 in (25 mm) through either a finished ceiling or a finished wall and at least 2 in (50 mm) through a floor.

6.12.3 Piping outlet location

A piping outlet shall be located as close as practicable to the appliance being served.

6.13 Drip and dirt pockets

6.13.1 Dirt pocket installation

A dirt pocket shall be installed at the bottom of any piping or tubing on the final drop serving an appliance other than

- a) an illuminating appliance;
- b) a range;

- c) a clothes dryer;
- d) an outdoor grill;
- e) a portable appliance or equipment;
- f) a decorative appliance;
- g) a gas log;
- h) a room heater; and
- i) an appliance incorporating a sediment trap.

6.13.2 Size

Both a drip pocket and a dirt pocket shall be sized as follows:

- a) The depth of the pocket shall be either 3 in (75 mm) or equal to the internal diameter of the piping it serves, whichever is greater.
- b) The diameter of the pocket shall be either NPS 2 or equal to the diameter of the piping it serves, whichever is less.

6.13.3 Location

A drip or dirt pocket shall be located where readily accessible to permit cleaning and emptying.

6.13.4 Termination

A drip or dirt pocket shall be capped.

6.13.5 Drip pocket installation

A drip pocket shall be provided at all points in a piping system where condensation can collect, such as points where the piping is exposed to either wide ranges or sudden changes in temperature.

6.13.6 Additional requirements

When a drip or dirt pocket is required by either Clause [6.13.1](#) or [6.13.5](#), it shall be connected to the piping it serves through the bottom opening of a tee, the other two openings of which shall serve as continuity for the piping system.

6.14 Prohibited practices

6.14.1 Defective piping

A defective section of piping or tubing shall be replaced and not repaired.

6.14.2 Nesting

Bushings shall not be nested.

6.14.3 Left and right hand threads

A pipe fitting containing both left- and right-hand threads, a thread protector, or running threads shall not be used.

6.14.4 Union or combination of fittings

A union or a combination of fittings designed and intended to act as a swing joint shall not be used where piping is concealed.

6.14.5 Field bending

Piping shall not be field bent.

6.14.6 Electrical

Piping or tubing shall not be used for an electrical ground. An electric circuit shall not utilize piping or tubing in lieu of wiring, except for a low-voltage control circuit, ignition circuit, or electronic flame-detection device circuit incorporated as part of an appliance.

6.14.7 Fittings

A close nipple, a street elbow, or a street tee shall not be used in a piping system.

6.14.8 Meter connection

CSST or copper tubing shall not be used to connect to a meter, unless the meter assembly is independently supported.

6.15 Underground piping and tubing

6.15.1 Prohibition

Piping having a nominal diameter of less than NPS 1/2 shall not be used underground.

6.15.2 Piping connection

Underground piping systems shall be joined or connected by welding, approved mechanical compression, or press-connect fittings.

6.15.3 Tubing connection

Underground tubing systems shall be joined or connected by brazing, approved mechanical compression, or approved flared or press-connect fittings.

6.15.4 Location depth

Piping and tubing shall be located neither less than 15 in (400 mm) underground nor less than 24 in (610 mm) under a commercial driveway or parking lot, except when it rises above ground at the point of supply to either a building or an outdoor appliance. Additional depth of cover shall be required where the piping is located in areas where physical damage is likely to occur, such as farm operations.

6.15.5 Trench

A trench for underground piping or tubing shall be properly graded to prevent a sag in the piping or tubing.

6.15.6 Backfill

Backfill material shall be free of sharp objects, large stones, or any other material that can damage the piping or tubing.

6.15.7 Location

Underground piping or tubing shall not pass below a foundation or wall, or under a building.

6.15.8 Building entrance

Piping or tubing entering a building shall rise above grade before entry, unless otherwise permitted by the authority having jurisdiction.

6.15.9 Outside wall

A watertight seal shall be provided at any point where piping or tubing passes through an outside wall below ground level.

6.15.10 Under pavement

When piping or tubing is laid under pavement and an entry to a building is made above ground level, a sleeve shall be inserted to protect the piping or tubing where it comes through the pavement to permit free movement of the soil and covering without placing strain on the piping or tubing.

6.15.11 Vent pipe

A vent pipe inspection point shall be installed adjacent to a building, either when the entire piping or tubing is covered with paving or when paving extends 25 ft (7.5 m) or more from the building, unless other approved provisions have been made for venting the area surrounding the piping.

6.15.12 Connection to plastic

An approved transition fitting shall be used for connecting piping or tubing of either steel or copper to plastic.

6.15.13 Gas supply to plastic piping or tubing

The gas supply to underground plastic piping or tubing shall be controlled by a shut-off valve situated above ground.

6.15.14 Plastic piping or tubing

Plastic piping or tubing shall be accompanied by a tracing wire or other electronically detectable tracing media.

6.15.15 Common trench

When underground piping or tubing share a common trench with underground electrical systems, a minimum separation of 12 in (300 mm) shall be maintained between the gas system and the electrical system.

6.16 Protection of piping and tubing

6.16.1 Corrosive atmospheres

Outdoor piping or indoor piping and tubing that is exposed to atmospheres that are corrosive to the piping or tubing shall be protected by either painting or coating.

6.16.2 Underground

Piping, tubing, or fittings laid underground shall be protected against corrosion in accordance with good engineering practice or in accordance with the manufacturer's instructions.

6.16.3 Movement

Piping or tubing shall be mounted, braced, and supported to provide for expansion, contraction, jarring, vibration, and settling, and shall be protected against either damage or breakage due to strain, wear, and mechanical impact.

Note: *Moving snow loads and ice on sloped roofs have been known to damage or break pipes.*

6.16.4 Hollow walls or partitions

Tubing run inside hollow walls or partitions within 1.75 in (43 mm) of the surface shall be protected against physical damage and puncture at the joists, studs, and plates by the use of No. 16 USG (1.59 mm) plates or sleeves. This provision shall not apply to tubing that passes directly through walls or partitions.

6.16.5 Appliance doors or covers

Piping or tubing shall be located in a position free from the arc of movement of all appliance doors or covers.

6.16.6 Galvanic corrosion

To avoid galvanic corrosion, metallic piping or tubing shall be installed in such a manner that it is not in contact with any other dissimilar metallic pipe or structure.

6.16.7 Exterior wall above ground

When piping or tubing passes through an exterior wall above ground, it shall be sealed watertight and the portion of piping or tubing that runs through the wall shall be sleeved or double wrapped with a pipe wrap tape.

6.16.8 Masonry or concrete

When piping or tubing passes through masonry or concrete, the portion of piping or tubing that runs through this material shall be sleeved or double wrapped with a pipe wrap tape.

6.16.9 Sleeve use

A sleeve shall enclose the entire length of piping or tubing that passes through an exterior wall to an unheated, inaccessible building element, sealed watertight and the portion of piping or tubing that runs through the sleeve shall be double wrapped with a pipe wrap tape.

6.16.10 Sleeve material

When piping or tubing is run in a sleeve, the sleeve shall be of such material and so installed as to protect the piping or tubing from damage and galvanic action.

6.16.11 Plastic material

Care shall be exercised to protect plastic materials from excessive heat and harmful chemicals.

6.16.12 Plastic pipe and tubing support

Plastic pipe and tubing shall be adequately supported during storage.

6.16.13 Plastic direct sunlight exposure

Plastic pipe and tubing shall be protected from exposure to direct sunlight.

6.16.14 CSST

CSST and fittings shall be protected against physical damage in accordance with the manufacturer's certified installation instructions and with this Code.

6.16.15 Vehicle protection

The portions of gas piping systems installed above grade in locations that do not afford protection from damage from vehicles on any street, highway, avenue, alley, or a parking lot shall be protected by posts or guardrails in compliance with Clause [6.16.16](#) unless otherwise approved by the authority having jurisdiction.

6.16.16 Vehicle protection requirements

Gas piping systems shall be protected from vehicular damage by one of the following means:

- a) posts that
 - i) are not less than 12 in (300 mm) from the riser, regulator, or equipment;
 - ii) are spaced not more than 42 in (1050 mm) apart;
 - iii) are buried not less than 36 in (900 mm) below grade;
 - iv) extend at least 30 in (760 mm) above grade; and
 - v) are one of the following:
 - 1) 4 in (100 mm) capped steel pipe;
 - 2) 4 in (100 mm) tubing filled with concrete;
 - 3) 8 in (200 mm) pressure-treated wood, either square or round; or
 - 4) 6 in (150 mm) minimum dimension reinforced concrete; or
- b) guardrails that are
 - i) not less than 12 in (300 mm) from the riser, regulator, or equipment;
 - ii) of the steel deep beam type, 12 in (300 mm);
 - iii) supported by 6 in (150 mm) minimum pressure-treated wooden posts, either square or round, located not more than 42 in (1050 mm) apart, centre to centre, and buried not less than 36 in (900 mm) below grade; and
 - iv) located so that the top of the beam is not less than 24 in (610 mm) nor more than 30 in (760 mm) above grade.

6.17 Identification of piping or tubing

6.17.1 General

At every care or detention occupancy, commercial, industrial, and assembly building, piping or tubing shall be identified by one of the following:

- a) the entire piping or tubing system shall be painted yellow;
- b) the piping or tubing system shall be provided with yellow banding that has a minimum width of 1 in (25 mm); or
- c) the piping or tubing system shall be labelled or marked "GAS"* or "PROPANE"†, as applicable, utilizing yellow labels or markings.

When identified in accordance with Item b) or c), the identification intervals shall not exceed 20 ft (6 m).

* The equivalent French wording is "GAZ".

† The equivalent French wording is "PROPANE".

6.17.2 Pressure in excess of 14 in w.c. (3.5 kPa)

At every care or detention occupancy, commercial, industrial, and assembly building, where the piping or tubing pressure is in excess of 14 in w.c. (3.5 kPa), both the piping or tubing and the pressure shall be identified at the following locations:

- a) shut-off valves; and
- b) wall, ceiling, and floor penetrations.

6.17.3 Residential buildings

Tubing systems for residential buildings shall be identified in accordance with Clause [6.17.1](#), except that identification intervals shall not exceed 6 ft (2 m) along their entire length.

6.17.4 Two or more gas meters

Every piping or tubing system that enters a building that has two or more gas meters shall be permanently identified with the room number, apartment number, or the area of the building it serves.

6.18 Manual shut-off valves

6.18.1 Certification or approval

A manual shut-off valve shall be certified to CSA 3.11, CSA 3.16, or CSA/ANSI Z21.15/CSA 9.1, or approved for use with gas, and it shall not be subjected to either a temperature or a pressure outside of its certified rating range.

6.18.2 Appliance manual shut-off

A readily accessible manual shut-off valve for each appliance shall be installed

- a) in either the drop or riser as close as possible to the valve train of a commercial- and industrial-type appliance;
- b) in either the drop or riser of a residential appliance;
- c) in the horizontal piping between the drop or riser and the appliance valve train. It may be the same size as the appliance connection when located within 2 ft (600 mm) of the appliance; or
- d) within a maximum of 50 ft (15 m) of the individual residential appliance that it serves and shall be identified as to the appliance it serves by means of either
 - i) a metal tag attached to the valve; or
 - ii) a clear legible sign displayed in a permanent manner adjacent to the valve.

6.18.3 Appliance manual shut-off exception

The requirements for an individual manual shut-off valve specified in Clause [6.18.2](#) may be waived when a readily accessible single manual shut-off valve is

- a) installed for commercial cooking appliances manifolded in line; or
- b) installed in common supply piping to more than one direct-vent room heater that is part of a heating system in a dwelling unit, provided that it is less than 50 ft (15 m) away from each appliance.

6.18.4 Type and size

A readily accessible manual shut-off valve shall be of either the ball, eccentric, or lubricated-plug-type where

- a) the piping is larger than NPS 1;
- b) the tubing is 1 in (25.4 mm) OD or larger; or

- c) the pressure exceeds 0.5 psig (3.5 kPa).

6.18.5 Several piping systems

When a shut-off valve controls several piping systems, it shall be

- a) readily accessible for operation at all times;
- b) provided with an installed handle;
- c) installed to provide protection from damage; and
- d) clearly marked with an enamelled metal, substantial fibre, or other permanent tag, so that the piping system it controls can be readily identified.

6.18.6 Spring-loaded valve

A spring-loaded valve shall be installed in a manner that will prevent its plug from being accidentally lifted off the valve seat.

6.18.7 Quick-disconnect

A quick-disconnect device shall not be substituted for a manual shut-off valve.

6.18.8 Adjoining buildings

Piping or tubing that extends from one building to another shall have a shut-off valve at the point of exit from the first building and one at the point of entry to the adjoining building.

6.18.9 Propane

Where propane is distributed from a storage tank to more than one riser by an underground system, a valve shall be installed on each riser.

6.18.10 Underground propane storage tank

Where a building is supplied from an underground propane storage tank, a manual gas shutoff valve shall be installed and readily accessible in the gas supply piping system or riser before it enters the building(s). It shall be identified on the piping or tubing to its use "Propane" by a prominent displayed sign immediately visible upstream of the valve and having red lettering of a minimum 1 in (25 mm) high on a white background.

6.18.11 Classroom, laboratory, or similar facility

When multiple outlets are installed in a classroom, laboratory, or similar facility, they shall be controlled by a clearly identified master shut-off valve in a readily accessible location within that room.

6.18.12 Classroom, laboratory, or similar facility out of service

When a facility as described in Clause [6.18.11](#) is taken out of service, the supply of gas to the facility shall be either plugged or capped.

6.19 Manual-reset valves

A safety control device external to a piping system shall include an automatic valve of the manual-reset type to shut off the gas supply.

6.20 Gas hose and fittings

6.20.1 Vented appliance

Except as required in Clause [7.23.3](#), a gas hose connection to a vented appliance shall be prohibited.

6.20.2 Vibration

A gas hose may be used with an unvented appliance when such an appliance is mobile during operation, is portable, or requires isolation from vibration.

6.20.3 Installation

When a gas hose is used

- a) for permanent installation, it shall not exceed 10 ft (3 m) in length and shall neither extend from one room to another nor pass through any wall, partition, ceiling, or floor;
- b) for a portable appliance, it shall not exceed 30 ft (9.5 m) unless otherwise stated in the appliance standard;
- c) for connecting a construction heater, it shall be neither less than 15 ft (4.5 m) nor more than 75 ft (22.9 m) in total length;
- d) for cutting, welding, and preheating equipment, it shall not exceed 100 ft (30 m) in length;
- e) it shall not be subject to a temperature in excess of 125 °F (50 °C);
- f) a slip-on end shall not be permitted;
- g) the shut-off valve for the appliance shall be in the gas supply piping or tubing and is immediately upstream to the gas hose;
- h) it shall be protected from damage;
- i) the handle or hand wheel of a shut-off valve on an independent connection shall not be closer than 6 in (150 mm) from the handle or hand wheel of any other shut-off valve;
- j) a shut-off valve shall not be placed at floor level or in any other location where it can be turned on by accident; and
- k) from a wall outlet, it shall be located where the passage of persons, vehicles, or equipment across the gas hose is at a minimum.

6.20.4 Replacement

When a sign of wear, deterioration, or other damage is apparent in the reinforcement material of a gas hose, the gas hose shall be replaced immediately.

6.20.5 Metallic gas hose

A metallic gas hose

- a) may be used to connect an appliance in commercial, industrial, or process applications when vibration, expansion, contraction, or other circumstances of an appliance installation warrant its use;
- b) shall not be used in a concealed location;
- c) shall neither extend from one room to another nor pass through any wall, partition, ceiling, or floor;
- d) when used to connect an appliance to rigid supply piping, shall have a shut-off valve in the piping immediately upstream of the metallic gas hose; and
- e) shall comply with CGA CR96-001 or ANSI/CAN/UL 536.

6.20.6 Interconnected equipment

When tanks or pieces of equipment are interconnected, provisions shall be made to compensate for vibration and differential settling of the tanks, equipment, and interconnecting piping. Where a gas hose is used for this purpose, it shall be a metallic gas hose complying with ANSI/CAN/UL 536 or a Type II or Type III gas hose complying with CSA 8.1.

6.20.7 Transferring liquid propane

When a gas hose is used for transferring liquid propane from one container to another, the gas hose shall be protected by a hydrostatic relief valve located between the shut-off valves on the gas hose.

6.20.8 Propane applications

In propane applications, a gas hose shall not be run from a tank and/or vapourizer that is installed outdoors to an appliance located within a building, except where the gas hose connects to an appliance used for temporary construction-heating purposes.

6.21 Gas connectors

6.21.1 Certification for gas connectors

A gas connector shall be certified to CSA/ANSI Z21.24/CSA 6.10, ANSI Z21.69/CSA 6.16, ANSI Z21.75/CSA 6.27, CSA/ANSI Z21.54/CSA 8.4, or CSA/ANSI Z21.101/CSA 8.5.

6.21.2 Installation for gas connectors

A gas connector shall

- a) be protected from damage;
- b) not pass through a wall, floor, ceiling, or partition;
- c) be connected to rigid piping or tubing located in the same area as the appliance; and
- d) comply with Clauses [4.5.2](#) and [4.9.1](#).

6.21.3 Corrugated metal gas connector

Except as specified in Clause [7.23.3](#), a corrugated metal gas connector certified to CSA/ANSI Z21.24/CSA 6.10 may be used to connect

- a) a range, refrigerator, clothes dryer, or built-in counter appliance to piping or tubing, provided that the connector length does not exceed 6 ft (2 m);
- b) a suspended appliance to piping or tubing, provided that the gas connector length does not exceed 2 ft (600 mm); or
- c) a decorative appliance, a room heater, or a direct-vent wall furnace to piping or tubing, except when the appliance, heater, or furnace is installed as a free-standing unit, provided that the gas connector length does not exceed 2 ft (600 mm).

6.21.4 Additional application

A gas connector certified to ANSI Z21.101/CSA 8.5 may be used to connect an appliance such as a range, refrigerator, or clothes dryer to the building piping.

6.21.5 Large gas utilization equipment

Commercial cooking appliances certified for use with casters or otherwise subject to movement during cleaning and other large gas utilization equipment that can be moved shall be connected by a gas connector that is certified to either ANSI Z21.69/CSA 6.16 or CSA/ANSI Z21.101/CSA 8.5.

6.21.6 Commercial cooking appliance

When the gas connector described in Clause [6.21.5](#) is used with a commercial cooking appliance installed on wheels or rollers, a noncombustible restraining device shall be provided to protect the gas connector, and the installation shall be in accordance with Clause [7.32.4](#).

6.21.7 Vented appliance

A gas connector not exceeding 2 ft (600 mm) may be used on a vented appliance, such as a free-standing space heater, provided that the appliance is secured to prevent dislodgement of the vent.

6.21.8 Outdoor gas appliance and manufactured homes

A gas connector certified to ANSI Z21.75/CSA 6.27 may be used to connect

- a) an appliance for outdoor use that is not frequently moved after installation; or
- b) a mobile home that is not installed on a permanent foundation.

These connectors are not intended for use with wheeled, caster mounted, or portable appliances.

6.21.9 Portable outdoor appliances

A gas connector certified to CSA/ANSI Z21.54/CSA 8.4 may be used to connect a portable outdoor appliance in an unconcealed space.

6.22 Testing of piping, tubing, hose, and fittings

6.22.1 Requirements

The source of test pressure shall be isolated while the piping or tubing system is under test, and the system shall retain the test pressure for the minimum duration required in Table [6.3](#) without showing any drop in pressure.

6.22.2 Before connection requirements

Except as required in Clause [6.22.5](#), before an appliance is connected to, and before fuel gas is introduced to, a new gas piping system or a replacement, modification, or addition to an in-service gas piping system, the new, replacement, or modified gas piping system not yet introduced to the fuel gas shall be pressure tested using either air, inert gas, or carbon dioxide in the following manner:

- a) Appliance shut-off valves not rated for the test pressure being used and meters and pressure regulators shall not be connected to the piping or tubing system under test.
- b) The test pressure shall be measured by either a pressure gauge or equivalent device and, if a gauge is used, the minimum diameter shall be 3 in (75 mm) and the maximum range shall exceed the test pressure by at least 15% but not more than 300%. The pressure gauge or equivalent device shall be calibrated to read in increments of not more than 2 psig (14 kPa) or 2% of the maximum dial reading of the pressure gauge, whichever is less.
- c) A pressure recorder when used for this test shall be calibrated to the requirements of Item b).
- d) The pressure and duration of the test shall be in accordance with Table [6.3](#).

Table 6.3
Pressure test requirements
 (See Clauses [6.22.1](#) and [6.22.2](#).)

Working pressure, psig (kPa)	Diameter of pipe or tubing	Length of pipe or tubing, ft (m)	Test pressure, psig (kPa)	Test duration, min
Up to and including 2 (14)	All sizes	200 (60) or less	15 (100)	15
Up to and including 2 (14)	All sizes	> 200 (60)	15 (100)	60
Over 2 (14) but not more than 33 (230)	All sizes	200 (60) or less	50 (340)	60
Over 2 (14) but not more than 33 (230)	All sizes	> 200 (60)	50 (340)	180
Over 33 (230)*	All sizes	All lengths	1.5 times the maximum operating pressure	180
All welded pipe	All sizes	All lengths	The greater of 50 psig (340 kPa) or 1.5 times the maximum operating pressure	180

* Propane maximum operating pressure is defined as

- 250 psig (1725 kPa) for piping and tubing operating at container pressure;
- 350 psig (2400 kPa) when connected to the outlet of a pump or compressor; or
- 375 psig (2570 kPa) minimum or the setting of the hydrostatic relief valve in piping that can contain liquid propane, that can be isolated by valves, and that requires hydrostatic relief valves as specified in [Clause 5.4.1](#) of this Code or [Clause 5.6.1](#) of CSA B149.2.

Notes:

- These test pressures and test durations are minimum requirements. Circumstances can require test pressures and test durations in excess of those shown in the table.
- All wrapped and/or factory-coated piping and tubing systems (except for CSST and copper) of all sizes and lengths shall be tested at a minimum pressure of 100 psig (700 kPa) in accordance with the time duration in the table.

6.22.3 After connection requirements

Before an appliance is first activated after its installation or after gas is reintroduced to part or all of a gas piping system after an interruption of service, one or more valves shall be selected to isolate the necessary portion of the gas piping system and all of the gas piping system downstream of these valves, through to the inlet of the appliance's valve train, shall be tested for leakage in the following manner:

- Before turning on the gas for the test, a check shall be made to ensure that any opening from which gas can escape is closed.
- Immediately after allowing the gas into the piping or tubing system, a test shall be made to determine that no gas is escaping by carefully watching the test dial of the meter or by using a pressure gauge or equivalent device.

- c) Where a meter is not provided, the pressure shall be measured with either a pressure gauge or equivalent device calibrated to read in increments not greater than those specified in Clause [6.22.2](#) b), with the following exceptions:
 - i) for a system where the working pressure is 0.5 psig (3.5 kPa) or less, the pressure gauge or equivalent device shall be calibrated to read in increments of not greater than 1 in w.c. (250 Pa); and
 - ii) for a system where the working pressure exceeds 0.5 psig (3.5 kPa) but does not exceed 5 psig (35 kPa), the pressure gauge or equivalent device shall be calibrated to read in increments of not greater than 2 in w.c. of pressure (0.5 kPa).
- d) The test described in Item b) shall be of a 10 min duration.
- e) When leakage is observed, the gas supply to the appliance shall be shut off until the necessary repairs have been made.

6.22.4 Valve train leak test

Subsequent to a leak test prescribed in Clause [6.22.3](#), each appliance connection, valve, valve train, and system component shall be checked for leaks at the threaded or flanged connection to gas piping, tubing, or hose, while under normal operating pressure with either a liquid solution or a leak-detection device. See Annex [K](#).

6.22.5 Addition to existing piping or tubing

An addition to an existing piping or tubing system shall be tested as an individual system in accordance with Clause [6.22](#), except that

- a) where the addition is 20 ft (6 m) or less in length and the normal working pressure is less than 0.5 psig (3.5 kPa), the addition shall be leak tested in accordance with Clause [6.22.3](#) e); and
- b) where the addition is accomplished using a welded tie-in and the new system has been tested in compliance with Clause [6.22](#), the tie-in weld shall be tested in accordance with Clause [6.22.3](#) e).

6.22.6 Enclosed or concealed systems

When any part of a piping or tubing system is to be enclosed or concealed, the tests specified in Clause [6.22.2](#) shall precede the work of closing in.

6.23 Purging of gas piping systems

6.23.1 Purging outdoors

A gas piping system shall be purged to the outdoors using approved engineering practices or in accordance with Clauses [6.23.2](#) through [6.23.4](#) where it meets either of the following:

- a) the gas pressure in the gas piping system is greater than 2 psig (14 kPa); or
- b) the gas piping system being purged contains one or more sections of pipe, tubing, or gas hose meeting the size and length described in Table [6.4](#).

Table 6.4
Gas piping systems which require purging
 (See Clauses [6.23.1](#) to [6.23.3](#).)

Nominal piping, tubing, or gas hose size, in	Length of piping, tubing, or gas hose, ft (m)
≥ to 2-1/2 but < 3	> 50 (15.2)
≥ to 3 but < 4	> 30 (9)
≥ to 4 but < 6	> 15 (5)
≥ to 6 but < 8	> 10 (3)
≥ 8	Any length

Note: Corrugated stainless steel tubing of size 2 in or less are to be sized accordingly to the same as piping.

6.23.2 Containing air

When a gas piping system meeting the description of Table [6.4](#) and containing air is placed in service, it shall be first purged with an inert gas and then purged with fuel gas in accordance with Clause [6.23.4](#).

6.23.3 Removal of existing gas piping system

Where an existing gas piping system is removed from service for the purpose of repair, alteration, or abandonment, the section that is opened shall be isolated from the gas supply and the line purged in accordance Clause [6.23.4](#). Where the piping, tubing, or gas hose meets the description of Table [6.4](#), the residual fuel gas in the piping, tubing, or gas hose shall be purged with an inert gas.

6.23.4 Purge discharging and operations

The open end of a gas piping system being purged shall be discharged directly to an outdoor location. Purging operations shall comply with all of the following requirements:

- a) During purging operations, the open point of discharge shall be continuously attended by a qualified person.
- b) The point of discharge shall be directly controlled during the purging operation by means of a quarter-turn shut-off valve having an attached operating handle within 5 ft (1.5 m) of the open end. No other valve shall be used to control or terminate the purge. The person doing the purging shall be in direct control of the purging gas supply by means of this valve, and the point of discharge shall be monitored by a combustible gas indicator. The purge line shall not be smaller than NPS 1/2.
- c) The point of discharge shall be located
 - i) at least 10 ft (3 m) from a source of ignition, and precautions shall be taken to either remove or shut off any source of ignition prior to commencing the purge;
 - ii) at least 10 ft (3 m) from a building or a building opening; and
 - iii) at least 25 ft (7.5 m) from a mechanical air intake opening.
- d) Purging operations introducing fuel gas shall be maintained by a continuously burning flame at the burner port(s) until a stable gas flame is established.
- e) The device used to ignite the pressure vented or purged gas shall be as follows:
 - i) a burner, not located in a combustion chamber, with a continuous or intermittent pilot;
 - ii) two independent burners, not located in a combustion chamber, which reliably ignite one from the other; or
 - iii) an approved burner for such purpose.

- f) Persons not involved in the purging operations and persons smoking shall be prohibited from all areas within 10 ft (3 m) of the point of discharge.

6.23.5 Flaring

When flaring is used to purge a gas piping system, an approved purge burner shall be used.

6.23.6 Additional purging conditions

A gas piping system not meeting the conditions specified in Clause [6.23.1](#) may be purged either

- a) to the outdoors, in accordance with Clause [6.23.4](#);
- b) to the indoors, in accordance with Clause [6.23.7](#); or
- c) in accordance with good engineering practice.

6.23.7 Purging indoors

When the conditions in Clause [6.23.6](#) allow it, a gas piping system shall be purged in an indoor space only in accordance with one of the following options:

- a) the gas in the piping system shall be ignited at an appliance having an input rating up to and including 400 000 Btu/h (120 kW) with a readily accessible burner not located in a combustion chamber. Such burner shall be provided with a continuous source of ignition and a continuously burning flame shall be maintained at the burner port(s) until a stable gas flame is established; or
- b) for an appliance not equipped with a continuous pilot, in accordance with the procedure described in Annex [H](#).

When there is an open point of discharge during the purge, it shall be continuously attended by a qualified person, and the point of discharge shall be monitored by a combustible gas indicator.

6.23.8 Appliance purging

When all gas piping systems have been purged and placed in service, the appliance or equipment piping shall be purged prior to being placed in service and the pilot lighted.

6.24 Purging gas from a piping or tubing system

Carbon dioxide or nitrogen, or a mixture of these, or air shall be used when purging gas from a piping or tubing system for the purpose of repair, alteration, or abandonment. The applicable procedures in Clause [6.23](#) shall apply.

6.25 Rooftop gas piping systems

6.25.1 Piping support

Piping on a rooftop may be supported with treated wood blocks or material having characteristics at least equivalent to treated wood blocks and protection against outdoor exposure. The support spacing of piping NPS 1 and greater shall comply with Table [6.2](#), and support shall be provided for every fitting.

Piping less than NPS 1 shall be supported

- a) vertically according to Table [6.2](#); and
- b) horizontally every 4 ft (1.2 m).

6.25.2 Tubing support

The support spacing on a rooftop for all tubing shall be supported

- a) vertically according to Table [6.2](#); and
- b) continuously with treated wood and planks when it is laid horizontally.

6.25.3 Expansion and flexibility

Piping and tubing shall be installed in accordance with Clauses [6.16.1](#) and [6.16.3](#), and means for expansion shall be provided. See Annex [G](#).

6.26 Inspection

6.26.1 Plastic piping and tubing

Plastic piping and tubing shall be inspected before and after installation for defects such as cuts, scratches, and gouges. Damaged cylindrical pieces shall be cut out and replaced. Inspection shall be adequate to confirm that sound joints have been made.

6.26.2 Plastic piping and tubing joints

Joints in plastic piping and tubing shall be visually checked for evidence of poor bonding. Where inspection reveals defective joints, they shall be cut out and replaced.

6.27 Highway vehicles, recreational vehicles, mobile outdoor food service units, wash-mobiles, and mobile homes

6.27.1 Hose connector

A hose connector rated at not less than 350 psig (2500 kPa) shall be provided between

- a) the cylinder valve outlet and the inlet of the regulator when the regulator is rigidly mounted on a support bracket; or
- b) the regulator outlet and the main propane piping or tubing when the regulator is rigidly fixed to the cylinder valve outlet.

6.27.2 Propane piping

A propane line supplying a furnace in a mobile home may be routed through a combustion-air or outdoor air make-up opening in the furnace base, provided that

- a) the propane piping through the furnace base consists of a single piece; and
- b) any connection to this piping is made above the level of the top of the furnace base and below the lowest level of the floor or furnace base extension.

6.27.3 Shut-off valve

An individual shut-off valve shall not be required for each appliance in a recreational vehicle.

6.27.4 Hose

Hose shall not be used in lieu of piping or tubing but may be used in conjunction with piping or tubing.

6.27.5 Connections

Except for the final connection of piping, tubing, or hose to an appliance or for connection to a valve as described in Clause [4.19.2](#), there shall be no connections in the piping or tubing within a vehicle.

Note: This requirement does not apply to the piping systems installed at the factory and certified in accordance with CSA Z240 RV.

7 Installation of specific types of appliances

7.1 Boilers

7.1.1 Provincial requirement

A boiler shall conform to the requirements of the provincial boiler and pressure vessel regulations as applicable.

7.1.2 Installation location

A boiler shall be installed on a

- a) firm and level base; and
- b) noncombustible floor or support, except where the boiler is
 - i) certified for installation on a combustible floor;
 - ii) installed using the boiler manufacturer's certified special base; or
 - iii) installed as specified in Clause [4.13.3](#).

7.1.3 Clearances

Except where permitted by Clause [4.13.2](#), a boiler shall be installed with the following minimum clearances from combustible material:

- a) vertical — 18 in (450 mm);
- b) sides and rear — 18 in (450 mm); and
- c) front — 48 in (1200 mm).

See also Clause [4.14.2](#).

7.2 Generators, compressors/pressure boosters, engines, and turbines

7.2.1 General requirements

7.2.1.1 Certification body acceptance

The installation of appliances and associated equipment covered by Clause [7.2](#) shall comply with this Code, applicable Standards, manufacturer's instructions, and local requirements, including fire regulations, building codes, and zoning requirements.

7.2.1.2 Installation access

Appliances and associated equipment covered by Clause [7.2](#) shall be installed such that all service, maintenance, inspection, and repair as required by the manufacturer can be accomplished.

7.2.1.3 Venting and air minimum requirements

Venting and air-supply requirements, as a minimum, shall be provided in accordance with applicable clauses of Clause [8](#) (excluding Clause [8.5.4](#)) or designed in accordance with any special requirements contained in the manufacturer's instructions.

This requirement shall apply to engines and turbines of all inputs, and the ventilation openings shall be arranged to minimize short-circuiting of the exhaust and make-up airstreams.

7.2.1.4 Housing codes

Rooms or enclosures housing appliances and associated equipment covered by Clause [7.2](#) shall be constructed in accordance with national or local building codes.

7.2.1.5 Housing ventilation

Rooms or structures housing appliances and associated equipment covered by Clause [7.2](#) shall have ventilation designed to minimize the possibility of hazardous accumulation of flammable vapours or gases during appliance or equipment operation or when shut down.

7.2.1.6 Foundation and framework

Appliances and associated equipment covered by Clause [7.2](#) shall be installed on a firm level foundation, set on a suitable framework supplied by the manufacturer, or field fabricated in accordance with the manufacturer's instructions.

7.2.1.7 Rooftop installation

Appliances and associated equipment covered by Clause [7.2](#) that are located on rooftops shall comply with Clauses [4.14.5](#), [4.14.6](#), and [6.25](#), and provisions shall be made for oil spill containment.

7.2.1.8 Installation protection

The installation of appliances covered by Clause [7.2](#) and associated equipment shall be protected by approved means against impact, ice build-up, flooding, and blockage of ventilation.

7.2.1.9 Installation protection

The ventilation required by Clause [7.2.1.5](#) shall be interconnected with a gas detector that

- a) is installed in accordance with the detector manufacturer's instructions for the type of gas;
- b) is set to activate at gas detection levels at and above one-fifth of the lower limit of flammability;
- c) upon activation, produces an audible and visual alarm;
- d) is interlocked with the mechanical ventilation system; and
- e) is interlocked to shut off the appliance.

7.2.2 Compressors/pressure boosters

7.2.2.1 Compliance and approval

A compressor/pressure booster shall comply with Clause 17.2 of CSA B149.3 and shall be approved by the authority having jurisdiction.

7.2.2.2 Ventilation

Ventilation air shall be provided to the space occupied by a pressure booster to prevent any accumulation of gas in the event that leakage occurs.

7.2.2.3 Installation requirements

The following requirements shall apply to all piping, tubing, hose, compressor/pressure boosters, and components operating at an outlet pressure higher than those permitted by Table [5.1](#):

- a) the installation of all components shall be approved by the authority having jurisdiction or shall be tested in accordance with the manufacturer's instructions;
- b) the service line to the inlet of a compressor shall be tested in accordance with the requirements of Clause [6.22](#); and

- c) discharge piping from a remote compressor that is not directly connected by the manufacturer from the outlet of a compressor shall be installed in accordance with the requirements of the applicable clauses of CSA B51 for pressure piping.

7.2.2.4 Installation location

The compressor shall be installed outdoors unless approved and labelled for indoor installation.

7.2.2.5 Fittings

The number of fittings used in a supply line, discharge line, or hose shall be minimized to reduce the possibility of leakage.

7.2.2.6 Discharge piping

The discharge piping shall be supported in accordance with the manufacturer's instructions or the requirements of CSA B51 for pressure piping.

7.2.2.7 Pressure booster

It is not necessary to meet the requirements of Clause [7.2.2.2](#) where a hermetically sealed pressure booster is installed.

7.2.2.8 Pressure boosting compressors

Hermetically sealed compressors used for pressure boosting for the purpose of supplying burners, torches, or cylinder-filling applications shall comply with the requirements of Clause [7.2](#) and shall not have a capacity greater than 500 scf/h (14.15 m³/h).

7.2.2.9 Cylinder filling

Cylinder-filling applications are covered in Clause [9](#).

7.2.3 Emergency generators

7.2.3.1 Independent piping

Piping or tubing serving a generator that provides power for safety purposes shall be independent of any other gas supply to the building or any other generator that provides power for safety purposes, and shall be provided with a manual valve identified with a permanent sign in accordance with CSA C282 as applicable. This valve shall be at the point of entry of the piping system to the building.

7.2.3.2 Dedicated gas supply

When a gas supply to a generator is required for safety purposes, the supply of gas to the generator shall be arranged such that the gas supply to the piping serving other appliances can be shut off without interrupting the gas supply to the generator in accordance with CSA C282 as applicable. The valve used for this purpose shall be at the point of entry of the piping system to the building.

7.2.3.3 Overpressure protection

An overpressure protection device used in conjunction with a pressure regulator on piping serving a generator that provides emergency electrical power supply to a building shall be either a monitoring

regulator or an overpressure relief device only. Overpressure shut-off devices are not permitted in this application.

Note: Refer to CSA C282 for additional piping requirements for fuel systems serving emergency power supplies.

7.2.4 Non-motive engines and turbines

7.2.4.1 Certification

An engine or turbine shall be certified to ANSI/CAN UL/ULC 2200 or comply with CSA B149.3.

7.2.4.2 Supply piping sizing

Supply piping to a non-motive engine or turbine shall be sized according to both the maximum rate of gas consumption and the requirements of Clause 6. When precise information regarding the maximum rate of gas consumption is not available from the manufacturer, the size of piping shall be computed for normal operation. Normal operation shall be considered the provision of 10 000 Btu/h (3 kW) per brake horsepower for 4-cycle engines and 13 000 Btu/h (4 kW) per brake horsepower for 2-cycle engines. When either overloads or high starting loads are likely to be encountered, the size of piping shall be computed on a basis proportional to the increased input required.

7.2.4.3 Exhaust gas piping

Engine or turbine exhaust gases shall be piped so they can neither be trapped nor be drawn into a building through a window, door, or other opening.

7.2.4.4 Venting

Pressure regulators and overpressure relief devices installed in the valve train of an engine or turbine shall be vented in accordance with Clause 5.5, with outdoor vent terminations in accordance with Clause 5.6.

The venting of combination regulating and vapourizing equipment shall not be required where a solenoid valve is installed upstream of this equipment.

7.2.5 Additional requirements for gas engines and turbines in buildings

7.2.5.1 Ventilation location

A room containing an engine or turbine shall be ventilated

- a) at the floor level for propane fuel; or
- b) as close as practicable to the ceiling for natural gas.

7.2.5.2 Ventilation operation

When a propane-fuelled engine or turbine is installed within a room that is below grade, mechanical exhaust ventilation shall be provided and operated continuously, and approval of the authority having jurisdiction shall be obtained. The mechanical ventilation shall be electrically interlocked to shut down the engine in the event of failure of the exhaust system, or means shall be provided to purge the room at the floor level before the engine is started. The mechanical ventilation shall be in operation when the engine is running, and the room shall be ventilated before and during any repairs to the engine.

7.2.5.3 Fire protection

An engine or turbine shall be installed in a room that has a minimum 2 h fire separation from the remainder of the building, and the room shall have a fire door having a 1.5 h fire protection rating on every opening that communicates with other sections of the building. The door shall not have a glass panel or vent and shall be of the swinging automatic-closing type, gasketed on all sides, including the top and bottom, to prevent natural gas or propane from entering other sections of the building.

7.2.5.4 Exhaust gas discharge

Exhaust gases shall be discharged from the building to the outdoors in a manner that will not create a fire or any other hazard. Clearances to combustible materials shall be acceptable for future and potential uses and applications of the engines or turbines and shall adhere to the following requirements:

- a) exhaust systems with exhaust gases temperatures of less than 1400 °F (760 °C) shall be in compliance with Clauses [7.2.5.7](#) to [7.2.5.10](#); or
- b) exhaust systems with exhaust gases temperatures of 1400 °F (760 °C) or greater shall be constructed in accordance with national or local building codes.

7.2.5.5 Exhaust gas piping

Exhaust gases shall be piped

- a) by the most direct route to an outdoor location where they can neither be trapped nor be drawn into a building through a window, door, or other opening;
- b) to a chimney conforming to Clause [7.2.5.8](#) b); or
- c) in accordance with the engine or turbine manufacturer's instructions.

7.2.5.6 Exhaust gas discharge

An exhaust pipe shall not discharge into

- a) a gas vent;
- b) a chimney that has a temperature rating less than the maximum temperature of the exhaust gas; or
- c) a chimney that serves another appliance.

7.2.5.7 Exhaust pipe termination

An exhaust pipe passing through a combustible roof shall be insulated at the point of passage by a ventilated metal thimble that shall extend not less than 9 in (225 mm) above and 9 in (225 mm) below the roof construction and that is 12 in (300 mm) larger in diameter than the pipe. If a double-ventilated metal thimble is used, the same requirements shall apply, except that the thimble shall be at least 6 in (150 mm) larger in diameter than the exhaust pipe.

7.2.5.8 Exhaust pipe guarding

An exhaust pipe that passes through a combustible wall or partition shall be guarded at the point of passage by either

- a) a double-ventilated metal thimble not less than 12 in (300 mm) larger in diameter than the exhaust pipe; or
- b) a metal or burned fire clay thimble built into either the brickwork or another noncombustible material. The brickwork or other noncombustible material shall provide not less than 8 in (200 mm) of insulation between the thimble and the combustible material.

7.2.5.9 Exhaust piping limitations

An exhaust pipe shall not be installed in a floor, ceiling, attic, or concealed space but may pass through such spaces, provided that it is installed inside either a masonry chimney or a factory-built chimney of appropriate temperature rating and provided that the chimney does not serve any other type of appliance and meets the requirements of Clause [8.12.1](#).

7.2.5.10 Exhaust pipe clearances

An exhaust pipe shall have a clearance of at least 9 in (225 mm) to adjacent combustible materials, except as specified in Clauses [7.2.5.7](#) to [7.2.5.9](#).

7.3 Carbon dioxide generators

7.3.1 Certification

A generator used in a greenhouse shall be certified for the application.

7.3.2 Combustion air

A generator used in a produce storage area shall take its combustion air from outside the storage area.

7.3.3 Carbon dioxide production

A generator installed for the production of carbon dioxide in a greenhouse may take its combustion air from inside the greenhouse when the rate of combustion does not exceed 20 Btu/h/ft³ (0.2 kW/m³) of greenhouse volume, the concentration of carbon dioxide does not exceed 5000 parts per million (ppm), and the concentration of carbon monoxide in the atmosphere does not exceed 35 ppm. The concentration of carbon dioxide and carbon monoxide shall be verified upon initial start-up.

7.4 Commercial-type clothes dryers

7.4.1 Hot surface protection

A dryer used in a laundromat-type of installation shall be installed such that access to the top of the dryer is screened or otherwise protected to prevent any material coming into contact with the hot surface.

7.4.2 Minimum clearances

Except where permitted by Clause [4.13.2](#), a dryer shall have the following minimum clearances from combustible material:

- a) above — 18 in (450 mm);
- b) front — 18 in (450 mm); and
- c) back and sides — 18 in (450 mm).

7.4.3 Flexible exhaust ducting

A UL 2158A certified flexible foil noncombustible-type duct may be used as a transition connection between the dryer exhaust and a rigid moisture duct, if there is no requirement in the dryer installation instructions as for which type of duct material is to be used.

7.4.4 Exhaust terminations

A dryer shall be connected to a metal moisture-exhaust duct that terminates outdoors not less than 3 ft (0.9 m) from any pressure regulator vent termination and not less than 10 ft (3 m) from an outdoor air intake.

7.4.5 Exhaust duct limitations

A moisture-exhaust duct shall not be secured with screws and shall not be connected into any vent connector, vent, or chimney.

7.4.6 Make-up air

Provision shall be made for make-up air to the area where the dryer is installed.

7.4.7 Exhaust duct clearances

A moisture-exhaust duct shall have a clearance of at least 6 in (150 mm) to combustible material but may be installed with a reduced clearance, provided that the combustible material is protected as specified in Table [4.1](#).

7.5 Domestic-type clothes dryers

7.5.1 Exhaust duct requirements

A dryer shall be equipped with a moisture-exhaust duct that terminates outside the building, and the duct shall be constructed of a material that is

- a) noncombustible; or
- b) certified as meeting the requirements for Class 1 air ducts contained in CAN/ULC-S110.

7.5.2 Exhaust duct termination

A moisture-exhaust duct shall not terminate within 3 ft (0.9 m) in any direction of any pressure regulator vent termination or outdoor air intake.

7.5.3 Exhaust duct limitations

A moisture-exhaust duct shall not be secured with screws and shall not be connected into any vent connector, vent, or chimney.

7.5.4 Minimum clearances

Except where permitted by Clause [4.13.2](#), a dryer shall have the following minimum clearances from combustible material:

- a) above — 6 in (150 mm);
- b) front — 24 in (610 mm); and
- c) back and sides — 6 in (150 mm).

7.6 Conversions

7.6.1 Minimum clearances

The minimum clearances from combustible material for a boiler or a furnace converted to gas shall be

- a) for a boiler, as specified in Clause [7.1.3](#);
- b) for a forced-air furnace
 - i) from top (casing, bonnet, or plenum) — 1 in (25 mm);

- ii) the jacket sides and rear — 6 in (150 mm); and
- iii) front — 24 in (610 mm); and
- c) for a gravity furnace
 - i) vertical — 6 in (150 mm);
 - ii) sides and rear — 6 in (150 mm); and
 - iii) front — 24 in (610 mm), except as specified in Clause [4.13.2](#).

7.6.2 Conversion requirements

An appliance to be converted shall be thoroughly cleaned, leak tested, and examined for serviceability. Any unserviceable parts shall be repaired or replaced.

7.6.3 Chimney requirements

When an existing vented appliance is to be converted from a solid or liquid fuel, the chimney shall be examined and shall meet the requirements of Clauses [8.12.2](#) to [8.12.11](#).

7.7 Conversion burners

7.7.1 Conversion requirements

A conversion burner designed by the manufacturer of the appliance to be converted, or a burner compatible with the appliance to be converted, shall be used.

7.7.2 Mobile homes

When an oil-fired furnace in a mobile home is converted to gas, the conversion shall be by means of

- a) the furnace manufacturer's certified burner conversion package; or
- b) a certified conversion burner, provided that
 - i) it is compatible with the furnace and its installation;
 - ii) the basic design of the furnace is not altered; and
 - iii) when adjusted, the burner firing rate does not exceed the furnace manufacturer's specified firing rate.

7.7.3 Flame impingement

A conversion burner shall be correctly positioned and firmly secured to eliminate direct flame impingement on any surface other than a flame spreader.

7.8 Conversion of warm air furnaces

7.8.1 Installation

A revertible flue furnace shall not be converted by the installation of

- a) a natural-draft burner, unless
 - i) the centreline of the flue collar is at least 12 in (300 mm) above the burner port;
 - ii) the flue outlet is extended to permit installation of the draft hood or draft regulator so that the relief opening is at least 12 in (300 mm) above the highest flue pass; and
 - iii) a bypass at least 1 in (25 mm) in diameter is connected to the top of the highest flue pass and extends through the outer casing, terminating in the vent connector. A direct draft damper may be used as an alternative to the bypass; or
- b) a fan-assisted burner, unless
 - i) there is compliance with the requirements of Items a) i) to iii); or

- ii) there is compliance with the requirements of Items a) i) and ii), and the burner incorporates spark ignition, a pre-purge, and an automatic valve with an integral dual safety shut-off feature.

7.8.2 Bypass requirements

The bypass referred to in Item a) iii) of Clause [7.8.1](#) shall be gas-tight and shall be constructed of metal at least equivalent in strength and corrosion resistance to the metal from which it is extended.

7.8.3 Secondary heating surfaces

A forced-air furnace with a secondary heating surface located on the suction side of the circulating air blower shall not be converted to gas except where such a heating surface consists of a single cylindrical flue pipe passing directly from the primary heat exchanger to the flue collar and having a single continuously gas-tight welded joint.

7.8.4 Markings

An easily readable caution sign or label of durable material having black letters on a yellow background shall be either on or attached to each side of the circulating air blower compartment access door of a fuel converted forced-air furnace and shall be worded as follows: "WARNING: THIS COMPARTMENT MUST BE CLOSED EXCEPT WHEN SERVICING." and "AVERTISSEMENT : CE COMPARTIMENT EST OBLIGÉ DE RESTER FERMÉ, SAUF POUR L'ENTRETIEN."

The lettering of "WARNING" and "AVERTISSEMENT" shall be a minimum of 3/8 in (9.6 mm) in height, and the remainder of the lettering shall be a minimum of 3/16 in (4.8 mm) in height.

7.8.5 Flue outlet material

The flue outlet referred to in Clause [7.8.1](#) shall be made of a material at least equivalent in strength and corrosion resistance to No. 24 GSG (0.60 mm) galvanized steel.

7.8.6 Temperature limits

An automatically controlled gravity or forced-air furnace shall be equipped with a high-temperature limit control, and the maximum setting of the control shall be

- a) 350 °F (175 °C) for a gravity furnace; and
- b) 250 °F (120 °C) for a forced-air furnace.

7.9 Conversion of ranges

7.9.1 Control valve access

A range burner with a control valve located at the rear of the range shall not be installed unless such a valve is readily accessible.

7.9.2 Burner location

A burner shall be located so that proper flame characteristics are maintained at all times.

7.9.3 Draft regulator

A manually operated damper shall be replaced with a draft regulator, and the vent size may be reduced to a minimum of 4 in (100 mm).

7.10 Counter appliances

Except where permitted by Clause [4.13.2](#), a counter appliance shall have the following minimum clearances from combustible material:

- a) above — 36 in (900 mm);
- b) front — 6 in (150 mm); and
- c) back and sides — 6 in (150 mm).

7.11 Direct-vent appliances

A vent terminal of a direct-vent appliance shall be located in accordance with the requirements of Clause [8.14.8](#).

7.12 Furnaces used with cooling units

7.12.1 Cooling coil condensate

When a furnace is installed in conjunction with a refrigeration coil, the arrangement shall be such that condensate is prevented from dripping onto the heating surface either by disposal of the condensate or by other means.

7.12.2 Parallel damper control

When a furnace is installed in parallel with a refrigeration coil, a damper or other means used to control the air supply shall effectively prevent the circulation of

- a) cooled air through the furnace; and
- b) heated air over the refrigeration coil.

7.12.3 Furnace upstream requirements

When a forced-air furnace is installed in series with a cooling unit and is upstream from a refrigeration coil, the coil shall either be designed to withstand the high pressure that might develop or be equipped with a suitable device to prevent excessive pressure.

7.12.4 Furnace downstream requirements

A forced-air furnace installed in series with a cooling unit and downstream of a refrigeration coil shall be designed for such use.

7.13 Central furnaces

7.13.1 Installation

A central furnace shall be installed on

- a) a firm, level base; and
- b) a noncombustible floor or support, except where the furnace is
 - i) certified for installation on a combustible floor;
 - ii) installed using the central furnace manufacturer's certified special base; or
 - iii) installed as specified in Clause [4.13.3](#).

7.13.2 Minimum clearances

Except where permitted by Clause [4.13.2](#), a central furnace shall be installed with the following minimum clearances from combustible material:

- a) above — 1 in (25 mm);

- b) the jacket sides and rear — 6 in (150 mm); and
- c) front — 24 in (610 mm).

See also Clause [4.14.2](#).

7.13.3 Return air inlets

Return-air inlets shall not be installed in an enclosure or crawl space that provides combustion air to a furnace.

7.13.4 Floor furnaces

Floor furnaces shall not be installed.

7.13.5 Construction heat

A furnace that is used to heat a residence under construction shall be

- a) installed
 - i) on a finished concrete floor;
 - ii) on a poured concrete slab that is at least
 - 1) 4 in (100 mm) thick;
 - 2) 1 in (25 mm) in height above the height of the basement floor when complete; and
 - 3) 6 in (150 mm) longer and wider than the base of the furnace; or
 - iii) by suspending it in a safe manner;
- b) piped in accordance with Clause [6](#);
- c) vented in accordance with Clause [8](#);
- d) fitted with a warm-air plenum and a return-air plenum, both of which are protected against an influx of waste or discarded material; and
- e) provided with a thermostat that is installed on a solid bracket or wall and located in the furnace area or a location supplied with heat through ductwork from the furnace.

7.13.6 Return air ducting

Furnace return-air ducting installed in an enclosure in which any spillage-susceptible appliances are located (including the furnace) shall be sealed to the furnace casing, and joints in the ducting shall be sealed to prevent infiltration of air from the enclosure into the return-air ducting.

7.14 Downflow furnaces

7.14.1 Marking

A downflow furnace that has a downward warm-air discharge and is marked “FOR INSTALLATION ON NONCOMBUSTIBLE FLOORS ONLY”* shall be provided with a separate base certified as a component part of the furnace when installed either on or passing through a combustible floor.

* The equivalent French wording is “INSTALLER SUR UN PLANCHER INCOMBUSTIBLE SEULEMENT”.

7.14.2 Floor openings

A downflow furnace shall be installed so that there is not an open passage in the floor through which flame or hot gases from a fire originating in the area below the floor can travel to the room above.

7.14.3 Enclosure air limitations

When a downflow furnace is located in an enclosure, circulating air and combustion air shall not be taken from the same space.

7.15 Duct furnaces

7.15.1 Location

A duct furnace shall not be installed on the negative pressure side of the air-circulating blower.

7.15.2 Duct access

A duct furnace shall have a removable access panel located in the duct connected to the furnace on both the upstream and downstream sides.

7.15.3 Circulating air limitations

Circulating air shall not be taken from an enclosure containing a duct furnace.

7.15.4 Control locations

All controls shall be located outside the duct except for the sensing element of a control.

7.15.5 Structural support

A duct furnace shall be supported in such a manner that no weight is carried by the supply piping.

7.15.6 Minimum clearances

Except where permitted by Clause [4.13.2](#), a duct furnace shall have the following minimum clearances from combustible material:

- a) above — 6 in (150 mm);
- b) front (burner side) — 24 in (610 mm);
- c) back and sides — 6 in (150 mm); and
- d) bottom — 6 in (150 mm).

See also Clause [4.14.2](#).

7.16 Horizontal furnaces

7.16.1 Installation options

A horizontal furnace may be installed in a crawl space or suspended from a floor or ceiling.

7.16.2 Attic space walkway

When a horizontal furnace is installed in an attic space, a permanent substantial walkway shall be provided to the control side of a horizontal furnace.

7.16.3 Attic limitations

A horizontal furnace shall not be installed in an attic space containing either exposed combustible insulation or wrapping for such insulation.

7.16.4 Minimum clearances

Except where permitted by Clause [4.13.2](#), a horizontal furnace shall have the following minimum clearances from combustible material:

- a) above — 6 in (150 mm);
- b) front (burner side) — 24 in (610 mm);
- c) back and sides — 6 in (150 mm); and

d) bottom — 6 in (150 mm).

See also Clause [4.14.2](#).

7.17 Wall furnaces (recessed heaters)

7.17.1 Combustible wall approval

A wall furnace installed in a combustible wall shall be approved for such installation.

7.17.2 Service access

A panel, grille, or access door that must be removed for normal servicing shall not be attached to the building.

7.17.3 Partition venting

When a vent is required to be concealed in a partition, a Type BW vent kit shall be used.

7.18 Construction heaters and torches

7.18.1 Location

A construction heater shall be located to minimize the danger of damage and upset.

7.18.2 Installation

A construction heater shall be installed on a solid, level, noncombustible base or, where so designed, suspended in accordance with the manufacturer's certified installation instructions.

7.18.3 Clearance to combustibles

Combustible material such as straw, canvas, wood, and debris shall be kept clear of a construction heater in accordance with the clearances specified on the heater's instruction plate.

7.18.4 Air supply

When a construction heater is operating, an adequate air supply shall be provided.

7.18.5 Piping

Piping, tubing, hose, and fittings shall be supported, secured, and protected from damage and strain.

7.18.6 Temporary piping

When a construction heater is connected to temporary piping, the piping and connections shall be in accordance with the requirements of Clause [6](#). In addition, a shut-off valve shall be provided with a lever or hand wheel when a branch line is in service. Hose may be used as a connector to a heater if the shut-off valve is installed immediately upstream from the hose.

7.18.7 Lessor responsibility

It shall be the responsibility of the lessor of a construction heater, at the time of delivery to the lessee, to ensure that

- a) the construction heater is approved and in safe operating condition; and
- b) the lessee is instructed in the safe installation and use of the construction heater and components in accordance with the requirements of Clause [7.18](#).

7.18.8 User responsibility

It shall be the responsibility of the user of a construction heater and its components to ensure that

- a) the construction heater and its components are installed and used in accordance with the requirements of Clause [7.18](#);
- b) the maintenance of the construction heater and its components is performed by a qualified installer;
- c) the handling and operation of the construction heater and its components are done by persons who have been instructed in such handling and operation; and
- d) a malfunctioning or damaged construction heater is removed from service.

7.18.9 Installation limitations

A construction heater shall

- a) only be installed in a building under construction, repair, or improvement; and
- b) not be installed in any inhabited dwelling unit or inhabited sections of a building.

7.18.10 Torch attendance

A torch intended for manual operation shall not be left unattended while in operation.

7.19 Direct-fired door air heaters

Note: This clause is for direct-fired door air heaters that were certified to CAN1-3.12 or earlier editions.

7.19.1 Installation references

A direct-fired door air heater shall be installed in accordance with the requirements of Clauses [7.20.7](#) to [7.20.11](#).

7.19.2 Interlocking

A direct-fired door air heater shall be interlocked with an associated door so that the heater can operate only when the door served is

- a) at least 80% open; or
- b) at least 15% open and an interlocked time relay is provided to prove the door is 80% open within 60 s from heater start-up.

7.19.3 Heated air direction

Heated air and combustion products from a direct-fired door air heater shall be directed outdoors.

7.19.4 Installation references

A combination direct-fired make-up air heater for use in a doorway shall be installed in accordance with Clauses [7.19](#) and [7.20](#).

7.20 Direct-fired make-up air heaters (DFMAH)

Note: This clause is for DFMAHs that were certified to CGA 3.7 or CAN1-3.7 or earlier editions.

7.20.1 Exhaust requirements

Except as specified in Clauses [7.20.4](#), [7.20.12](#), and [7.20.13](#), a DFMAH shall not be installed unless it is necessary to exhaust inside air and replace it with preheated air.

7.20.2 Sleeping areas

A DFMAH shall not supply make-up air to an area where sleeping accommodation is provided.

7.20.3 Exhaust air balance

Except as specified in Clauses [7.20.4](#), [7.20.12](#), and [7.20.13](#), a DFMAH shall

- a) have an air discharge capacity that does not exceed the total discharge capacity of the exhaust system by more than 10%; and
- b) be interlocked in such a way as to prevent its operation unless the exhaust system is functioning.

7.20.4 System air balance

The total air replacement of an installation in a spray booth shall not exceed the total exhaust capacity of the booth.

7.20.5 Mine shafts

A DFMAH that is either installed in or ducted to a mine shaft and that is provided with an exhaust opening from the shaft that has an adequate free cross-sectional area need not comply with Clause [7.20.3](#).

7.20.6 Interlocking

When the exhaust system referred to in Clause [7.20.3](#) can affect the satisfactory venting of other gas appliances, the DFMAH shall have the exhaust system interlocked in such a way that it will only function when the DFMAH blower is operating.

7.20.7 Outside air

All air handled by a DFMAH shall be brought directly from outdoors, except that primary combustion air not exceeding 0.5% of the total air handled may be taken from inside the building.

7.20.8 Automatic louver interlocking

Automatically operated louvers for inlet or discharge air shall have the louvers interlocked to ensure maximum designed opening prior to the start-up of the DFMAH.

7.20.9 Intake location

The outdoor intake of a DFMAH shall be located not less than 20 ft (6 m) horizontally from a vertical plane in which combustible gas, vapour, or dust is present.

7.20.10 Clearance to combustibles

A DFMAH shall be installed with clearance from combustible material of not less than that marked on the rating plate. See also Clause [4.14.2](#).

7.20.11 Duct trapping

Supply or discharge ducting in which accumulations of gas or combustion products can be trapped shall not be attached to a DFMAH.

7.20.12 Elevator shaft or stairwell

A DFMAH, when installed for the purpose of ventilating or pressurizing an elevator shaft or stairwell, shall be

- a) activated only by a fire alarm system under normal operation; and
- b) equipped with a normally open momentary manual switch that will permit testing of the DFMAH without activating the fire alarm system. This switch shall be either installed on or located directly adjacent to the DFMAH and shall be identified as to its function.

7.20.13 CO Sensing

When a DFMAH is installed for the ventilation of a storage garage and the DFMAH is solely actuated by a carbon monoxide sensor, the DFMAH shall be installed with a normally open momentary manual switch to permit testing of the DFMAH. This switch shall be located either on or directly adjacent to the DFMAH and shall be identified as to its function.

7.20.14 Kitchens

When a DFMAH is either installed in or ducted to a kitchen, it shall be installed in accordance with the following procedure:

- a) Where the food preparation area and the area frequented by the public is interconnected by means
 - i) of either normally closed doors or a permanent opening(s) and the total free area of the opening(s) does not exceed 16 ft² (1.5 m²), the DFMAH shall be interlocked with a mechanical exhaust system and the proven airflow capacity of the interlocked exhaust system shall not be less than 90% of the make-up air supply; or
 - ii) other than those indicated in Item i), the DFMAH shall be interlocked with a mechanical exhaust system and the proven airflow capacity of the interlocked exhaust system shall in no case be less than 100% or more than 105% of the make-up air supply.
 - iii) of an approved engineering design
- b) All the other requirements of Clause [7.20](#) shall apply, where applicable.

7.21 Non-recirculating and recirculating direct gas-fired industrial air heaters (DFIAH)

7.21.1 Installation limitations

A DFIAH shall be installed only in industrial buildings, except as specified in Clauses [7.21.11](#) to [7.21.13](#).

7.21.2 Sleeping areas

A DFIAH shall not supply air to an area where sleeping accommodation is provided.

7.21.3 Functionality references

A DFIAH may be used as a door air heater, a make-up air heater, or a space heater, or for a combination of these functions. When a DFIAH is used as a door air heater, a make-up air heater, or space heater, the requirements of Clause [7.21](#) shall supersede the requirements of Clause [7.19](#) for door air heaters and Clause [7.20](#) for make-up heaters.

7.21.4 Clearance to combustibles

A DFIAH shall be installed with a clearance from combustible material not less than that marked on the rating plate. See also Clause [4.14.2](#).

7.21.5 Supply air

All supply air to a

- a) non-recirculating DFIAH shall be ducted directly from outdoors.
- b) recirculating DFIAH shall be ducted directly from outdoors or from a combination of outdoor and return air from the space served by the recirculating DFIAH. The minimum ventilation rate (percent of outdoor air) shall be as indicated on the recirculating DFIAH rating plate.

7.21.6 Duct purge

Inlet ducting, when used, shall be purged with at least four air changes prior to an ignition attempt.

7.21.7 Damper interlocks

Outside air dampers or closing louvres not certified as an integral part of the appliance shall be interlocked so that the main burners do not operate until the air dampers are fully open.

7.21.8 Exhaust options and exfiltration

The design of the installation shall include adequate provisions to permit a DFIAH to operate at its rated capacity, taking into account the structure's designed exfiltration rate, by providing properly designed relief openings or an interlocked powered exhaust system, or a combination of these methods.

Where the installation is dependent upon exfiltration through the building envelope or through relief openings, the structure's designed exfiltration rate and the size of relief openings shall be determined by a professional engineer.

Relief openings shall be louvres or counterbalanced gravity dampers. Motorized dampers or closable louvres may be used, provided that they are interlocked so that the main burners do not operate until the air dampers are fully open.

7.21.9 Exhaust interlocking

When a DFIAH is interlocked with a powered exhaust system, it shall be installed in accordance with Clause [7.21.13](#) where applicable or to the following:

- a) At startup, the DFIAH and powered exhaust system shall have their fans powered simultaneously;
- b) Subject to compliance with Clause [7.21.7](#), the DFIAH's integral airflow proving system must prove sufficient airflow before the burner can be enabled;
- c) The powered exhaust system operation shall be proven within 2 min of the DFIAH fan start. Failure to prove operation of the powered exhaust system shall disable the heating; and
- d) When a variable air volume (VAV) DFIAH is interlocked to multiple powered exhaust air fans, the operating airflow capacity of the DFIAH and powered exhaust air shall be matched to provide satisfactory venting of other gas fired appliances.

Note: For VAV systems, a space static pressure sensing system that controls the system and the space served is an acceptable interlock.

7.21.10 Intake location

The outdoor air intake of a DFIAH shall be located not less than 20 ft (6 m) horizontally from a vertical plane in which combustible gas, vapour, or dust is present.

7.21.11 Fire alarm

A DFIAH, when installed for the purpose of ventilating or pressurizing an elevator shaft or stairwell, shall be

- a) activated only by a fire alarm system under normal operation;
- b) of the non-recirculating type DFIAH; and
- c) equipped with a normally open momentary manual switch that will permit testing of the DFIAH without activating the fire alarm system. This switch shall be either installed on or located directly adjacent to the DFIAH and shall be identified as to its function.

7.21.12 Storage garages

- a) When a DFIAH is installed for the ventilation of a storage garage and the DFIAH is solely actuated by a carbon monoxide sensor, the DFIAH shall be installed with a normally open momentary manual switch to permit testing of the DFIAH. This switch shall be located either on or directly adjacent to the DFIAH and shall be identified as to its function; and
- b) DFIAH shall provide adequate outside air to positively pressurize the storage space.

7.21.13 Kitchens

When a DFIAH is either installed in or ducted to a kitchen, it shall be installed in accordance with the following procedure:

- a) Where the food preparation area and the area frequented by the public is interconnected by means
 - i) of either normally closed doors or a permanent opening(s) and the total free area of the opening(s) does not exceed 16 ft² (1.5 m²), the DFIAH shall be interlocked with a mechanical exhaust system and the proven airflow capacity of the interlocked exhaust system shall not be less than 90% of the make-up air supply
 - ii) other than those indicated in Item a) i), the DFIAH shall be interlocked with a mechanical exhaust system and the proven airflow capacity of the interlocked exhaust system shall in no case be less than 100% or more than 105% of the make-up air supply; or
 - iii) of an approved engineering design.
- b) All the other requirements of Clause [7.21](#) shall apply, where applicable.

7.22 Direct gas-fired process air heaters (DFPAH)

7.22.1 Location

A DFPAH shall be installed only in industrial buildings.

7.22.2 Sleeping areas

A DFPAH shall not supply air to an area where sleeping accommodation is provided.

7.22.3 Standard reference

A DFPAH shall be certified to be in compliance with ANSI Z83.25/CSA 3.19.

7.22.4 Clearance to combustibles

A DFPAH shall be installed with a clearance from combustible material not less than that marked on the rating plate. See also Clause [4.14.2](#).

7.22.5 Operating modes

A DFPAH can have two operating modes, ventilation and/or process:

- a) In ventilation mode, the space supplied by the DFPAH can be occupied, and all make-up supply air to a DFPAH capable of operating in ventilation mode shall be ducted directly from outdoors when in ventilation mode.
- b) In process mode, the space supplied by the DFPAH shall not be occupied and recirculation of air from the space served is allowed.

7.22.6 Exhaust and relief — Ventilation mode

7.22.6.1 Design and Installation

The design of the installation shall include adequate provisions to permit a non-recirculating DFPAH in ventilation mode to operate at its rated capacity, taking into account the structure's designed exfiltration rate, by providing properly designed relief openings or an interlocked powered exhaust system, or a combination of these methods.

7.22.6.2 Exfiltration

Where the installation is dependent upon exfiltration through the building envelope or through relief openings, the structure's designed exfiltration rate and the size of relief openings shall be determined by a professional engineer.

7.22.6.3 Relief openings

Relief openings shall be louvres or counterbalanced gravity dampers. Motorized dampers or closable louvres may be used, provided that they are interlocked so that the main burners do not operate until the air dampers are fully open.

7.22.7 Exhaust and relief — Process mode

7.22.7.1 Design and installation

The design of the installation shall include adequate provisions to permit non-recirculating airflow of a DFPAH in the process mode to operate at its rated capacity by providing properly designed ducted relief openings determined by a professional engineer, or an interlocked powered exhaust system, or a combination of these methods. Relief opening(s), powered exhaust system(s), or a combination of both shall terminate outdoors.

7.22.7.2 Relief openings

Relief openings shall be louvres or counterbalanced gravity dampers. Motorized dampers or closable louvres may be used, provided that they are interlocked so that the main burner do not operate until the air dampers are fully open. The opening area of the exhaust fan used in the ventilation mode may also be considered for open relief area for the process mode.

7.22.7.3 Powered exhaust limitations

Where the installation is dependent solely on an interlocked powered exhaust system, the air flow capacity of the interlocked exhaust system shall in no case be less than 95% or more than 105% of the outside air supply to the DFPAH in the process mode. This exhaust fan may be a different exhaust fan than that provided in Clause [7.22.6](#) depending on the ability to turn down the airflow accurately.

7.22.8 Spray area exhaust interlocking

For a spray area installation, an interlocked exhaust system shall be utilized to exhaust the ventilation air supply to the DFPAH in the ventilation mode and an interlocked exhaust system shall be utilized to exhaust the outside air supply to the DFPAH in the process mode.

7.22.9 Spray booth mode

In a spray booth application, an interlock shall be provided to lock out the spraying equipment unless the DFPAH is operated in the ventilation mode.

7.22.10 Duct purge

In a spray booth application, the inlet ducting, when used, shall be purged with at least four air changes prior to an ignition attempt.

7.22.11 Process mode

In a DFPAH application that has a process mode, an interlock shall be provided to lockout the spray area lighting and the spraying equipment while in process mode.

7.22.12 Ventilation mode purge

In a DFPAH application, an interlock shall be provided to ensure the DFPAH is operated in the ventilation mode for a minimum of 3 min or a minimum of four air changes of the enclosure volume including any inlet ducting, whichever is greater, at the start of the process.

7.22.13 Access interlocking

For the space served by a DFPAH operating in the process mode, an interlock for access points, such as a door or partition, shall be provided to immediately shut down the process mode if entry is made. The following warning marking shall be posted at each access opening:

“Do not enter this space until the post purge cycle is complete.” and “Ne pas pénétrer dans cet espace tant que le cycle de post-purge n’est pas terminée”.

7.22.14 Post process purge

In process applications that are intended to be entered by personnel following the process mode operation, the controls shall include a post-purge timer to purge contaminants from the space and cool the products to avoid a burn hazard. The length of the post-purge period shall be field adjustable to accommodate the specifics of that particular application.

7.22.15 Remote interlocking

Outside air dampers or closing louvers not certified as an integral part of the appliance shall be interlocked so that the main burners do not operate until the air dampers are fully open.

7.22.16 Air intake location

The outdoor air intake of a DFPAH shall be located not less than 20 ft (6 m) horizontally from a vertical plane in which combustible gas, vapour, or dust is present except for a DFPAH serving a spray area that it is interlocked with in accordance with

- a) Clauses [7.22.8](#), [7.22.9](#), and [7.22.11](#);
- b) Clause [7.22.12](#); or
- c) Clause [7.22.13](#).

7.22.17 Process minimum ventilation

In the process mode, a DFPAH that recirculates process area air must provide the minimum ventilation airflow to support complete combustion [200 cfm (95 dm³/s) per 1000 ft³/h (28 m³/h)] of natural gas based on the maximum capacity of the heater) plus an allowance to sufficiently dilute the VOCs from the drying or curing to maintain the lower explosive level (LEL) below a 25% threshold value, if applicable. The duct systems shall be designed to prevent recirculation of insufficiently diluted products of combustion.

7.22.18 Inspection and maintenance

In spray applications or other applications with flammable airborne particulate matter, the duct system of a DFPAH that recirculates process area air shall be provided with doors, panels, or other means to facilitate inspection, maintenance, cleaning, and access to fire protection devices. Filters that are suitable for their intended service shall be installed in the return air duct system.

7.23 Infrared heaters

7.23.1 Unvented installation requirements

Where an infrared heater is of the unvented type, it shall

- a) be protected against physical damage;
- b) comply with Clause [4.9.2](#);
- c) not be installed in a residential or care or detention occupancy building;
- d) when certified for use as a construction heater, comply with Clause [7.18](#) and be exempted from the requirements of Items e) to g);
- e) be provided with mechanical ventilation that
 - i) is so located that the products of combustion from each heater are effectively removed outdoors;
 - ii) has a ventilation volume of at least 300 cfm (142 dm³/s) for each 100 000 Btu/h (30 kW) input or fraction thereof; and
 - iii) is sufficient to maintain the level of carbon dioxide at less than 5000 ppm measured 6 ft (2 m) above the work area;
- f) have the ventilation system so interlocked that any reduction of the volume of airflow required by Item e) for a heater or group of heaters will cause the shutdown of that heater or group of heaters;
- g) be provided with combustion and ventilation air compatible with Item e);
- h) not be installed in an exit passageway or stairway within 8 ft (2.5 m) measured horizontally from an exit door; and
- i) be provided with clearance from combustible material as certified and indicated on the appliance.

7.23.2 CO₂ detection

Where it is not possible to interlock heaters due to the use of portable infrared heaters, the ventilation system described in Clause [7.23.1](#) e) shall be connected to a carbon dioxide monitor with an audible and visual alarm.

7.23.3 Vented installation requirements

An infrared heater of the vented type shall be installed in accordance with the certified markings and the manufacturer's certified installation instructions. It shall be protected against physical damage. A tube-type infrared heater shall only be connected with a Type I gas hose that is

- a) certified as being in compliance with CSA 8.1; and
- b) of a length of 36 ± 6 in (90 ± 15 cm).

7.23.4 Vehicles

When an infrared heater is installed in an area of a garage or a car wash where vehicles can be exposed to radiant heat, it shall

- a) comply with Clause [7.23.1](#) if of the unvented type; or
- b) comply with Clause [7.23.3](#) if of the vented type.

The minimum clearance from the radiant face of the infrared heater to the upper surface of the highest vehicle shall not be less than the certified clearance from combustible material as indicated on the appliance. Provision shall be made to maintain these minimum specified clearances by an interlock that shuts off the gas supply until the required clearances have been re-established, a protective bar, or a device that provides a warning when the clearances are not being maintained.

7.23.5 Aircraft repair areas

When an infrared heater is installed in either a repair or shop area that communicates with an aircraft hangar, the minimum clearance from the floor to the infrared heater shall be 8 ft (2.4 m).

7.23.6 Aircraft hangars

An infrared heater shall not be located in an area of an aircraft hangar where it can be subjected to physical damage by aircraft, cranes, movable scaffolding, or other objects.

7.23.7 Garages

Where an infrared heater is installed in a garage and the clearance from combustible material specified in Clause [7.23.4](#) cannot be maintained when the vehicle is raised on a hoist, an electrical interlock shall be provided to shut off the burner and prevent its operation until the required clearance has been re-established.

7.24 Room heaters

7.24.1 Bathroom heaters

A room heater in a bathroom shall be of the direct-vent type.

7.24.2 Fireplace installation

A room heater installed in a fireplace shall be designed for installation in a fireplace and marked "FOR USE IN NONCOMBUSTIBLE FIREPLACES"*. Its installation shall comply with the local building code or, in the absence of such, with the *National Building Code of Canada*.

* The equivalent French wording is "À UTILISER DANS UN FOYER INCOMBUSTIBLE".

7.24.3 Venting

The products of combustion from a radiant-type room heater shall enter either a vent or a chimney.

7.24.4 Fireplace damper

When a radiant-type room heater is installed in a fireplace that has a manual damper, a permanent stop shall be provided on the damper control to prevent complete closure of the flue outlet.

7.24.5 Sleeping or public area installation

A room heater installed either in a room where sleeping accommodation is provided or in a public area shall

- a) be of the automatic temperature-controlled type;
- b) be equipped with a pressure regulator;
- c) have a 100% safety shut-off control; and
- d) be vented.

7.24.6 General Installation

A room heater shall be placed so as not to cause a hazard to a wall, floor, curtain, furniture, or door when open, or hinder the free movement of persons.

7.24.7 Circulating type minimum clearances

Except where permitted by Clause [4.13.2](#), a free-standing circulating-type room heater shall have the following minimum clearances from combustible material:

- a) above — 36 in (900 mm);
- b) front — 24 in (610 mm); and
- c) back and sides — 6 in (150 mm).

7.24.8 Radiant type minimum clearances

Except where permitted by Clause [4.13.2](#), a free-standing radiant-type room heater shall have the following minimum clearances from combustible material:

- a) above — 36 in (900 mm);
- b) front — 24 in (610 mm); and
- c) back and sides — 6 in (150 mm).

7.25 Decorative appliances and gas logs

7.25.1 Manufacture instructions

Decorative appliances and gas logs shall be installed in accordance with the manufacturer's certified installation instructions.

7.25.2 Fireplace installation

A decorative appliance installed in a fireplace shall be designed for installation in a fireplace and marked "FOR USE IN NONCOMBUSTIBLE FIREPLACES"*. Its installation shall comply with the local building code or, in the absence of such, with the *National Building Code of Canada*.

* The equivalent French wording is "À UTILISER DANS UN FOYER INCOMBUSTIBLE".

7.25.3 Venting

The products of combustion from a decorative appliance shall enter either a vent or a chimney.

7.25.4 Bathroom and bedroom installation

A decorative appliance shall not be installed either in a bathroom or in a room where sleeping accommodation is provided, except that an appliance certified to ANSI Z21.50/CSA 2.22 may be installed in

- a) a bathroom if the appliance is of the direct-vent type;
- b) a bedroom or a bed-sitting room if the appliance is of the direct-vent type; and

- c) a bedroom or a bed-sitting room if the appliance is not of the direct-vent type where the appliance is identified by a marking on or adjacent to the rating plate stating “ALSO CERTIFIED FOR INSTALLATION IN A BEDROOM OR BED-SITTING ROOM”*.

* The equivalent French wording is “ÉGALEMENT CERTIFIÉ POUR INSTALLATION DANS UNE CHAMBRE À COUCHER OU UN STUDIO”.

7.25.5 Chimney damper

A gas log shall be certified for the application, and when a gas log is installed, the chimney damper shall be permanently secured in an open position to effectively vent the appliance.

7.26 Pool heaters

7.26.1 Installation location

A pool heater shall be installed on a

- a) firm and level base; and
- b) non-combustible floor or support, except where the pool heater is
 - i) certified for installation on a combustible floor;
 - ii) installed using the pool heater manufacturer’s certified special base; or
 - iii) installed as specified in Clause [4.13.3](#).

7.26.2 Minimum clearances

Except where permitted by Clause [4.13.2](#), a pool heater shall be installed with the following minimum clearances from combustible material:

- a) vertical — 18 in (450 mm);
- b) sides and rear — 18 in (450 mm); and
- c) front — 48 in (1200 mm).

7.26.3 Outdoor installation

A pool heater installed outdoors shall be located a minimum of 18 in (450 mm) from any property line.

7.26.4 Wall-mounted type

A wall-mounted pool heater shall be securely attached.

7.26.5 Outdoor type

An outdoor pool heater shall not be installed beneath any structure, including a deck.

7.26.6 Finned-tube outdoor installation

A pool heater of the finned-tube type shall be installed outdoors or in an enclosure that is not normally occupied and does not directly communicate with occupied areas. Direct-vent pool heaters of the finned-tube type are exempt from this requirement.

7.26.7 Indoor installation

When a finned-tube indoor gas-fired pool heater that had prior approval for indoor installation that communicates with an occupied area is being replaced with a new finned-tube-type heater, the heater shall be of the direct-vent type.

7.26.8 Maintenance

Where a heater is installed in accordance with Clause [7.26.7](#), it shall be the responsibility of the owner of the heater to provide maintenance for the heater in accordance with the manufacturer's instructions, but in no case less than once annually.

7.26.9 Outdoor flue discharge

An outdoor pool heater shall be installed so that the flue discharge is in accordance with Clause [8.14.10](#).

7.27 Water heaters

7.27.1 Installation location

A water heater, unless of the direct-vent type, shall not be installed in a bathroom, bedroom, or any enclosure where sleeping accommodation is provided.

7.27.2 Pressure relief termination

The temperature and pressure relief device on a tank-type water heater or the pressure relief device provided for an instantaneous (tankless) water heater shall have a discharge pipe of a size at least equal to the nominal size of the device outlet. The discharge pipe shall terminate not more than 12 in (300 mm) above the floor.

7.27.3 Combustibles protection

An instantaneous-type (tankless) water heater, unless certified for installation on a combustible wall, shall be provided with appropriate protection as specified in Table [4.1](#). Such protection shall extend the full length and width of the heater and its draft hood.

7.27.4 Minimum clearances

Except where permitted by Clause [4.13.2](#), the minimum clearance from combustible material for an underfired storage-type water heater shall be 2 in (50 mm), and the minimum clearance for any other type of water heater shall be 6 in (150 mm). See also Clause [4.14.2](#).

7.27.5 Water supply

Before installing an instantaneous-type (tankless) water heater, the installer shall ensure that there is sufficient water supply for proper operation of the heater.

7.27.6 Return and combustion air

Except for direct-vent water heaters, when the water heater is used in a combo heating system, return-air inlets shall not be installed in the same enclosure that contains both an air-handling unit and the water heater. Adequate combustion air shall be provided for the water heater.

7.27.7 Return air infiltration

When the return air duct(s) of an air-handling unit in a combo heating system is installed in an enclosure in which any spillage-susceptible appliances are located, it shall be sealed to the air-handling unit casing, and joints in the ducting shall be sealed to prevent infiltration of air from the enclosure into the return-air ducting.

7.28 Unit heaters

7.28.1 Suspended installation

A suspended unit heater shall be firmly supported with metal hangers or brackets.

7.28.2 Negative pressure

The location of either a suspended unit heater or duct attached to a suspended unit heater shall be such that a negative pressure will not be created in the room in which the unit heater is located.

7.28.3 Garage installation

When a unit heater is installed in a garage, a minimum clearance of 8 ft (2.4 m) shall be maintained between the base of the heater and the garage floor. The minimum clearance may be reduced when a substantial guard is provided to prevent vehicles from striking the heater. See also Clause [4.16](#).

7.28.4 Clearance to combustibles

Except where permitted by Clause [4.13.2](#), all clearances from combustible material shall be a minimum of 18 in (450 mm). See also Clause [4.14.2](#).

7.29 Hotplates

7.29.1 Piping

A hotplate shall be connected with rigid piping and secured to prevent movement.

7.29.2 Installation locations

A hotplate shall not be installed in a bedroom but may be installed in a bed-sitting room, provided that it is not required to be used for space-heating purposes.

7.29.3 Clearance to combustibles

Except where permitted by Clause [4.13.2](#), a hotplate shall have the following minimum clearances from combustible material:

- a) above — 30 in (760 mm);
- b) front — 6 in (150 mm); and
- c) back and sides — 6 in (150 mm).

7.30 Incinerators

7.30.1 Chimney location

An incinerator shall be installed as close as practicable to a chimney.

7.30.2 Wall-type installations

An incinerator of the wall type shall be installed in a noncombustible wall that communicates directly with a chimney flue.

7.30.3 Draft regulator

An incinerator shall not be equipped with a draft hood. When draft control is required, a draft regulator of the single-acting type shall be used.

7.30.4 Chimney connection

A vent connector shall be directly connected to a chimney through a separate thimble.

7.30.5 Vent guarding

When a vent connector passes through either a wall or partition of combustible material, it shall be guarded at the point of passage as specified in Clause [8.18.12](#).

7.30.6 Clearance to combustibles

Except where permitted by Clause [4.13.2](#), an incinerator shall be installed with a minimum clearance of 12 in (300 mm) from all combustible material.

7.31 Lighting

7.31.1 Pendant type support

A pendant-type light fixture shall be supported to remove the direct weight of the fixture from the gas piping systems.

7.31.2 Wall installation

A wall bracket fixture shall be firmly supported, and if of the swing type, shall be provided with a stop to prevent contact of the globe with combustible material.

7.31.3 Clearance to combustibles

Except where permitted by Clause [4.13.2](#), a bracket or pendant fixture shall have the following minimum clearances from combustible material:

- a) above — 18 in (450 mm); and
- b) all sides — 5 in (125 mm).

7.31.4 Ventilation

Ventilation shall be provided when a lighting appliance is installed or used in an enclosure. The ventilation openings shall be at the top of the enclosure or as high as possible at the top of it and at the bottom of the enclosure, not less than 12 in (300 mm) above the outside grade level. The openings shall be a minimum of 4 in² (25.8 cm²) for each lighting appliance installed or used.

7.32 Commercial cooking appliances

7.32.1 Floor mounting

A commercial cooking appliance shall be installed level on a firm foundation.

7.32.2 Floor mount clearances

When installed on an unprotected combustible material, the appliance shall have legs that provide a minimum of 4 in (100 mm) clearance between the metal base and the material.

7.32.3 Floor mount construction

When the appliance does not have legs at least 4 in (100 mm) high, it shall not be installed on combustible material, except when the material under the appliance is protected with sheet metal and 2 in (50 mm) of air space is maintained between the base of the appliance and the sheet metal.

7.32.4 Side clearances

A noncombustible fixed means shall be provided to maintain a minimum clearance of 6 in (150 mm) between combustible material and the sides and rear of an appliance. See also Clause [6.21.5](#).

7.32.5 Top clearances

Any portion of combustible material located within 18 in (450 mm) horizontally of a cooking top section shall be protected for a vertical distance of 36 in (900 mm) above the surface of the cooking top.

7.32.6 Fryer separation

A clearance of not less than 16 in (400 mm) shall be provided between a deep fat fryer and an open flame of an adjacent appliance unless a noncombustible divider extending not less than 7 in (175 mm) above the fryer and the open flame of the adjacent appliance is provided.

7.33 Residential-type ranges

7.33.1 Installation limitations

A range shall not be installed in a bedroom but may be installed in a bed-sitting room, provided that it is not required to be used for space-heating purposes.

7.33.2 Floor mounting

A residential-type range shall be installed level. When installed on a combustible floor, it shall be set on its own legs or baffled base and shall be certified for such mounting.

7.33.3 Venting

When a vent connector is attached to a range, suitable provision shall be made by the installer for the protection of adjacent combustible material.

7.33.4 Clearance to combustibles

Except where permitted by Clauses [4.13.2](#) and [7.33.5](#) to [7.33.7](#), a residential-type range shall have the following minimum clearances from combustible material:

- a) above — 30 in (760 mm); and
- b) back and sides — 6 in (150 mm).

7.33.5 Zero clearance requirements

A residential-type range certified for zero side and rear clearances may be installed with zero side clearance from combustible material, provided that there is a 1 in (25 mm) clearance at the back and a 28 MSG (0.3 mm) sheet metal shield on the side walls spaced out a minimum of 0.25 in (6 mm). The shield shall extend from the rear wall to the front edge of the range and from 5 in (125 mm) below to 30 in (760 mm) above the cooking surface.

7.33.6 New appliance clearances

A new residential-type range shall be installed in accordance with its certified clearances from combustible material as specified on the appliance rating plate.

7.33.7 Clearance reductions

When the underside of the combustible material above a residential-type range is protected with insulating millboard at least 0.25 in (6 mm) thick covered with sheet metal not less than 28 MSG

(0.3 mm), the clearance above the range as specified in Clause [7.33.4](#) may be reduced to 24 in (610 mm).

7.33.8 Exhaust hoods

An exhaust hood, an exhaust appliance, or a combination thereof installed above a range shall be installed in accordance with instructions for such an exhaust hood or appliance.

7.34 Refrigerators

7.34.1 Minimum clearances

Except where permitted by Clause [4.13.2](#), a refrigerator shall have the following minimum clearances from combustible material, disregarding any guard or baffle on the refrigerator:

- a) above — 12 in (300 mm); and
- b) back and sides — 2 in (50 mm).

7.34.2 Direct vent

A refrigerator installed in a dwelling unit shall be of the direct-vent type.

7.34.3 Unvented

An unvented refrigerator shall be installed in an area that is not normally occupied and does not directly communicate with occupied areas.

7.35 Operation of appliances at shows, exhibitions, or other similar events

The operation of appliances at shows, exhibitions, or other similar events shall comply with Annex [M](#) and meet any additional requirements of the authority having jurisdiction.

8 Venting systems and air supply for appliances

8.1 General

8.1.1 Requirements

The requirements of Clause [8.2](#) shall apply to central-heating furnaces, boilers, and hot-water heaters.

8.1.2 Direct vent exceptions

The requirements of Clauses [8.2](#) to [8.5](#) shall not apply to a direct-vent appliance.

Note: *Regardless of the category type, when an appliance utilizes indoor combustion air, it is not considered a direct-vent appliance.*

8.1.3 Air supply interference

Interference with the air supply for an appliance shall be prohibited.

8.1.4 Input reference

Air supply shall be provided in accordance with

- a) Clauses [8.2](#) and [8.3](#) when either an appliance or a combination of appliances has a total input of up to and including 400 000 Btu/h (120 kW); or

- b) Clause [8.4](#) when either an appliance or a combination of appliances has a total input exceeding 400 000 Btu/h (120 kW).

8.1.5 Combustion air

When an appliance other than a central-heating appliance or a domestic water heater is installed in a location where there is insufficient air for combustion, provisions shall be made to provide an air supply sized in accordance with Table [8.1](#) or [8.2](#).

8.2 Air-supply determination for central-heating furnaces, boilers, and hot-water heaters

8.2.1 Requirements

An outdoor air supply sized in accordance with Clause [8.2.2](#) shall be provided for an enclosure or a structure in which an appliance is installed when the enclosure or structure

- a) has windows and doors of either close-fitting or sealed construction, and the exterior walls are covered by a continuous, sealed vapour barrier and gypsum wallboard (drywall) or plywood or a similar material having sealed joints; or
- b) has an equivalent leakage area of 78 in² (0.05 m²) or less at a differential pressure of 0.00145 psig (10 Pa) as determined by a recognized Canadian fan depressurization test procedure.

8.2.2 Sizing

Except as required in Clause [8.2.3](#), the free area of the outdoor air supply required by Clause [8.2.1](#) shall be determined from Table [8.1](#) for an appliance having a draft-control device and from Table [8.2](#) for an appliance not having a draft-control device, using the total input of all appliances in the structure or enclosure. If an appliance with a draft-control device and an appliance without a draft-control device are installed within the same structure or enclosure, the required free area of the air-supply opening shall be the greater of

- a) that required by Table [8.1](#), using the total input of only those appliances having draft-control devices; or
- b) that required by Table [8.2](#), using the total input of all appliances.

8.2.3 Input exceptions

An outdoor air supply shall not be required for a single water heater with an input of 50 000 Btu/h (15 kW) or less within an enclosure or structure where there are no other appliances that require an air supply. Except for direct-vent water heaters, when the water heater is contained in an enclosure, permanent openings shall be provided as described in Clause [8.2.6](#).

8.2.4 Other exceptions

An outdoor air supply, if required, shall be sized in accordance with Clause [8.2.5](#) and shall be provided for an enclosure or a structure in which an appliance is installed when the enclosure or structure is neither constructed as described in Clause [8.2.1](#) a) nor complies with Clause [8.2.1](#) b).

Note: The determination of an outdoor air supply requirement is made using Table [8.3](#) for appliances having draft control devices and Table [8.4](#) for appliances not having draft control devices. (Refer to Clause [8.2.5](#).)

8.2.5 Free area sizing

The free area of the outdoor air supply, if required by Clause [8.2.4](#), shall be determined from Table [8.3](#) for an appliance having a draft-control device and Table [8.4](#) for an appliance not having a draft-control device, using the total input of all appliances in the structure or enclosure. If an appliance with a draft-

control device and an appliance without a draft-control device are installed within the same structure or enclosure, the required free area of the air supply opening shall be the greater of

- a) that required by Table 8.3, using the total input of only those appliances having draft-control devices; or
- b) that required by Table 8.4, using the total input of all appliances.

Table 8.1

Combustion/dilution air requirements for appliances having draft-control devices when the combined input is up to and including 400 000 Btu/h (120 kW) and the structure complies with Clause 8.2.1 a) or b)

(See Clauses 8.1.5, 8.2.2, and 8.21.2, and Tables 8.3 and 8.4.)

Total input of appliances*, thousands of Btu/h (kW)	Required free area of air-supply opening or duct, in² (mm²)	Acceptable approximate round duct equivalent† diameter, in (mm)
25 (8)	7 (4 500)	3 (75)
50 (15)	7 (4 500)	3 (75)
75 (23)	11 (7 000)	4 (100)
100 (30)	14 (9 000)	4 (100)
125 (37)	18 (12 000)	5 (125)
150 (45)	22 (14 000)	5 (125)
175 (53)	25 (16 000)	6 (150)
200 (60)	29 (19 000)	6 (150)
225 (68)	32 (21 000)	6 (150)
250 (75)	36 (23 000)	7 (175)
275 (83)	40 (26 000)	7 (175)
300 (90)	43 (28 000)	7 (175)
325 (98)	47 (30 000)	8 (200)
350 (105)	50 (32 000)	8 (200)
375 (113)	54 (35 000)	8 (200)
400 (120)	58 (37 000)	9 (225)

* For total inputs falling between listed figures, the next largest listed input shall be used.

† These figures are based on a maximum equivalent duct length of 20 ft (6 m). For equivalent duct lengths in excess of 20 ft (6 m) up to and including a maximum of 50 ft (15 m), the round duct diameter shall be increased by one size.

Table 8.2
Combustion air requirements for appliances not having draft-control devices when
the combined input is up to and including 400 000 Btu/h (120 kW) and the
structure complies with Clause 8.2.1 a) or b)
 (See Clauses 8.1.5, 8.2.2, and 8.21.2.)

Total input of appliances*, thousands of Btu/h (kW)	Required free area of air- supply opening or duct, in² (mm²)	Acceptable approximate round duct equivalent† diameter, in (mm)
25 (8)	4 (2 600)	2 (50)
50 (15)	4 (2 600)	2 (50)
75 (23)	5 (3 200)	3 (75)
100 (30)	7 (4 500)	3 (75)
125 (37)	9 (5 800)	4 (100)
150 (45)	11 (7 000)	4 (100)
175 (53)	13 (8 400)	4 (100)
200 (60)	14 (9 000)	5 (125)
225 (68)	16 (10 300)	5 (125)
250 (75)	18 (12 000)	5 (125)
275 (83)	20 (13 000)	5 (125)
300 (90)	22 (14 000)	6 (150)
325 (98)	23 (15 000)	6 (150)
350 (105)	25 (16 000)	6 (150)
375 (113)	27 (17 400)	6 (150)
400 (120)	29 (19 000)	6 (150)

* For total inputs falling between listed figures, the next largest listed input shall be used.

† These figures are based on a maximum equivalent duct length of 20 ft (6 m). For equivalent duct lengths in excess of 20 ft (6 m) up to and including a maximum of 50 ft (15 m), the round duct diameter shall be increased by one size.

Table 8.3

Combustion/dilution air requirements for appliances having draft-control devices when the combined input is up to and including 400 000 Btu/h (120 kW) and the structure does not comply with Clause 8.2.1 a) or b)

(See Clauses [8.2.4](#) and [8.2.5](#).)

	Volume of the enclosure or structure, ft ³ (m ³)														
	500 (14)	1000 (28)	2000 (57)	3000 (85)	4000 (113)	5000 (142)	6000 (170)	7000 (198)	8000 (227)	9000 (255)	10 000 (283)	12 500 (354)	15 000 (425)	17 500 (495)	20 000 (566)
Total input of appliances*, thousands of Btu/h (kW)	Required air supply opening†, in ² (thousands of mm ²)														
25 (8)	4 (3)	4 (3)	0	0	0	0	0	0	0	0	0	0	0	0	0
50 (15)	7 (5)	7 (5)	7 (5)	0	0	0	0	0	0	0	0	0	0	0	0
75 (23)	11 (7)	11 (7)	11 (7)	11 (7)	11 (7)	0	0	0	0	0	0	0	0	0	0
100 (30)	14 (9)	14 (9)	14 (9)	14 (9)	14 (9)	14 (9)	0	0	0	0	0	0	0	0	0
125 (37)	18 (12)	18 (12)	18 (12)	18 (12)	18 (12)	18 (12)	18 (12)	18 (12)	0	0	0	0	0	0	0
150 (45)	22 (14)	22 (14)	22 (14)	22 (14)	22 (14)	22 (14)	22 (14)	22 (14)	22 (14)	0	0	0	0	0	0
175 (53)	25 (16)	25 (16)	25 (16)	25 (16)	25 (16)	25 (16)	25 (16)	25 (16)	25 (16)	25 (16)	25 (16)	0	0	0	0
200 (60)	29 (19)	29 (19)	29 (19)	29 (19)	29 (19)	29 (19)	29 (19)	29 (19)	29 (19)	29 (19)	29 (19)	0	0	0	0
225 (68)	32 (21)	32 (21)	32 (21)	32 (21)	32 (21)	32 (21)	32 (21)	32 (21)	32 (21)	32 (21)	32 (21)	32 (21)	0	0	0
250 (75)	36 (23)	36 (23)	36 (23)	36 (23)	36 (23)	36 (23)	36 (23)	36 (23)	36 (23)	36 (23)	36 (23)	36 (23)	0	0	0

(Continued)

Table 8.3 (Continued)

	Volume of the enclosure or structure, ft ³ (m ³)														
	500 (14)	1000 (28)	2000 (57)	3000 (85)	4000 (113)	5000 (142)	6000 (170)	7000 (198)	8000 (227)	9000 (255)	10 000 (283)	12 500 (354)	15 000 (425)	17 500 (495)	20 000 (566)
Total input of appliances*, thousands of Btu/h (kW)	Required air supply opening†, in ² (thousands of mm ²)														
275 (83)	40 (26)	40 (26)	40 (26)	40 (26)	40 (26)	40 (26)	40 (26)	40 (26)	40 (26)	40 (26)	40 (26)	40 (26)	40 (26)	0	0
300 (90)	43 (28)	43 (28)	43 (28)	43 (28)	43 (28)	43 (28)	43 (28)	43 (28)	43 (28)	43 (28)	43 (28)	43 (28)	43 (28)	43 (28)	0
325 (98)	47 (30)	47 (30)	47 (30)	47 (30)	47 (30)	47 (30)	47 (30)	47 (30)	47 (30)	47 (30)	47 (30)	47 (30)	47 (30)	47 (30)	0
350 (105)	50 (32)	50 (32)	50 (32)	50 (32)	50 (32)	50 (32)	50 (32)	50 (32)	50 (32)	50 (32)	50 (32)	50 (32)	50 (32)	50 (32)	50 (32)
375 (113)	54 (35)	54 (35)	54 (35)	54 (35)	54 (35)	54 (35)	54 (35)	54 (35)	54 (35)	54 (35)	54 (35)	54 (35)	54 (35)	54 (35)	54 (35)
400 (120)	58 (37)	58 (37)	58 (37)	58 (37)	58 (37)	58 (37)	58 (37)	58 (37)	58 (37)	58 (37)	58 (37)	58 (37)	58 (37)	58 (37)	58 (37)

* For total inputs falling between listed figures, the next largest input shall be used.

† See Table 8.1 for acceptable approximate round duct equivalent.

Table 8.4
Combustion air requirements for appliances not having draft-control devices when the combined input is up to and including 400 000 Btu/h (120 kW) and the structure does not comply with Clause 8.2.1 a) or b)
 (See Clauses [8.2.4](#) and [8.2.5](#).)

Total input of appliances*, thousands of Btu/h (kW)	Volume of the enclosure or structure, ft ³ (m ³)														
	500 (14)	1000 (28)	2000 (57)	3000 (85)	4000 (113)	5000 (142)	6000 (170)	7000 (198)	8000 (227)	9000 (255)	10 000 (283)	12 500 (354)	15 000 (425)	17 500 (495)	20 000 (566)
	Required air supply opening†, in ² (thousands of mm ²)														
25 (8)	2 (1)	0	0	0	0	0	0	0	0	0	0	0	0	0	0
50 (15)	4 (3)	4 (3)	0	0	0	0	0	0	0	0	0	—	—	—	—
75 (23)	5 (3)	5 (3)	5 (3)	0	0	0	0	0	0	0	0	—	—	—	—
100 (30)	7 (5)	7 (5)	7 (5)	0	0	0	0	0	0	0	0	—	—	—	—
125 (37)	9 (6)	9 (6)	9 (6)	9 (6)	0	0	0	0	0	0	0	—	—	—	—
150 (45)	11 (7)	11 (7)	11 (7)	11 (7)	11 (7)	0	0	0	0	0	0	—	—	—	—
175 (53)	13 (8)	13 (8)	13 (8)	13 (8)	13 (8)	13 (8)	0	0	0	0	0	—	—	—	—
200 (60)	14 (9)	14 (9)	14 (9)	14 (9)	14 (9)	14 (9)	0	0	0	0	0	—	—	—	—
225 (68)	16 (10)	16 (10)	16 (10)	16 (10)	16 (10)	16 (10)	16 (10)	0	0	0	0	—	—	—	—
250 (75)	18 (12)	18 (12)	18 (12)	18 (12)	18 (12)	18 (12)	18 (12)	18 (12)	0	0	0	—	—	—	—
275 (83)	20 (13)	20 (13)	20 (13)	20 (13)	20 (13)	20 (13)	20 (13)	20 (13)	20 (13)	0	0	—	—	—	—
300 (90)	22 (14)	22 (14)	22 (14)	22 (14)	22 (14)	22 (14)	22 (14)	22 (14)	22 (14)	0	0	—	—	—	—
325 (98)	23 (15)	23 (15)	23 (15)	23 (15)	23 (15)	23 (15)	23 (15)	23 (15)	23 (15)	23 (15)	0	—	—	—	—

(Continued)

Table 8.4 (Continued)

Total input of applian- ces*, thousands of Btu/h (kW)	Volume of the enclosure or structure, ft ³ (m ³)														
	500 (14)	1000 (28)	2000 (57)	3000 (85)	4000 (113)	5000 (142)	6000 (170)	7000 (198)	8000 (227)	9000 (255)	10 000 (283)	12 500 (354)	15 000 (425)	17 500 (495)	20 000 (566)
	Required air supply opening†, in ² (thousands of mm ²)														
350 (105)	25 (16)	25 (16)	25 (16)	25 (16)	25 (16)	25 (16)	25 (16)	25 (16)	25 (16)	25 (16)	25 (16)	—	—	—	—
375 (113)	27 (17)	27 (17)	27 (17)	27 (17)	27 (17)	27 (17)	27 (17)	27 (17)	27 (17)	27 (17)	27 (17)	—	—	—	—
400 (120)	29 (19)	29 (19)	29 (19)	29 (19)	29 (19)	29 (19)	29 (19)	29 (19)	29 (19)	29 (19)	29 (19)	—	—	—	—

* For total inputs falling between listed figures, the next largest input shall be used.

† See Table [8.1](#) for acceptable approximate round duct equivalent.

8.2.6 Enclosure volume

When an appliance(s) is located within an enclosure and permanent openings sized and located in accordance with Items a) and b) of this clause are supplied to allow communication between the enclosure and the rest of the structure, the total volume of the structure may be used to determine air supply requirements, provided that the structure is not constructed as described in Clause [8.2.1](#) a) and does not comply with Clause [8.2.1](#) b). Otherwise, the volume of the enclosure shall be used. The openings shall be as follows:

- a) In all cases, an opening shall be provided that shall
 - i) have a free area of not less than 1 in² per 1000 Btu/h (2225 mm² per kW) of the total input of all appliances within the enclosure; and
 - ii) be located not more than 18 in (450 mm) or less than 6 in (150 mm) above floor level.
- b) When one or more appliances are equipped with draft-control devices, an additional opening shall be supplied having the same free area as the opening required in Item a), and the opening shall be located as near the ceiling as practicable, but in no case lower than the relief opening of the lowest draft-control device.

8.3 Air-supply openings and ducts

8.3.1 General requirements

Except as specified in Clauses [8.3.3](#) and [8.3.4](#), a duct shall be used to provide the outside air supply required by Clauses [8.2.1](#) and [8.2.4](#). The duct shall

- a) be of either metal or a material meeting the Class I requirements of CAN/ULC-S110;
- b) communicate directly with the outdoors;
- c) be of at least the same cross-sectional area as the free area of the air-supply inlet opening to which it connects; and
- d) terminate within 1 ft (300 mm) above, and within 2 ft (600 mm) horizontally from, the burner level of the appliance having the largest input.

8.3.2 Duct type

A square- or rectangular-shaped duct shall only be used when the required free area of the air-supply opening is 9 in² (5800 mm²) or larger, and when used, its smaller dimension shall not be less than 3 in (75 mm).

8.3.3 Non-ducted openings

An opening may be used in lieu of a duct to provide the outside air supply to an appliance as required by Clauses [8.2.1](#) and [8.2.4](#). The opening shall be located within 1 ft (300 mm) above, and 2 ft (600 mm) horizontally from, the burner level of the appliance having the largest input.

8.3.4 Combustion-air supply devices

A certified combustion-air supply device may be used in lieu of a duct to provide the outside air supply to an appliance as required in Clauses [8.2.1](#) and [8.2.4](#). The combustion-air supply system shall have air flow proving interlocked to the appliance(s) served, and sufficient airflow shall be demonstrated.

8.3.5 Inlet opening

An air-supply inlet opening from the outdoors shall be equipped with a means to prevent the direct entry of rain and wind, and such means shall not reduce the required free area of the air-supply opening.

8.3.6 Inlet location

An air-supply inlet opening from the outdoors shall be located not less than 12 in (300 mm) above the outside grade level.

8.3.7 Inlet clearances

An air-supply opening shall not be located within 3 ft (0.9 m) of a moisture exhaust duct.

Note: A moisture-exhaust duct (e.g., a gas or electric clothes dryer discharge; spa exhaust) is considered to interfere with the combustion air intake when located within 3 ft (0.9 m) of the air intake.

8.4 Air-supply requirements for appliances having a total input exceeding 400 000 Btu/h (120 kW) (see Clause [8.1.4](#))

8.4.1 Location

Ventilation of the space occupied by an appliance or equipment shall be provided by an opening for ventilation air at the highest practicable point communicating with the outdoors, and this opening shall not terminate within 12 in (300 mm) of any combustion air opening. The total cross-sectional area of such an opening shall be at least 10% of the area required in Clauses [8.4.2](#) and [8.4.3](#), but in no case shall the cross-sectional area be less than 10 in² (6500 mm²).

8.4.2 With draft control device

When the air supply is provided by natural airflow from the outdoors for an appliance and a venting system with a draft control device is installed in the same space or enclosure, there shall be a permanent air-supply opening having a cross-sectional area of not less than 1 in²/7000 Btu/h (310 mm²/kW) up to and including 1 000 000 Btu/h (300 kW), plus 1 in²/14 000 Btu/h (155 mm²/kW) in excess of 1 000 000 Btu/h (300 kW). This opening shall be either located at or ducted to a point not more than 18 in (450 mm) or less than 6 in (150 mm) above the floor level. This air-supply opening requirement shall be in addition to the opening for ventilation air required in Clause [8.4.1](#). See also Clauses [4.1.3](#) and [4.1.5](#).

8.4.3 Without draft control device

When air supply is provided by natural airflow from outdoors for an appliance and a venting system without a draft-control device is installed in the same space or enclosure, there shall be a permanent air-supply opening having a total cross-sectional area of not less than 1 in² for each 30 000 Btu/h (70 mm² for each kW) of the total rated input. The location of the opening(s) shall not interfere with the intended purpose of the opening(s) for the ventilation air referred to in Clause [8.4.1](#). The air supply opening requirement shall be in addition to the opening for ventilation air required in Clause [8.4.1](#). See also Clauses [4.1.3](#) and [4.1.5](#).

8.4.4 Combination

When air is provided by natural airflow from outdoors into a space or enclosure containing both types of appliances or venting systems described in Clauses [8.4.2](#) and [8.4.3](#), the cross-sectional area of the opening shall not be less than the sum of the required cross-sectional areas for all types of appliances or venting systems when calculated in accordance with Clauses [8.4.2](#) and [8.4.3](#), as applicable. The location of the opening(s) shall not interfere with the intended purpose of the opening(s) for the ventilation air referred to in Clause [8.4.1](#). This air-supply opening requirement shall be in addition to the opening for ventilation air required in Clause [8.4.1](#). See also Clauses [4.1.3](#) and [4.1.5](#).

8.4.5 Discharge opening

When an air-supply duct is used to meet the requirements of either Clause [8.4.2](#) or [8.4.3](#), its discharge opening shall be located where there is no possibility of cold air affecting steam or water pipes and electrical or mechanical equipment.

8.5 Air-supply dampers, louvres, and grilles

8.5.1 Free area calculation

The free area of an air-supply opening required in Clauses [8.2](#) and [8.3](#) shall be calculated by subtracting the blockage area of all fixed louvres, grilles, or screens from the gross area of the opening.

8.5.2 Minimum aperture size

Apertures in a fixed louver, grille, or screen shall have no dimension smaller than 0.25 in (6 mm).

8.5.3 Manual dampers and louvres

Neither a manually operated damper nor manually adjustable louvres shall be used.

8.5.4 Automatic dampers and louvres

Except as permitted by Clause [8.5.6](#), an automatically operated damper or automatically adjustable louver shall be interlocked so that the main burner cannot operate unless either the damper or louver is in the fully open position.

8.5.5 Automatic combustion air damper

An automatic combustion air damper installed in the air supply within a dwelling unit shall be certified.

8.5.6 Emergency generators, stand-by generators, and firewater pumps

For emergency generators, stand-by generators, or firewater pumps, the combustion air damper interlock is not required, provided

- a) the intake combustion air dampers and the ventilation air dampers, if provided separately, are sized for a maximum face velocity of 500 fpm (2.5 m/s);
- b) the combustion air dampers and ventilation air dampers
 - i) fail open on loss of power;
 - ii) open on a generator run command from the generator control panel; and
 - iii) fail open fully in under 30 s;
- c) the combustion air damper remains open at all times while the generator is operating;
- d) the ventilation air damper remains open for at least the first 30 s after start of all dampers moving to the open position;* and
- e) the combustion air damper is equipped with a position switch that is set at least at 85% open to annunciate an alarm to a supervised location if the damper is not proved open after 30 s.

** The intent is for the ventilation air damper to go to its fully open position and not come under temperature control (if provided) until the combustion air damper is also fully open.*

8.6 Conditions created by exhaust fans, air-supply fans, circulating fans, or fireplaces

When it is determined that the operation of another appliance or other equipment, including an exhaust fan, air-supply fan, circulating fan, or fireplace adversely affects the venting, combustion, or

burning characteristics of a gas appliance, either the condition shall be corrected or the fuel supply to the affected appliance shall be discontinued.

Note: Refer to CSA F300 for assessment and possible solutions to mitigate the depressurization issue.

8.7 Engineered installations

Subject to the approval of the authority having jurisdiction, outdoor air-supply provisions other than those described in Clauses [8.2](#) and [8.3](#) may be used if designed in accordance with good engineering practice.

8.8 Air supply by mechanical means

8.8.1 Airflow sensing

When the air supply is provided by mechanical means, an airflow-sensing device shall be installed. It shall be wired into the safety limit circuit of the primary safety control to shut off the gas in the event of air-supply failure. When an appliance is not equipped with a combustion safety control, the restoration of the gas supply shall be by a manual-reset device.

8.8.2 Make-up air heater

When adequate air supply is provided for combustion air for all appliances by a make-up air heater and the appliances are interlocked with the heater, the requirements of Clauses [8.1](#) to [8.6](#) shall not apply.

8.9 Appliance venting

8.9.1 General exceptions

Every appliance shall be connected to either an effective chimney or a vent, except

- a) for a radiant heater installed in a masonry fireplace with a permanently opened flue;
- b) for an appliance that is approved for use without a vent;
- c) for an appliance installed in accordance with Clause [8.24.5](#) in a building (other than a residential or care or detention occupancy building) where adequate ventilation is provided by an exhaust fan, by a natural-draft ventilator, or by other means acceptable to the authority having jurisdiction;
- d) for an appliance installed for the production of carbon dioxide in a greenhouse where the rate of combustion does not exceed 3 Btu/h/ft³ (30 W/m³) of greenhouse volume and the concentration of carbon dioxide in the atmosphere does not exceed 5000 ppm (0.5%); and
- e) as provided in Clauses [8.14.12](#), [8.24](#), and [8.30](#).

8.9.2 Appliance location

Except for a direct-vent appliance, an appliance requiring venting shall be located as close as practicable to either a chimney or vent.

8.9.3 Fasteners

A venting system shall be firmly attached to either a draft-hood outlet or flue collar by sheet metal screws or mechanical fasteners, or in accordance with the manufacturer's instructions.

Note: A moisture-exhaust duct from a clothes dryer is not considered to be a connector. See Clauses [7.4](#) and [7.5](#).

8.9.4 Hangers

A venting system shall be securely supported by noncombustible hangers suitable for the weight and design of the materials employed. A slip joint in the horizontal section of a venting system shall be

secured either with sheet metal screws or in accordance with the manufacturer's instructions to prevent sagging.

8.9.5 Plastic vent inspection

Venting systems that employ plastic vents shall be installed such that the first 3 ft (0.9 m), or total vent run if less than 3 ft (0.9 m) from the appliance flue outlet, is readily accessible for visual inspection.

8.9.6 Plastic vent type

Vents constructed using plastic piping shall be certified to ULC S636.

8.10 Methods of venting appliances

8.10.1 Effective venting

A vent or chimney shall provide effective venting and shall be designed and constructed to remove all flue gases to the outdoors.

8.10.2 Chimney applications

A chimney suitable to the application shall be used for venting the following appliances:

- a) an incinerator, except as provided in Clause [8.10.11](#);
- b) an appliance that may be readily converted to the use of solid or liquid fuels;
- c) a combination gas- and oil-burning appliance; and
- d) an appliance approved for use with a chimney only.

8.10.3 Venting system

The type of venting system to be used shall be in accordance with Table [8.5](#).

8.10.4 Special venting system

A special venting system or a BH venting system shall be installed in accordance with the terms of its listing and the appliance and vent manufacturer's instructions.

8.10.5 Draft control device location

When used on an appliance having a special venting system, a draft-control device shall be located in a position where positive vent pressures will not occur.

Table 8.5
Type of venting system to be used
(See Clause [8.10.3](#).)

Gas utilization appliance	Type of venting system
Certified Category I appliance	Type B gas vent*
Certified equipment with draft hood	Chimney, special venting system*
Equipment certified for use with Type B gas vent	Certified chimney lining system for gas venting, Type BH or special venting system certified for this appliance*

(Continued)

Table 8.5 (Continued)

Gas utilization appliance	Type of venting system
Certified vented wall furnace	Type BW gas vent (Clauses 7.17.3 , 8.10.8 , 8.11) if designed for use with a BW vent
Category II appliance	Type BH vent or special venting system (Clause 8.10.4)
Category III appliance	Type BH vent or special venting system (Clause 8.10.4)
Category IV appliance	Type BH vent or special venting system (Clause 8.10.4)
Incinerators, indoors	Chimney (Clauses 8.18.2 , 8.18.12)
Incinerators, outdoors	Single-wall metal pipe (Clauses 8.10.11 , 8.18.2 , 8.18.12)
Appliance that may be converted to use of solid fuel	1202°F (650 °C) chimney (Clauses 8.12.3 , 8.12.6)
Certified combination gas- and oil-burning appliance	Type L vent or chimney (Clauses 8.10.2 , 8.12.4)
Combination gas- and solid-fuel-burning appliance	1202°F (650 °C) chimney (Clauses 8.10.2 , 8.12.2)
Appliance certified for use with chimneys only	Chimney (Clause 8.12)
Decorative appliance in vented fireplace	Chimney (Clauses 7.23.3 to 7.23.5)
Direct-vent appliance	See Clause 8.10.4
Appliance with integral vent	See Clause 8.10.4
Appliance in commercial and industrial installations	Chimney, ventilating hood, and exhaust system (Clause 8.30)

* See Clause [8](#).

8.10.6 Positive venting limitations

An appliance that operates at a positive vent pressure shall not be connected to a venting system serving any other appliance (common vent), except for cases in which all of the following conditions are met:

- All the appliances sharing the vent are of the same type and manufacturer.
- All the appliances are certified by the manufacturer for application in common vent configurations.
- The venting system is sized appropriately by the appliance manufacturer or in accordance with its instructions and as such is considered a special venting system as defined in Clause [3](#).
- The venting components are either supplied or as recommended by the appliance manufacturer.
- A method of flue backflow prevention is present in either the appliance or vent for each appliance and is installed in accordance with the appliance manufacturer's instructions.
- The manufacturer's operation and installation instructions include common venting specific instructions.
- The special venting system or unique feature(s) necessary for common venting application are in accordance with the applicable ANSI or CSA standards for the specific appliance type, CSA/ANSI Z21.13/CSA 4.9 being the case for boilers.

8.10.7 Type B vent

A Type B vent shall only be used with an appliance that is

- a) certified with a draft hood; or
- b) certified and marked for use with a Type B vent.

8.10.8 Type BW vent

A Type BW vent shall only be used with a recessed wall furnace.

8.10.9 Type L vent

Except as specified in Clause [8.10.10](#), a Type L vent shall only be used with an appliance certified for use with Type L vents.

8.10.10 Type L in lieu of type B

A Type L vent may be used in lieu of a Type B vent.

8.10.11 Incinerator venting

Pipe that is either metal of not less than No. 20 GSG (0.8 mm) galvanized sheet steel or an equivalent noncombustible, corrosion-resistant material shall be used for venting an incinerator installed in locations such as an open shed, a breezeway, or a carport, provided that the metal pipe is exposed for its full length and suitable clearances are maintained. See Table [8.6](#).

8.10.12 Vent location limitations

A flue gas vent or a vent connector shall not be installed in either a duct or a shaft used for return air, hot air, ventilating air, or combustion air.

8.10.13 Vent joints and seams

Joints and seams in vents or vent connectors installed in space used to convey return air, such as the space in a false ceiling, shall be sealed.

8.10.14 Special venting limitations

A special venting system shall neither pass through an unheated space nor be installed outdoors unless insulated in accordance with the appliance and/or vent manufacturer's instructions.

8.11 Vent and chimney requirements

A Type B, BH, BW, or L vent or a factory-built chimney used for venting an appliance shall be certified.

8.12 Chimneys

8.12.1 Code requirements

A masonry, concrete, or metal chimney shall be built and installed in accordance with the local building code or, in the absence of such, in accordance with the *National Building Code of Canada*.

8.12.2 Chimney flue examination

Except as provided in Clause [8.21.6](#), before replacing an existing appliance or connecting a vent connector to a chimney, the chimney flue shall be examined to ascertain that the chimney

- a) is properly constructed;

- b) is lined with a tile or a metal liner;
- c) is clear and free of soot, creosote, or obstructions;
- d) will effectively conduct the products of combustion outdoors; and
- e) is sized in accordance with Clause [8.13](#).

8.12.3 Connection to solid fuel flues

An appliance installed in a dwelling unit shall not be connected to a flue serving

- a) a solid-fuelled appliance; or
- b) a solid-fuelled fireplace unless the opening from the fireplace to the flue is permanently closed.

8.12.4 Connection to liquid fuel flues

When a chimney flue serving an appliance that burns a liquid fuel also serves a gas-fuelled appliance, the gas-fuelled appliance vent connector shall be

- a) through a separate flue opening above the flue pipe connection from the liquid-fuelled appliance; or
- b) connected into a shop-fabricated branch fitting that is located
 - i) in a dwelling unit, not more than 30 in (760 mm) from the flue entrance; and
 - ii) in other than a dwelling unit, as close as practicable to the chimney.

8.12.5 Two or more openings

When two or more openings are provided into one chimney flue, they shall be at different levels.

8.12.6 Connection to gas fuel flues

When a chimney flue serving an appliance that burns a solid fuel also serves a gas-fuelled appliance in other than a dwelling unit, the venting of the gas-fuelled appliance shall be through a separate flue opening above the flue pipe connection from the other appliance.

8.12.7 Chimney clean-outs

A chimney shall be provided with a clean-out opening. The clean-out shall be of such construction that it will remain tightly closed when not in use. A tee fitting used as either a clean-out or condensate drain shall have a tight-fitting cap to prevent entrance of air into the chimney at that point.

8.12.8 Masonry chimneys

The flue of a masonry chimney that vents one or more gas-fuelled space-heating appliances with combined inputs of 400 000 Btu/h (120 kW) or less shall be lined in accordance with Clause [8.12.10](#), except where the flue has an existing clay-tile or transite liner

- a) that is capable of removing the total volume of flue gases to the outdoors;
- b) in which the capacity of the liner, as determined by the use of Table [C.5](#), [C.6](#), [C.7](#), or [C.8](#) in Annex [C](#), does not exceed the total input of the appliances to be connected thereto by more than 25%; and
- c) that has been inspected and accepted by an installer in accordance with Clause [8.12.2](#).

8.12.9 Unsafe chimneys

Where inspection reveals that an existing masonry, concrete, or metal chimney is not safe for the intended application, it shall be

- a) repaired or rebuilt;
- b) replaced with a chimney of the same type that complies with Clause [8.12.1](#); or
- c) replaced by either an approved vent or approved factory-built chimney that complies with Clause [8.13](#).

8.12.10 Chimney liner

A metal chimney liner shall provide a continuous lining from the base inside the space where the appliance is located to the top of the masonry chimney flue, and it shall comply with the requirements of ULC S635. It shall be installed in accordance with the manufacturer's instructions.

8.12.11 Semi-detached house venting

The venting of a gas-fired appliance into a chimney flue common to both halves of a semi-detached house shall not be permitted.

8.13 Vent and chimney sizing

8.13.1 Sizing

A vent or a chimney serving a single appliance shall provide effective venting and shall be sized

- a) so that its effective area is not less than that of the draft-control device outlet or the flue outlet; or
- b) in accordance with good engineering practice, such as by the use of
 - i) Table [C.1](#), [C.2](#), [C.5](#), or [C.6](#) of Annex [C](#) for a draft-hood-equipped or a fan-assisted Category I appliance; or
 - ii) engineered venting tables acceptable to the authority having jurisdiction.

8.13.2 Multiple appliance venting

A vent or a chimney serving more than one appliance shall provide effective venting and shall be sized

- a) so that its effective flue area is not less than that of the largest draft-control device outlet or the largest flue outlet, plus 50% of the sum of the outlet areas of the additional appliances; or
- b) in accordance with good engineering practice, such as by the use of
 - i) Table [C.3](#), [C.4](#), [C.7](#), or [C.8](#) of Annex [C](#) for a draft-hood-equipped or a fan-assisted Category I appliance; or
 - ii) engineered venting tables acceptable to the authority having jurisdiction.

8.13.3 Vent shape

A vent may be any shape, provided that its venting capacity is equal to the capacity of the round pipe for which it is substituted, and its minimum internal dimension is not less than 2 in (50 mm). In no case shall the area be less than the area of a 3 in (75 mm) inside diameter pipe. See Table [C.10](#) of Annex [C](#).

8.14 Vent and chimney termination

8.14.1 Wind consideration

A vent or chimney shall extend high enough above either a building or a neighbouring obstruction so that wind from any direction will not create a positive pressure in the vicinity of either the vent termination or the chimney termination.

8.14.2 Vent minimum distances

Except for a special venting system with positive vent pressure, a vent shall extend not less than 2 ft (600 mm) above the highest point where it passes through a flat roof of a building and not less than 2 ft (600 mm) higher than any portion of a building within a horizontal distance of 10 ft (3 m).

8.14.3 Positive vent pressure minimum distances

A vent used in a special venting system with positive vent pressure and passing through a roof shall extend at least 18 in (450 mm) above the highest point where it passes through the roof surface and the same distance above any other obstruction within a horizontal distance of 18 in (450 mm).

8.14.4 Chimney minimum distances

A chimney shall extend not less than 3 ft (0.9 m) above the highest point where it passes through the roof of a building and not less than 2 ft (600 mm) higher than any portion of a building within a horizontal distance of 10 ft (3 m).

8.14.5 Pitched roof distances

Except for a special venting system with positive vent pressure, a vent passing through a pitched roof shall extend above the highest point where it passes through a roof surface in accordance with Figure [8.1](#) and shall extend not less than 2 ft (600 mm) above any obstruction within a horizontal distance of 10 ft (3 m).

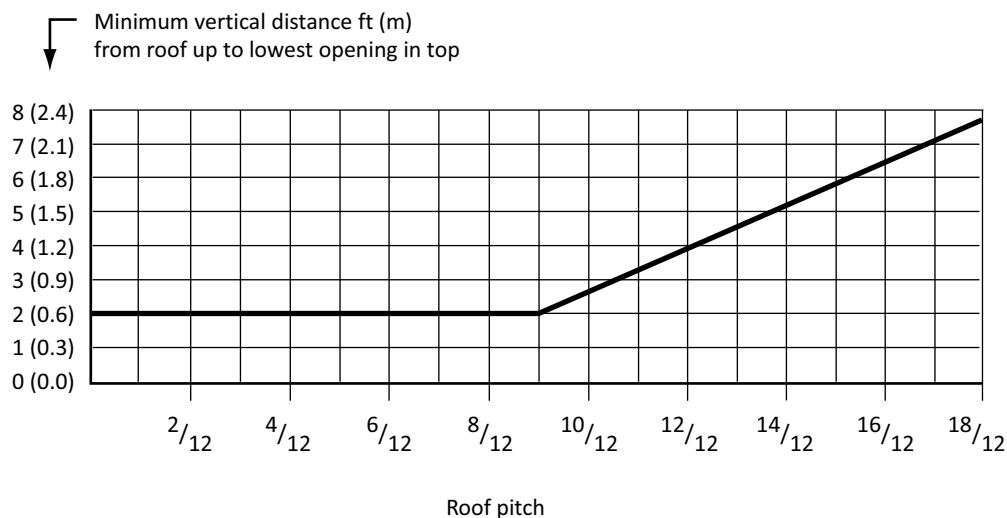
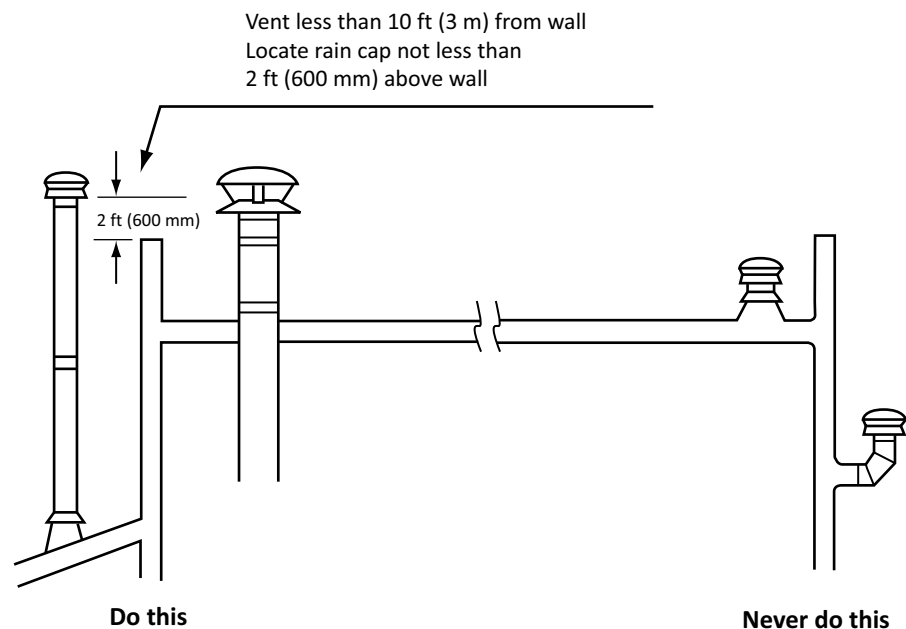
8.14.6 Height above appliance

A vent or chimney shall extend not less than 5 ft (1.5 m) in height above either the highest connected appliance draft-hood outlet or flue collar except as provided in Clause [8.24.1](#).

8.14.7 Wall furnace vent height

A vent serving a wall furnace shall extend not less than 12 ft (3.6 m) in height above the bottom of the furnace and shall contain neither lateral nor horizontal sections unless the furnace is certified for room-heater-type venting and is so marked.

Figure 8.1
Vent passing through a pitched roof
 (See Clause 8.14.5.)



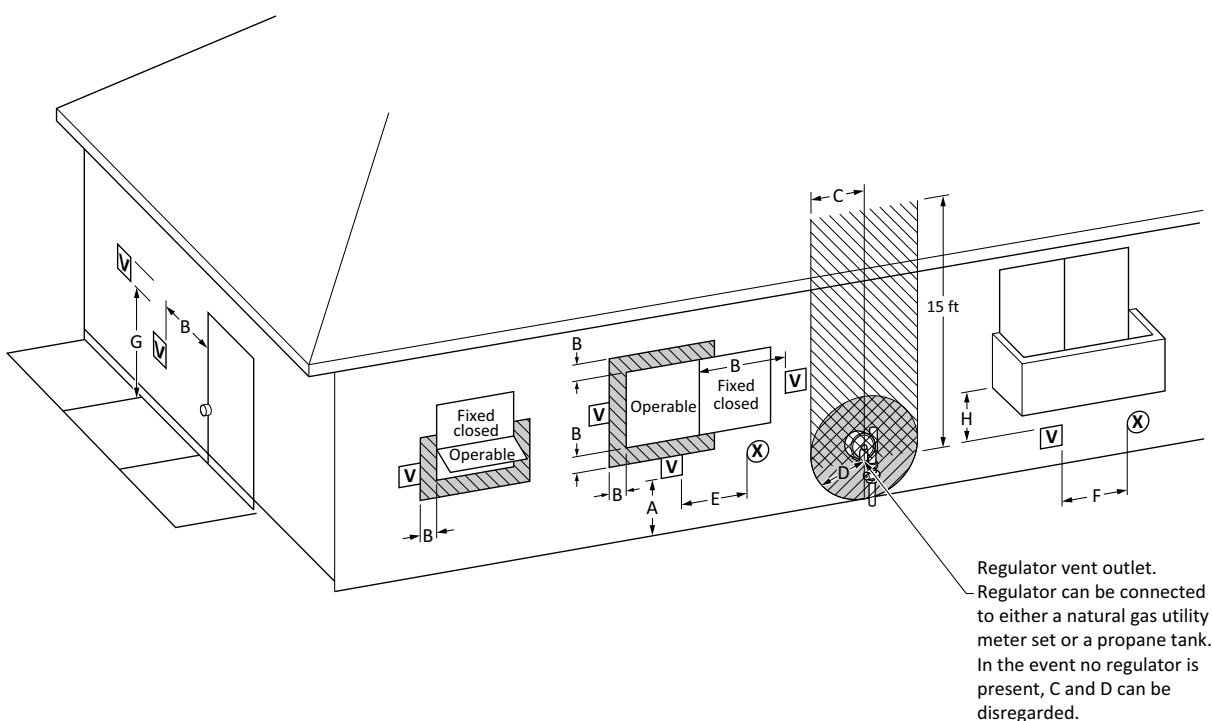
8.14.8 Vent termination limitations

A vent shall not terminate

- where it could cause hazardous frost or ice accumulations on adjacent property surfaces;
- less than 7 ft (2.1 m) above a paved sidewalk or a paved driveway that is located on public property;
- within 6 ft (1.8 m) of a mechanical air-supply inlet to any building;
- above a regulator within 3 ft (0.9 m) horizontally of the vertical centreline of the regulator vent outlet to a maximum vertical distance of 15 ft (4.5 m);

- e) except as required by Clause 8.14.8 d), any distance less than that of any gas pressure regulator vent outlet as detailed in Table 5.3;
- f) less than 1 ft (300 mm) above grade level;
- g) within the following distances of a window or door that can be opened in any building, or of the combustion air inlet of any other appliance:
 - i) 6 in (150 mm) for inputs up to and including 10 000 Btu/h (3 kW);
 - ii) 12 in (300 mm) for inputs from 10 000 Btu/h (3 kW) up to and including 100 000 Btu/h (30 kW); and
 - iii) 3 ft (0.9 m) for inputs exceeding 100 000 Btu/h (30 kW); and
- h) underneath a veranda, porch, or deck unless
 - i) the veranda, porch, or deck is fully open on a minimum of two sides beneath the floor; and
 - ii) the distance between the top of the vent termination and the underside of the veranda, porch, or deck is greater than 1 ft (300 mm).

Figure 8.2
Vent termination
 (See Clause 8.14.8.)



Legend:

- V = Vent terminal
- X = Air supply inlet
- = Area where terminal is not permitted

(Continued)

Figure 8.2 (Continued)

Dimension	Clause
A =	8.14.8 f)
B =	8.14.8 g)
C =	8.14.8 d)
D =	8.14.8 e) and Table 5.2 (applies to any natural gas and propane pressure regulators)
E =	8.14.8 g)
F =	8.14.8 c)
G =	8.14.8 b)
H =	8.14.8 h)

Note: This figure is intended to illustrate the clearance requirements in Clause [8.14.8](#).

See also Clauses [7.4.4](#), [7.5.2](#), and [8.14.10](#) for vent termination requirements.

8.14.9 Direct-vent clearances

When more than one direct-vent appliance of the same make and model are installed, the clearances between the air-intake and exhaust vent terminals may be reduced from the clearances required by this Code, provided that the appliances have been tested and certified for such reduced clearances. The manufacturer's instructions shall specify and illustrate the reduced clearances.

8.14.10 Outdoor pool heater flue

The flue gas discharge opening for an outdoor pool heater shall terminate not less than 10 ft (3 m) from any building opening.

8.14.11 Termination cap

A terminus of a vent shall be fitted with a cap either in accordance with the vent manufacturer's installation instructions or in accordance with the manufacturer's installation instructions for a special venting system.

8.14.12 Wall termination vent limitations

A vent from an appliance shall not extend through an exterior wall and terminate adjacent to the exterior wall unless

- a) the appliance is a direct-vent appliance;
- b) the appliance is intended for connection to a special venting system;
- c) the appliance and its complete vent assembly are specifically certified to be installed in this manner;
- d) the venting system is equipped with a certified power venter that complies with Clause [8.29.2](#); or
- e) the venting system is equipped with a certified power venter that complies with Clause [8.24.2](#).

8.14.13 Dormant vent installation

The vent from one or more gas-fired appliances may be installed vertically inside a dormant masonry flue, a dormant certified chimney, or a dormant vent, provided that

- a) each appliance is equipped with its own individual vent or common stack(s), as is the case with common vented appliance sets meeting the requirements of Clause [8.10.6](#), installed in accordance with the requirements of Clauses [8.12](#) to [8.14](#), as applicable; and
- b) spacers are installed to maintain a minimum clearance of 1 in (25 mm) between an active vent and a combustible vent.

The space surrounding any vent shall not be used to convey flue gases for any other appliance.

8.15 Vent and chimney support

8.15.1 Independent support

A vent or chimney shall be adequately supported independent of the appliance being served.

8.15.2 Manufacturer's instructions

A vent or factory-built chimney shall be installed in accordance with the manufacturer's instructions.

8.15.3 Direct connection to flue outlet or draft-hood outlet

A vent may be directly connected to the flue outlet or draft-hood outlet of the appliance that it serves, provided that the vent is independently supported, and the connection is made in accordance with the manufacturer's instructions.

8.15.4 Double to single wall adapter

When the vent referred to in Clause [8.15.3](#) is of double-wall construction, the connection shall be made by the use of a certified double-wall to single-wall adapter, and by

- a) an adjustable telescopic-type fitting, fabricated by the vent manufacturer; or
- b) a single-wall vent connector having a maximum length not exceeding 18 in (450 mm) and a minimum length not less than 12 in (300 mm).

8.15.5 Single-wall vent connector

When a single-wall vent connector connects an appliance to a Type B vent

- a) the base fitting shall be accessible for inspection;
- b) the connection shall be by means of a certified double-wall to single-wall adapter; and
- c) the clearance from combustibles as specified in Table [8.6](#) shall be maintained.

8.16 Vents and chimneys serving two or more appliances

Note: Clauses [8.16.1](#) and [8.16.2](#) apply to common venting of non-positive vent Category I and II appliances. Refer to Clause [8.10.6](#) for common vent application of Category III and IV positive vent pressure appliances.

8.16.1 Connection locations

When two or more vent connectors enter either a common vent or a common chimney, they shall not enter at the same horizontal plane. The smallest vent connector shall enter at the highest level consistent with the available headroom and clearance from combustible material, except that a vent connector from an incinerator shall be installed at the lowest level. When the vent connectors are of the same diameter, the vent connector serving the appliance with the lowest British thermal unit input shall be connected at the highest level.

8.16.2 Common vent connector

When two or more appliances are vented through a common vent connector, the common vent connector shall be located at the highest level consistent with the available headroom and clearance from combustible material.

8.17 Vents outside buildings

8.17.1 Natural-draft type

When the installation of a vent used in a natural-draft venting system is impracticable inside a building, it may be done outdoors, provided that the vent is

- a) certified for outside installation; and
- b) installed in accordance with the vent manufacturer's instructions.

8.17.2 Type B or Type L

The portion of an indoor-installed Type B or Type L vent that extends above the roof line shall be certified for exterior use, and each length shall be so identified.

8.18 Vent connectors

8.18.1 Materials without dilution air

A vent connector that serves an appliance without flue gas dilution air shall be constructed of materials having corrosion resistance at least equivalent to that of No. 24 GSG (0.60 mm) galvanized steel.

8.18.2 Materials for incinerators

A vent connector that serves an incinerator shall be constructed of materials suitable for the intended use of the incinerator and shall have corrosion resistance and durability to heat at least equivalent to that of No. 20 GSG (0.90 mm) galvanized steel.

8.18.3 Materials with dilution air

Unless certified, a vent connector that serves an appliance with a draft hood shall not be made of material less than

- a) No. 28 GSG (0.38 mm) for all sizes up to and including 5 in (127 mm) diameter;
- b) No. 26 GSG (0.47 mm) for sizes over 5 in (127 mm) up to and including 8 in (203 mm) diameter;
- c) No. 24 GSG (0.60 mm) for sizes over 8 in (203 mm) up to and including 16 in (406 mm) diameter; and
- d) No. 20 GSG (0.90 mm) for sizes over 16 in (406 mm) up to and including 30 in (760 mm) diameter.

For sizes over 30 in (760 mm), the vent connector shall be subject to good engineering practice.

Unless certified, an aluminum vent connector shall be not less than No. 22 B&SG (0.64 mm), except that a Type B vent may be used for the connector between the draft hood and the chimney.

8.18.4 Minimum size

A vent connector that serves a single appliance with a single flue outlet shall not be smaller than the minimum size of either the vent or chimney as determined in Clause [8.13.1](#).

8.18.5 Size change

When a vent connector is required to have a size other than that of either the appliance flue collar or draft-hood outlet, the change in size shall be made as follows:

- a) If the size is increased, the change in size shall be made at either the appliance flue collar or draft-hood outlet, except that when either the flue collar or draft-hood outlet is inside the casing of the appliance, the increase shall be made immediately external to the appliance casing.
- b) If the size is reduced, the change in size shall be made at the appliance flue collar.

8.18.6 Size change calculations

When the size of either a draft hood or a flue collar is changed in accordance with Clause [8.18.5](#), the revised size shall be considered the size of either the draft hood or flue-collar outlet when making calculations in accordance with Clauses [8.13.1](#) to [8.13.3](#) and [8.18.7](#).

8.18.7 Multiple appliances

A vent connector that serves two or more appliances, or a vent connector that serves a single appliance with two or more flue outlets, shall

- a) have
 - i) an area not less than the total area of all vent connectors, with the size of each individual vent connector determined separately in accordance with Clause [8.18.5](#); and
 - ii) a total length not exceeding the values specified in Table [C.9](#). The total length of the connector in Table [C.9](#) shall include manifold length and each individual appliance length; or
- b) be sized in accordance with an engineering table acceptable to the authority having jurisdiction.

8.18.8 Size change location

When either a chimney or a vent has been sized in accordance with the requirements of Clauses [8.13.1](#) and [8.13.2](#), and the flue area is less than the area of the vent connector, the change in size shall be made where the vent connector is attached to either the chimney or the vent.

8.18.9 Special venting system minimum clearances

A special venting system shall be installed with a minimum clearance from combustible material in accordance with its certification.

8.18.10 Type B minimum clearances

Except as provided in Clause [8.18.13](#), a vent connector material of Type B vent shall be installed with a minimum clearance from combustible material. This clearance shall be maintained if it passes through either a combustible wall or partition, in accordance with Table [8.6](#).

8.18.11 Appliance minimum clearances

Except as provided in Clause [8.18.13](#), the minimum clearance of a vent connector of other than Type B vent material from a combustible wall or partition shall be in accordance with Table [8.6](#).

Table 8.6
Appliance vent connector clearances
 (See Clauses [8.10.11](#), [8.15.5](#), [8.18.10](#), and [8.18.11](#).)

Appliance connector	Minimum distance from combustible material, in (mm)	
	Type B vent connector	Other than Type B vent connector
Boiler	1 (25)*	6 (150)*
Warm-air furnace	1 (25)*	6 (150)*
Service water heater	1 (25)*	6 (150)*
Space heater	1 (25)*	6 (150)*
Floor furnace	3 (75)†	9 (225)
Incinerator	Not permitted	18 (450)
Conversion burner (with draft hood)	6 (150)	9 (225)

* Except as otherwise certified.

† 3 in (75 mm) for a distance of not less than 3 ft (0.9 m) from the outlet of the draft hood. Beyond 3 ft (0.9 m), the minimum clearance shall be 1 in (25 mm).

8.18.12 Combustible wall or partition penetrations

When a vent connector of other than Type B vent material passes through either a combustible wall or partition, the combustible material shall be guarded at the point of passage with

- a) a ventilated metal thimble not smaller than the following:
 - i) for an appliance other than a floor furnace and incinerator, 4 in (100 mm) larger in diameter than the vent connector, unless there is not less than 6 ft (1.8 m) of vent connector in the open between the draft hood and thimble, in which case the thimble may be 2 in (50 mm) larger in diameter than the vent connector;
 - ii) for a floor furnace, 6 in (150 mm) larger in diameter than the vent connector; and
 - iii) for an incinerator, 12 in (300 mm) larger in diameter than the vent connector; or
- b) noncombustible insulation and clearances that will prevent the surface temperature of the combustible material from exceeding 194 °F (90 °C).

8.18.13 Combustible wall protection

Except for passage through either a combustible wall or combustible partition, the clearance from a vent connector to combustible material may be reduced when the combustible material is protected as specified in Table [8.7](#).

Table 8.7
Vent connector clearance reductions with specified forms of protection
 (See Clause [8.18.13.](#))

Type of protection	Where the required clearance with no protection is		
	6 in (150 mm)	9 in (225 mm)	18 in (450 mm)
	Clearance may be reduced by, in (mm)		
0.25 in (6 mm) insulating millboard spaced 1 in (25 mm) apart with noncombustible spacers	3 (75)	3 (75)	6 (150)
No. 28 MSG (0.30 mm) metal on 0.25 in (6 mm) insulating millboard	4 (100)	5 (125)	6 (150)
No. 28 MSG (0.30 mm) metal spaced 1 in (25 mm) apart with noncombustible spacers	4 (100)	5 (125)	9 (225)

8.18.14 General installation

A vent connector shall be installed so as to avoid unnecessary turns and other features that create added resistance to the flow of flue gases.

8.18.15 Fasteners

A vent connector shall be firmly attached to either a draft-hood outlet or flue collar by sheet metal screws or mechanical fasteners, or in accordance with the manufacturer's instructions.

Note: A moisture-exhaust duct from a clothes dryer is not considered to be a vent connector. See Clauses [7.4](#) and [7.5](#).

8.18.16 General installation for type B

A vent connector of Type B vent material shall be securely assembled in accordance with the manufacturer's instructions.

8.18.17 No dips or sags

A vent connector shall be installed without dips or sags.

8.18.18 Converted furnace or boiler

A vent connector of either Type B or Type L vent material shall not be used between the flue outlet and the draft-control device of either a converted furnace or converted boiler.

8.18.19 Horizontal design

The horizontal run of a vent connector shall be as short as practicable, and the appliance shall be located as near the vent or chimney as practicable.

8.18.20 Hanger support

A vent connector shall be securely supported by noncombustible hangers suitable for the weight and design of the materials employed. A slip joint in the horizontal section of a vent connector shall be secured with either sheet metal screws or in accordance with the manufacturer's instructions to prevent sagging.

8.18.21 Unheated spaces

When a vent connector from a Category I appliance passes through an unheated space, that portion of the vent connector shall be either a Type B or Type L vent.

8.18.22 General construction

A vent connector shall be of either metal or other noncombustible material capable of withstanding the flue gas temperatures involved. It shall have sufficient strength to withstand damage likely to occur under the conditions of use and shall be securely supported.

8.18.23 Floor or ceiling limitations

A single-wall vent connector shall not pass through either a floor or a ceiling.

8.19 Chimney connections

8.19.1 Location above clean-out

The entrance of a vent connector into a chimney shall be above the chimney clean-out opening.

8.19.2 Connection to chimney

The vent connector shall not protrude into the chimney and obstruct the chimney flue.

8.19.3 Cleaning sleeve

A sleeve shall be used to facilitate removal of the vent connector for cleaning.

8.19.4 Chimney liner connection

The space between a chimney and chimney liner shall be sealed at the point of entry of a vent connector.

8.20 Size and height of interconnected vent connectors

Two or more vent connectors may be joined through a common vent connector, provided that

- each vent connector has the greatest possible rise consistent with the headroom available and has the required clearances from combustible material;
- the size of each vent connector is in accordance with Clauses [8.18.5](#) and [8.18.7](#); and
- in the case of positive pressure appliances, the appliance or common vent system meets the requirements of Clause [8.10.6](#), as applicable.

8.21 Multi-storey venting

8.21.1 Common gas vents

A common gas vent is permissible in multi-storey installations to vent appliances

- certified with a draft hood; or
- certified and marked for use with a Type B vent

and located on one or more floor levels, provided that the venting system is installed in accordance with the requirements of Clause [8.21](#).

8.21.2 Appliance location and combustion air

When a common gas vent is used in accordance with Clause [8.21.1](#), the gas appliances shall

- a) be installed in an enclosure having access solely from an unoccupied space such as a hallway, service area, or outdoor balcony; and
- b) have combustion air supplied to the enclosure by means of grilles or ducts that communicate directly with the outdoors, are sized in accordance with the requirements of Table [8.1](#) or [8.2](#), and are installed in accordance with Clause [8.3](#). (See Figure [C.13](#) of Annex [C](#) for a practical method of meeting this requirement.) Combustion air shall not be taken from inhabited or occupied spaces within the building.

8.21.3 Location exemption for one dwelling unit

The requirements of Clause [8.21.2](#) shall not apply if all appliances served by a common venting system are located within one dwelling unit.

8.21.4 Return air requirements

When a forced-air furnace is installed in an enclosure in accordance with Clause [8.21.2](#), no opening shall be located in the furnace return-air system within the enclosure, and means shall be provided on the return-air system to prevent the infiltration of air from inside the enclosure.

8.21.5 Vent size and type

When a common gas vent is used in accordance with Clause [8.21.1](#), the common gas vent shall be

- a) of a Type B or Type L vent;
- b) sized in accordance with the requirements indicated in Tables [C.3](#) and [C.4](#) and illustrated in Figures [C.11](#) and [C.12](#) of Annex [C](#); and
- c) installed in accordance with the manufacturer's instructions.

8.21.6 Replacement appliances

With approval of the authority having jurisdiction, a common chimney is permissible in multi-storey installations to vent appliances replacing existing appliances, provided that

- a) new appliances are of the same size (input capacity with similar efficiency range) and venting type (fan-assisted or natural draft) as the original;
- b) the venting system does not include significant changes; and
- c) the venting system provides effective venting and shows no signs of condensation or deterioration.

Evidence shall be provided to authenticate the characteristics of the replaced appliance to the authority having jurisdiction.

8.22 Dampers and attachments

A device or attachment that might in any way impair either the combustion or safe venting of the combustion products shall be prohibited.

8.23 Draft hoods

8.23.1 Zero over-fire draft appliances

Except for an incinerator, a dual oven-type combination range, and a direct-vent appliance, an appliance requiring zero over-fire draft for operation shall be installed with a draft hood.

8.23.2 Positive over-fire or induced draft appliances

A draft hood shall not be used on an appliance with either positive over-fire draft or an induced draft.

8.23.3 Appliance supplied draft hood

A draft hood either supplied with or forming part of an appliance shall be installed without alteration.

8.23.4 Approved design

When a draft hood is required and it is not supplied by the appliance manufacturer, it shall be supplied by the installer, and it shall be of an approved design. See Annex [F](#).

8.23.5 Outlet size

The draft-hood outlet shall be of the same size as the appliance flue collar unless otherwise sized in the appliance manufacturer's instructions.

8.23.6 Location to combustion air

A draft hood shall be in the same room as the combustion air opening of the appliance. A draft hood shall not be installed in a false ceiling space, in a room other than the one the appliance serves, or in any manner that could permit a difference in pressure between the draft-hood relief opening and the combustion air supply. The draft hood supplied for a conversion burner shall be located so that the burner is capable of safe and efficient operation.

8.23.7 Installation orientation

A draft hood shall be installed in the position for which it was designed with reference to the horizontal and vertical planes, and shall be located so that the relief opening is not obstructed by any part of the appliance or adjacent construction. The appliance and its draft hood shall be located so that the relief opening is accessible for checking vent operation.

8.23.8 Draft regulators

A draft regulator shall not be used as a substitute for a draft hood.

8.24 Venting arrangements

8.24.1 Exclusions

The requirements for a draft-control device in [Clauses 8.8 to 8.24](#) shall not apply either to a direct-vent appliance or to an appliance requiring a special venting system.

8.24.2 Power venters

A power venter may be used in place of a natural-draft vent, provided that means are provided to prevent the flow of gas to the main burner in the event of failure of the power venter. Such a venter shall not be used in conjunction with an incinerator.

8.24.3 Vent connectors

A vent connector that serves an appliance designed for natural-draft venting shall not be connected to any portion of a venting system that is under positive pressure or serves an appliance requiring a special venting system.

8.24.4 Exhaust systems

An exhaust hood or canopy for an industrial appliance may be used in place of a direct flue connection in certain cases, for instance, when the process itself requires fume disposal. The design of the venting system shall conform to the requirements of the authority having jurisdiction.

8.24.5 Venting directly into space

When located in a large and adequately ventilated space, an appliance may be operated by discharging the combustion products directly into the space, subject to the approval of the authority having jurisdiction and provided that the maximum input of the appliance does not exceed 20 Btu/h/ft³ (0.2 kW/m³) of the space in which the appliance is located. An appliance designed to produce a controlled atmosphere need not be subject to these conditions.

8.24.6 Multiple appliances

A venting system that serves one or more appliances shall provide adequate venting and shall be sized in accordance with approved engineering design.

8.25 Draft regulators

A draft regulator, when used, shall be located so that the relief opening is not obstructed by any part of the appliance or adjacent construction. When used with an incinerator, a draft regulator shall be of the single-acting type. In all other installations, it shall be of the double-acting type.

8.26 Automatic vent damper or automatic flue damper

8.26.1 Integral component

An electrically operated automatic vent damper device shall not be used in a dwelling unit except where the device is provided as an integral component of a certified appliance.

8.26.2 Vent types

A residential appliance certified with an automatically operated vent damper device shall be connected to

- a) a Type B vent complying with CAN/ULC-S605;
- b) a Type L vent complying with CAN/ULC-S609;
- c) a factory-built chimney complying with CAN/ULC-S604, CAN/ULC-S629, or UL 959; or
- d) either a masonry or a concrete chimney lined with a certified chimney liner.

8.26.3 Thermally actuated vent damper

A thermally actuated automatic vent damper device shall be installed

- a) in accordance with the manufacturer's instructions; and
- b) only in the vent connector of an appliance equipped with a draft hood that is connected to
 - i) a certified factory-built chimney, Type B vent, or Type L vent; or
 - ii) a masonry chimney flue that is lined in accordance with Clause [8.12.10](#).

8.26.4 Appliance installation

An automatic flue damper

- a) shall not be installed on an appliance within a dwelling unit except when provided as an integral component of a certified appliance;
- b) shall not be installed on an appliance equipped with a draft hood or a draft regulator;
- c) when installed on an appliance equipped with a continuous pilot, shall be designed or constructed to provide a fixed minimum opening of 20% of the flue area and shall be interlocked with the burner control system so that
 - i) the damper is proven to be in the fully open position before ignition of the main burner; and
 - ii) the fuel supply to the appliance will be shut off in the event of failure except where the damper is provided with means to ensure that it will fall fully open upon loss of driving medium; and
- d) when installed on an appliance equipped with an intermittent pilot, an interrupted pilot, direct-spark ignition, or hot surface ignition of the main burner, shall be interlocked with the burner control system so that
 - i) the damper is proven to be in the fully open position before ignition of the fuel; and
 - ii) the fuel supply to the appliance will be shut off in the event of damper failure, except where the damper is provided with means to ensure that it will fall fully open upon loss of driving medium.

8.27 Manually operated flue dampers

8.27.1 Limitations for use

A manually operated flue damper shall not be used with

- a) a residential appliance;
- b) an appliance equipped with a draft-control device; or
- c) a commercial- or industrial-type appliance, unless the damper is
 - i) provided with a fixed opening; and
 - ii) designed, constructed, and field tested to ensure safe operation at a fixed minimum opening.

8.27.2 Pressure point adjuster

When a baffle or neutral pressure point adjuster is used, it shall

- a) be located upstream of the draft regulator; and
- b) have a fixed safe minimum opening.

8.28 Installation of draft-control devices

8.28.1 Installation requirements

When a draft-control device is either part of an appliance or supplied by the appliance manufacturer, it shall be installed in accordance with the appliance manufacturer's installation instructions.

8.28.2 Installation orientation

A draft-control device shall be installed in the position for which it is designed with reference to the horizontal and vertical planes and shall be so located that relief openings are not obstructed.

8.29 Induced- or forced-draft devices

8.29.1 Failure provisions

When an induced- or forced-draft device is used, provision shall be made to prevent the flow of gas to the burner on failure of the device.

8.29.2 Water heater size limitation

A power venter certified for use as an add-on to a water heater may be used on a water heater that has an input of 50 000 Btu/h (15 kW) or less.

8.30 Venting of appliances into canopies

8.30.1 General requirements

An appliance may be vented through an exhaust canopy installed in a location other than a dwelling unit, provided that

- a) the canopy complies with the requirements of the local building code or, in the absence of such, with the requirements of the *National Building Code of Canada*;
- b) the exhaust volume of the canopy is sufficient to provide for capture and removal of grease-laden vapours and products of combustion;
- c) the appliance has an input not exceeding 400 000 Btu/h (120 kW) and its flue outlet is directly under the canopy; and
- d) the appliance is interlocked with the exhaust in accordance with Clause [8.30.2](#), except where it is approved to ANSI Z83.11/CSA 1.8.

8.30.2 Appliance interlocking

An approved appliance not identified in Clause [8.30.1](#) d) may be installed under a canopy, provided that the appliance is interlocked so that operation is permitted only when exhaust airflow is proven.

8.30.3 Multiple appliance interlocking

Where more than one appliance is installed under the same canopy in accordance with Clause [8.30.2](#), a single system installed in accordance with Clause [8.30.2](#) may be used to interlock the appliances.

8.30.4 Booster water heater

Clause [8.30.2](#) shall not apply to a booster water heater with an input of 50 000 Btu/h (15 kW) or less that supplies water to an automatic dishwasher.

8.31 Heat reclaimers

8.31.1 Installation requirements

A heat reclaimer shall not be used with a gas-fired appliance installed in a dwelling unit, in mobile housing, or in a recreational vehicle unless it is

- a) approved for the application; and
- b) installed in accordance with the manufacturer's instructions.

8.31.2 Commercial or industrial requirements

A heat reclaimer installed on a commercial or industrial gas-fired appliance shall be

- a) approved for the application; and

- b) installed in accordance with the manufacturer's instructions.

8.32 Overpressure protection for engine venting systems

8.32.1 Overpressure forces

Combustion exhaust systems for engine-driven appliances shall be designed and constructed to withstand forces caused by the ignition of unburned fuel or shall have provisions to relieve those forces without damaging the exhaust system.

8.32.2 Protection of fabricated venting systems

Unlisted exhaust venting systems shall be deemed to meet the requirements of Clause [8.32.1](#) provided that

- a) the unlisted chimney conforms to Clause [8.18.2](#) and NFPA 211; and
- b) are constructed of material that is at least equivalent to Schedule 10 stainless steel pipe*, fabricated with welded joints, and pressure tested to 50 psig (340 kPa).

* Such as stainless steel pipe complying with ASTM A312/A312M.

8.32.3 Commercial or industrial requirements

Where over-pressure relief devices that form part of a certified chimney system are used to meet the requirement of Clause [8.32.1](#), the over-pressure relief device may terminate inside a dedicated service room containing the engine-driven appliance or engine enclosure provided

- a) they are readily visible for inspection; and
- b) there is no visual evidence that the relief device is leaking combustion products.

Note: Refer to manufacturer instructions for certified chimneys and vent connectors.

9 Natural gas compressors and cylinders

9.1 Installation of compressors

9.1.1 Applicability

Manifold applications shall comply with the requirements of Clause [7.2](#), and cylinder-filling applications shall comply with the requirements of Clause [9](#).

9.1.2 Protection

Compressors, cylinders, valves, regulators, gauges, piping, tubing, hose, and other equipment shall be protected against damage.

9.1.3 Installation occupancy

The compressor shall only be installed in industrial occupancies as defined in the *National Building Code of Canada*, unless otherwise approved.

9.1.4 Exposure

The compressor shall be installed indoors or outdoors in accordance with its certification.

9.1.5 Vibration isolation

Where required, a compressor shall be isolated from vibration at the inlet and outlet by a gas hose.

9.1.6 Component pressure rating

Piping, hose, and fittings utilized in the cylinder manifold shall be rated for a minimum of 1.5 times the maximum operating pressure.

9.1.7 Supply capacity

The gas supplier or distributor shall be notified by the installer prior to installing a compressor to analyze capacity considerations.

9.2 Requirements for cylinders

9.2.1 Qualification

Refillable cylinders shall be manufactured, tested, inspected, and marked in accordance with the requirements of the *Transportation of Dangerous Goods Regulations (TDG Regulations)* of Transport Canada.

9.2.2 Maximum capacity

Cylinders used indoors shall have a water capacity not greater than 175 lb (80 kg).

9.2.3 Marking

In addition to the marking requirements, the cylinder shall be legibly marked "FOR NATURAL GAS ONLY"*.

* The equivalent French wording is "GAZ NATUREL SEULEMENT".

9.2.4 Requalification

Re-examination and re-marking of a refillable cylinder shall be done in accordance with the *TDG Regulations* of Transport Canada.

9.2.5 Protection

Cylinders shall be protected as required by the *TDG Regulations* of Transport Canada.

9.3 Cylinder filling and cylinders

9.3.1 Visual inspection

Cylinders shall be selected and filled in accordance with the requirements of the *TDG Regulations* of Transport Canada. Prior to filling, a cylinder shall be given a thorough visual inspection, and it shall be rejected and taken out of service if evidence is found that it has a bad dent, damaged foot ring or protective collar, corroded area, leak, or another condition that indicates a possible weakness that would render it unfit for service.

9.3.2 Maximum pressure

Cylinders used indoors shall not be filled in excess of 300 psig (2100 kPa) at 59 °F (15 °C).

9.3.3 Filling operations

A cylinder connected for filling shall be located outdoors, unless indoor filling is approved by the authority having jurisdiction, and shall be

- a) filled through
 - i) an approved manifold; and
 - ii) an approved connector equipped with a back-check valve at the point of cylinder connection;
- b) located not less than 3 ft (0.9 m) from a building opening or air intake; and
- c) located not less than 10 ft (3 m) from a mechanical air intake.

9.3.4 Filling manifold

The manifold specified in Clause [9.3.3 a\) i\)](#) shall be located outdoors, unless indoor filling is approved by the authority having jurisdiction, and shall be

- a) connected to the compressor with rigid piping or tubing; and
- b) installed in accordance with Clause [9.1.1](#).

9.3.5 Indoor filling

Where the authority having jurisdiction permits indoor filling, the room in which filling takes place shall be vented to the outdoors by a mechanical ventilation system equipped with a proven airflow interlock to stop the operation of the compressor in the event that the ventilation rate is less than 10 times the flow capacity of the compressor.

9.3.6 Signage

Signs marked “NO SMOKING” and “DÉFENSE DE FUMER” or signs using symbols shall be prominently displayed in the filling area.

9.4 Cylinder storage

9.4.1 Clearance to fuels

Any cylinder in storage shall be located not less than 20 ft (6 m) from any flammable liquid, oxidizing agent, or combustible gas other than natural gas, or shall be separated in a manner acceptable to the authority having jurisdiction.

9.4.2 Storage cabinet

Cylinders that have been in service and are not connected for use or filling shall be stored outdoors in a storage cabinet in accordance with Clause [9.4.3](#).

9.4.3 Cabinet design

The storage cabinet shall

- a) be supplied with a top cover;
- b) be made of noncombustible material and be structurally sound, with no openings greater than 4 in² (25.8 cm²);
- c) have at least 2 sides constructed to provide equal ventilation through openings at the top and bottom of the side, providing, as a minimum, the equivalent total opening of 15% open area on each side panel;
- d) not restrict the dispersion of any fuel gas leak to ensure it is well ventilated;
- e) have its base on a firm level footing in an upright position; and
- f) not be located against other objects, or have objects attached, that restrict ventilation.

9.4.4 Clearance to property line

Cylinders stored outdoors shall be located

- a) 5 ft (1.5 m) from a property line if the aggregate capacity of expanded gas* is up to and including 6000 ft³ (170 000 L);
- b) 25 ft (7.5 m) from a property line if the aggregate of expanded gas* is more than 6000 ft³ (170 000 L) and up to and including 18 000 ft³ (500 000 L); and
- c) 50 ft (15 m) from a property line if the aggregate of expanded gas* exceeds 18 000 ft³ (500 000 L).

* This is evaluated at standard pressure of 14.7 psi absolute (101.3 kPa) and temperature of 59 °F (15 °C).

9.4.5 Protection

Each side of a cylinder storage area exposed to vehicular traffic shall be protected by barriers, posts, or guardrails.

9.4.6 Cylinder valve

Any cylinder in storage shall have the cylinder valve closed.

9.5 Cylinders connected for use

9.5.1 Indoor safety

A cylinder may be used indoors to supply natural gas for welding, cutting, and preheating when portability is necessary, provided that

- a) a pressure regulator is employed and directly connected to the equipment or cylinder valve, or located on a manifold that is connected to the cylinder valve;
- b) the total number of cylinders connected to other industrial gas cylinders does not exceed four, and their aggregate volume of expanded gas to the absolute pressure of 14.696 psi (101.325 kPa) and a temperature of 59 °F (15 °C) does not exceed 1000 scf (28.3 m³). If there is more than one manifold of cylinders, the manifolds may be located in the same area, provided that they are separated by a distance of at least 15 ft (4.5 m);
- c) the cylinder, regulating equipment, and manifold are not located where they are subject to damage or to temperatures in excess of 125 °F (50 °C); and
- d) the cylinder is equipped with an excess-flow valve that is either integral to the cylinder valve or connected to the cylinder valve outlet. In either case, it shall be installed in such a manner that undue strain beyond the excess-flow valve will not cause breakage between the cylinder and the excess flow valve.

9.5.2 Indoor location

A cylinder in use inside a building shall not be located near an exit, a stairway, or an area normally used or intended for the safe evacuation of people.

9.5.3 Connection location

Connection and disconnection of cylinders shall be done in a well-ventilated area with no source of ignition within 10 ft (3 m) from the point of connection.

10 Residential fuelling appliances (RFAs) and vehicle fuelling appliances (VFAs) used for natural gas without storage

10.1 Installation

10.1.1 Vehicle refuelling appliance (VRA) installation

A VRA that was certified to CSA 12.6 shall be installed in accordance with the manufacturer's installation instructions and local requirements, including fire regulations, building codes, and zoning requirements.

Note: This clause is for VRAs that were certified to CGA 12.6-M94 and CSA 12.6-2004.

10.1.2 RFA installation

An RFA shall be certified to CSA/ANSI NGV 5.1 and shall be installed in accordance with the manufacturer's installation instructions and local requirements, including fire regulations, building codes, and zoning requirements.

10.1.3 VFA installation

A VFA shall be certified to CSA/ANSI NGV 5.2 and shall be installed in accordance with the manufacturer's installation instructions and local requirements, including fire regulations, building codes, and zoning requirements.

Note: For VFAs connected to storage vessels, see CSA B108.1.

10.1.4 RFA or VFA exposure

An RFA or a VFA shall be installed outdoors unless certified and labelled for indoor installation. An indoor installation of an RFA or a VFA shall be in a non-living space.

10.1.5 RFA or VFA support

An RFA or a VFA shall be installed on a firm support to prevent undue stress on the gas piping system and electrical conduits.

10.1.6 Protection

The outdoor or indoor installation of an RFA or a VFA, and associated equipment, shall be protected by approved means against vehicle impact, ice build-up, flooding, and blockage of ventilation, where required.

10.1.7 Indoor fuelling

When a vehicle is fuelled indoors, a gas detector shall

- a) be installed within 6 in (150 mm) of the ceiling or highest point indoors above the fuelling area, or be supplied by the RFA or VFA as mounted directly to the RFA or VFA;
- b) be set to activate at natural gas detection levels at and above one-fifth of the lower limit of flammability of natural gas;
- c) upon activation, produce an audible and visual alarm;
- d) be interlocked with a mechanical ventilation system (see Clause [10.1.8](#)); and
- e) be interlocked to shut off the RFA or the VFA.

10.1.8 Ventilation

The mechanical ventilation system referred to in Clause [10.1.7 d\)](#) shall

- a) vent the fuelling area to the outdoors at a flow rate of 25 times the flow rate of the VFA; and
- b) provide for minimum clearances from the discharge as specified in Table [5.3](#)

Alternatively, the mechanical ventilation system may be determined by the RFA or VFA installation instructions.

10.2 Pressure relief devices and other vents and vent lines

10.2.1 Vent discharge

The discharge from pressure relief devices, and other vents and vent lines, shall be vented to the outdoors

- a) at least 3 ft (0.9 m) away from sources of ignition;
- b) a minimum of 3 ft (0.9 m) horizontally distant from, and 6 in (150 mm) above, openings or vents into a building or space where gas is likely to accumulate; and
- c) away from an area occupied by the public, e.g., a sidewalk.

10.2.2 Vent termination

The outdoor vent termination shall be equipped with a means to prevent the entry of water, insects, foreign material, and ice build-up.

10.2.3 Vent pipe sizing

The vent piping and tubing shall be sized so that the capacity of the relief device is not restricted more than is allowed in the manufacturer's instructions.

10.3 Piping, tubing, and hose

10.3.1 Requirements

The following requirements shall apply to all gas piping systems required to install an RFA or a VFA and its associated equipment:

- a) The RFA or the VFA and associated equipment shall be installed in accordance with the manufacturer's instructions.
- b) The gas supply line to the inlet of the RFA or the VFA shall be installed in accordance with the requirements of Clause [6](#) of this Code or the authority having jurisdiction.
- c) Discharge piping or tubing, not connected at the factory, from the outlet of the RFA or the VFA shall be installed in accordance with the requirements of CSA B108.1.

10.3.2 Restrictions

The use of a gas hose in an installation shall be restricted to the following:

- a) a fuelling hose shall
 - i) meet CSA/ANSI NGV 4.2/CSA 12.52;
 - ii) be limited to a maximum length of 25 ft (7.6 m);
 - iii) be supported above the floor or ground level or otherwise protected from mechanical damage from abrasion and being driven over; and
 - iv) be equipped with a breakaway quick closing device; and

- b) a gas hose installed on the gas supply line to an RFA or a VFA to prevent abrasion damage resulting from vibration shall
 - i) meet CSA 8.1 or CSA 8.3, and
 - ii) be limited to a maximum length of 3 ft (0.9 m).

10.3.3 Fittings

The number of fittings used in a supply line, discharge line, or gas hose shall be minimized to reduce the possibility of leakage.

10.3.4 Support

The discharge piping or tubing shall be supported so that there is no span greater than 4 ft (1.2 m).

10.4 Testing of piping, tubing, gas hose, and fittings

The following requirements shall apply to a gas piping system, RFA or VFA associated equipment, and components:

- a) They shall be tested in accordance with the manufacturer's instructions.
- b) The supply line to the inlet of an RFA or a VFA shall be tested in accordance with the requirements of Clause [6.22](#).
- c) Discharge piping or tubing and gas hose, not connected at the factory, from the outlet of an RFA or a VFA shall be installed in accordance with the test pressure requirement of CSA B108.1.

10.5 Installation of safety equipment, signs, and/or symbols

There shall be a prominently displayed, readily visible sign within 10 ft (3 m) of a point of transfer at an NGV dispensing location

- a) with the words "NO SMOKING — TURN IGNITION OFF DURING VEHICLE REFUELLING"* in black letters not less than 1/2 in (12 mm) in height on a safety yellow background; or
- b) with the international symbols for "NO SMOKING"† and "IGNITION OFF"‡ at least 2 in (50 mm) in diameter, coloured red and black on a white background.

* The equivalent French wording is "DÉFENSE DE FUMER — ÉTEINDRE LE MOTEUR AVANT LE REMPLISSAGE".

† The equivalent French wording is "DÉFENSE DE FUMER".

‡ The equivalent French wording is "ÉTEINDRE LE MOTEUR".

10.6 Refuelling of vehicles

A vehicle being refuelled with NGV shall have its engine turned off.

Annex A (informative)

Sizing and capacities of piping and tubing for natural gas

Note: This informative annex has been written in mandatory language to facilitate adoption where users of the Code or regulatory authorities wish to adopt it formally as additional requirements to this Code.

A.1 General

In order to determine the size of piping or tubing to be used in designing a gas piping system, the following factors shall be considered:

- a) allowable loss in pressure from point of delivery to equipment;
- b) maximum gas demand;
- c) length of piping and number of fittings;
- d) relative density of the gas; and
- e) diversity factor.

A.2 Description of tables

A.2.1 Outlet capacity

The total energy to be provided at each outlet should be determined directly from the manufacturer's Btu/h (kW) input rating of the equipment to be installed.

A.2.2 Natural gas quality

The tables listed in this annex assume that a gas having a calorific value of 1000 Btu/ft³ (38 MJ/m³) and a relative density of 0.6 for natural gas will be used.

A.2.3 Table applicability

Capacities for gas in Btu/h (kW) for different sizes and lengths are shown in Tables [A.1](#) to [A.7](#) for iron pipe or equivalent rigid pipe and in Tables [A.8](#) to [A.14](#) for tubing.

A.2.4 Fitting allowance

The formula used to derive Tables [A.1](#) to [A.4](#) and [A.8](#) to [A.11](#) contains a factor ($F = 1.2$), which is used to multiply the piping or tubing length, to allow for a reasonable number of fittings. Tables [A.5](#) to [A.7](#) and [A.12](#) to [A.14](#) do not contain a fitting allowance ($F = 1$), and Table [A.16](#) should be utilized to determine equivalent lengths of fittings.

A.2.5 Tubing type

Tables [A.8](#) to [A.14](#) covering tubing are based on Type K tubing. Type K was selected since it has a smaller inside diameter than Types G and L.

A.2.6 Relative density

The application of the multipliers in Table [A.15](#) converts the capacities given in Tables [A.1](#) to [A.14](#) to capacities with another gas of different relative density. Such application is accomplished by multiplying the capacities given in Tables [A.1](#) to [A.14](#) by the multipliers shown in Table [A.15](#). If the exact relative density does not appear in the table, the next higher value of relative density shown shall be chosen.

A.3 Use of capacity tables

A.3.1 Size determination

To determine the size of each section of piping or tubing in a system within the range of the capacity tables, the procedure outlined in Clauses [A.3.2](#) to [A.3.9](#) shall be followed.

A.3.2 Demand determination

The gas demand of each appliance to be attached to the piping or tubing system shall be determined. The gas demand in terms of Btu/h (kW) for each piping system outlet shall be calculated.

A.3.3 Pressure determination

The design system pressure, the allowable loss in pressure (pressure drop), and the relative density of the gas to be used in the piping or tubing system shall be determined.

A.3.4 Length determination

The length of piping or tubing from the point of delivery to the most remote outlet in the building shall be measured.

A.3.5 Formulae

In the appropriate capacity table, the row showing the measured length, or the next larger length if the table does not give the exact length shall be selected. This is the only length used in determining the size of any section of piping or tubing. If the relative density factor is to be applied, the values in the selected column of the table shall be multiplied by the appropriate multiplier from Table [A.15](#).

Capacities may also be determined by using the following formulas, which were used to generate the tables*:

for high pressure [1.5 psig (10.3 kPa) and above]:

$$Q = (K_1) D^{2.623} \left[\frac{(P_1^2 - P_2^2) Y}{C_r \times L \times F} \right]^{0.541} (b \times z)$$

for low pressure [less than 1.5 psig (10.3 kPa)]:

$$Q = (K_2) D^{2.623} \left[\frac{\Delta H}{C_r \times L \times F} \right]^{0.541} (b \times z)$$

where

Q = rate, Btu/h (kW) at 60 °F (15.6 °C) and 30 in Hg (101.3 kPa)

K_1 = 2237 for imperial measurements

= 0.3576 for SI (metric) measurements

D = inside diameter of pipe, in (cm)

P_1 = upstream pressure, absolute, psi (kPa)

P_2 = downstream pressure, absolute, psi (kPa)

Y = superexpansibility factor†

= 1/supercompressibility factor

= 1 for intents and purposes

b = calorific value of the gas, Btu/ft³ (MJ/m³)

- z = factor to convert MJ/h to kW
 = 1 for imperial measurements
 = 0.2778 for SI (metric) measurements
 Cr = factor for viscosity, density, and temperature
 = $K_3 S T \left[\frac{Z}{S} \right]^{0.152}$
 = 0.6094 for natural gas
 where
 K_3 = 0.00354 for imperial measurements
 = 0.00638 for SI (metric) measurements
 S = relative density of gas at 60 °F (15.6 °C) and 30 in Hg (101.3 kPa), (natural gas = 0.6)
 T = absolute temperature, °R = °F + 460 (imperial)
 = absolute temperature, K = °C + 273 (metric)
 Z = viscosity of gas, centipoise (0.012 for natural gas)

Legend:

- L = length of pipe, ft (m)
 F = fitting allowance factor
 = 1.2 for Tables [A.1](#) to [A.4](#) and [A.8](#) to [A.11](#)
 = 1 for Tables [A.5](#) to [A.7](#) and [A.12](#) to [A.14](#)
 K_2 = 2313 for imperial measurements
 = 0.1509 for SI (metric) measurements
 ΔH = pressure drop, in w.c. (Pa)

* For further details on the formulas, refer to B.C. Shebeko, *Polyflo Flow Computer*, 1974, available from Polyflo Computer Company, Box 50126, Dallas, Texas 75250, USA.

† For values for natural gas, refer to *Manual for Determination of Supercompressibility Factors for Natural Gas*, available from the American Gas Association, 1515 Wilson Boulevard, Arlington, Virginia 22209, USA.

For values for liquefied petroleum gases, refer to Engineering Data Book, available from Gas Processors Association.

A.3.6 Row choice

The horizontal row specified in Clause [A.3.5](#) shall be used to locate all gas demand figures for this particular system of piping.

A.3.7 Column choice

Starting at the most remote outlet, in the horizontal row selected in accordance with Clause [A.3.6](#), the gas demand for that outlet shall be located. If the exact figure of demand is not shown, the next larger figure to the right in the row shall be chosen.

A.3.8 Correct size

In the top row above this demand figure, the correct size of gas piping will be found.

A.3.9 Repeat

Proceed in a similar manner for each outlet and each section of gas piping. For each section of piping, the total gas demand supplied by that section shall be determined.

A.4 Bends, fittings, and valves expressed in equivalent lengths in feet of straight pipe

The equivalent lengths in feet shown in Table [A.16](#)* have been computed on the basis that the inside diameter corresponds to that of Schedule 40 (standard-weight) steel pipe, which is close enough for most purposes to that of other schedules of pipe. Where a more specific equivalent length is desired, this may be computed by multiplying the actual inside diameter of the pipe in inches by $n/12$, or the actual inside diameter in feet by “n” (“n” is a heading in Table [A.16](#)). The equivalent length values can be used with reasonable accuracy for copper or brass fittings and bends. For copper or brass valves, however, the equivalent length of pipe should be taken as 45% longer than the values in Table [A.16](#), which are for steel pipe. Resistance per foot of copper or brass pipe is less than that of steel.

* From Crocker (1945), Table XIV, pp. 100–101. Used by permission of McGraw-Hill Book Company.

Table A.1 a)
Maximum capacity of natural gas in thousands of Btu/h for Schedule 40 pipe and plastic pipe, including fittings, for pressures of less than 7 in w.c. based on a pressure drop of 0.5 in w.c.
 (See Clauses [6.3.2](#), [A.2.3](#), [A.2.4](#), [A.2.6](#), [A.3.5](#), [E.1.3](#), and [E.2.3](#).)

Length of pipe, ft	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
10	156	326	614	1 261	1 890	3 639	5 800	10 253	20 914
20	107	224	422	867	1 299	2 501	3 986	7 047	14 374
30	86	180	339	696	1 043	2 008	3 201	5 659	11 543
40	74	154	290	596	893	1 719	2 740	4 843	9 879
50	65	137	257	528	791	1 524	2 428	4 293	8 756
60	59	124	233	478	717	1 380	2 200	3 889	7 933
70	54	114	214	440	659	1 270	2 024	3 578	7 299
80	51	106	199	409	613	1 181	1 883	3 329	6 790
90	48	99	187	384	576	1 109	1 767	3 123	6 371
100	45	94	177	363	544	1 047	1 669	2 950	6 018
125	40	83	157	322	482	928	1 479	2 615	5 333
150	36	75	142	291	437	841	1 340	2 369	4 832
175	33	69	131	268	402	774	1 233	2 180	4 446
200	31	64	121	249	374	720	1 147	2 028	4 136
250	27	57	108	221	331	638	1 017	1 797	3 666
300	25	52	98	200	300	578	921	1 628	3 321
350	23	48	90	184	276	532	847	1 498	3 056
400	21	44	83	171	257	495	788	1 394	2 843
450	20	42	78	161	241	464	740	1 308	2 667
500	19	39	74	152	228	438	699	1 235	2 519

(Continued)

Table A.1 a) (Continued)

Length of pipe, ft	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
600	17	36	67	138	206	397	633	1 119	2 283
700	16	33	62	127	190	365	582	1 030	2 100
800	15	30	57	118	177	340	542	958	1 954
900	14	29	54	111	166	319	508	899	1 833
1 000	13	27	51	104	156	301	480	849	1 732
1 200	12	24	46	95	142	273	435	769	1 569
1 400	11	23	42	87	130	251	400	708	1 443
1 600	10	21	39	81	121	234	372	658	1 343
1 800	9	20	37	76	114	219	349	618	1 260
2 000	9	19	35	72	108	207	330	583	1 190

Table A.1 b)
Maximum capacity of natural gas in kW for Schedule 40 pipe and plastic pipe, including fittings, for pressures of less than 1.75 kPa based on a pressure drop of 125 Pa

Length of pipe, m	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
3	46	97	182	374	560	1078	1718	3038	6196
6	32	66	125	257	385	741	1181	2088	4259
9	25	53	100	206	309	595	948	1677	3420
12	22	46	86	176	264	509	812	1435	2927
15	19	40	76	156	234	451	719	1272	2594
18	18	37	69	142	212	409	652	1152	2350
21	16	34	64	130	195	376	600	1060	2162
24	15	31	59	121	182	350	558	986	2012
27	14	29	55	114	171	328	523	925	1887
30	13	28	52	108	161	310	494	874	1783
35	12	26	48	99	148	285	455	804	1640
40	11	24	45	92	138	266	423	748	1526
45	11	22	42	86	129	249	397	702	1432
50	10	21	40	82	122	235	375	663	1352
60	9	19	36	74	111	213	340	601	1225
70	8	18	33	68	102	196	313	553	1127
80	8	16	31	63	95	182	291	514	1049
90	7	15	29	59	89	171	273	482	984
100	7	14	27	56	84	162	258	456	929
125	6	13	24	50	74	143	228	404	824

(Continued)

Table A.1 b) (Continued)

Length of pipe, m	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
150	6	12	22	45	67	130	207	366	746
175	5	11	20	41	62	119	190	337	687
200	5	10	19	39	58	111	177	313	639
250	4	9	17	34	51	99	157	278	566
300	4	8	15	31	46	89	142	252	513
350	4	7	14	28	43	82	131	231	472
400	3	7	13	26	40	76	122	215	439
500	3	6	11	23	35	68	108	191	389
600	3	5	10	21	32	61	98	173	353

Table A.2 a)
Maximum capacity of natural gas in thousands of Btu/h for Schedule 40 pipe and plastic pipe, including fittings, for pressures of 7 in w.c. up to 14 in w.c. based on a pressure drop of 1 in w.c.
 (See Clauses [6.3.2](#), [6.3.4](#), [A.2.3](#), [A.2.4](#), [A.2.6](#), [A.3.5](#), [E.1.2](#), and [E.2.2](#).)

Length of pipe, ft	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
10	227	474	894	1 835	2 749	5 295	8 439	14 919	30 429
20	156	326	614	1 261	1 890	3 639	5 800	10 253	20 914
30	125	262	493	1 013	1 517	2 922	4 658	8 234	16 795
40	107	224	422	867	1 299	2 501	3 986	7 047	14 374
50	95	199	374	768	1 151	2 217	3 533	6 246	12 739
60	86	180	339	696	1 043	2 008	3 201	5 659	11 543
70	79	166	312	640	959	1 848	2 945	5 206	10 619
80	74	154	290	596	893	1 719	2 740	4 843	9 879
90	69	145	272	559	837	1 613	2 571	4 544	9 269
100	65	137	257	528	791	1 524	2 428	4 293	8 756
125	58	121	228	468	701	1 350	2 152	3 805	7 760
150	52	110	207	424	635	1 223	1 950	3 447	7 031
175	48	101	190	390	584	1 126	1 794	3 171	6 469
200	45	94	177	363	544	1 047	1 669	2 950	6 018
250	40	83	157	322	482	928	1 479	2 615	5 333
300	36	75	142	291	437	841	1 340	2 369	4 832
350	33	69	131	268	402	774	1 233	2 180	4 446
400	31	64	121	249	374	720	1 147	2 028	4 136
450	29	61	114	234	351	675	1 076	1 903	3 881
500	27	57	108	221	331	638	1 017	1 797	3 666

(Continued)

Table A.2 a) (Continued)

Length of pipe, ft	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
600	25	52	98	200	300	578	921	1 628	3 321
700	23	48	90	184	276	532	847	1 498	3 056
800	21	44	83	171	257	495	788	1 394	2 843
900	20	42	78	161	241	464	740	1 308	2 667
1 000	19	39	74	152	228	438	699	1 235	2 519
1 200	17	36	67	138	206	397	633	1 119	2 283
1 400	16	33	62	127	190	365	582	1 030	2 100
1 600	15	30	57	118	177	340	542	958	1 954
1 800	14	29	54	111	166	319	508	899	1 833
2 000	13	27	51	104	156	301	480	849	1 732

Table A.2 b)
Maximum capacity of natural gas in kW for Schedule 40 pipe and plastic pipe, including fittings, for pressures of 1.75 kPa up to 3.50 kPa based on a pressure drop of 250 Pa

Length of pipe, m	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
3	67	141	265	544	815	1569	2500	4420	9015
6	46	97	182	374	560	1078	1718	3038	6196
9	37	78	146	300	450	866	1380	2439	4976
12	32	66	125	257	385	741	1181	2088	4259
15	28	59	111	228	341	657	1047	1850	3774
18	25	53	100	206	309	595	948	1677	3420
21	23	49	92	190	284	547	873	1542	3146
24	22	46	86	176	264	509	812	1435	2927
27	20	43	81	166	248	478	762	1346	2746
30	19	40	76	156	234	451	719	1272	2594
35	18	37	70	144	216	415	662	1170	2386
40	17	35	65	134	201	386	616	1088	2220
45	16	32	61	126	188	362	578	1021	2083
50	15	31	58	119	178	342	546	965	1968
60	13	28	52	108	161	310	494	874	1783
70	12	26	48	99	148	285	455	804	1640
80	11	24	45	92	138	266	423	748	1526
90	11	22	42	86	129	249	397	702	1432
100	10	21	40	82	122	235	375	663	1352
125	9	19	35	72	108	209	332	588	1199

(Continued)

Table A.2 b) (Continued)

Length of pipe, m	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
150	8	17	32	65	98	189	301	532	1086
175	7	16	29	60	90	174	277	490	999
200	7	14	27	56	84	162	258	456	929
250	6	13	24	50	74	143	228	404	824
300	6	12	22	45	67	130	207	366	746
350	5	11	20	41	62	119	190	337	687
400	5	10	19	39	58	111	177	313	639
500	4	9	17	34	51	99	157	278	566
600	4	8	15	31	46	89	142	252	513

Table A.3 a)
Maximum capacity of natural gas in thousands of Btu/h for Schedule 40 pipe and plastic pipe, including fittings, for
pressures of 2 psig based on a pressure drop of 1 psig
 (See Clauses [6.3.4](#), [A.2.3](#), [A.2.4](#), [A.2.6](#), [A.3.5](#), [E.1.3](#), and [E.2.3](#).)

Length of pipe, ft	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
10	1 440	3 012	5 673	11 648	17 452	33 610	53 570	94 701	193 161
20	990	2 070	3 899	8 005	11 994	23 100	36 818	65 088	132 758
30	795	1 662	3 131	6 429	9 632	18 550	29 566	52 268	106 610
40	680	1 423	2 680	5 502	8 244	15 877	25 305	44 734	91 244
50	603	1 261	2 375	4 876	7 306	14 071	22 427	39 647	80 868
60	546	1 142	2 152	4 418	6 620	12 749	20 321	35 923	73 272
70	503	1 051	1 980	4 065	6 090	11 729	18 695	33 049	67 409
80	468	978	1 842	3 782	5 666	10 912	17 392	30 746	62 711
90	439	917	1 728	3 548	5 316	10 238	16 318	28 848	58 840
100	414	867	1 632	3 351	5 022	9 671	15 414	27 249	55 580
125	367	768	1 447	2 970	4 451	8 571	13 661	24 151	49 260
150	333	696	1 311	2 691	4 032	7 766	12 378	21 882	44 633
175	306	640	1 206	2 476	3 710	7 145	11 388	20 131	41 062
200	285	596	1 122	2 303	3 451	6 647	10 594	18 728	38 200
250	252	528	994	2 042	3 059	5 891	9 389	16 599	33 856
300	229	478	901	1 850	2 772	5 338	8 507	15 040	30 676
350	210	440	829	1 702	2 550	4 911	7 827	13 836	28 221
400	196	409	771	1 583	2 372	4 568	7 281	12 872	26 255
450	184	384	724	1 485	2 226	4 286	6 832	12 077	24 634
500	174	363	683	1 403	2 102	4 049	6 453	11 408	23 269

(Continued)

Table A.3 a) (Continued)

Length of pipe, ft	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
600	157	329	619	1 271	1 905	3 669	5 847	10 337	21 083
700	145	302	570	1 170	1 752	3 375	5 379	9 510	19 396
800	135	281	530	1 088	1 630	3 140	5 004	8 847	18 045
900	126	264	497	1 021	1 530	2 946	4 695	8 301	16 931
1 000	119	249	470	964	1 445	2 783	4 435	7 841	15 993
1 200	108	226	426	874	1 309	2 521	4 019	7 104	14 490
1 400	99	208	392	804	1 204	2 320	3 697	6 536	13 331
1 600	92	193	364	748	1 120	2 158	3 439	6 080	12 402
1 800	87	181	342	702	1 051	2 025	3 227	5 705	11 636
2 000	82	171	323	663	993	1 913	3 048	5 389	10 992

Table A.3 b)
Maximum capacity of natural gas in kW for Schedule 40 pipe and plastic pipe, including fittings, for pressures of 14 kPa based on a pressure drop of 7 kPa

Length of pipe, m	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
3	430	899	1 693	3 475	5 207	10 028	15 983	28 255	57 632
6	295	618	1 163	2 388	3 579	6 892	10 985	19 420	39 610
9	237	496	934	1 918	2 874	5 535	8 821	15 595	31 808
12	203	424	800	1 642	2 460	4 737	7 550	13 347	27 224
15	180	376	709	1 455	2 180	4 198	6 691	11 829	24 128
18	163	341	642	1 318	1 975	3 804	6 063	10 718	21 862
21	150	314	591	1 213	1 817	3 500	5 578	9 861	20 112
24	140	292	550	1 128	1 690	3 256	5 189	9 173	18 711
27	131	274	516	1 059	1 586	3 055	4 869	8 607	17 556
30	124	259	487	1 000	1 498	2 885	4 599	8 130	16 583
35	114	238	448	920	1 378	2 655	4 231	7 480	15 256
40	106	221	417	856	1 282	2 470	3 936	6 958	14 193
45	99	208	391	803	1 203	2 317	3 693	6 529	13 317
50	94	196	369	759	1 136	2 189	3 489	6 167	12 579
60	85	178	335	687	1 030	1 983	3 161	5 588	11 397
70	78	163	308	632	947	1 824	2 908	5 141	10 485
80	73	152	286	588	881	1 697	2 705	4 782	9 755
90	68	143	269	552	827	1 593	2 538	4 487	9 152
100	64	135	254	521	781	1 504	2 398	4 239	8 645
125	57	119	225	462	692	1 333	2 125	3 757	7 662

(Continued)

Table A.3 b) (Continued)

Length of pipe, m	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
150	52	108	204	419	627	1 208	1 925	3 404	6 943
175	48	100	188	385	577	1 111	1 771	3 131	6 387
200	44	93	175	358	537	1 034	1 648	2 913	5 942
250	39	82	155	318	476	916	1 460	2 582	5 266
300	36	74	140	288	431	830	1 323	2 339	4 772
350	33	68	129	265	397	764	1 217	2 152	4 390
400	30	64	120	246	369	711	1 133	2 002	4 084
500	27	56	106	218	327	630	1 004	1 775	3 619
600	24	51	96	198	296	571	910	1 608	3 279

Table A.4 a)
Maximum capacity of natural gas in thousands of Btu/h for Schedule 40 pipe and plastic pipe, including fittings, for pressures of 2 psig based on a pressure drop of 1.5 psig
 (See Clauses [6.3.4](#), [A.2.3](#), [A.2.4](#), [A.2.6](#), and [A.3.5](#).)

Length of pipe, ft	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
10	1 779	3 719	7 006	14 383	21 550	41 503	66 150	116 941	238 522
20	1 222	2 556	4 815	9 885	14 811	28 525	45 464	80 373	163 935
30	982	2 053	3 866	7 938	11 894	22 907	36 509	64 542	131 646
40	840	1 757	3 309	6 794	10 180	19 605	31 247	55 240	112 672
50	745	1 557	2 933	6 022	9 022	17 376	27 694	48 958	99 859
60	675	1 411	2 657	5 456	8 175	15 744	25 093	44 360	90 479
70	621	1 298	2 445	5 019	7 521	14 484	23 085	40 810	83 240
80	577	1 207	2 274	4 670	6 996	13 474	21 476	37 966	77 439
90	542	1 133	2 134	4 381	6 565	12 643	20 150	35 622	72 658
100	512	1 070	2 016	4 139	6 201	11 942	19 034	33 649	68 632
125	454	948	1 787	3 668	5 496	10 584	16 869	29 822	60 828
150	411	859	1 619	3 323	4 979	9 590	15 285	27 021	55 114
175	378	791	1 489	3 057	4 581	8 823	14 062	24 859	50 704
200	352	735	1 385	2 844	4 262	8 208	13 082	23 126	47 171
250	312	652	1 228	2 521	3 777	7 274	11 594	20 497	41 806
300	282	591	1 113	2 284	3 422	6 591	10 505	18 571	37 880
350	260	543	1 024	2 101	3 149	6 064	9 665	17 085	34 849
400	242	505	952	1 955	2 929	5 641	8 991	15 895	32 420
450	227	474	893	1 834	2 748	5 293	8 436	14 913	30 419
500	214	448	844	1 733	2 596	5 000	7 969	14 087	28 733

(Continued)

Table A.4 a) (Continued)

Length of pipe, ft	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
600	194	406	765	1 570	2 352	4 530	7 220	12 764	26 034
700	179	373	703	1 444	2 164	4 168	6 642	11 743	23 951
800	166	347	654	1 344	2 013	3 877	6 180	10 924	22 282
900	156	326	614	1 261	1 889	3 638	5 798	10 250	20 907
1 000	147	308	580	1 191	1 784	3 436	5 477	9 682	19 748
1 200	133	279	526	1 079	1 617	3 113	4 962	8 773	17 893
1 400	123	257	483	993	1 487	2 864	4 565	8 071	16 462
1 600	114	239	450	923	1 384	2 665	4 247	7 508	15 314
1 800	107	224	422	866	1 298	2 500	3 985	7 045	14 369
2 000	101	212	399	818	1 226	2 362	3 764	6 654	13 573

Table A.4 b)
Maximum capacity of natural gas in kW for Schedule 40 pipe and plastic pipe, including fittings, for pressures of 14 kPa based on a pressure drop of 10 kPa

Length of pipe, m	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
3	517	1 082	2 038	4 184	6 269	12 074	19 244	34 020	69 389
6	356	744	1 401	2 876	4 309	8 298	13 226	23 381	47 691
9	286	597	1 125	2 309	3 460	6 664	10 621	18 776	38 297
12	244	511	963	1 976	2 961	5 703	9 090	16 070	32 778
15	217	453	853	1 752	2 625	5 055	8 057	14 242	29 050
18	196	410	773	1 587	2 378	4 580	7 300	12 905	26 321
21	181	378	711	1 460	2 188	4 214	6 716	11 872	24 215
24	168	351	662	1 358	2 035	3 920	6 248	11 045	22 528
27	158	330	621	1 275	1 910	3 678	5 862	10 363	21 137
30	149	311	586	1 204	1 804	3 474	5 537	9 789	19 966
35	137	286	539	1 108	1 660	3 196	5 094	9 006	18 368
40	127	266	502	1 030	1 544	2 973	4 739	8 378	17 088
45	120	250	471	967	1 449	2 790	4 447	7 861	16 033
50	113	236	445	913	1 368	2 635	4 200	7 425	15 145
60	102	214	403	827	1 240	2 388	3 806	6 728	13 722
70	94	197	371	761	1 141	2 197	3 501	6 189	12 625
80	88	183	345	708	1 061	2 044	3 257	5 758	11 745
90	82	172	324	664	996	1 917	3 056	5 403	11 020
100	78	162	306	628	940	1 811	2 887	5 103	10 409
125	69	144	271	556	833	1 605	2 558	4 523	9 225

(Continued)

Table A.4 b) (Continued)

Length of pipe, m	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
150	62	130	246	504	755	1 454	2 318	4 098	8 359
175	57	120	226	464	695	1 338	2 133	3 770	7 690
200	53	112	210	431	646	1 245	1 984	3 507	7 154
250	47	99	186	382	573	1 103	1 758	3 109	6 341
300	43	90	169	346	519	1 000	1 593	2 817	5 745
350	39	82	155	319	478	920	1 466	2 591	5 285
400	37	77	144	296	444	856	1 364	2 411	4 917
500	32	68	128	263	394	758	1 209	2 137	4 358
600	29	62	116	238	357	687	1 095	1 936	3 949

Table A.5 a)
Maximum capacity of natural gas in thousands of Btu/h for Schedule 40 pipe and plastic pipe for pressures of 5 psig
based on a pressure drop of 2.5 psig
 (See Clauses [A.2.3](#), [A.2.4](#), [A.2.6](#), and [A.3.5](#).)

Length of pipe, ft	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
10	2 800	5 855	11 028	22 643	33 925	65 337	104 137	184 096	375 496
20	1 924	4 024	7 580	15 562	23 317	44 906	71 573	126 528	258 077
30	1 545	3 231	6 087	12 497	18 724	36 061	57 475	101 606	207 245
40	1 323	2 766	5 210	10 696	16 026	30 864	49 192	86 962	177 375
50	1 172	2 451	4 617	9 479	14 203	27 354	43 598	77 073	157 204
60	1 062	2 221	4 183	8 589	12 869	24 785	39 503	69 833	142 438
70	977	2 043	3 849	7 902	11 839	22 801	36 342	64 246	131 041
80	909	1 901	3 581	7 351	11 014	21 212	33 809	59 768	121 909
90	853	1 783	3 359	6 897	10 334	19 903	31 722	56 079	114 383
100	806	1 685	3 173	6 515	9 762	18 800	29 964	52 972	108 045
125	714	1 493	2 812	5 774	8 652	16 662	26 557	46 948	95 759
150	647	1 353	2 548	5 232	7 839	15 097	24 062	42 538	86 764
175	595	1 245	2 344	4 813	7 212	13 889	22 137	39 135	79 822
200	554	1 158	2 181	4 478	6 709	12 921	20 594	36 407	74 259
250	491	1 026	1 933	3 969	5 946	11 452	18 252	32 267	65 814
300	445	930	1 751	3 596	5 388	10 376	16 538	29 236	59 633
350	409	855	1 611	3 308	4 957	9 546	15 215	26 897	54 861
400	381	796	1 499	3 078	4 611	8 881	14 154	25 022	51 038
450	357	747	1 406	2 888	4 327	8 332	13 281	23 478	47 887
500	337	705	1 329	2 728	4 087	7 871	12 545	22 177	45 234

(Continued)

Table A.5 a) (Continued)

Length of pipe, ft	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
600	306	639	1 204	2 471	3 703	7 131	11 366	20 094	40 985
700	281	588	1 107	2 274	3 407	6 561	10 457	18 486	37 706
800	262	547	1 030	2 115	3 169	6 104	9 728	17 198	35 078
900	245	513	967	1 985	2 974	5 727	9 128	16 136	32 912
1 000	232	485	913	1 875	2 809	5 410	8 622	15 242	31 089
1 200	210	439	827	1 699	2 545	4 901	7 812	13 810	28 169
1 400	193	404	761	1 563	2 341	4 509	7 187	12 705	25 915
1 600	180	376	708	1 454	2 178	4 195	6 686	11 820	24 109
1 800	169	353	664	1 364	2 044	3 936	6 273	11 090	22 621
2 000	159	333	628	1 288	1 930	3 718	5 926	10 476	21 367

Table A.5 b)
Maximum capacity of natural gas in kW for Schedule 40 pipe and plastic pipe for pressures of 34 kPa based on a pressure drop of 17 kPa

Length of pipe, m	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
3	821	1 716	3 232	6 635	9 942	19 147	30 518	53 950	110 041
6	564	1 179	2 221	4 561	6 833	13 160	20 975	37 080	75 631
9	453	947	1 784	3 662	5 487	10 568	16 843	29 776	60 734
12	388	810	1 527	3 134	4 696	9 045	14 416	25 485	51 980
15	344	718	1 353	2 778	4 162	8 016	12 776	22 587	46 069
18	311	651	1 226	2 517	3 771	7 263	11 576	20 465	41 742
21	286	599	1 128	2 316	3 470	6 682	10 650	18 828	38 402
24	266	557	1 049	2 154	3 228	6 216	9 908	17 515	35 726
27	250	523	985	2 021	3 029	5 833	9 296	16 434	33 520
30	236	494	930	1 909	2 861	5 509	8 781	15 524	31 663
35	217	454	856	1 757	2 632	5 069	8 079	14 282	29 130
40	202	423	796	1 634	2 448	4 715	7 516	13 286	27 100
45	190	396	747	1 533	2 297	4 424	7 052	12 466	25 427
50	179	374	705	1 448	2 170	4 179	6 661	11 775	24 018
60	162	339	639	1 312	1 966	3 787	6 035	10 669	21 762
70	149	312	588	1 207	1 809	3 484	5 552	9 816	20 021
80	139	290	547	1 123	1 683	3 241	5 165	9 132	18 625
90	130	272	513	1 054	1 579	3 041	4 847	8 568	17 476
100	123	257	485	995	1 491	2 872	4 578	8 093	16 507
125	109	228	430	882	1 322	2 546	4 057	7 173	14 630

(Continued)

Table A.5 b) (Continued)

Length of pipe, m	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
150	99	207	389	799	1 198	2 307	3 676	6 499	13 256
175	91	190	358	735	1 102	2 122	3 382	5 979	12 195
200	85	177	333	684	1 025	1 974	3 146	5 562	11 345
250	75	157	295	606	908	1 750	2 789	4 930	10 055
300	68	142	268	549	823	1 585	2 527	4 467	9 111
350	62	131	246	505	757	1 458	2 325	4 109	8 382
400	58	122	229	470	705	1 357	2 163	3 823	7 798
500	52	108	203	417	624	1 203	1 917	3 388	6 911
600	47	98	184	378	566	1 090	1 737	3 070	6 262

Table A.6 a)
Maximum capacity of natural gas in thousands of Btu/h for Schedule 40 pipe and plastic pipe for pressures of 10 psig
based on a pressure drop of 5 psig
 (See Clauses [A.2.3](#), [A.2.4](#), [A.2.6](#), and [A.3.5](#).)

Length of pipe, ft	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
10	4 503	9 415	17 736	36 414	54 559	105 075	167 473	296 063	603 873
20	3 095	6 471	12 190	25 027	37 498	72 218	115 103	203 482	415 039
30	2 485	5 197	9 789	20 098	30 112	57 993	92 432	163 403	333 291
40	2 127	4 448	8 378	17 201	25 772	49 635	79 110	139 852	285 254
50	1 885	3 942	7 425	15 245	22 841	43 990	70 114	123 948	252 815
60	1 708	3 572	6 728	13 813	20 696	39 858	63 528	112 306	229 069
70	1 571	3 286	6 190	12 708	19 040	36 669	58 445	103 320	210 740
80	1 462	3 057	5 758	11 822	17 713	34 114	54 372	96 120	196 053
90	1 372	2 868	5 403	11 092	16 620	32 008	51 015	90 186	183 950
100	1 296	2 709	5 103	10 478	15 699	30 234	48 189	85 189	173 758
125	1 148	2 401	4 523	9 286	13 914	26 796	42 709	75 501	153 999
150	1 040	2 176	4 098	8 414	12 607	24 279	38 697	68 410	139 534
175	957	2 001	3 770	7 741	11 598	22 337	35 601	62 936	128 370
200	890	1 862	3 508	7 201	10 790	20 780	33 120	58 550	119 423
250	789	1 650	3 109	6 382	9 563	18 417	29 353	51 892	105 843
300	715	1 495	2 817	5 783	8 665	16 687	26 596	47 018	95 901
350	658	1 376	2 591	5 320	7 971	15 352	24 468	43 256	88 228
400	612	1 280	2 411	4 949	7 416	14 282	22 763	40 241	82 079
450	574	1 201	2 262	4 644	6 958	13 400	21 358	37 757	77 012
500	542	1 134	2 137	4 387	6 572	12 658	20 174	35 665	72 745

(Continued)

Table A.6 a) (Continued)

Length of pipe, ft	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
600	491	1 028	1 936	3 975	5 955	11 469	18 280	32 315	65 912
700	452	945	1 781	3 657	5 479	10 551	16 817	29 729	60 638
800	421	880	1 657	3 402	5 097	9 816	15 645	27 657	56 412
900	395	825	1 555	3 192	4 782	9 210	14 679	25 950	52 930
1 000	373	780	1 468	3 015	4 517	8 700	13 866	24 512	49 997
1 200	338	706	1 331	2 732	4 093	7 882	12 563	22 210	45 301
1 400	311	650	1 224	2 513	3 765	7 252	11 558	20 433	41 676
1 600	289	605	1 139	2 338	3 503	6 746	10 753	19 009	38 772
1 800	271	567	1 068	2 194	3 287	6 330	10 089	17 835	36 378
2 000	256	536	1 009	2 072	3 105	5 979	9 530	16 847	34 363

Table A.6 b)
Maximum capacity of natural gas in kW for Schedule 40 pipe and plastic pipe for pressures of 70 kPa based on a pressure drop of 35 kPa

Length of pipe, m	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
3	1 346	2 815	5 303	10 887	16 312	31 415	50 070	88 515	180 542
6	925	1 935	3 644	7 482	11 211	21 591	34 413	60 836	124 085
9	743	1 554	2 927	6 009	9 003	17 338	27 635	48 853	99 645
12	636	1 330	2 505	5 143	7 705	14 839	23 652	41 812	85 283
15	564	1 178	2 220	4 558	6 829	13 152	20 962	37 057	75 585
18	511	1 068	2 011	4 130	6 188	11 917	18 993	33 577	68 485
21	470	982	1 851	3 799	5 692	10 963	17 473	30 890	63 006
24	437	914	1 722	3 534	5 296	10 199	16 256	28 737	58 615
27	410	857	1 615	3 316	4 969	9 569	15 252	26 963	54 996
30	387	810	1 526	3 133	4 694	9 039	14 407	25 469	51 949
35	356	745	1 404	2 882	4 318	8 316	13 254	23 431	47 793
40	332	693	1 306	2 681	4 017	7 736	12 331	21 798	44 462
45	311	650	1 225	2 516	3 769	7 259	11 569	20 453	41 717
50	294	614	1 157	2 376	3 560	6 857	10 928	19 319	39 406
60	266	557	1 049	2 153	3 226	6 213	9 902	17 505	35 704
70	245	512	965	1 981	2 968	5 716	9 110	16 104	32 848
80	228	476	898	1 843	2 761	5 317	8 475	14 982	30 558
90	214	447	842	1 729	2 590	4 989	7 952	14 057	28 672
100	202	422	795	1 633	2 447	4 713	7 511	13 278	27 083
125	179	374	705	1 447	2 169	4 177	6 657	11 768	24 003

(Continued)

Table A.6 b) (Continued)

Length of pipe, m	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
150	162	339	639	1 311	1 965	3 784	6 032	10 663	21 749
175	149	312	588	1 207	1 808	3 482	5 549	9 810	20 009
200	139	290	547	1 122	1 682	3 239	5 162	9 126	18 614
250	123	257	485	995	1 491	2 871	4 575	8 088	16 497
300	111	233	439	901	1 351	2 601	4 146	7 329	14 948
350	103	214	404	829	1 242	2 393	3 814	6 742	13 752
400	95	199	376	771	1 156	2 226	3 548	6 272	12 793
500	85	177	333	684	1 024	1 973	3 145	5 559	11 339
600	77	160	302	619	928	1 788	2 849	5 037	10 274

Table A.7 a)
Maximum capacity of natural gas in thousands of Btu/h for Schedule 40 pipe and plastic pipe for pressures of 20 psig
based on a pressure drop of 10 psig
 (See Clauses [6.3.5](#), [A.2.3](#), [A.2.4](#), [A.2.6](#), and [A.3.5](#).)

Length of pipe, ft	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
10	7 669	16 036	30 207	62 018	92 922	178 959	285 232	504 240	1 028 489
20	5 271	11 021	20 761	42 625	63 865	122 998	196 039	346 561	706 875
30	4 233	8 851	16 672	34 229	51 286	98 771	157 426	278 301	567 645
40	3 623	7 575	14 269	29 296	43 894	84 536	134 736	238 190	485 831
50	3 211	6 714	12 646	25 964	38 902	74 922	119 414	211 103	430 583
60	2 909	6 083	11 459	23 525	35 248	67 885	108 198	191 275	390 139
70	2 676	5 596	10 542	21 643	32 428	62 453	99 541	175 970	358 923
80	2 490	5 206	9 807	20 135	30 168	58 101	92 603	163 706	333 909
90	2 336	4 885	9 202	18 892	28 306	54 514	86 887	153 600	313 296
100	2 207	4 614	8 692	17 845	26 737	51 494	82 073	145 090	295 937
125	1 956	4 089	7 703	15 816	23 697	45 638	72 740	128 591	262 284
150	1 772	3 705	6 980	14 330	21 471	41 351	65 907	116 512	237 648
175	1 630	3 409	6 421	13 184	19 753	38 043	60 634	107 190	218 633
200	1 517	3 171	5 974	12 265	18 377	35 391	56 408	99 720	203 396
250	1 344	2 811	5 294	10 870	16 287	31 367	49 993	88 380	180 266
300	1 218	2 547	4 797	9 849	14 757	28 420	45 298	80 078	163 334
350	1 120	2 343	4 413	9 061	13 576	26 146	41 673	73 671	150 265
400	1 042	2 180	4 106	8 430	12 630	24 324	38 769	68 537	139 793
450	978	2 045	3 852	7 909	11 850	22 823	36 376	64 306	131 163
500	924	1 932	3 639	7 471	11 194	21 558	34 360	60 743	123 896

(Continued)

Table A.7 a) (Continued)

Length of pipe, ft	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
600	837	1 750	3 297	6 769	10 142	19 533	31 133	55 037	112 259
700	770	1 610	3 033	6 228	9 331	17 970	28 642	50 634	103 277
800	716	1 498	2 822	5 794	8 681	16 718	26 646	47 105	96 079
900	672	1 406	2 648	5 436	8 145	15 686	25 001	44 197	90 148
1 000	635	1 328	2 501	5 135	7 693	14 817	23 616	41 748	85 153
1 200	575	1 203	2 266	4 652	6 971	13 425	21 397	37 827	77 155
1 400	529	1 107	2 085	4 280	6 413	12 351	19 685	34 800	70 981
1 600	492	1 030	1 939	3 982	5 966	11 490	18 313	32 375	66 034
1 800	462	966	1 820	3 736	5 598	10 781	17 183	30 376	61 958
2 000	436	913	1 719	3 529	5 288	10 183	16 231	28 693	58 525

Table A.7 b)
Maximum capacity of natural gas in kW for Schedule 40 pipe and plastic pipe for pressures of 140 kPa based on a pressure drop of 70 kPa

Length of pipe, m	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
3	2 296	4 801	9 044	18 568	27 820	53 578	85 396	150 964	307 919
6	1 578	3 300	6 216	12 761	19 121	36 824	58 692	103 757	211 631
9	1 267	2 650	4 991	10 248	15 354	29 571	47 132	83 320	169 947
12	1 085	2 268	4 272	8 771	13 141	25 309	40 339	71 311	145 453
15	961	2 010	3 786	7 773	11 647	22 431	35 751	63 202	128 912
18	871	1 821	3 431	7 043	10 553	20 324	32 393	57 266	116 804
21	801	1 675	3 156	6 480	9 709	18 698	29 801	52 684	107 458
24	745	1 559	2 936	6 028	9 032	17 395	27 724	49 012	99 969
27	699	1 462	2 755	5 656	8 474	16 321	26 013	45 986	93 797
30	661	1 381	2 602	5 343	8 005	15 417	24 572	43 438	88 600
35	608	1 271	2 394	4 915	7 364	14 183	22 606	39 963	81 511
40	565	1 182	2 227	4 573	6 851	13 195	21 030	37 178	75 831
45	531	1 109	2 090	4 290	6 428	12 380	19 732	34 883	71 149
50	501	1 048	1 974	4 053	6 072	11 694	18 639	32 950	67 207
60	454	949	1 789	3 672	5 502	10 596	16 888	29 855	60 895
70	418	873	1 645	3 378	5 062	9 748	15 537	27 466	56 022
80	389	813	1 531	3 143	4 709	9 069	14 454	25 552	52 118
90	365	762	1 436	2 949	4 418	8 509	13 562	23 975	48 901
100	344	720	1 357	2 785	4 173	8 037	12 810	22 646	46 191
125	305	638	1 202	2 469	3 699	7 123	11 353	20 071	40 938

(Continued)

Table A.7 b) (Continued)

Length of pipe, m	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
150	277	578	1 089	2 237	3 351	6 454	10 287	18 186	37 093
175	254	532	1 002	2 058	3 083	5 938	9 464	16 731	34 125
200	237	495	932	1 914	2 868	5 524	8 804	15 565	31 747
250	210	439	826	1 697	2 542	4 896	7 803	13 795	28 137
300	190	397	749	1 537	2 303	4 436	7 070	12 499	25 494
350	175	366	689	1 414	2 119	4 081	6 505	11 499	23 454
400	163	340	641	1 316	1 971	3 797	6 051	10 697	21 819
500	144	302	568	1 166	1 747	3 365	5 363	9 481	19 338
600	131	273	515	1 057	1 583	3 049	4 859	8 590	17 522

Table A.8 a)
Maximum capacity of natural gas in thousands of Btu/h for copper tubing, including fittings, for pressures of less than 7 in w.c. based on a pressure drop of 0.5 in w.c.
 (See Clauses [6.3.2](#), [A.2.3](#), [A.2.4](#), [A.2.5](#), [A.2.6](#), and [A.3.5](#).)

Length of tube, ft	Outside diameter (in)						
	3/8	1/2	5/8	3/4	7/8	1-1/8	1-3/8
10	24	50	101	176	250	535	963
20	17	34	69	121	172	368	662
30	13	27	56	97	138	295	531
40	11	23	48	83	118	253	455
50	10	21	42	74	105	224	403
60	9	19	38	67	95	203	365
70	8	17	35	62	87	187	336
80	8	16	33	57	81	174	313
90	7	15	31	54	76	163	293
100	7	14	29	51	72	154	277
125	6	13	26	45	64	136	245
150	6	11	23	41	58	124	222
175	5	11	21	38	53	114	205
200	5	10	20	35	50	106	190
250	4	9	18	31	44	94	169
300	4	8	16	28	40	85	153
350	4	7	15	26	37	78	141
400	3	7	14	24	34	73	131
450	3	6	13	23	32	68	123
500	3	6	12	21	30	64	116

(Continued)

Table A.8 a) (Continued)

Length of tube, ft	Outside diameter (in)						
	3/8	1/2	5/8	3/4	7/8	1-1/8	1-3/8
600	3	5	11	19	27	58	105
700	2	5	10	18	25	54	97
800	2	5	9	16	23	50	90
900	2	4	9	15	22	47	84
1000	2	4	8	15	21	44	80
1200	2	4	8	13	19	40	72
1400	2	3	7	12	17	37	66
1600	2	3	6	11	16	34	62
1800	1	3	6	11	15	32	58
2000	1	3	6	10	14	30	55

Table A.8 b)
Maximum capacity of natural gas in kW for copper tubing, including fittings, for pressures of less than 1.75 kPa based on a pressure drop of 125 Pa

Length of tube, m	Outside diameter (mm)						
	9.5	13	16	19	22	29	35
3	7	15	30	52	74	158	285
6	5	10	21	36	51	109	196
9	4	8	17	29	41	87	157
12	3	7	14	25	35	75	135
15	3	6	13	22	31	66	119
18	3	6	11	20	28	60	108
21	2	5	10	18	26	55	100
24	2	5	10	17	24	51	93
27	2	4	9	16	23	48	87
30	2	4	9	15	21	46	82
35	2	4	8	14	20	42	75
40	2	4	7	13	18	39	70
45	2	3	7	12	17	37	66
50	2	3	7	11	16	35	62
60	1	3	6	10	15	31	56
70	1	3	5	10	13	29	52
80	1	2	5	9	13	27	48
90	1	2	5	8	12	25	45
100	1	2	4	8	11	24	43
125	1	2	4	7	10	21	38
150	1	2	4	6	9	19	34

(Continued)

Table A.8 b) (Continued)

Length of tube, m	Outside diameter (mm)						
	9.5	13	16	19	22	29	35
175	1	2	3	6	8	18	32
200	1	2	3	5	8	16	29
250	1	1	3	5	7	14	26
300	1	1	2	4	6	13	24
350	1	1	2	4	6	12	22
400	1	1	2	4	5	11	20
500	—	1	2	3	5	10	18
600	—	1	2	3	4	9	16

Table A.9 a)
Maximum capacity of natural gas in thousands of Btu/h for copper tubing, including fittings, for pressures of 7 in w.c. to 14 in w.c. based on a pressure drop of 1 in w.c.
 (See Clauses [6.3.2](#), [A.2.3](#), [A.2.4](#), [A.2.5](#), [A.2.6](#), and [A.3.5](#).)

Length of tube, ft	Outside diameter (in)						
	3/8	1/2	5/8	3/4	7/8	1-1/8	1-3/8
10	35	72	147	257	364	778	1401
20	24	50	101	176	250	535	963
30	19	40	81	142	201	429	773
40	17	34	69	121	172	368	662
50	15	30	61	107	152	326	586
60	13	27	56	97	138	295	531
70	12	25	51	90	127	272	489
80	11	23	48	83	118	253	455
90	11	22	45	78	111	237	427
100	10	21	42	74	105	224	403
125	9	18	37	65	93	198	357
150	8	17	34	59	84	180	324
175	7	15	31	55	77	165	298
200	6	14	29	51	72	154	277
250	6	13	26	45	64	136	245
300	5	11	23	41	58	124	222
350	5	11	21	38	53	114	205
400	4	10	20	35	50	106	190
450	4	9	19	33	46	99	179
500	4	9	18	31	44	94	169

(Continued)

Table A.9 a) (Continued)

Length of tube, ft	Outside diameter (in)						
	3/8	1/2	5/8	3/4	7/8	1-1/8	1-3/8
600	3	7	16	28	40	85	153
700	3	7	15	26	37	78	141
800	3	7	14	24	34	73	131
900	3	6	13	23	32	68	123
1000	3	6	12	21	30	64	116
1200	3	5	11	19	27	58	105
1400	2	5	10	18	25	54	97
1600	2	5	9	16	23	50	90
1800	2	4	9	15	22	47	84
2000	2	4	8	15	21	44	80

Table A.9 b)
**Maximum capacity of natural gas in kW for copper tubing, including fittings, for pressures of 1.75 kPa up to 3.50 kPa
based on a pressure drop of 250 Pa**

Length of tube, m	Outside diameter (mm)						
	9.5	13	16	19	22	29	35
3	10	21	44	76	108	231	415
6	7	15	30	52	74	158	285
9	6	12	24	42	60	127	229
12	5	10	21	36	51	109	196
15	4	9	18	32	45	97	174
18	4	8	17	29	41	87	157
21	4	7	15	27	38	80	145
24	3	7	14	25	35	75	135
27	3	7	13	23	33	70	126
30	3	6	13	22	31	66	119
35	3	6	12	20	29	61	110
40	3	5	11	19	27	57	102
45	2	5	10	18	25	53	96
50	2	5	9	17	24	50	91
60	2	4	9	15	21	46	82
70	2	4	8	14	20	42	76
80	2	4	7	13	18	39	70
90	2	3	7	12	17	37	66
100	2	3	7	11	16	35	62
125	1	3	6	10	14	31	55
150	1	3	5	9	13	28	50

(Continued)

Table A.9 b) (Continued)

Length of tube, m	Outside diameter (mm)						
	9.5	13	16	19	22	29	35
175	1	2	5	8	12	26	46
200	1	2	4	8	11	24	43
250	1	2	4	7	10	21	38
300	1	2	4	6	9	19	34
350	1	2	3	6	8	18	32
400	1	2	3	5	8	16	29
500	1	1	3	5	7	14	26
600	1	1	2	4	6	13	24

Table A.10 a)
Maximum capacity of natural gas in thousands of Btu/h for copper tubing, including fittings, for pressures of 2 psig
based on a pressure drop of 1 psig
 (See Clauses [6.3.4](#), [A.2.3](#), [A.2.4](#), [A.2.5](#), [A.2.6](#), and [A.3.5](#).)

Length of tube, ft	Outside diameter (in)						
	3/8	1/2	5/8	3/4	7/8	1-1/8	1-3/8
10	222	458	933	1630	2312	4939	8891
20	153	315	641	1120	1589	3395	6111
30	123	253	515	899	1276	2726	4907
40	105	217	440	770	1092	2333	4200
50	93	192	390	682	968	2068	3722
60	84	174	354	618	877	1873	3373
70	78	160	325	569	807	1724	3103
80	72	149	303	529	751	1603	2887
90	68	140	284	496	704	1504	2708
100	64	132	268	469	665	1421	2558
125	57	117	238	416	590	1260	2267
150	51	106	215	377	534	1141	2055
175	47	97	198	346	492	1050	1890
200	44	91	184	322	457	977	1758
250	39	80	163	286	405	866	1558
300	35	73	148	259	367	784	1412
350	32	67	136	238	338	722	1299
400	30	62	127	222	314	671	1209
450	28	58	119	208	295	630	1134
500	27	55	112	196	279	595	1071

(Continued)

Table A.10 a) (Continued)

Length of tube, ft	Outside diameter (in)						
	3/8	1/2	5/8	3/4	7/8	1-1/8	1-3/8
600	24	50	102	178	252	539	970
700	22	46	94	164	232	496	893
800	21	43	87	152	216	461	831
900	19	40	82	143	203	433	779
1000	18	38	77	135	191	409	736
1200	17	34	70	122	173	370	667
1400	15	32	64	112	160	341	614
1600	14	29	60	105	148	317	571
1800	13	28	56	98	139	298	536
2000	13	26	53	93	132	281	506

Table A.10 b)
Maximum capacity of natural gas in kW for copper tubing, including fittings, for pressures of 14 kPa based on a pressure drop of 7 kPa

Length of tube, m	Outside diameter (mm)						
	9.5	13	16	19	22	29	35
3	66	137	278	486	690	1474	2653
6	46	94	191	334	474	1013	1824
9	37	75	154	268	381	813	1464
12	31	65	131	230	326	696	1253
15	28	57	116	204	289	617	1111
18	25	52	106	184	262	559	1006
21	23	48	97	170	241	514	926
24	22	44	90	158	224	478	861
27	20	42	85	148	210	449	808
30	19	39	80	140	199	424	763
35	18	36	74	129	183	390	702
40	16	34	69	120	170	363	653
45	15	32	64	112	159	341	613
50	14	30	61	106	151	322	579
60	13	27	55	96	136	291	525
70	12	25	51	88	126	268	483
80	11	23	47	82	117	249	449
90	11	22	44	77	110	234	421
100	10	21	42	73	103	221	398
125	9	18	37	65	92	196	353
150	8	16	34	59	83	178	320

(Continued)

Table A.10 b) (Continued)

Length of tube, m	Outside diameter (mm)						
	9.5	13	16	19	22	29	35
175	7	15	31	54	76	163	294
200	7	14	29	50	71	152	274
250	6	12	25	44	63	135	242
300	5	11	23	40	57	122	220
350	5	10	21	37	53	112	202
400	5	10	20	34	49	104	188
500	4	9	17	31	43	93	167
600	4	8	16	28	39	84	151

Table A.11 a)
Maximum capacity of natural gas in thousands of Btu/h for copper tubing, including fittings, for pressures of 2 psig
based on a pressure drop of 1.5 psig
 (See Clauses [6.3.4](#), [A.2.3](#), [A.2.4](#), [A.2.5](#), [A.2.6](#), and [A.3.5](#).)

Length of tube, ft	Outside diameter (in)						
	3/8	1/2	5/8	3/4	7/8	1-1/8	1-3/8
10	274	566	1 151	2 012	2 855	6 099	10 980
20	189	389	791	1 383	1 962	4 192	7 546
30	151	312	636	1 111	1 576	3 366	6 060
40	130	267	544	951	1 349	2 881	5 186
50	115	237	482	843	1 195	2 553	4 597
60	104	215	437	763	1 083	2 313	4 165
70	96	198	402	702	996	2 128	3 832
80	89	184	374	653	927	1 980	3 565
90	84	172	351	613	870	1 858	3 345
100	79	163	331	579	822	1 755	3 159
125	70	144	294	513	728	1 555	2 800
150	63	131	266	465	660	1 409	2

(Continued)

Table A.11 a) (Continued)

Length of tube, ft	Outside diameter (in)						
	3/8	1/2	5/8	3/4	7/8	1-1/8	1-3/8
							537
175	58	120	245	428	607	1 296	2 334
200	54	112	228	398	565	1 206	2 171
250	48	99	202	353	500	1 069	1 924
300	44	90	183	320	453	969	1 744
350	40	83	168	294	417	891	1 604
400	37	77	157	274	388	829	1 492
450	35	72	147	257	364	778	1 400
500	33	68	139	242	344	735	1 323
600	30	62	126	220	312	666	1 198
700	28	57	116	202	287	612	1 102
800	26	53	108	188	267	570	1 026
900	24	50	101	176	250	535	962
1 000	23	47	95	167	236	505	909
1 200	21	42	86	151	214	458	824

(Continued)

Table A.11 a) (Continued)

Length of tube, ft	Outside diameter (in)						
	3/8	1/2	5/8	3/4	7/8	1-1/8	1-3/8
1 400	19	39	79	139	197	421	758
1 600	18	36	74	129	183	392	705
1 800	17	34	69	121	172	367	661
2 000	16	32	66	115	162	347	625

Table A.11 b)
Maximum capacity of natural gas in kW for copper tubing, including fittings, for pressures of 14 kPa based on a pressure drop of 10 kPa

Length of tube, m	Outside diameter (mm)						
	9.5	13	16	19	22	29	35
3	80	165	335	586	831	1774	3194
6	55	113	230	402	571	1220	2196
9	44	91	185	323	458	979	1763
12	38	78	158	277	392	838	1509
15	33	69	140	245	348	743	1337
18	30	62	127	222	315	673	1212
21	28	57	117	204	290	619	1115
24	26	53	109	190	270	576	1037
27	24	50	102	178	253	541	973
30	23	47	96	168	239	511	919
35	21	44	89	155	220	470	846
40	20	41	83	144	205	437	787
45	18	38	77	135	192	410	738
50	17	36	73	128	181	387	697
60	16	33	66	116	164	351	632
70	15	30	61	107	151	323	581
80	14	28	57	99	141	300	541
90	13	26	53	93	132	282	507
100	12	25	50	88	125	266	479
125	11	22	45	78	110	236	425
150	10	20	40	71	100	214	385

(Continued)

Table A.11 b) (Continued)

Length of tube, m	Outside diameter (mm)						
	9.5	13	16	19	22	29	35
175	9	18	37	65	92	197	354
200	8	17	35	60	86	183	329
250	7	15	31	53	76	162	292
300	7	14	28	48	69	147	264
350	6	13	26	45	63	135	243
400	6	12	24	41	59	126	226
500	5	10	21	37	52	111	201
600	5	9	19	33	47	101	182

Table A.12 a)
Maximum capacity of natural gas in thousands of Btu/h for copper tubing for pressures of 5 psig based on a pressure drop of 2.5 psig
 (See Clauses [A.2.3](#), [A.2.4](#), [A.2.5](#), [A.2.6](#), and [A.3.5](#).)

Length of tube, ft	Outside diameter (in)						
	3/8	1/2	5/8	3/4	7/8	1-1/8	1-3/8
10	432	891	1 813	3 168	4 495	9 601	17 285
20	297	612	1 246	2 177	3 089	6 599	11 880
30	238	492	1 000	1 749	2 481	5 299	9 540
40	204	421	856	1 497	2 123	4 535	8 165
50	181	373	759	1 326	1 882	4 020	7 236
60	164	338	688	1 202	1 705	3 642	6 557
70	151	311	633	1 106	1 569	3 351	6 032
80	140	289	589	1 029	1 459	3 117	5 612
90	132	271	552	965	1 369	2 925	5 265
100	124	256	522	912	1 293	2 763	4 973
125	110	227	462	808	1 146	2 448	4 408
150	100	206	419	732	1 039	2 218	3 994
175	92	189	385	673	955	2 041	3 674
200	85	176	358	627	889	1 899	3 418
250	76	156	318	555	788	1 683	3 030
300	69	142	288	503	714	1 525	2 745
350	63	130	265	463	657	1 403	2 525
400	59	121	246	431	611	1 305	2 349
450	55	114	231	404	573	1 224	2 204
500	52	107	218	382	541	1 157	2 082

(Continued)

Table A.12 a) (Continued)

Length of tube, ft	Outside diameter (in)						
	3/8	1/2	5/8	3/4	7/8	1-1/8	1-3/8
600	47	97	198	346	491	1 048	1 887
700	43	89	182	318	451	964	1 736
800	40	83	169	296	420	897	1 615
900	38	78	159	278	394	842	1 515
1 000	36	74	150	262	372	795	1 431
1 200	32	67	136	238	337	720	1 297
1 400	30	61	125	219	310	663	1 193
1 600	28	57	116	203	289	616	1 110
1 800	26	54	109	191	271	578	1 041
2 000	25	51	103	180	256	546	984

Table A.12 b)
Maximum capacity of natural gas in kW for copper tubing for pressures of 34 kPa based on a pressure drop of 17 kPa

Length of tube, m	Outside diameter (mm)						
	9.5	13	16	19	22	29	35
3	127	261	531	929	1317	2814	5066
6	87	179	365	638	905	1934	3482
9	70	144	293	512	727	1553	2796
12	60	123	251	439	622	1329	2393
15	53	109	222	389	552	1178	2121
18	48	99	202	352	500	1067	1922
21	44	91	185	324	460	982	1768
24	41	85	172	301	428	914	1645
27	39	80	162	283	401	857	1543
30	36	75	153	267	379	810	1458
35	34	69	141	246	349	745	1341
40	31	64	131	229	324	693	1248
45	29	60	123	215	304	650	1171
50	28	57	116	203	288	614	1106
60	25	52	105	184	261	556	1002
70	23	48	97	169	240	512	922
80	21	44	90	157	223	476	857
90	20	41	84	147	209	447	805
100	19	39	80	139	198	422	760
125	17	35	71	123	175	374	674
150	15	31	64	112	159	339	610

(Continued)

Table A.12 b) (Continued)

Length of tube, m	Outside diameter (mm)						
	9.5	13	16	19	22	29	35
175	14	29	59	103	146	312	561
200	13	27	55	96	136	290	522
250	12	24	49	85	120	257	463
300	10	22	44	77	109	233	419
350	10	20	40	71	100	214	386
400	9	19	38	66	93	199	359
500	8	16	33	58	83	177	318
600	7	15	30	53	75	160	288

Table A.13 a)
Maximum capacity of natural gas in thousands of Btu/h for copper tubing for pressures of 10 psig based on a
pressure drop of 5 psig
 (See Clauses [A.2.3](#), [A.2.4](#), [A.2.5](#), [A.2.6](#), and [A.3.5](#).)

Length of tube, ft	Outside diameter (in)						
	3/8	1/2	5/8	3/4	7/8	1-1/8	1-3/8
10	695	1 433	2 915	5 095	7 228	15 440	27 797
20	477	985	2 004	3 502	4 968	10 612	19 105
30	383	791	1 609	2 812	3 989	8 522	15 342
40	328	677	1 377	2 407	3 414	7 294	13 131
50	291	600	1 220	2 133	3 026	6 464	11 637
60	263	544	1 106	1 933	2 742	5 857	10 544
70	242	500	1 017	1 778	2 523	5 388	9 701
80	225	465	946	1 654	2 347	5 013	9 025
90	212	437	888	1 552	2 202	4 703	8 467
100	200	412	839	1 466	2 080	4 443	7 998
125	177	365	743	1 299	1 843	3 938	7 089
150	160	331	674	1 177	1 670	3 568	6 423
175	148	305	620	1 083	1 537	3 282	5 909
200	137	283	577	1 008	1 429	3 054	5 497
250	122	251	511	893	1 267	2 706	4 872
300	110	228	463	809	1 148	2 452	4 414
350	101	209	426	744	1 056	2 256	4 061
400	94	195	396	693	982	2 099	3 778
450	89	183	372	650	922	1 969	3 545
500	84	173	351	614	871	1 860	3 349

(Continued)

Table A.13 a) (Continued)

Length of tube, ft	Outside diameter (in)						
	3/8	1/2	5/8	3/4	7/8	1-1/8	1-3/8
600	76	156	318	556	789	1 685	3 034
700	70	144	293	512	726	1 550	2 791
800	65	134	272	476	675	1 442	2 597
900	61	126	256	447	634	1 353	2 436
1 000	58	119	241	422	598	1 278	2 301
1 200	52	108	219	382	542	1 158	2 085
1 400	48	99	201	352	499	1 066	1 918
1 600	45	92	187	327	464	991	1 785
1 800	42	86	176	307	435	930	1 675
2 000	40	82	166	290	411	879	1 582

Table A.13 b)

Maximum capacity of natural gas in kW for copper tubing for pressures of 70 kPa based on a pressure drop of 35 kPa

Length of tube, m	Outside diameter (mm)						
	9.5	13	16	19	22	29	35
3	208	428	872	1523	2161	4617	8311
6	143	294	599	1047	1485	3173	5712
9	115	236	481	841	1193	2548	4587
12	98	202	412	720	1021	2181	3926
15	87	179	365	638	905	1933	3480
18	79	163	331	578	820	1751	3153
21	72	150	304	532	754	1611	2901
24	67	139	283	495	702	1499	2698
27	63	131	266	464	658	1406	2532
30	60	123	251	438	622	1328	2392
35	55	113	231	403	572	1222	2200
40	51	106	215	375	532	1137	2047
45	48	99	201	352	499	1067	1920
50	45	94	190	333	472	1008	1814
60	41	85	172	301	427	913	1644
70	38	78	159	277	393	840	1512
80	35	73	148	258	366	781	1407
90	33	68	138	242	343	733	1320
100	31	64	131	229	324	693	1247
125	28	57	116	203	287	614	1105
150	25	52	105	184	260	556	1001

(Continued)

Table A.13 b) (Continued)

Length of tube, m	Outside diameter (mm)						
	9.5	13	16	19	22	29	35
175	23	47	97	169	240	512	921
200	21	44	90	157	223	476	857
250	19	39	80	139	197	422	759
300	17	35	72	126	179	382	688
350	16	33	66	116	165	352	633
400	15	30	62	108	153	327	589
500	13	27	55	96	136	290	522
600	12	24	50	87	123	263	473

Table A.14 a)
Maximum capacity of natural gas in thousands of Btu/h for copper tubing for pressures of 20 psig based on a
pressure drop of 10 psig
 (See Clauses [A.2.3](#), [A.2.4](#), [A.2.5](#), [A.2.6](#), and [A.3.5](#).)

Length of tube, ft	Outside diameter (in)						
	3/8	1/2	5/8	3/4	7/8	1-1/8	1-3/8
10	1 183	2 441	4 965	8 677	12 311	26 297	47 342
20	813	1 677	3 412	5 964	8 461	18 074	32 538
30	653	1 347	2 740	4 789	6 795	14 514	26 129
40	559	1 153	2 345	4 099	5 815	12 422	22 363
50	495	1 022	2 079	3 633	5 154	11 010	19 820
60	449	926	1 883	3 292	4 670	9 975	17 958
70	413	852	1 733	3 028	4 296	9 177	16 522
80	384	792	1 612	2 817	3 997	8 538	15 370
90	360	743	1 512	2 643	3 750	8 011	14 421
100	340	702	1 429	2 497	3 542	7 567	13 622
125	302	622	1 266	2 213	3 139	6 706	12 073
150	273	564	1 147	2 005	2 845	6 076	10 939
175	251	519	1 055	1 845	2 617	5 590	10 064
200	234	483	982	1 716	2 435	5 201	9 363
250	207	428	870	1 521	2 158	4 609	8 298
300	188	388	789	1 378	1 955	4 176	7 518
350	173	357	725	1 268	1 799	3 842	6 917
400	161	332	675	1 179	1 673	3 574	6 435
450	151	311	633	1 107	1 570	3 354	6 038
500	142	294	598	1 045	1 483	3 168	5 703

(Continued)

Table A.14 a) (Continued)

Length of tube, ft	Outside diameter (in)						
	3/8	1/2	5/8	3/4	7/8	1-1/8	1-3/8
600	129	266	542	947	1 344	2 870	5 167
700	119	245	499	871	1 236	2 641	4 754
800	111	228	464	811	1 150	2 457	4 423
900	104	214	435	761	1 079	2 305	4 150
1 000	98	202	411	718	1 019	2 217	3 920
1 200	89	183	372	651	924	1 973	3 551
1 400	82	168	343	599	850	1 815	3 267
1 600	76	157	319	557	790	1 688	3 040
1 800	71	147	299	523	742	1 584	2 852
2 000	67	139	283	494	701	1 496	2 694

Table A.14 b)
Maximum capacity of natural gas in kW for copper tubing for pressures of 140 kPa based on a pressure drop of 70 kPa

Length of tube, m	Outside diameter (mm)						
	9.5	13	16	19	22	29	35
3	354	731	1 487	2 598	3 686	7 874	14 175
6	243	502	1 022	1 786	2 533	5 412	9 742
9	195	403	820	1 434	2 034	4 346	7 823
12	167	345	702	1 227	1 741	3 719	6 696
15	148	306	622	1 088	1 543	3 296	5 934
18	134	277	564	986	1 398	2 987	5 377
21	124	255	519	907	1 286	2 748	4 947
24	115	237	483	844	1 197	2 556	4 602
27	108	223	453	791	1 123	2 398	4 318
30	102	210	428	748	1 061	2 266	4 079
35	94	193	394	688	976	2 084	3 752
40	87	180	366	640	908	1 939	3 491
45	82	169	344	600	852	1 819	3 275
50	77	159	324	567	805	1 719	3 094
60	70	145	294	514	729	1 557	2 803
70	64	133	270	473	671	1 433	2 579
80	60	124	252	440	624	1 333	2 399
90	56	116	236	413	585	1 250	2 251
100	53	110	223	390	553	1 181	2 126
125	47	97	198	345	490	1 047	1 885
150	43	88	179	313	444	949	1 708

(Continued)

Table A.14 b) (Continued)

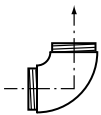
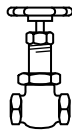
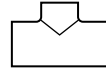
Length of tube, m	Outside diameter (mm)						
	9.5	13	16	19	22	29	35
175	39	81	165	288	409	873	1 571
200	37	75	153	268	380	812	1 461
250	32	67	136	237	337	719	1 295
300	29	60	123	215	305	652	1 173
350	27	56	113	198	281	600	1 080
400	25	52	105	184	261	558	1 004
500	22	46	93	163	231	494	890
600	20	42	85	148	210	448	807

Table A.15
Multipliers to be used with Tables A.1 to A.14 when the relative density of the gas is other than 0.60

(See Clauses [6.3.6](#), [A.2.6](#), [A.3.5](#), [E.1.4](#), and [E.2.4](#).)

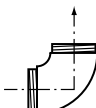
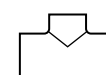
Relative density	Multiplier
0.35	1.31
0.40	1.23
0.45	1.16
0.50	1.10
0.55	1.04
0.60	1.00
0.65	0.962
0.70	0.926
0.75	0.895
0.80	0.867
0.85	0.841
0.90	0.817
1.00	0.775
1.10	0.740
1.20	0.707
1.30	0.680
1.80	0.577
1.90	0.565
2.00	0.547
2.10	0.535

Table A.16 a)
Resistance of bends, fittings, and valves for natural gas and propane expressed in equivalent length of straight pipe in feet*
 (See Clauses [6.3.7](#), [A.2.4](#), and [A.4](#).)

		Threaded fittings†		Valves (threaded, flanged, or welded)					90° welding elbows and smooth bends‡	Welding tees	
		Elbows									
		45°	90°	Tee	Plug	Globe	Angle	Swing check	R/d§ = 1-1/2	Forged	Mitre**
k factor		0.42	0.9	1.8	0.9	10	5	2.5	0.36	1.35	1.8
n (L/D ratio††)		14	30	60	30	333	167	83	12	45	60
Nominal pipe size, in (Schedule 40)	Inside diameter (d), in										
3/8	0.493	0.58	1.23	2.46	1.23	13.7	6.85	3.42	0.49	1.85	2.46
1/2	0.622	0.73	1.55	3.10	1.55	17.3	8.65	4.32	0.62	2.33	3.10
3/4	0.824	0.96	2.06	4.12	2.06	22.9	11.4	5.72	0.82	3.09	4.12
1	1.049	1.22	2.62	5.24	2.62	29.1	14.6	7.27	1.05	3.93	5.24
1-1/4	1.380	1.61	3.45	6.90	3.45	38.3	19.1	9.58	1.38	5.17	6.90
1-1/2	1.610	1.88	4.02	8.04	4.02	44.7	22.4	11.2	1.61	6.04	8.04
2	2.067	2.41	5.17	10.3	5.17	57.4	28.7	14.4	2.07	7.75	10.3
2-1/2	2.469	2.88	6.16	12.3	6.16	68.5	34.3	17.1	2.47	9.25	12.3

(Continued)

Table A.16 a) (Continued)

										90° welding elbows and smooth bends‡	
		Threaded fittings†		Valves (threaded, flanged, or welded)					Welding tees		
		Elbows									
		45°	90°	Tee	Plug	Globe	Angle	Swing check	R/d§ = 1-1/2	Forged	Mitre**
k factor		0.42	0.9	1.8	0.9	10	5	2.5	0.36	1.35	1.8
n (L/D ratio††)		14	30	60	30	333	167	83	12	45	60
Nominal pipe size, in (Schedule 40)	Inside diameter (d), in										
3	3.068	3.58	7.67	15.3	7.67	85.2	42.6	21.3	3.07	11.5	15.3
4	4.026	4.70	10.1	20.2	10.1	112	56.0	28.0	4.03	15.1	20.2
5	5.047	5.88	12.6	25.2	12.6	140	70.0	35.0	5.05	18.9	25.2

* Values for welded fittings are for conditions where the bore is not obstructed by weld spatter or backing rings. If appreciably obstructed, values for threaded fittings shall be used.

† Flanged fittings have three-fourths the resistance of threaded fittings.

‡ Tabular figures give the extra resistance due to curvature alone, to which should be added the actual length of the bend.

§ R/d is the ratio of the elbow or bend radius to the inside diameter of the pipe.

** Small socket-welding fittings are equivalent to mitre elbows and mitre tees.

†† n = resistance in equivalent number of diameters of straight pipe, computed from the relation $n = k / 4f$, where the friction factor, f , is assumed to be 0.0075;

$n = L / D$, where L = equivalent length of Schedule 40 straight pipe in feet, and D = inside diameter of pipe in feet. For pipe having other inside diameters, the resistance expressed in equivalent length in feet can be computed from the above n values.

Note: The equivalent lengths in feet shown in the table have been computed on the basis that the inside diameter corresponds to that of Schedule 40 (standard-weight) steel pipe, which is close enough for most purposes involving other schedules of pipe. Where a more specific solution for equivalent length is desired, this may be made by multiplying the actual inside diameter of the pipe in feet by $n / 12$, or the actual diameter in feet by n , read from the table heading. The

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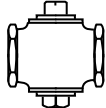
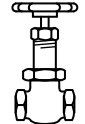
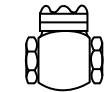
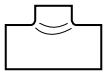
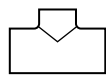
Table A.16 a) (Continued)

equivalent length values can be used with reasonable accuracy for copper or brass fittings and bends. For copper or brass values, however, the equivalent length of pipe should be taken as 45% longer than the values in the table, which are for steel pipe. Resistance per foot of copper or brass pipe is less than that of steel.

Example of calculation: To obtain the total equivalent length, add to the actual length of pipe involved the length in feet to allow for various fittings as shown in this table. Thus, if the problem involves 300 ft of NPS 4 pipe having three standard 90° elbows and two plug valves, the total equivalent length will be as follows:

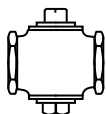
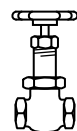
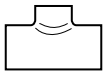
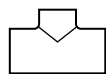
$$300 + (3 \times 10.1) + (2 \times 10.1) = 350.5 \quad \text{ft}$$

Table A.16 b)
Resistance of bends, fittings, and valves for natural gas and propane expressed in equivalent length of straight pipe in metres*

									90° welding elbows and smooth bends‡		
		Threaded fittings†			Valves (threaded, flanged, or welded)					Welding tees	
		Elbows									
		45°	90°	Tee	Plug	Globe	Angle	Swing check	R/d§ = 1-1/2	Forged	Mitre**
k factor		0.42	0.9	1.8	0.9	10	5	2.5	0.36	1.35	1.8
n (L/D ratio††)		14	30	60	30	333	167	83	12	45	60
Nominal pipe size, in (Schedule 40)	Inside diameter (d), mm										
3/8	12.52	0.18	0.37	0.75	0.37	4.18	2.09	1.04	0.15	0.56	0.75
1/2	15.80	0.22	0.47	0.94	0.47	5.27	2.64	1.29	0.19	0.71	0.94
3/4	20.93	0.29	0.63	1.26	0.63	6.98	3.47	1.74	0.25	0.94	1.26
1	26.64	0.37	0.80	1.60	0.80	8.87	4.45	2.22	0.32	1.20	1.60
1-1/4	35.05	0.49	1.05	2.10	1.05	11.67	5.82	2.92	0.42	1.58	2.10
1-1/2	40.89	0.49	1.23	2.45	1.23	13.62	6.83	3.41	0.49	1.84	2.45
2	52.50	0.73	1.58	3.14	1.58	17.50	8.75	4.39	0.63	2.36	3.14
2-1/2	62.71	0.88	1.88	3.75	1.88	20.88	10.45	5.21	0.75	2.82	3.75
3	77.93	1.09	2.34	4.66	2.34	25.97	12.98	6.49	0.94	3.51	4.66

(Continued)

Table A.16 b) (Continued)

		Threaded fittings†			Valves (threaded, flanged, or welded)				90° welding elbows and smooth bends‡	Welding tees	
		Elbows									
		45°	90°	Tee	Plug	Globe	Angle	Swing check	R/d§ = 1-1/2	Forged	Mitre**
k factor		0.42	0.9	1.8	0.9	10	5	2.5	0.36	1.35	1.8
n (L/D ratio††)		14	30	60	30	333	167	83	12	45	60
Nominal pipe size, in (Schedule 40)	Inside diameter (d), mm										
4	102.3	1.23	3.08	6.16	3.08	34.14	17.07	8.53	1.23	4.60	6.16
5	128.2	1.79	3.84	7.68	3.84	42.67	21.33	10.67	1.54	5.76	7.68

* Values for welded fittings are for conditions where the bore is not obstructed by weld spatter or backing rings. If appreciably obstructed, values for threaded fittings shall be used.

† Flanged fittings have three-fourths the resistance of threaded fittings.

‡ Tabular figures give the extra resistance due to curvature alone, to which should be added the actual length of the bend.

§ R/d is the ratio of the elbow or bend radius to the inside diameter of the pipe.

** Small socket-welding fittings are equivalent to mitre elbows and mitre tees.

†† n = resistance in equivalent number of diameters of straight pipe, computed from the relation $n = k / 4f$, where the friction factor, f , is assumed to be 0.0075;

$n = L / D$, where L = equivalent length of Schedule 40 straight pipe in metres and D = inside diameter of pipe in metres. For pipe having other inside diameters, the resistance expressed in equivalent length in metres can be computed from the above n values.

Note: The equivalent lengths in metres shown in the table have been computed on the basis that the inside diameter corresponds to that of Schedule 40 (standard-weight) steel pipe, which is close enough for most purposes involving other schedules of pipe. Where a more specific solution for equivalent length is desired, this may be made by multiplying the actual inside diameter of the pipe in metres by $n / 12$, or the actual diameter in metres by n , read from the table heading. The equivalent length values can be used with reasonable accuracy for copper or brass fittings and bends. For copper or brass values, however, the

(Continued)

Table A.16 b) (Continued)

equivalent length of pipe should be taken as 45% longer than the values in the table, which are for steel pipe. Resistance per metre of copper or brass pipe is less than that of steel.

Example of calculation: To obtain the total equivalent length, add to the actual length of pipe involved the length in metres to allow for various fittings as shown in this table. Thus, if the problem involves 100 m of NPS 4 pipe having three standard 90° elbows and two plug valves, the total equivalent length will be as follows:

$$100 + (3 \times 3.08) + (2 \times 3.08) = 115.4 \text{ m}$$

Table A.17
Use of capacity tables for sizing plastic pipe for natural gas and propane
 (See Clause [6.3.5.](#))

Nominal pipe size	Pipe inside diameters, in									
	Schedule 40 steel pipe		SDR 21 plastic pipe		SDR 13.5 plastic pipe		SDR 11 plastic pipe		SDR 8.8 plastic pipe	
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
1/2	0.64	0.67	0.68	0.72	0.62*	0.66*	0.61*	0.66*	0.6*	0.65*
3/4	0.83	0.87	0.83	0.88	0.83	0.88	0.81*	0.87*	0.76*	0.82*
1	1.07	1.1	1.09	1.14	1.08	1.13	1.02*	1.08*	0.95*	1.02*
1-1/4	1.4	1.43	1.44	1.49	1.37*	1.42*	1.3*	1.37*	1.21*	1.29*
1-1/2	1.63	1.66	1.67	1.73	1.57*	1.62*	1.48*	1.55*	1.4*	1.48*
2	2.1	2.14	2.1	2.15	1.98*	2.03*	1.89*	1.95*	1.77*	1.84*
2-1/2	2.5	2.55	2.55	2.61	2.39*	2.46*	2.29*	2.36*	2.14*	2.23*

* Denotes inside diameter of plastic pipe that may be less than the inside diameter of Schedule 40 steel pipe. The tables in this annex should be used only for sizing plastic pipe with inside diameters equal to or greater than steel pipe. Sometimes plastic pipe with inside diameters less than the inside diameter of Schedule 40 steel pipe sized using the tables in this annex does not supply gas at a sufficient volume and pressure. Plastic pipe with inside diameters less than the inside diameter of Schedule 40 steel pipe should be sized using the calculation method detailed in this annex.

Notes:

- 1) Dimensions of Schedule 40 steel pipe are derived from ASTM A53/A53M and ASME B36.10.
- 2) Dimensions of plastic pipe are derived from CSA B137.4.

Annex B (informative)

Sizing and capacities of piping and tubing for propane

Note: This informative annex has been written in mandatory language to facilitate adoption where users of the Code or regulatory authorities wish to adopt it formally as additional requirements to this Code.

B.1 General

In order to determine the size of piping or tubing to be used in designing a gas piping system, the following factors shall be considered:

- a) allowable loss in pressure from point of delivery to equipment;
- b) maximum gas demand;
- c) length of piping and number of fittings;
- d) relative density of the gas; and
- e) diversity factor.

B.2 Description of tables

B.2.1 Outlet capacity

The total energy to be provided at each outlet should be determined directly from the manufacturer's Btu/h (kW) input rating of the equipment to be installed.

B.2.2 Propane quality

The tables listed in this annex assume that a gas having a calorific value of 2520 Btu/ft³ (94 MJ/m³) and a relative density of 1.52 for propane shall be used.

B.2.3 Table applicability

Capacities for propane in Btu/h (kW) for different sizes and lengths are shown in Tables [B.1](#) to [B.5](#) for iron pipe or equivalent rigid pipe and in Tables [B.6](#) to [B.10](#) for tubing.

B.2.4 Fitting allowance

The formula used to derive Tables [B.1](#), [B.2](#), [B.6](#), and [B.7](#) contains a factor ($F = 1.2$), which is used to multiply the piping or tubing length to allow for a reasonable number of fittings. Tables [B.3](#) to [B.5](#) and [B.8](#) to [B.10](#) do not contain a fitting allowance ($F = 1$), and Table [B.11](#) should be utilized to determine equivalent lengths of fittings.

B.2.5 Tubing type

Tables [B.6](#) to [B.10](#) covering tubing are based on Type K tubing. Type K was selected since it has a smaller inside diameter than Types G and L.

B.3 Use of capacity tables

B.3.1 Size determination

To determine the size of each section of propane piping or tubing in a system within the range of the capacity tables, the procedure outlined in Clauses [B.3.2](#) to [B.3.9](#) shall be followed.

B.3.2 Demand determination

The propane demand shall be determined for each appliance to be attached to the piping or tubing system. The propane demand shall be calculated in terms of Btu/h (kW) for each piping system outlet.

B.3.3 Pressure determination

The design system pressure, the allowable loss in pressure (pressure drop), and the relative density of propane to be used in the piping or tubing system shall be determined.

B.3.4 Length determination

The length of piping or tubing shall be measured from the point of delivery to the most remote outlet in the building.

B.3.5 Formulae

In the appropriate capacity table, the row showing the measured length, or the next larger length if the table does not give the exact length, shall be selected. This is the only length used in determining the size of any section of propane piping or tubing.

Capacities may also be determined by using the following formulas, which were used to generate the tables*:

for high pressure [1.5 psig (10.3 kPa) and above]:

$$Q = (K_1)D^{2.623} \left[\frac{(P_1^2 - P_2^2)Y}{Cr \times L \times F} \right]^{0.541} (b \times z)$$

for low pressure [less than 1.5 psig (10.3 kPa)]:

$$Q = (K_2)D^{2.623} \left[\frac{\Delta H}{Cr \times L \times F} \right]^{0.541} (b \times z)$$

where

Q = rate, Btu/h (kW) at 60°F (15.6 °C) and 30 in Hg (101.3 kPa)

K_1 = 2237 for imperial measurements

= 0.3576 for SI (metric) measurements

D = inside diameter of pipe, in (cm)

P_1 = upstream pressure, absolute, psi (kPa)

P_2 = downstream pressure, absolute, psi (kPa)

Y = superexpansibility factor†

= 1/supercompressibility factor

= 1 for intents and purposes

b = calorific value of the propane, Btu/ft³ (MJ/m³)

z = factor to convert MJ/h to kW

= 1 for imperial measurements

= 0.2778 for SI (metric) measurements

Cr = factor for viscosity, density, and temperature

$$= K_3 ST \left[\frac{Z}{S} \right]^{0.152}$$

- = 1.260 for propane
- where
- K_3 = 0.00354 for imperial measurements
 = 0.00638 for SI (metric) measurements
- S = relative density of propane at 60 °F (15.6 °C) and 30 in Hg (101.3 kPa), (propane = 1.52)
- T = absolute temperature, °R = °F + 460 (imperial)
 = absolute temperature, K = °C + 273 (metric)
- Z = viscosity of propane, centipoise (0.008 for propane)
- L = length of pipe, ft (m)
- F = fitting allowance factor
 = 1.2 for Tables [B.1](#), [B.2](#), [B.6](#), and [B.7](#)
 = 1 for Tables [B.3](#) to [B.5](#) and [B.8](#) to [B.10](#)
- K_2 = 2313 for imperial measurements
 = 0.1509 for SI (metric) measurements
- ΔH = pressure drop, in w.c. (Pa)

* For further details on the formulas, refer to B.C. Shebeko, *Polyflo Flow Computer*, 1974, available from Polyflo Computer Company, Box 50126, Dallas, Texas 75250, USA.

† For values for natural gas, refer to *Manual for Determination of Supercompressibility Factors for Natural Gas*, available from the American Gas Association, 1515 Wilson Boulevard, Arlington, Virginia 22209, USA.

For values for liquefied petroleum gases, refer to *Engineering Data Book*, available from Gas Processors Association, 1812 First Place, Tulsa, Oklahoma 74102, USA.

B.3.6 Row choice

The horizontal row specified in Clause [B.3.5](#) shall be used to locate all propane demand figures for this particular system of piping.

B.3.7 Column choice

Starting at the most remote outlet, in the horizontal row selected in accordance with Clause [B.3.6](#), the propane demand for that outlet shall be located. If the exact figure of demand is not shown, the next larger figure to the right in the row shall be chosen.

B.3.8 Correct size

In the top row above this demand figure, the correct size of propane piping will be found.

B.3.9 Repeat

The procedure shall be repeated in a similar manner for each outlet and each section of propane piping. For each section of piping, the total propane demand supplied by that section shall be determined.

B.4 Bends, fittings, and valves expressed in equivalent lengths in feet of straight pipe

The equivalent lengths in feet shown in Table [B.11](#)* have been computed on the basis that the inside diameter corresponds to that of Schedule 40 (standard-weight) steel pipe, which is close enough for most purposes to that of other schedules of pipe. Where a more specific solution for equivalent length

is desired, this may be computed by multiplying the actual inside diameter of the pipe in inches by $n/12$, or the actual inside diameter in feet by “n” (“n” is a heading in Table [B.11](#)). The equivalent length values can be used with reasonable accuracy for copper or brass fittings and bends. For copper or brass valves, however, the equivalent length of pipe should be taken as 45% longer than the values in Table [B.11](#), which are for steel pipe. Resistance per foot of copper or brass pipe is less than that of steel.

* From Crocker (1945). Table XIV, pp. 100–101. Used by permission of McGraw-Hill Book Company.

Table B.1 a)
Maximum capacity of propane in thousands of Btu/h for Schedule 40 pipe and plastic pipe, including fittings, for pressures of 11 in w.c. based on a pressure drop of 1 in w.c.
 (See Clauses [6.3.2](#), [6.3.5](#), [B.2.3](#), [B.2.4](#), and [B.3.5](#).)

Length of pipe, ft	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
10	386	807	1 520	3 121	4 676	9 005	14 353	25 374	51 754
20	265	555	1 045	2 145	3 214	6 189	9 865	17 439	35 571
30	213	445	839	1 722	2 581	4 970	7 922	14 004	28 564
40	182	381	718	1 474	2 209	4 254	6 780	11 986	24 447
50	162	338	636	1 307	1 958	3 770	6 009	10 623	21 667
60	146	306	577	1 184	1 774	3 416	5 445	9 625	19 632
70	135	282	530	1 089	1 632	3 143	5 009	8 855	18 061
80	125	262	493	1 013	1 518	2 924	4 660	8 238	16 803
90	118	246	463	951	1 424	2 743	4 372	7 729	15 765
100	111	232	437	898	1 345	2 591	4 130	7 301	14 892
125	98	206	388	796	1 192	2 297	3 660	6 471	13 198
150	89	186	351	721	1 080	2 081	3 317	5 863	11 959
175	82	172	323	663	994	1 914	3 051	5 394	11 002
200	76	160	301	617	925	1 781	2 839	5 018	10 235
250	68	141	266	547	820	1 578	2 516	4 447	9 071
300	61	128	241	496	743	1 430	2 279	4 030	8 219
350	56	118	222	456	683	1 316	2 097	3 707	7 561
400	52	110	207	424	636	1 224	1 951	3 449	7 035
450	49	103	194	398	596	1 148	1 830	3 236	6 600
500	46	97	183	376	563	1 085	1 729	3 057	6 235

(Continued)

Table B.1 a) (Continued)

Length of pipe, ft	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
600	42	88	166	341	510	983	1 567	2 770	5 649
700	39	81	153	313	470	904	1 441	2 548	5 197
800	36	75	142	292	437	841	1 341	2 370	4 835
900	34	71	133	274	410	789	1 258	2 224	4 536
1 000	32	67	126	258	387	746	1 188	2 101	4 285
1 200	29	61	114	234	351	676	1 077	1 903	3 882
1 400	27	56	105	215	323	622	991	1 751	3 572
1 600	25	52	98	200	300	578	922	1 629	3 323
1 800	23	49	92	188	282	542	865	1 529	3 118
2 000	22	46	86	178	266	512	817	1 444	2 945

Table B.1 b)
**Maximum capacity of propane in kW for Schedule 40 pipe and plastic pipe, including fittings, for pressures of 2.7 kPa
based on a pressure drop of 250 Pa**

Length of pipe, m	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
3	114	239	450	925	1 385	2 668	4 253	7 518	15 334
6	79	164	310	636	952	1 834	2 923	5 167	10 539
9	63	132	249	510	765	1 473	2 347	4 149	8 463
12	54	113	213	437	654	1 260	2 009	3 551	7 243
15	48	100	189	387	580	1 117	1 780	3 147	6 420
18	43	91	171	351	526	1 012	1 613	2 852	5 817
21	40	83	157	323	483	931	1 484	2 624	5 351
24	37	78	146	300	450	866	1 381	2 441	4 978
27	35	73	137	282	422	813	1 295	2 290	4 671
30	33	69	130	266	399	768	1 224	2 163	4 412
35	30	63	119	245	367	706	1 126	1 990	4 059
40	28	59	111	228	341	657	1 047	1 851	3 776
45	26	55	104	214	320	617	983	1 737	3 543
50	25	52	98	202	302	582	928	1 641	3 347
60	23	47	89	183	274	528	841	1 487	3 032
70	21	43	82	168	252	485	774	1 368	2 790
80	19	40	76	157	234	452	720	1 272	2 595
90	18	38	72	147	220	424	675	1 194	2 435
100	17	36	68	139	208	400	638	1 128	2 300
125	15	32	60	123	184	355	565	1 000	2 039

(Continued)

Table B.1 b) (Continued)

Length of pipe, m	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
150	14	29	54	111	167	321	512	906	1 847
175	13	26	50	102	154	296	471	833	1 699
200	12	25	46	95	143	275	438	775	1 581
250	10	22	41	84	127	244	389	687	1 401
300	9	20	37	77	115	221	352	622	1 270
350	9	18	34	70	106	203	324	573	1 168
400	8	17	32	66	98	189	301	533	1 087
500	7	15	28	58	87	168	267	472	963
600	7	14	26	53	79	152	242	428	873

Table B.2 a)
Maximum capacity of propane in thousands of Btu/h for Schedule 40 pipe and plastic pipe, including fittings, for
pressures of 2 psig based on a pressure drop of 1 psig
 (See Clauses [6.3.4](#), [6.3.5](#), [B.2.3](#), [B.2.4](#), and [B.3.5](#).)

Length of pipe, ft	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
10	2 450	5 122	9 649	19 810	29 682	57 165	91 112	161 070	328 531
20	1 684	3 521	6 632	13 616	20 400	39 289	62 621	110 702	225 798
30	1 352	2 827	5 326	10 934	16 382	31 551	50 287	88 898	181 323
40	1 157	2 420	4 558	9 358	14 021	27 003	43 039	76 085	155 189
50	1 026	2 145	4 040	8 294	12 427	23 932	38 145	67 433	137 541
60	929	1 943	3 660	7 515	11 259	21 685	34 562	61 099	124 623
70	855	1 788	3 367	6 913	10 359	19 950	31 796	56 210	114 651
80	795	1 663	3 133	6 432	9 637	18 559	29 580	52 293	106 661
90	746	1 560	2 939	6 035	9 042	17 413	27 754	49 065	100 076
100	705	1 474	2 776	5 700	8 541	16 449	26 217	46 346	94 531
125	625	1 306	2 461	5 052	7 570	14 578	23 235	41 076	83 782
150	566	1 184	2 230	4 578	6 859	13 209	21 053	37 218	75 912
175	521	1 089	2 051	4 211	6 310	12 152	19 368	34 240	69 838
200	484	1 013	1 908	3 918	5 870	11 305	18 018	31 853	64 971
250	429	898	1 691	3 472	5 202	10 019	15 969	28 231	57 583
300	389	813	1 532	3 146	4 714	9 078	14 469	25 579	52 174
350	358	748	1 410	2 894	4 337	8 352	13 312	23 533	47 999
400	333	696	1 312	2 693	4 034	7 770	12 384	21 893	44 654
450	312	653	1 231	2 526	3 785	7 290	11 619	20 541	41 898
500	295	617	1 162	2 386	3 576	6 886	10 976	19 403	39 576

(Continued)

Table B.2 a) (Continued)

Length of pipe, ft	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
600	267	559	1 053	2 162	3 240	6 240	9 945	17 581	35 859
700	246	514	969	1 989	2 981	5 740	9 149	16 174	32 990
800	229	479	901	1 851	2 773	5 340	8 511	15 047	30 691
900	215	449	846	1 736	2 602	5 011	7 986	14 118	28 796
1 000	203	424	799	1 640	2 458	4 733	7 544	13 336	27 200
1 200	184	384	724	1 486	2 227	4 288	6 835	12 083	24 646
1 400	169	354	666	1 367	2 049	3 945	6 288	11 116	22 674
1 600	157	329	620	1 272	1 906	3 670	5 850	10 342	21 093
1 800	148	309	581	1 193	1 788	3 444	5 489	9 703	19 791
2 000	139	291	549	1 127	1 689	3 253	5 185	9 166	18 695

Table B.2 b)
**Maximum capacity of propane in kW for Schedule 40 pipe and plastic pipe, including fittings, for pressures of 14 kPa
based on a pressure drop of 7 kPa**

Length of pipe, m	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
3	731	1 528	2 879	5 911	8 857	17 057	27 186	48 060	98 026
6	502	1 050	1 979	4 063	6 087	11 723	18 685	33 031	67 373
9	403	844	1 589	3 262	4 888	9 414	15 004	26 525	54 103
12	345	722	1 360	2 792	4 184	8 057	12 842	22 702	46 305
15	306	640	1 205	2 475	3 708	7 141	11 381	20 120	41 039
18	277	580	1 092	2 242	3 360	6 470	10 312	18 231	37 185
21	255	533	1 005	2 063	3 091	5 952	9 487	16 772	34 209
24	237	496	935	1 919	2 875	5 538	8 826	15 603	31 825
27	223	466	877	1 801	2 698	5 196	8 281	14 640	29 860
30	210	440	828	1 701	2 548	4 908	7 822	13 829	28 206
35	193	405	762	1 565	2 344	4 515	7 197	12 722	25 949
40	180	376	709	1 456	2 181	4 201	6 695	11 836	24 141
45	169	353	665	1 366	2 046	3 941	6 282	11 105	22 650
50	160	334	628	1 290	1 933	3 723	5 934	10 490	21 395
60	145	302	569	1 169	1 751	3 373	5 376	9 504	19 386
70	133	278	524	1 075	1 611	3 103	4 946	8 744	17 835
80	124	259	487	1 000	1 499	2 887	4 601	8 134	16 592
90	116	243	457	939	1 407	2 709	4 317	7 632	15 568
100	110	229	432	887	1 329	2 559	4 078	7 209	14 705
125	97	203	383	786	1 177	2 268	3 614	6 390	13 033

(Continued)

Table B.2 b) (Continued)

Length of pipe, m	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
150	88	184	347	712	1 067	2 055	3 275	5 789	11 809
175	81	169	319	655	982	1 890	3 013	5 326	10 864
200	75	158	297	609	913	1 759	2 803	4 955	10 107
250	67	140	263	540	809	1 559	2 484	4 392	8 957
300	61	127	238	489	733	1 412	2 251	3 979	8 116
350	56	116	219	450	675	1 299	2 071	3 661	7 467
400	52	108	204	419	628	1 209	1 926	3 406	6 946
500	46	96	181	371	556	1 071	1 707	3 018	6 156
600	42	87	164	336	504	971	1 547	2 735	5 578

Table B.3 a)
Maximum capacity of propane in thousands of Btu/h for Schedule 40 pipe and plastic pipe for pressures of 5 psig
based on a pressure drop of 2.5 psig
 (See Clauses [6.3.5](#), [B.2.3](#), [B.2.4](#), and [B.3.5](#).)

Length of pipe, ft	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
10	4 762	9 958	18 757	38 511	57 701	111 127	177 118	313 113	638 651
20	3 273	6 844	12 892	26 468	39 658	76 377	121 732	215 201	438 941
30	2 628	5 496	10 353	21 255	31 846	61 333	97 755	172 814	352 485
40	2 249	4 704	8 861	18 191	27 256	52 493	83 666	147 906	301 682
50	1 994	4 169	7 853	16 123	24 157	46 524	74 151	131 087	267 375
60	1 806	3 777	7 115	14 608	21 888	42 154	67 187	118 774	242 261
70	1 662	3 475	6 546	13 440	20 137	38 781	61 811	109 271	222 877
80	1 546	3 233	6 090	12 503	18 733	36 078	57 503	101 655	207 344
90	1 451	3 033	5 714	11 731	17 577	33 851	53 953	95 380	194 544
100	1 370	2 865	5 397	11 081	16 603	31 976	50 964	90 095	183 765
125	1 214	2 539	4 783	9 821	14 715	28 339	45 168	79 850	162 868
150	1 100	2 301	4 334	8 899	13 333	25 677	40 926	72 350	147 570
175	1 012	2 117	3 987	8 187	12 266	23 623	37 651	66 561	135 763
200	942	1 969	3 710	7 616	11 411	21 977	35 027	61 922	126 301
250	835	1 745	3 288	6 750	10 113	19 477	31 044	54 880	111 938
300	756	1 581	2 979	6 116	9 163	17 648	28 128	49 725	101 424
350	696	1 455	2 741	5 627	8 430	16 236	25 877	45 747	93 309
400	647	1 353	2 550	5 234	7 843	15 104	24 074	42 559	86 806
450	607	1 270	2 392	4 911	7 359	14 172	22 588	39 931	81 447
500	574	1 200	2 260	4 639	6 951	13 387	21 336	37 719	76 934

(Continued)

Table B.3 a) (Continued)

Length of pipe, ft	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
600	520	1 087	2 047	4 203	6 298	12 129	19 332	34 176	69 708
700	478	1 000	1 884	3 867	5 794	11 159	17 785	31 442	64 131
800	445	930	1 752	3 598	5 390	10 381	16 546	29 250	59 661
900	417	873	1 644	3 375	5 058	9 740	15 524	27 445	55 978
1 000	394	824	1 553	3 188	4 777	9 201	14 664	25 924	52 877
1 200	357	747	1 407	2 889	4 329	8 336	13 287	23 489	47 910
1 400	329	687	1 295	2 658	3 982	7 669	12 224	21 610	44 077
1 600	306	639	1 204	2 473	3 705	7 135	11 372	20 104	41 005
1 800	287	600	1 130	2 320	3 476	6 694	10 670	18 862	38 473
2 000	271	567	1 067	2 191	3 283	6 324	10 079	17 817	36 342

Table B.3 b)
Maximum capacity of propane in kW for Schedule 40 pipe and plastic pipe for pressures of 34 kPa based on a pressure drop of 17 kPa

Length of pipe, m	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
3	1 396	2 918	5 497	11 286	16 910	32 568	51 908	91 764	187 169
6	959	2 006	3 778	7 757	11 622	22 384	35 676	63 069	128 640
9	770	1 611	3 034	6 229	9 333	17 975	28 649	50 647	103 303
12	659	1 379	2 597	5 331	7 988	15 384	24 520	43 347	88 414
15	584	1 222	2 301	4 725	7 080	13 635	21 732	38 418	78 359
18	529	1 107	2 085	4 281	6 415	12 354	19 690	34 809	70 999
21	487	1 018	1 918	3 939	5 901	11 366	18 115	32 024	65 319
24	453	947	1 785	3 664	5 490	10 573	16 852	29 792	60 766
27	425	889	1 675	3 438	5 151	9 921	15 812	27 953	57 015
30	402	840	1 582	3 248	4 866	9 371	14 936	26 404	53 856
35	369	773	1 455	2 988	4 476	8 621	13 741	24 291	49 547
40	344	719	1 354	2 779	4 165	8 020	12 783	22 599	46 094
45	322	674	1 270	2 608	3 907	7 525	11 994	21 203	43 248
50	305	637	1 200	2 463	3 691	7 108	11 330	20 029	40 852
60	276	577	1 087	2 232	3 344	6 441	10 265	18 147	37 015
70	254	531	1 000	2 053	3 077	5 925	9 444	16 695	34 053
80	236	494	930	1 910	2 862	5 512	8 786	15 532	31 680
90	222	463	873	1 792	2 686	5 172	8 243	14 573	29 724
100	209	438	825	1 693	2 537	4 886	7 787	13 766	28 077
125	186	388	731	1 501	2 248	4 330	6 901	12 200	24 884

(Continued)

Table B.3 b) (Continued)

Length of pipe, m	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
150	168	352	662	1 360	2 037	3 923	6 253	11 054	22 547
175	155	323	609	1 251	1 874	3 609	5 753	10 170	20 743
200	144	301	567	1 164	1 743	3 358	5 352	9 461	19 297
250	128	267	502	1 031	1 545	2 976	4 743	8 385	17 103
300	116	242	455	934	1 400	2 696	4 298	7 598	15 497
350	106	222	419	860	1 288	2 481	3 954	6 990	14 257
400	99	207	390	800	1 198	2 308	3 678	6 502	13 263
500	88	183	345	709	1 062	2 045	3 260	5 763	11 755
600	79	166	313	642	962	1 853	2 954	5 222	10 651

Table B.4 a)
Maximum capacity of propane in thousands of Btu/h for Schedule 40 pipe and plastic pipe for pressures of 10 psig
based on a pressure drop of 5 psig
 (See Clauses [6.3.5](#), [B.2.3](#), [B.2.4](#), and [B.3.5](#).)

Length of pipe, ft	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
10	7 658	16 014	30 166	61 933	92 795	178 714	284 841	503 549	1 027 079
20	5 264	11 006	20 733	42 566	63 777	122 829	195 770	346 086	705 905
30	4 227	8 838	16 649	34 182	51 216	98 636	157 210	277 919	566 867
40	3 618	7 565	14 249	29 256	43 834	84 420	134 551	237 863	485 165
50	3 206	6 704	12 629	25 929	38 849	74 820	119 250	210 814	429 993
60	2 905	6 075	11 443	23 493	35 200	67 792	108 049	191 012	389 604
70	2 673	5 589	10 527	21 613	32 384	62 368	99 404	175 729	358 431
80	2 486	5 199	9 794	20 107	30 127	58 021	92 476	163 482	333 451
90	2 333	4 878	9 189	18 866	28 267	54 439	86 768	153 390	312 866
100	2 204	4 608	8 680	17 821	26 701	51 423	81 960	144 891	295 531
125	1 953	4 084	7 693	15 794	23 664	45 575	72 640	128 414	261 924
150	1 770	3 700	6 970	14 311	21 442	41 295	65 817	116 353	237 322
175	1 628	3 404	6 413	13 166	19 726	37 990	60 551	107 043	218 333
200	1 515	3 167	5 966	12 248	18 351	35 343	56 331	99 583	203 117
250	1 342	2 807	5 287	10 855	16 264	31 324	49 925	88 258	180 019
300	1 216	2 543	4 791	9 836	14 737	28 381	45 236	79 968	163 110
350	1 119	2 340	4 407	9 049	13 558	26 111	41 616	73 570	150 059
400	1 041	2 177	4 100	8 418	12 613	24 291	38 716	68 443	139 601
450	977	2 042	3 847	7 898	11 834	22 791	36 326	64 217	130 983
500	923	1 929	3 634	7 461	11 178	21 529	34 313	60 659	123 726

(Continued)

Table B.4 a) (Continued)

Length of pipe, ft	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
600	836	1 748	3 293	6 760	10 128	19 506	31 090	54 962	112 105
700	769	1 608	3 029	6 219	9 318	17 946	28 603	50 564	103 135
800	715	1 496	2 818	5 786	8 669	16 695	26 609	47 040	95 947
900	671	1 404	2 644	5 428	8 134	15 664	24 966	44 136	90 024
1 000	634	1 326	2 498	5 128	7 683	14 796	23 583	41 691	85 036
1 200	575	1 201	2 263	4 646	6 961	13 407	21 368	37 775	77 049
1 400	529	1 105	2 082	4 274	6 404	12 334	19 658	34 753	70 884
1 600	492	1 028	1 937	3 976	5 958	11 474	18 288	32 331	65 944
1 800	461	965	1 817	3 731	5 590	10 766	17 159	30 335	61 873
2 000	436	911	1 717	3 524	5 280	10 170	16 209	28 654	58 445

Table B.4 b)
Maximum capacity of propane in kW for Schedule 40 pipe and plastic pipe for pressures of 70 kPa based on a pressure drop of 35 kPa

Length of pipe, m	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
3	2 290	4 788	9 019	18 517	27 745	53 433	85 164	150 555	307 085
6	1 574	3 291	6 199	12 727	19 069	36 724	58 533	103 476	211 058
9	1 264	2 643	4 978	10 220	15 313	29 491	47 004	83 095	169 487
12	1 082	2 262	4 260	8 747	13 106	25 240	40 229	71 118	145 059
15	959	2 005	3 776	7 752	11 615	22 370	35 654	63 031	128 563
18	869	1 816	3 421	7 024	10 524	20 269	32 306	57 111	116 487
21	799	1 671	3 148	6 462	9 682	18 647	29 721	52 541	107 167
24	743	1 554	2 928	6 012	9 008	17 348	27 649	48 879	99 698
27	697	1 458	2 747	5 641	8 451	16 277	25 943	45 862	93 543
30	659	1 378	2 595	5 328	7 983	15 375	24 505	43 321	88 361
35	606	1 267	2 388	4 902	7 344	14 145	22 544	39 855	81 291
40	564	1 179	2 221	4 560	6 833	13 159	20 973	37 077	75 625
45	529	1 106	2 084	4 279	6 411	12 347	19 678	34 788	70 957
50	500	1 045	1 969	4 042	6 056	11 663	18 588	32 861	67 025
60	453	947	1 784	3 662	5 487	10 567	16 842	29 774	60 730
70	417	871	1 641	3 369	5 048	9 722	15 495	27 392	55 871
80	388	810	1 527	3 134	4 696	9 044	14 415	25 483	51 977
90	364	760	1 432	2 941	4 406	8 486	13 525	23 910	48 768
100	343	718	1 353	2 778	4 162	8 016	12 776	22 585	46 066
125	304	637	1 199	2 462	3 689	7 104	11 323	20 017	40 827

(Continued)

Table B.4 b) (Continued)

Length of pipe, m	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
150	276	577	1 086	2 231	3 342	6 437	10 259	18 136	36 993
175	254	531	1 000	2 052	3 075	5 922	9 438	16 685	34 033
200	236	494	930	1 909	2 861	5 509	8 781	15 522	31 661
250	209	438	824	1 692	2 535	4 883	7 782	13 757	28 060
300	190	396	747	1 533	2 297	4 424	7 051	12 465	25 425
350	174	365	687	1 410	2 113	4 070	6 487	11 468	23 391
400	162	339	639	1 312	1 966	3 786	6 035	10 669	21 760
500	144	301	566	1 163	1 742	3 356	5 349	9 455	19 286
600	130	272	513	1 054	1 579	3 041	4 846	8 567	17 474

Table B.5 a)
Maximum capacity of propane in thousands of Btu/h for Schedule 40 pipe and plastic pipe for pressures of 20 psig
based on a pressure drop of 10 psig
 (See Clauses [6.3.5](#), [B.2.3](#), [B.2.4](#), and [B.3.5](#).)

Length of pipe, ft	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
10	13 043	27 274	51 377	105 482	158 044	304 377	485 128	857 621	1 749 272
20	8 965	18 745	35 311	72 497	108 623	209 196	333 426	589 438	1 202 265
30	7 199	15 053	28 356	58 218	87 228	167 992	267 753	473 339	965 461
40	6 161	12 884	24 269	49 827	74 656	143 780	229 162	405 117	826 310
50	5 461	11 418	21 509	44 160	66 166	127 429	203 102	359 048	732 343
60	4 948	10 346	19 489	40 013	59 951	115 460	184 025	325 323	663 556
70	4 552	9 518	17 930	36 811	55 154	106 222	169 301	299 293	610 463
80	4 235	8 855	16 680	34 246	51 310	98 819	157 501	278 435	567 918
90	3 973	8 308	15 650	32 132	48 143	92 719	147 778	261 246	532 859
100	3 753	7 848	14 783	30 351	45 476	87 581	139 591	246 772	503 335
125	3 326	6 955	13 102	26 900	40 304	77 622	123 717	218 709	446 097
150	3 014	6 302	11 871	24 373	36 518	70 331	112 096	198 166	404 196
175	2 773	5 798	10 922	22 423	33 597	64 704	103 127	182 310	371 855
200	2 579	5 394	10 160	20 860	31 255	60 194	95 940	169 605	345 940
250	2 286	4 780	9 005	18 488	27 701	53 349	85 030	150 318	306 600
300	2 071	4 331	8 159	16 752	25 099	48 338	77 043	136 199	277 802
350	1 906	3 985	7 506	15 411	23 091	44 470	70 879	125 301	255 574
400	1 773	3 707	6 983	14 337	21 481	41 371	65 939	116 568	237 762
450	1 663	3 478	6 552	13 452	20 155	38 817	61 868	109 372	223 085
500	1 571	3 286	6 189	12 707	19 039	36 666	58 440	103 312	210 724

(Continued)

Table B.5 a) (Continued)

Length of pipe, ft	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
600	1 424	2 977	5 608	11 513	17 250	33 222	52 951	93 609	190 932
700	1 310	2 739	5 159	10 592	15 870	30 564	48 715	86 119	175 655
800	1 218	2 548	4 799	9 854	14 764	28 434	45 319	80 117	163 413
900	1 143	2 391	4 503	9 246	13 853	26 679	42 522	75 171	153 325
1 000	1 080	2 258	4 254	8 733	13 085	25 201	40 166	71 006	144 830
1 200	978	2 046	3 854	7 913	11 856	22 834	36 393	64 337	131 226
1 400	900	1 882	3 546	7 280	10 907	21 007	33 481	59 189	120 726
1 600	837	1 751	3 299	6 772	10 147	19 543	31 148	55 064	112 313
1 800	786	1 643	3 095	6 354	9 521	18 336	29 225	51 665	105 379
2 000	742	1 552	2 924	6 002	8 993	17 320	27 606	48 802	99 541

Table B.5 b)
Maximum capacity of propane in kW for Schedule 40 pipe and plastic pipe for pressures of 140 kPa based on a pressure drop of 70 kPa

Length of pipe, m	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
3	3 905	8 166	15 382	31 582	47 319	91 132	145 250	256 776	523 740
6	2 684	5 612	10 572	21 706	32 522	62 634	99 829	176 480	359 964
9	2 155	4 507	8 490	17 431	26 116	50 298	80 166	141 720	289 064
12	1 845	3 857	7 266	14 918	22 352	43 048	68 612	121 294	247 401
15	1 635	3 419	6 440	13 222	19 810	38 153	60 810	107 501	219 267
18	1 481	3 098	5 835	11 980	17 950	34 569	55 098	97 403	198 672
21	1 363	2 850	5 368	11 021	16 513	31 803	50 689	89 610	182 776
24	1 268	2 651	4 994	10 253	15 363	29 587	47 157	83 365	170 037
27	1 190	2 488	4 686	9 620	14 414	27 760	44 246	78 218	159 540
30	1 124	2 350	4 426	9 087	13 616	26 222	41 794	73 885	150 701
35	1 034	2 162	4 072	8 360	12 526	24 124	38 450	67 973	138 643
40	962	2 011	3 788	7 778	11 653	22 443	35 770	63 236	128 981
45	902	1 887	3 554	7 297	10 934	21 057	33 562	59 332	121 018
50	852	1 782	3 357	6 893	10 328	19 891	31 703	56 045	114 313
60	772	1 615	3 042	6 246	9 358	18 022	28 725	50 780	103 576
70	711	1 486	2 799	5 746	8 609	16 580	26 426	46 717	95 289
80	661	1 382	2 604	5 345	8 009	15 425	24 585	43 461	88 648
90	620	1 297	2 443	5 015	7 515	14 473	23 067	40 778	83 175
100	586	1 225	2 308	4 738	7 098	13 671	21 789	38 519	78 567
125	519	1 086	2 045	4 199	6 291	12 116	19 311	34 139	69 632

(Continued)

Table B.5 b) (Continued)

Length of pipe, m	Pipe size (NPS)								
	1/2	3/4	1	1-1/4	1-1/2	2	2-1/2	3	4
150	470	984	1 853	3 804	5 700	10 978	17 497	30 932	63 092
175	433	905	1 705	3 500	5 244	10 100	16 097	28 457	58 044
200	403	842	1 586	3 256	4 879	9 396	14 975	26 474	53 998
250	357	746	1 406	2 886	4 324	8 327	13 272	23 463	47 858
300	323	676	1 274	2 615	3 918	7 545	12 026	21 260	43 363
350	297	622	1 172	2 406	3 604	6 941	11 064	19 559	39 893
400	277	579	1 090	2 238	3 353	6 458	10 293	18 195	37 113
500	245	513	966	1 983	2 972	5 723	9 122	16 126	32 892
600	222	465	875	1 797	2 693	5 186	8 265	14 612	29 803

Table B.6 a)
Maximum capacity of propane in thousands of Btu/h for copper tubing, including fittings, for pressures of 11 in w.c.
based on a pressure drop of 1 in w.c.
 (See Clauses [6.3.2](#), [B.2.3](#), [B.2.4](#), [B.2.5](#), and [B.3.5](#).)

Length of tube, ft	Outside diameter (in)						
	3/8	1/2	5/8	3/4	7/8	1-1/8	1-3/8
10	60	123	250	437	619	1323	2382
20	41	84	172	300	426	909	1637
30	33	68	138	241	342	730	1315
40	28	58	118	206	293	625	1125
50	25	51	105	183	259	554	997
60	23	47	95	166	235	502	904
70	21	43	87	152	216	462	831
80	19	40	81	142	201	430	773
90	18	37	76	133	189	403	726
100	17	35	72	126	178	381	685
125	15	31	64	111	158	337	608
150	14	28	58	101	143	306	550
175	13	26	53	93	132	281	506
200	12	24	49	86	123	262	471
250	10	22	44	77	109	232	418
300	9	20	40	69	98	210	378
350	9	18	37	64	91	193	348
400	8	17	34	59	84	180	324
450	8	16	32	56	79	169	304
500	7	15	30	53	75	159	287

(Continued)

Table B.6 a) (Continued)

Length of tube, ft	Outside diameter (in)						
	3/8	1/2	5/8	3/4	7/8	1-1/8	1-3/8
600	6	13	27	48	68	144	260
700	6	12	25	44	62	133	239
800	6	11	23	41	58	124	223
900	5	11	22	38	54	116	209
1000	5	10	21	36	51	110	197
1200	4	9	19	33	46	99	179
1400	4	8	17	30	43	91	164
1600	4	8	16	28	40	85	153
1800	4	7	15	26	37	80	144
2000	3	7	14	25	35	75	136

Table B.6 b)
Maximum capacity of propane in kW for copper tubing, including fittings, for pressures of 2.7 kPa based on a pressure drop of 250 Pa

Length of tube, m	Outside diameter (mm)						
	9.5	13	16	19	22	29	35
3	18	36	74	129	184	392	706
6	12	25	51	89	126	269	485
9	10	20	41	71	101	216	390
12	8	17	35	61	87	185	333
15	7	15	31	54	77	164	296
18	7	14	28	49	70	149	268
21	6	13	26	45	64	137	246
24	6	12	24	42	60	127	229
27	5	11	23	39	56	119	215
30	5	10	21	37	53	113	203
35	5	10	20	34	49	104	187
40	4	9	18	32	45	97	174
45	4	8	17	30	42	91	163
50	4	8	16	28	40	86	154
60	3	7	15	26	36	78	140
70	3	7	13	24	33	71	128
80	3	6	13	22	31	66	119
90	3	6	12	21	29	62	112
100	3	5	11	19	28	59	106
125	2	5	10	17	24	52	94
150	2	4	9	16	22	47	85

(Continued)

Table B.6 b) (Continued)

Length of tube, m	Outside diameter (mm)						
	9.5	13	16	19	22	29	35
175	2	4	8	14	20	43	78
200	2	4	8	13	19	40	73
250	2	3	7	12	17	36	64
300	1	3	6	11	15	32	58
350	1	3	6	10	14	30	54
400	1	3	5	9	13	28	50
500	1	2	5	8	12	25	44
600	1	2	4	7	10	22	40

Table B.7 a)
Maximum capacity of propane in thousands of Btu/h for copper tubing, including fittings, for pressures of 2 psig
based on a pressure drop of 1 psig
 (See Clauses [6.3.4](#), [B.2.3](#), [B.2.4](#), [B.2.5](#), and [B.3.5](#).)

Length of tube, ft	Outside diameter (in)						
	3/8	1/2	5/8	3/4	7/8	1-1/8	1-3/8
10	378	780	1 586	2 772	3 932	8 400	15 122
20	260	536	1 090	1 905	2 703	5 773	10 394
30	209	430	875	1 530	2 170	4 636	8 346
40	178	368	749	1 309	1 858	3 968	7 143
50	158	326	664	1 160	1 646	3 517	6 331
60	143	296	602	1 051	1 492	3 186	5 736
70	132	272	553	967	1 372	2 931	5 277
80	123	253	515	900	1 277	2 727	4 910
90	115	237	483	844	1 198	2 559	4 607
100	109	224	456	798	1 132	2 417	4 351
125	96	199	404	707	1 003	2 142	3 857
150	87	180	366	640	909	1 941	3 494
175	80	166	337	589	836	1 786	3 215
200	75	154	314	548	778	1 661	2 991
250	66	137	278	486	689	1 472	2 651
300	60	124	252	440	625	1 334	2 402
350	55	114	232	405	575	1 227	2 209
400	51	106	216	377	534	1 142	2 055
450	48	99	202	353	501	1 071	1 929
500	46	94	191	334	474	1 012	1 822

(Continued)

Table B.7 a) (Continued)

Length of tube, ft	Outside diameter (in)						
	3/8	1/2	5/8	3/4	7/8	1-1/8	1-3/8
600	41	85	173	303	429	917	1 651
700	38	78	159	278	395	843	1 519
800	35	73	148	259	367	785	1 413
900	33	68	139	243	345	736	1 325
1 000	31	65	131	229	326	695	1 252
1 200	28	58	119	208	295	630	1 134
1 400	26	54	109	191	271	580	1 044
1 600	24	50	102	178	252	539	971
1 800	23	47	96	167	237	506	911
2 000	22	44	90	158	224	478	861

Table B.7 b)
Maximum capacity of propane in kW for copper tubing, including fittings, for pressures of 14 kPa based on a pressure drop of 7 kPa

Length of tube, m	Outside diameter (mm)						
	9.5	13	16	19	22	29	35
3	113	233	473	827	1173	2506	4512
6	77	160	325	568	806	1723	3101
9	62	128	261	456	648	1383	2490
12	53	110	224	391	554	1184	2131
15	47	97	198	346	491	1049	1889
18	43	88	180	314	445	951	1712
21	39	81	165	289	409	875	1575
24	37	76	154	269	381	814	1465
27	34	71	144	252	357	763	1374
30	32	67	136	238	338	721	1298
35	30	62	125	219	311	663	1194
40	28	57	117	204	289	617	1111
45	26	54	109	191	271	579	1043
50	25	51	103	181	256	547	985
60	22	46	94	164	232	496	892
70	21	42	86	150	213	456	821
80	19	39	80	140	199	424	764
90	18	37	75	131	186	398	717
100	17	35	71	124	176	376	677
125	15	31	63	110	156	333	600
150	14	28	57	100	141	302	544

(Continued)

Table B.7 b) (Continued)

Length of tube, m	Outside diameter (mm)						
	9.5	13	16	19	22	29	35
175	12	26	52	92	130	278	500
200	12	24	49	85	121	258	465
250	10	21	43	76	107	229	412
300	9	19	39	68	97	208	374
350	9	18	36	63	89	191	344
400	8	16	34	59	83	178	320
500	7	15	30	52	74	157	283
600	6	13	27	47	67	143	257

Table B.8 a)
Maximum capacity of propane in thousands of Btu/h for copper tubing for pressures of 5 psig based on a pressure drop of 2.5 psig
 (See Clauses [B.2.3](#), [B.2.4](#), [B.2.5](#), and [B.3.5](#).)

Length of tube, ft	Outside diameter (in)						
	3/8	1/2	5/8	3/4	7/8	1-1/8	1-3/8
10	735	1 516	3 083	5 388	7 644	16 329	29 397
20	505	1 042	2 119	3 703	5 254	11 223	20 205
30	405	836	1 702	2 974	4 219	9 013	16 225
40	347	716	1 456	2 545	3 611	7 714	13 887
50	308	634	1 291	2 256	3 200	6 836	12 307
60	279	575	1 170	2 044	2 900	6 194	11 151
70	256	529	1 076	1 880	2 668	5 699	10 259
80	238	492	1 001	1 749	2 482	5 301	9 544
90	224	462	939	1 641	2 329	4 974	8 955
100	211	436	887	1 550	2 200	4 699	8 459
125	187	386	786	1 374	1 949	4 164	7 497
150	170	350	712	1 245	1 766	3 773	6 793
175	156	322	655	1 145	1 625	3 471	6 249
200	145	300	610	1 066	1 512	3 229	5 814
250	129	266	540	944	1 340	2 862	5 153
300	117	241	490	856	1 214	2 593	4 669
350	107	221	450	787	1 117	2 386	4 295
400	100	206	419	732	1 039	2 220	3 996
450	94	193	393	687	975	2 082	3 749
500	88	183	371	649	921	1 967	3 541

(Continued)

Table B.8 a) (Continued)

Length of tube, ft	Outside diameter (in)						
	3/8	1/2	5/8	3/4	7/8	1-1/8	1-3/8
600	80	165	337	588	834	1 782	3 209
700	74	152	310	541	768	1 640	2 952
800	69	142	288	503	714	1 525	2 746
900	64	133	270	472	670	1 431	2 577
1 000	61	125	255	446	633	1 352	2 434
1 200	55	114	231	404	573	1 225	2 205
1 400	51	105	213	372	528	1 127	2 029
1 600	47	97	198	346	491	1 048	1 887
1 800	44	91	186	325	461	984	1 771
2 000	42	86	175	307	435	929	1 673

Table B.8 b)
Maximum capacity of propane in kW for copper tubing for pressures of 34 kPa based on a pressure drop of 17 kPa

Length of tube, m	Outside diameter (mm)						
	9.5	13	16	19	22	29	35
3	215	444	904	1579	2240	4786	8615
6	148	305	621	1085	1540	3289	5921
9	119	245	499	872	1236	2641	4755
12	102	210	427	746	1058	2261	4070
15	90	186	378	661	938	2004	3607
18	82	169	343	599	850	1815	3268
21	75	155	315	551	782	1670	3007
24	70	144	293	513	727	1554	2797
27	66	135	275	481	682	1458	2624
30	62	128	260	454	645	1377	2479
35	57	118	239	418	593	1267	2281
40	53	109	223	389	552	1179	2122
45	50	103	209	365	518	1106	1991
50	47	97	197	345	489	1045	1880
60	43	88	179	312	443	946	1704
70	39	81	164	287	408	871	1567
80	36	75	153	267	379	810	1458
90	34	71	143	251	356	760	1368
100	32	67	136	237	336	718	1292
125	29	59	120	210	298	636	1145
150	26	54	109	190	270	576	1038

(Continued)

Table B.8 b) (Continued)

Length of tube, m	Outside diameter (mm)						
	9.5	13	16	19	22	29	35
175	24	49	100	175	248	530	955
200	22	46	93	163	231	493	888
250	20	41	83	144	205	437	787
300	18	37	75	131	185	396	713
350	16	34	69	120	171	365	656
400	15	31	64	112	159	339	611
500	14	28	57	99	141	301	541
600	12	25	51	90	127	272	490

Table B.9 a)
Maximum capacity of propane in thousands of Btu/h for copper tubing for pressures of 10 psig based on a pressure drop of 5 psig
 (See Clauses [B.2.3](#), [B.2.4](#), [B.2.5](#), and [B.3.5](#).)

Length of tube, ft	Outside diameter (in)						
	3/8	1/2	5/8	3/4	7/8	1-1/8	1-3/8
10	1 181	2 437	4 958	8 665	12 294	26 261	47 277
20	812	1 675	3 408	5 956	8 449	18 049	32 493
30	652	1 345	2 737	4 783	6 785	14 494	26 093
40	558	1 151	2 342	4 093	5 807	12 405	22 332
50	495	1 020	2 076	3 628	5 147	10 994	19 793
60	448	925	1 881	3 287	4 663	9 962	17 934
70	412	851	1 730	3 024	4 290	9 165	16 499
80	384	791	1 610	2 813	3 991	8 526	15 349
90	360	742	1 510	2 640	3 745	8 000	14 401
100	340	701	1 427	2 493	3 537	7 556	13 603
125	301	622	1 264	2 210	3 135	6 697	12 056
150	273	563	1 146	2 002	2 841	6 068	10 924
175	251	518	1 054	1 842	2 613	5 582	10 050
200	234	482	981	1 714	2 431	5 193	9 350
250	207	427	869	1 519	2 155	4 603	8 286
300	188	387	787	1 376	1 952	4 170	7 508
350	173	356	724	1 266	1 796	3 837	6 907
400	161	331	674	1 178	1 671	3 569	6 426
450	151	311	632	1 105	1 568	3 349	6 029
500	142	294	597	1 044	1 481	3 163	5 695

(Continued)

Table B.9 a) (Continued)

Length of tube, ft	Outside diameter (in)						
	3/8	1/2	5/8	3/4	7/8	1-1/8	1-3/8
600	129	266	541	946	1 342	2 866	5 160
700	119	245	498	870	1 234	2 637	4 747
800	110	228	463	810	1 148	2 453	4 416
900	104	214	435	760	1 078	2 302	4 144
1 000	98	202	411	717	1 018	2 174	3 914
1 200	89	183	372	650	922	1 970	3 547
1 400	82	168	342	598	848	1 812	3 263
1 600	76	156	318	556	789	1 686	3 035
1 800	71	147	299	522	741	1 582	2 848
2 000	67	139	282	493	700	1 494	2 690

Table B.9 b)
Maximum capacity of propane in kW for copper tubing for pressures of 70 kPa based on a pressure drop of 35 kPa

Length of tube, m	Outside diameter (mm)						
	9.5	13	16	19	22	29	35
3	353	729	1 482	2 591	3 676	7 852	14 135
6	243	501	1 019	1 781	2 526	5 396	9 715
9	195	402	818	1 430	2 029	4 334	7 802
12	167	344	700	1 224	1 736	3 709	6 677
15	148	305	621	1 085	1 539	3 287	5 918
18	134	276	562	983	1 394	2 978	5 362
21	123	254	517	904	1 283	2 740	4 933
24	115	237	481	841	1 193	2 549	4 589
27	108	222	452	789	1 120	2 392	4 306
30	102	210	427	745	1 058	2 259	4 067
35	93	193	392	686	973	2 078	3 742
40	87	179	365	638	905	1 934	3 481
45	82	168	343	599	849	1 814	3 266
50	77	159	324	565	802	1 714	3 085
60	70	144	293	512	727	1 553	2 795
70	64	133	270	471	669	1 429	2 572
80	60	123	251	439	622	1 329	2 393
90	56	116	235	411	584	1 247	2 245
100	53	109	222	389	551	1 178	2 120
125	47	97	197	344	489	1 044	1 879
150	43	88	179	312	443	946	1 703

(Continued)

Table B.9 b) (Continued)

Length of tube, m	Outside diameter (mm)						
	9.5	13	16	19	22	29	35
175	39	81	164	287	407	870	1 567
200	36	75	153	267	379	810	1 457
250	32	67	135	237	336	717	1 292
300	29	60	123	215	304	650	1 170
350	27	56	113	197	280	598	1 077
400	25	52	105	184	260	556	1 002
500	22	46	93	163	231	493	888
600	20	41	84	147	209	447	804

Table B.10 a)
Maximum capacity of propane in thousands of Btu/h for copper tubing for pressures of 20 psig based on a pressure drop of 10 psig
 (See Clauses [B.2.3](#), [B.2.4](#), [B.2.5](#), and [B.3.5](#).)

Length of tube, ft	Outside diameter (in)						
	3/8	1/2	5/8	3/4	7/8	1-1/8	1-3/8
10	2 012	4 151	8 445	14 759	20 938	44 726	80 520
20	1 383	2 853	5 804	10 143	14 391	30 740	55 341
30	1 110	2 291	4 661	8 146	11 556	24 685	44 441
40	950	1 961	3 989	6 972	9 891	21 128	38 035
50	842	1 738	3 535	6 179	8 766	18 725	33 710
60	763	1 575	3 203	5 598	7 943	16 966	30 544
70	702	1 449	2 947	5 150	7 307	15 609	28 100
80	653	1 348	2 742	4 792	6 798	14 521	26 142
90	613	1 264	2 572	4 496	6 378	13 624	24 528
100	579	1 194	2 430	4 247	6 025	12 870	23 169
125	513	1 059	2 154	3 764	5 340	11 406	20 534
150	465	959	1 951	3 410	4 838	10 335	18 605
175	428	882	1 795	3 137	4 451	9 508	17 117
200	398	821	1 670	2 919	4 141	8 845	15 924
250	353	728	1 480	2 587	3 670	7 839	14 113
300	319	659	1 341	2 344	3 325	7 103	12 787
350	294	606	1 234	2 156	3 059	6 535	11 764
400	273	564	1 148	2 006	2 846	6 079	10 944
450	257	529	1 077	1 882	2 670	5 704	10 269
500	242	500	1 017	1 778	2 522	5 388	9 700

(Continued)

Table B.10 a) (Continued)

Length of tube, ft	Outside diameter (in)						
	3/8	1/2	5/8	3/4	7/8	1-1/8	1-3/8
600	220	453	922	1 611	2 285	4 882	8 789
700	202	417	848	1 482	2 103	4 491	8 085
800	188	388	789	1 379	1 956	4 178	7 522
900	176	364	740	1 294	1 835	3 920	7 058
1 000	167	344	699	1 222	1 734	3 703	6 667
1 200	151	311	633	1 107	1 571	3 355	6 040
1 400	139	286	583	1 019	1 445	3 087	5 557
1 600	129	267	542	948	1 344	2 872	5 170
1 800	121	250	509	889	1 261	2 694	4 851
2 000	114	236	481	840	1 191	2 545	4 582

Table B.10 b)

Maximum capacity of propane in kW for copper tubing for pressures of 140 kPa based on a pressure drop of 70 kPa

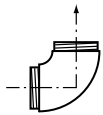
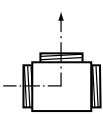
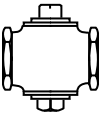
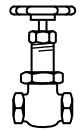
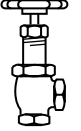
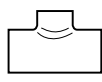
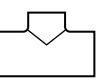
Length of tube, m	Outside diameter (mm)						
	9.5	13	16	19	22	29	35
3	602	1 243	2 528	4 419	6 269	13 391	24 108
6	414	854	1 738	3 037	4 309	9 204	16 569
9	332	686	1 395	2 439	3 460	7 391	13 306
12	285	587	1 194	2 087	2 961	6 326	11 388
15	252	520	1 059	1 850	2 625	5 606	10 093
18	228	471	959	1 676	2 378	5 080	9 145
21	210	434	882	1 542	2 188	4 673	8 413
24	196	404	821	1 435	2 035	4 348	7 827
27	183	379	770	1 346	1 910	4 079	7 344
30	173	358	728	1 271	1 804	3 853	6 937
35	159	329	669	1 170	1 660	3 545	6 382
40	148	306	623	1 088	1 544	3 298	5 937
45	139	287	584	1 021	1 449	3 094	5 571
50	131	271	552	964	1 368	2 923	5 262
60	119	246	500	874	1 240	2 648	4 768
70	110	226	460	804	1 141	2 436	4 386
80	102	210	428	748	1 061	2 267	4 080
90	96	197	402	702	996	2 127	3 829
100	90	186	379	663	940	2 009	3 616
125	80	165	336	587	833	1 780	3 205
150	73	150	305	532	755	1 613	2 904

(Continued)

Table B.10 b) (Continued)

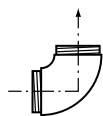
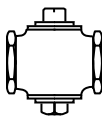
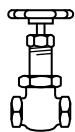
Length of tube, m	Outside diameter (mm)						
	9.5	13	16	19	22	29	35
175	67	138	280	490	695	1 484	2 672
200	62	128	261	456	646	1 381	2 486
250	55	114	231	404	573	1 224	2 203
300	50	103	209	366	519	1 109	1 996
350	46	95	193	337	478	1 020	1 836
400	43	88	179	313	444	949	1 708
500	38	78	159	278	394	841	1 514
600	34	71	144	251	357	762	1 372

Table B.11 a)
Resistance of bends, fittings, and valves for natural gas and propane expressed in equivalent length of straight pipe in feet*
 (See Clauses [6.3.7](#), [B.2.4](#), and [B.4](#).)

		Threaded fittings†		Valves (threaded, flanged, or welded)					90° welding elbows and smooth bends‡	Welding tees	
		Elbows									
		45°	90°	Tee	Plug	Globe	Angle	Swing check	R/d§ = 1-1/2	Forged	Mitre**
k factor		0.42	0.9	1.8	0.9	10	5	2.5	0.36	1.35	1.8
n (L/D ratio††)		14	30	60	30	333	167	83	12	45	60
Nominal pipe size, in (Schedule 40)	Inside diameter (d), in										
3/8	0.493	0.58	1.23	2.46	1.23	13.7	6.85	3.42	0.49	1.85	2.46
1/2	0.622	0.73	1.55	3.10	1.55	17.3	8.65	4.32	0.62	2.33	3.10
3/4	0.824	0.96	2.06	4.12	2.06	22.9	11.4	5.72	0.82	3.09	4.12
1	1.049	1.22	2.62	5.24	2.62	29.1	14.6	7.27	1.05	3.93	5.24
1-1/4	1.380	1.61	3.45	6.90	3.45	38.3	19.1	9.58	1.38	5.17	6.90
1-1/2	1.610	1.88	4.02	8.04	4.02	44.7	22.4	11.2	1.61	6.04	8.04
2	2.067	2.41	5.17	10.3	5.17	57.4	28.7	14.4	2.07	7.75	10.3
2-1/2	2.469	2.88	6.16	12.3	6.16	68.5	34.3	17.1	2.47	9.25	12.3

(Continued)

Table B.11 a) (Continued)

										90° welding elbows and smooth bends‡	
		Threaded fittings†		Valves (threaded, flanged, or welded)					Welding tees		
		Elbows									
		45°	90°	Tee	Plug	Globe	Angle	Swing check	R/d§ = 1-1/2	Forged	Mitre**
k factor		0.42	0.9	1.8	0.9	10	5	2.5	0.36	1.35	1.8
n (L/D ratio††)		14	30	60	30	333	167	83	12	45	60
Nominal pipe size, in (Schedule 40)	Inside diameter (d), in										
3	3.068	3.58	7.67	15.3	7.67	85.2	42.6	21.3	3.07	11.5	15.3
4	4.026	4.70	10.1	20.2	10.1	112	56.0	28.0	4.03	15.1	20.2
5	5.047	5.88	12.6	25.2	12.6	140	70.0	35.0	5.05	18.9	25.2

* Values for welded fittings are for conditions where the bore is not obstructed by weld spatter or backing rings. If appreciably obstructed, values for threaded fittings shall be used.

† Flanged fittings have three-fourths the resistance of threaded fittings.

‡ Tabular figures give the extra resistance due to curvature alone, to which should be added the actual length of the bend.

§ R/d is the ratio of the elbow or bend radius to the inside diameter of the pipe.

** Small socket-welding fittings are equivalent to mitre elbows and mitre tees.

†† n = resistance in equivalent number of diameters of straight pipe, computed from the relation $n = k / 4f$, where the friction factor, f , is assumed to be 0.0075;

$n = L / D$, where L = equivalent length of Schedule 40 straight pipe in feet and D = inside diameter of pipe in feet. For pipe having other inside diameters, the resistance expressed in equivalent length in feet can be computed from the above n values.

Note: The equivalent lengths in feet shown in the table have been computed on the basis that the inside diameter corresponds to that of Schedule 40 (standard-weight) steel pipe, which is close enough for most purposes involving other schedules of pipe. Where a more specific solution for equivalent length is desired, this may be made by multiplying the actual inside diameter of the pipe in feet by $n / 12$, or the actual diameter in feet by n , read from the table heading. The

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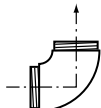
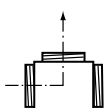
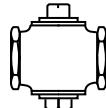
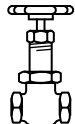
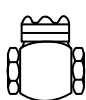
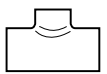
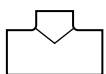
Table B.11 a) (Continued)

equivalent length values can be used with reasonable accuracy for copper or brass fittings and bends. For copper or brass values, however, the equivalent length of pipe should be taken as 45% longer than the values in the table, which are for steel pipe. Resistance per foot of copper or brass pipe is less than that of steel.

Example of calculation: To obtain the total equivalent length, add to the actual length of pipe involved the length in feet to allow for various fittings as shown in this table. Thus, if the problem involves 300 ft of NPS 4 pipe having three standard 90° elbows and two plug valves, the total equivalent length will be as follows:

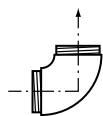
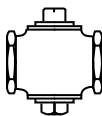
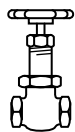
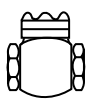
$$300 + (3 \times 10.1) + (2 \times 10.1) = 350.5 \quad \text{ft}$$

Table B.11 b)
Resistance of bends, fittings, and valves for natural gas and propane expressed in equivalent length of straight pipe in metres*

		Threaded fittings†		Valves (threaded, flanged, or welded)					90° welding elbows and smooth bends‡	Welding tees	
		Elbows									
		45°	90°	Tee	Plug	Globe	Angle	Swing check	R/d§ = 1-1/2	Forged	Mitre**
k factor		0.42	0.9	1.8	0.9	10	5	2.5	0.36	1.35	1.8
n (L/D ratio††)		14	30	60	30	333	167	83	12	45	60
Nominal pipe size, in (Schedule 40)	Inside diameter (d), mm										
3/8	12.52	0.18	0.37	0.75	0.37	4.18	2.09	1.04	0.15	0.56	0.75
1/2	15.80	0.22	0.47	0.94	0.47	5.27	2.64	1.29	0.19	0.71	0.94
3/4	20.93	0.29	0.63	1.26	0.63	6.98	3.47	1.74	0.25	0.94	1.26
1	26.64	0.37	0.80	1.60	0.80	8.87	4.45	2.22	0.32	1.20	1.60
1-1/4	35.05	0.49	1.05	2.10	1.05	11.67	5.82	2.92	0.42	1.58	2.10
1-1/2	40.89	0.49	1.23	2.45	1.23	13.62	6.83	3.41	0.49	1.84	2.45
2	52.50	0.73	1.58	3.14	1.58	17.50	8.75	4.39	0.63	2.36	3.14
2-1/2	62.71	0.88	1.88	3.75	1.88	20.88	10.45	5.21	0.75	2.82	3.75
3	77.93	1.09	2.34	4.66	2.34	25.97	12.98	6.49	0.94	3.51	4.66

(Continued)

Table B.11 b) (Continued)

										90° welding elbows and smooth bends‡		
Threaded fittings†			Valves (threaded, flanged, or welded)					Welding tees				
Elbows												
		45°	90°	Tee	Plug	Globe	Angle	Swing check	R/d§ = 1-1/2	Forged	Mitre**	
k factor		0.42	0.9	1.8	0.9	10	5	2.5	0.36	1.35	1.8	
n (L/D ratio††)		14	30	60	30	333	167	83	12	45	60	
Nominal pipe size, in (Schedule 40)		Inside diameter (d), mm										
4	102.3	1.23	3.08	6.16	3.08	34.14	17.07	8.53	1.23	4.60	6.16	
5	128.2	1.79	3.84	7.68	3.84	42.67	21.33	10.67	1.54	5.76	7.68	

* Values for welded fittings are for conditions where the bore is not obstructed by weld spatter or backing rings. If appreciably obstructed, values for threaded fittings shall be used.

† Flanged fittings have three-fourths the resistance of threaded fittings.

‡ Tabular figures give the extra resistance due to curvature alone, to which should be added the actual length of the bend.

§ R/d is the ratio of the elbow or bend radius to the inside diameter of the pipe.

** Small socket-welding fittings are equivalent to mitre elbows and mitre tees.

†† n = resistance in equivalent number of diameters of straight pipe, computed from the relation $n = k / 4f$, where the friction factor, f , is assumed to be 0.0075;

$n = L / D$, where L = equivalent length of Schedule 40 straight pipe in metres and D = inside diameter of pipe in metres. For pipe having other inside diameters, the resistance expressed in equivalent length in metres can be computed from the above n values.

Note: The equivalent lengths in metres shown in the table have been computed on the basis that the inside diameter corresponds to that of Schedule 40 (standard-weight) steel pipe, which is close enough for most purposes involving other schedules of pipe. Where a more specific solution for equivalent length is desired, this may be made by multiplying the actual inside diameter of the pipe in metres by $n / 12$, or the actual diameter in metres by n , read from the table heading. The equivalent length values can be used with reasonable accuracy for copper or brass fittings and bends. For copper or brass values, however, the

(Continued)

Table B.11 b) (Continued)

equivalent length of pipe should be taken as 45% longer than the values in the table, which are for steel pipe. Resistance per metre of copper or brass pipe is less than that of steel.

Example of calculation: To obtain the total equivalent length, add to the actual length of pipe involved the length in metres to allow for various fittings as shown in this table. Thus, if the problem involves 100 m of NPS 4 pipe having three standard 90° elbows and two plug valves, the total equivalent length will be as follows:

$$100 + (3 \times 3.08) + (2 \times 3.08) = 115.4 \quad m$$

Table B.12
Use of capacity tables for sizing plastic pipe for natural gas and propane
 (See Clause [6.3.5.](#))

Nominal pipe size	Pipe inside diameters, in									
	Schedule 40 steel pipe		SDR 21 plastic pipe		SDR 13.5 plastic pipe		SDR 11 plastic pipe		SDR 8.8 plastic pipe	
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
1/2	0.64	0.67	0.68	0.72	0.62*	0.66*	0.61*	0.66*	0.6*	0.65*
3/4	0.83	0.87	0.83	0.88	0.83	0.88	0.81*	0.87*	0.76*	0.82*
1	1.07	1.1	1.09	1.14	1.08	1.13	1.02*	1.08*	0.95*	1.02*
1-1/4	1.4	1.43	1.44	1.49	1.37*	1.42*	1.3*	1.37*	1.21*	1.29*
1-1/2	1.63	1.66	1.67	1.73	1.57*	1.62*	1.48*	1.55*	1.4*	1.48*
2	2.1	2.14	2.1	2.15	1.98*	2.03*	1.89*	1.95*	1.77*	1.84*
2-1/2	2.5	2.55	2.55	2.61	2.39*	2.46*	2.29*	2.36*	2.14*	2.23*

* Denotes inside diameter of plastic pipe that may be less than the inside diameter of Schedule 40 steel pipe. The tables in this annex should be used only for sizing plastic pipe with inside diameters equal to or greater than steel pipe. Sometimes plastic pipe with inside diameters less than the inside diameter of Schedule 40 steel pipe sized using the tables in this annex does not supply gas at a sufficient volume and pressure. Plastic pipe with inside diameters less than the inside diameter of Schedule 40 steel pipe should be sized using the calculation method detailed in this annex.

Notes:

- 1) Dimensions of Schedule 40 steel pipe are derived from ASTM A53/A53M and ASME B36.10.
- 2) Dimensions of plastic pipe are derived from CSA B137.4.

Annex C (informative)

Vent sizing tables for Category I natural gas and propane appliances

Note: This informative annex has been written in mandatory language to facilitate adoption where users of the Code or regulatory authorities wish to adopt it formally as additional requirements to this Code.

C.1 General

C.1.1 Table applicability

These tables were developed to provide a convenient method of determining vent sizes for venting natural-draft and fan-assisted Category I appliances. Additionally, the tables provide the vent sizes that apply to spillage-susceptible appliances such as natural-draft appliances in either tight structures (as defined in Clause 8.2.1) or structures where the mechanical ventilation is designed for no more than 5 Pa (0.02 in w.c.) depressurization.

C.1.2 Depressurization

The values applicable to spillage-susceptible appliances are listed in the DP (depressurization) column and are based upon building depressurization of 0.02 in w.c. (5 Pa). The DP column shall not be used for mechanically ventilated systems that allow more than 0.02 in w.c. (5 Pa) depressurization, nor shall spillage-susceptible appliances be installed in this type of environment.

C.1.3 Depressurization restrictions

Currently, this Code does not provide vent sizing for Category I fan-assisted appliances and for spillage-susceptible appliances under depressurization. These tables address this deficiency in the Code. Values under the DP column apply only to installations containing draft-hood-equipped appliances or combinations of Category I fan-assisted appliances and draft-hood-equipped appliances. These tables also provide vent sizes for several vented appliances that share a common vent, and vent sizes for tile-lined masonry chimneys.

Notes:

- 1) *The Installation Code Committee gratefully acknowledges the research conducted by the Canadian Gas Research Institute (CGRI) and the independent research sponsored by the Gas Research Institute to generate the tables in this annex. The research was conducted using a computer program, VENT-II, developed by Batelle. Recognition is given to both research organizations for allowing CSA Group to publish the tables in this annex.*
- 2) *For vent-sizing requirements not covered by the tables in this annex, please refer to the applicable installation codes.*
- 3) *Please refer to Clause 3 of this Code for definitions of terms used in this annex.*

C.2 General venting requirements (GVRs)

C.2.1 Vent category

All requirements contained in this annex apply to both Category I draft-hood-equipped and Category I fan-assisted combustion appliances. At no time shall a venting system for a certified Category II, III, or IV appliance be sized with these tables. The alternative sizing methods described in this Code may also be used to size the venting system for a draft-hood-equipped appliance. At this time, alternative sizing methods have not been developed for fan-assisted appliances; therefore, until engineering data is

developed to allow alternative sizing methods for Category I fan-assisted appliances, these vent tables shall be used.

C.2.2 Vent exposure

The vent tables included in this annex apply to vents and chimneys internal to the structure below the roof line. Exterior chimneys or vents not enclosed by the structure or a chase below the roof line can experience continuous condensation, depending on locality. A chimney with one or more sides exposed to the outside of the structure shall be considered to be an exterior chimney. A Type B vent or a certified chimney lining system passing through an unused masonry chimney flue shall not be considered to be exposed to the outdoors. The DP column shall be used to determine the capacity of a venting system within a building constructed in accordance with Clause [8.2.1](#).

C.2.3 Limitations

If the vent or vent connector size, determined from the tables, is smaller than the appliance draft-hood outlet or flue collar, the smaller size may be used, provided that

- the total vent height (H) is at least 10 ft (3 m);
- vents or vent connectors for appliance draft-hood outlets or flue collars 12 in (305 mm) in diameter or smaller are not reduced more than one table size [e.g., 12 in (305 mm) to 10 in (254 mm) is a one-size reduction];
- vents or vent connectors for appliance draft-hood outlets or flue collars above 12 in (305 mm) in diameter are not reduced more than two table sizes [e.g., 24 in (610 mm) to 20 in (508 mm) is a two-size reduction];
- the maximum capacity listed in the tables for a fan-assisted appliance is reduced by 10% (calculated as $0.90 \times$ maximum capacity); and
- the draft-hood outlet is greater than 4 in (102 mm) in diameter. For example, a 3 in (76 mm) diameter vent or vent connector shall not be connected to a 4 in (102 mm) diameter draft-hood outlet. This provision shall not apply to fan-assisted appliances.

C.2.4 Bend allowance

A single-appliance venting configuration with zero lateral lengths shall be assumed to have no elbows in the vent system in Tables [C.1](#) and [C.2](#). For all other vent configurations of single or multiple appliances, each vent connector shall be assumed to have two 90° turns. For each additional 90° turn or its equivalent*, the maximum capacity of each individual vent connector listed in the venting table should be reduced by 10% (calculated as $0.90 \times$ maximum listed capacity) for a natural draft appliance and 15% for a fan-assisted appliance. Except for manifolded common vents (see Figure [C.9](#)), the final turn into the vertical vent shall be counted as one 90° turn for each vent connector.

Note: *Two 45° turns are equivalent to one 90° turn.

C.2.5 Vent connector restrictions

If vent connectors are combined prior to entering the common vent, the maximum common vent capacity listed in the common venting tables shall be reduced by 10%, the equivalent of one 90° elbow ($0.90 \times$ maximum common vent capacity). See Figure [C.9](#). The horizontal length of the common vent connector manifold (L in Figure [C.9](#)) should not exceed 1-1/2 ft (457 mm) for each inch (25.4 mm) of common vent connector manifold diameter.

C.2.6 Vertical offset

If the common vertical vent is offset as shown in Figure [C.10](#), the maximum common vent capacity listed in the common venting tables should be reduced by 20%, the equivalent of two 90° elbows

($0.80 \times$ maximum common vent capacity). The horizontal length of the offset shall not exceed 1-1/2 ft (457 mm) for each inch (25.4 mm) of common vent diameter.

C.2.7 High altitude

Sea-level input rating shall be used when determining maximum capacity for high-altitude installation. Use actual input rating for determining minimum capacity for high-altitude installation.

C.2.8 Modulating inputs

For appliances with modulating input rates, the minimum vent or vent connector (FAN Min) capacity (determined from the tables) shall be less than the lowest appliance input rating, and the maximum vent or vent connector (FAN or NAT Max) capacity (determined from the tables) shall be greater than the highest appliance input rating.

C.2.9 Common venting

A fan-assisted appliance may be common-vented into an existing masonry chimney, provided that

- a) the chimney is currently serving at least one draft-hood-equipped appliance; and
- b) the vent connectors and chimney are sized in accordance with Tables [C.7](#) and [C.8](#).

C.2.10 Liner requirements

Single-appliance venting of a fan-assisted appliance into a tile-lined masonry chimney shall be prohibited. The chimney shall first be lined with either a Type B vent, sized in accordance with Table [C.1](#) or [C.2](#), or a certified lining system.

C.2.11 Liner capacity

Certified, corrugated metallic chimney liner systems in masonry chimneys shall be sized according to Table [C.1](#) or [C.2](#) for dedicated venting and Table [C.3](#) or [C.4](#) for common venting, with the maximum capacity reduced by 20% ($0.80 \times$ maximum capacity) and the minimum capacity as shown in the applicable table. Corrugated metal vent systems installed with bends or offsets require additional reduction of the vent maximum capacity (see Clause [C.2.6](#)).

C.2.12 Total height single storey

For multiple units of gas-utilization equipment all located on one floor, available total height shall be measured from the highest draft-hood outlet or flue collar up to the level of the cap or terminal. Vent connector rise shall be measured from the draft-hood outlet or flue collar to the level where the vent gas streams come together. (This is not applicable to multi-storey installations.)

C.2.13 Total height multiple storey

For multi-storey installations, available total height for each segment of the system shall be the vertical distance between the highest draft-hood outlet or flue collar entering that segment and the centreline of the next higher interconnection tee. (See Figure [C.11](#).)

C.2.14 Total height lowest storey

The size of the lowest vent connector and of the vertical vent leading to the lowest interconnection of a multi-storey system shall be in accordance with Table [C.1](#) or [C.2](#) for available total height up to the lowest interconnection. (See Figure [C.12](#).)

C.2.15 Interpolation and extrapolation

If the desired vent height and vent connector rise and/or lateral are between the table entries, linear interpolation may be used for calculation of the permissible appliance input ratings. Extrapolation beyond the table entries shall not be acceptable.

C.2.16 Preference

When the vent table permits more than one diameter of pipe to be used for a vent connector or vent, the smallest permitted diameter shall be preferred.

Table C.1
Capacity of Type B double-wall vents with Type B double-wall vent connectors serving a single Category I appliance
 (See Clauses [8.13.1](#), [C.2.4](#), [C.2.10](#), [C.2.11](#), and [C.2.14](#), Table [C.10](#), and Figure [C.1](#).)

Vent and vent connector diameter, D, in																					
3				4				5				6				7					
Appliance input rating limits, thousands of Btu/h																					
Height, H, ft	Lateral, L, ft	FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*	
		Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†
6	0	0	78	46	29	0	152	86	53	0	251	141	87	0	375	205	127	0	524	285	177
	2	13	51	36	22	18	97	67	41	27	157	105	64	32	232	157	96	44	321	217	132
	4	21	49	34	20	30	94	64	38	39	153	103	61	50	227	153	90	66	316	211	124
	6	25	46	32	19	36	91	61	36	47	149	100	59	59	223	149	88	78	310	205	121
8	0	0	84	50	37	0	165	94	69	0	276	155	113	0	415	235	172	0	583	320	234
	2	12	57	40	29	16	109	75	55	25	178	120	88	28	263	180	131	42	365	247	180
	5	23	53	38	27	32	103	71	50	42	171	115	82	53	255	173	123	70	356	237	168
	8	28	49	35	25	39	98	66	47	51	164	109	77	64	247	165	117	84	347	227	161
10	0	0	88	53	42	0	175	100	79	0	295	166	131	0	447	255	201	0	631	345	273
	2	12	61	42	34	17	118	81	65	23	194	129	103	26	289	195	156	40	402	273	218
	5	23	57	40	32	32	113	77	62	41	187	124	99	52	280	188	150	68	392	263	210
	10	30	51	36	28	41	104	70	55	54	176	115	90	67	267	175	137	88	376	245	191
15	0	0	94	58	50	0	191	112	96	0	327	187	161	0	502	285	245	0	716	390	335
	2	11	69	48	41	15	136	93	80	20	226	150	129	22	339	225	194	38	475	316	272
	5	22	65	45	38	30	130	87	74	39	219	142	121	49	330	217	184	64	463	300	255
	10	29	59	41	35	40	121	82	70	51	206	135	115	64	315	208	177	84	445	288	245
	15	35	53	37	31	48	112	76	65	61	195	128	109	76	301	198	168	98	429	275	234
20	0	0	97	61	55	0	202	119	107	0	349	202	182	0	540	307	276	0	776	430	387
	2	10	75	51	46	14	149	100	90	18	250	166	149	20	377	249	224	33	531	346	311

(Continued)

Table C.1 (Continued)

Vent and vent connector diameter, D, in																					
3				4				5				6				7					
Appliance input rating limits, thousands of Btu/h																					
Height, H, ft	Lateral, L, ft	FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*	
		Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†
	5	21	71	48	43	29	143	96	85	38	242	160	142	47	367	241	214	62	519	337	300
	10	28	64	44	39	38	133	89	79	50	229	150	134	62	351	228	203	81	499	321	286
	15	34	58	40	36	46	124	84	75	59	217	142	126	73	337	217	193	94	481	308	274
	20	48	52	35	31	55	116	78	69	69	206	134	119	84	322	206	183	107	464	295	263
25	0	0	99	63	58	0	208	124	113	0	362	211	194	0	564	322	294	0	815	453	415
	2	10	78	54	49	14	158	106	97	16	267	176	161	19	405	265	242	30	572	370	339
	5	21	74	51	47	29	152	102	93	37	259	168	153	46	394	257	234	60	560	361	329
	10	28	67	47	43	38	142	96	87	49	246	161	147	61	378	245	223	79	540	346	316
	15	34	61	NR	NR	45	133	90	82	58	233	153	139	72	363	233	213	92	521	333	303
	20	52	55	NR	NR	54	124	84	77	68	222	144	131	82	348	222	202	105	503	319	291
30	0	0	100	64	60	0	213	128	119	0	374	220	205	0	587	336	312	0	853	475	442
	2	9	81	56	52	13	166	112	104	14	283	185	172	18	432	280	260	27	613	394	366
	5	21	77	54	50	28	160	108	100	36	275	176	164	45	421	273	254	58	600	385	358
	10	27	70	50	47	37	150	102	95	48	262	171	159	59	405	261	243	77	580	371	345
	15	33	64	NR‡	NR‡	44	141	96	89	57	249	163	152	70	389	249	232	90	560	357	332
	20	56	58	NR‡	NR‡	53	132	90	84	66	237	154	143	80	374	237	220	102	542	343	319
	30	NR‡	NR‡	NR‡	NR‡	73	113	NR‡	NR‡	88	214	NR‡	NR‡	104	346	219	204	131	507	321	299
50	0	0	101	67	64	0	216	134	127	0	397	232	220	0	633	363	345	0	932	518	492
	2	8	86	61	58	11	183	122	116	14	320	206	196	15	497	314	298	22	715	445	423
	5	20	82	NR‡	NR‡	27	177	119	113	35	312	200	190	43	487	308	293	55	702	438	416
	10	26	76	NR‡	NR‡	35	168	114	108	45	299	190	181	56	471	298	283	73	681	426	405

(Continued)

Table C.1 (Continued)

Vent and vent connector diameter, D, in																					
3				4				5				6				7					
Appliance input rating limits, thousands of Btu/h																					
Height, H, ft	Lateral, L, ft	FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*	
		Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†
100	15	59	70	NR‡	NR‡	42	158	NR‡	NR‡	54	287	180	171	66	455	288	274	85	662	413	392
	20	NR‡	NR‡	NR‡	NR‡	50	149	NR‡	NR‡	63	275	169	161	76	440	278	264	97	642	401	381
	30	NR‡	NR‡	NR‡	NR‡	69	131	NR‡	NR‡	84	250	NR‡	NR‡	99	410	259	246	123	605	376	357
	0	NR‡	NR‡	NR‡	NR‡	0	218	NR‡	NR‡	0	407	NR‡	NR‡	0	665	400	380	0	997	560	532
	2	NR‡	NR‡	NR‡	NR‡	10	194	NR‡	NR‡	12	354	NR‡	NR‡	13	566	375	356	18	831	510	485
	5	NR‡	NR‡	NR‡	NR‡	26	189	NR‡	NR‡	33	347	NR‡	NR‡	40	557	369	351	52	820	504	479
	10	NR‡	NR‡	NR‡	NR‡	33	182	NR‡	NR‡	43	335	NR‡	NR‡	53	542	361	343	68	801	493	468
	15	NR‡	NR‡	NR‡	NR‡	40	174	NR‡	NR‡	50	321	NR‡	NR‡	62	528	353	335	80	782	482	458
	20	NR‡	NR‡	NR‡	NR‡	47	166	NR‡	NR‡	59	311	NR‡	NR‡	71	513	344	327	90	763	471	447
	30	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	78	290	NR‡	NR‡	92	483	NR‡	NR‡	115	726	449	427
	50	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	147	428	NR‡	NR‡	180	651	405	385

Vent and vent connector diameter, D, in																					
8				9				10				12				14					
Appliance input rating limits, thousands of Btu/h																					
Height, H, ft	Lateral, L, ft	FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*	
		Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†
6	0	0	698	370	229	0	897	470	291	0	1121	570	353	0	1645	850	527	0	2267	1170	725
	2	53	425	285	174	63	543	370	226	75	675	455	278	103	982	650	397	138	1346	890	543
	4	79	419	279	165	93	536	362	214	110	668	445	263	147	975	640	378	191	1338	880	519
	6	93	413	273	161	110	530	354	209	128	661	435	257	171	967	630	372	219	1330	870	513

(Continued)

Table C.1 (Continued)

Height, H, ft		Lateral, L, ft		Vent and vent connector diameter, D, in																			
				8				9				10				12				14			
				Appliance input rating limits, thousands of Btu/h																			
FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*	
Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†
8	0	0	780	415	303	0	1006	537	392	0	1261	660	482	0	1858	970	708	0	2571	1320	964		
	2	50	483	322	235	60	619	418	305	71	770	515	376	98	1124	745	544	130	1543	1020	745		
	5	83	473	313	222	99	607	407	289	115	758	503	357	154	1110	733	520	199	1528	1010	717		
	8	99	463	303	215	117	596	396	281	137	746	490	348	180	1097	720	511	231	1514	1000	710		
10	0	0	847	450	356	0	1096	585	462	0	1377	720	569	0	2036	1060	837	0	2825	1450	1146		
	2	48	533	355	284	57	684	457	366	68	852	560	448	93	1244	850	680	124	1713	1130	904		
	5	81	522	346	277	95	671	446	357	112	839	547	438	149	1229	829	663	192	1696	1105	884		
	10	104	504	330	257	122	651	427	333	142	817	525	410	187	1204	795	620	238	1669	1080	842		
15	0	0	970	525	452	0	1263	682	587	0	1596	840	722	0	2380	1240	1066	0	3323	1720	1479		
	2	45	633	414	356	53	815	544	468	63	1019	675	581	86	1495	985	847	114	2062	1350	1161		
	5	76	620	403	343	90	800	529	450	105	1003	660	561	140	1476	967	822	182	2041	1327	1128		
	10	99	600	386	328	116	777	507	431	135	977	635	540	177	1446	936	796	227	2009	1289	1096		
	15	115	580	373	317	134	755	491	417	155	953	610	519	202	1418	905	769	257	1976	1250	1063		
20	0	0	1057	575	518	0	1384	752	677	0	1756	930	837	0	2637	1350	1215	0	3701	1900	1710		
	2	41	711	470	423	50	917	612	551	59	1150	755	680	81	1694	1100	990	107	2343	1520	1368		
	5	73	697	460	409	86	902	599	533	101	1133	738	657	135	1674	1079	960	174	2320	1498	1333		
	10	95	675	443	394	112	877	576	513	130	1105	710	632	172	1641	1045	930	220	2282	1460	1299		
	15	111	654	427	380	129	853	557	496	150	1078	688	612	195	1609	1018	906	248	2245	1425	1268		
	20	125	634	410	365	145	830	537	478	167	1052	665	592	217	1578	990	881	273	2210	1390	1237		
25	0	0	1115	613	562	0	1466	804	736	0	1867	995	912	0	2821	1450	1329	0	3977	2035	1864		
	2	37	769	503	461	46	995	656	601	57	1251	810	742	78	1849	1205	1104	103	2565	1660	1521		

(Continued)

Table C.1 (Continued)

Vent and vent connector diameter, D, in																					
8				9				10				12				14					
Appliance input rating limits, thousands of Btu/h																					
Height, H, ft	Lateral, L, ft	FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*	
		Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†
	5	71	754	492	448	84	979	644	587	99	1233	795	724	131	1828	1184	1080	169	2540	1637	1492
	10	93	732	475	433	110	953	622	567	128	1203	770	702	168	1793	1150	1048	215	2499	1597	1456
	15	108	710	459	418	127	928	603	550	147	1175	748	682	191	1759	1119	1021	243	2460	1559	1421
	20	122	689	442	403	142	904	583	531	164	1148	725	661	212	1726	1088	992	267	2422	1520	1386
30	0	0	1173	650	605	0	1548	855	795	0	1977	1060	986	0	3004	1550	1442	0	4252	2170	2018
	2	33	826	535	498	42	1072	700	651	54	1351	865	804	74	2004	1310	1218	98	2786	1800	1674
	5	69	811	524	487	82	1055	688	640	96	1332	851	791	127	1981	1289	1199	164	2759	1775	1651
	10	91	788	507	472	107	1028	668	621	125	1301	829	771	164	1944	1254	1166	209	2716	1733	1612
	15	105	765	490	456	124	1002	648	603	143	1272	807	751	187	1908	1220	1135	237	2674	1692	1574
	20	119	743	473	440	139	977	628	584	160	1243	784	729	207	1873	1185	1102	260	2633	1650	1535
	30	149	702	444	413	171	929	594	552	195	1189	745	693	246	1807	1130	1051	305	2555	1585	1474
50	0	0	1297	708	673	0	1730	952	904	0	2231	1195	1135	0	3441	1825	1734	0	4934	2550	2423
	2	26	975	615	584	33	1276	813	772	41	1620	1010	960	66	2431	1513	1437	86	3409	2125	2019
	5	65	960	605	575	77	1259	798	758	90	1600	996	946	118	2406	1495	1420	151	3380	2102	1997
	10	86	935	589	560	101	1230	773	734	118	1567	972	923	154	2366	1466	1393	196	3332	2064	1961
	15	100	911	572	543	117	1203	747	710	136	1536	948	901	177	2327	1437	1365	222	3285	2026	1925
	20	113	888	556	528	131	1176	722	686	151	1505	924	878	195	2288	1408	1338	244	3239	1987	1888
	30	141	844	522	496	161	1125	670	637	183	1446	876	832	232	2214	1349	1282	287	3150	1910	1815
100	0	0	1411	770	732	0	1908	1040	988	0	2491	1310	1245	0	3925	2050	1948	0	5729	2950	2803
	2	21	1155	700	665	25	1536	935	888	30	1975	1170	1112	44	3027	1820	1729	72	4313	2550	2423
	5	60	1141	692	657	71	1519	926	880	82	1955	1159	1101	107	3002	1803	1713	136	4282	2531	2404

(Continued)

Table C.1 (Continued)

Vent and vent connector diameter, D, in																					
8				9				10				12				14					
Appliance input rating limits, thousands of Btu/h																					
Height, H, ft	Lateral, L, ft	FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*	
		Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†
	10	80	1118	679	645	94	1492	910	865	108	1923	1142	1085	142	2961	1775	1686	180	4231	2500	2375
	15	93	1095	666	633	109	1465	895	850	126	1892	1124	1068	163	2920	1747	1660	206	4182	2469	2346
	20	105	1073	653	620	122	1438	880	836	141	1861	1107	1052	181	2880	1719	1633	226	4133	2438	2316
	30	131	1029	627	596	149	1387	849	807	170	1802	1071	1017	215	2803	1663	1580	265	4037	2375	2256
	50	197	944	575	546	217	1288	787	748	241	1688	1000	950	292	2657	1550	1473	350	3856	2250	2138

Vent and vent connector diameter, D, in																					
16				18				20				22				24					
Appliance input rating limits, thousands of Btu/h																					
Height, H, ft	Lateral, L, ft	FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*	
		Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†
6	0	0	2983	1530	949	0	3802	1960	1215	0	4721	2430	1507	0	5737	2950	182-9	0	6853	3520	2182
	2	178	1769	1170	714	225	2250	1480	903	296	2782	1850	1129	360	3377	2220	135-4	426	4030	2670	1629
	4	242	1761	1160	684	300	2242	1475	870	390	2774	1835	1083	469	3370	2215	130-7	555	4023	2660	1569
	6	276	1753	1150	679	341	2235	1470	867	437	2767	1820	1074	523	3363	2210	130-4	618	4017	2650	1564
8	0	0	3399	1740	1270	0	4333	2220	1621	0	5387	2750	2008	0	6555	3360	245-3	0	7838	4010	2927
	2	168	2030	1340	978	212	2584	1700	1241	278	3196	2110	1540	336	3882	2560	186-9	401	4634	3050	2227
	5	251	2013	1330	944	311	2563	1685	1196	398	3180	2090	1484	476	3863	2545		562	4612	3040	2158

(Continued)

Table C.1 (Continued)

Heig- ht, H, ft		Vent and vent connector diameter, D, in																			
		16				18				20				22				24			
		Appliance input rating limits, thousands of Btu/h																			
Later- al, L, ft	FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*		
	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	
8																	180- 7				
	8	289	2000	1320	937	354	2552	1670	1186	450	3163	2070	1470	537	3850	2530	179- 6	630	4602	3030	2151
	0	0	3742	1925	1521	0	4782	2450	1936	0	5955	3050	2410	0	7254	3710	293- 1	0	8682	4450	3516
	2	161	2256	1480	1184	202	2868	1890	1512	264	3556	2340	1872	319	4322	2840	227- 2	378	5153	3390	2712
	5	243	2238	1461	1169	300	2849	1871	1497	382	3536	2318	1854	458	4301	2818	225- 4	540	5132	3371	2697
10	298	2209	1430	1115	364	2818	1840	1435	459	3504	2280	1778	546	4268	2780	216- 8	641	5099	3340	2605	
15	0	0	4423	2270	1952	0	5678	2900	2494	0	7099	3620	3113	0	8665	4410	379- 3	0	1039- 3	5300	4558
	2	147	2719	1770	1522	186	3467	2260	1944	239	4304	2800	2408	290	5232	3410	293- 3	346	6251	4080	3509
	5	229	2696	1748	1486	283	3442	2235	1900	355	4278	2777	2360	426	5204	3385	287- 7	501	6222	4057	3448
	10	283	2659	1712	1455	346	3402	2193	1864	432	4234	2739	2328	510	5159	3343	284- 2	599	6175	4019	3416
	15	318	2623	1675	1424	385	3363	2150	1828	479	4192	2700	2295	564	5115	3300	280- 5	665	6129	3980	3383
20	0	0	4948	2520	2268	0	6376	3250	2925	0	7988	4060	3654	0	9785	4980	448- 2	0	1753	6000	5400
	2	139	3097	2000	1800	175	3955	2570	2313	220	4916	3200	2880	269	5983	3910	351- 9	321	7154	4700	4230
	5	219	3071	1978	1760	270	3926	2544	2264	337	4885	3174	2825	403	5950	3880		475	7119	4662	4149

(Continued)

Table C.1 (Continued)

Heig- ht, H, ft		Later- al, L, ft		Vent and vent connector diameter, D, in																			
				16				18				20				22				24			
				Appliance input rating limits, thousands of Btu/h																			
		FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*			
		Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†		
																	345- 3						
10		273	3029	1940	1727	334	3880	2500	2225	413	4835	3130	2786	489	5 896	3830	340- 9	573	7063	4600	4094		
15		306	2988	1910	1700	372	3835	2465	2194	459	4786	3090	2750	541	5844	3795	337- 8	631	7007	4575	4072		
20		335	2948	1880	1673	404	3791	2430	2163	495	4737	3050	2715	585	5792	3760	334- 6	689	6953	4550	4050		
0		0	5337	2720	2492	0	6898	3510	3216	0	8665	4405	4036	0	1063- 4	5415	496- 2	0	1280- 1	6530	5983		
2		133	3397	2190	2007	167	4345	2810	2575	210	5408	3505	3212	255	6589	4280	392- 2	303	7886	5150	4719		
5		213	3369	2164	1973	261	4314	2782	2537	325	5374	3479	3172	388	6553	4251	387- 6	457	7847	5107	4656		
10		266	3323	2120	1933	325	4264	2735	2494	400	5319	3435	3132	473	6493	4202	383- 2	554	7784	5036	4591		
15		299	3279	2080	1897	363	4215	2693	2455	445	5265	3393	3093	524	6435	4161	379- 4	611	7722	4983	4543		
20		327	3236	2040	1860	394	4167	2650	2416	481	5212	3350	3055	567	6378	4120	375- 6	664	7662	4930	4494		
0		0	5725	2920	2716	0	7420	3770	3506	0	9341	4750	4418	0	1148- 3	5850	544- 1	0	1384- 8	7060	6566		
2		127	3696	2380	2213	159	4734	3050	2837	199	5900	3810	3543	241	7194	4650	432- 5	285	8617	5600	5208		
5		206	3666	2350	2186	252	4701	3020	2809	312	5863	3783	3518	373	7155	4622	429- 8	439	8574	5552	5163		
10		259	3617	2300	2139	316	4647	2970	2762	386	5803	3739	3477	456	7090	4574		535	8505	5471	5088		

(Continued)

Table C.1 (Continued)

Heig- ht, H, ft		Vent and vent connector diameter, D, in																			
		16				18				20				22				24			
		Appliance input rating limits, thousands of Btu/h																			
		FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*	
Later- al, L, ft	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	
50	15	292	3570	2250	2093	354	4594	2920	2716	431	5744	3695	3436	507	7026	4527	425-4 421-0	590	8437	5391	5014
	20	319	3523	2200	2046	384	4542	2870	2669	467	5686	3650	3395	548	6964	4480	416-6	639	8370	5310	4938
	30	369	3433	2130	1981	440	4442	2785	2590	540	5574	3565	3315	635	6842	4375	406-9	739	8239	5225	4859
	0	0	6711	3440	3268	0	8774	4460	4237	0	111-29	5635	5353	0	1376-7	6940	659-3	0	1669-4	8430	8009
	2	113	4554	2840	2698	141	5864	3670	3487	171	7339	4630	4399	209	8980	5695	541-0	251	1078-8	6860	6517
	5	191	4520	2813	2672	234	5826	3639	3457	283	7295	4597	4367	336	8933	5654	537-1	394	1073-7	6818	6477
	10	243	4464	2767	2629	295	5763	3585	3406	355	7224	4542	4315	419	8855	5585	530-6	491	1065-2	6749	6412
100	15	274	4409	2721	2585	330	5701	3534	3357	396	7155	4511	4285	465	8779	5546	526-9	542	1057-0	6710	6375
	20	300	4356	2675	2541	361	5641	3481	3307	433	7086	4479	4255	506	8704	5506	523-1	586	1048-8	6670	6337
	30	347	4253	2631	2499	412	5523	3431	3259	494	6953	4421	4200	577	8557	5444	517-2	672	1032-8	6603	6273
	0	0	7914	4050	3848	0	104-85	5300	5035	0	134-54	6700	6365	0	1681-7	8600	817-0	0	2057-8	103-00	9785
	2	95	5834	3500	3325	120	7591	4600	4370	138	9577	5800	5510	169	1180-3	7200	684-0	204	1426-4	8800	8360

(Continued)

Table C.1 (Continued)

Heig- ht, H, ft		Vent and vent connector diameter, D, in																			
		16				18				20				22				24			
		Appliance input rating limits, thousands of Btu/h																			
Later- al, L, ft	FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*		
	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	
5	172	5797	3475	3301	208	7548	4566	4338	245	9528	5769	5481	293	1174-8	7162	680-4	341	1420-4	8756	8318	
10	223	5737	3434	3262	268	7478	4509	4284	318	9447	5717	5431	374	1165-8	7100	674-5	436	1410-5	8683	8249	
15	252	5678	3392	3222	304	7409	4451	4228	358	9367	5665	5382	418	1156-9	7037	668-5	487	1400-7	8610	8180	
20	277	5619	3351	3183	330	7341	4394	4174	387	9289	5613	5332	452	1148-2	6975	662-6	523	1391-0	8537	8110	
30	319	5505	3267	3104	378	7209	4279	4065	446	9136	5509	5234	514	1131-0	6850	650-8	592	1372-0	8391	7971	
50	415	5289	3100	2945	486	6956	4050	3848	572	8841	5300	5035	659	1097-9	6600	627-0	752	1335-4	8100	7695	

* NAT = natural draft appliance

† DP = depressurization

‡ NR = not recommended

Table C.2
Capacity of Type B double-wall vents with single-wall metal vent connectors serving a single Category I appliance
 (See Clauses [8.13.1](#), [C.2.4](#), [C.2.10](#), [C.2.11](#), and [C.2.14](#), and Figures [C.1](#) and [C.2](#).)

Height, H, ft		Lateral, L, ft		Vent and vent connector diameter, D, in																			
				3				4				5				6				7			
				Appliance input rating limits, thousands of Btu/h																			
FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*	
Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†
6	0	38	77	45	28	59	151	85	53	85	249	140	87	126	373	204	126	165	522	284	176		
	2	39	51	36	22	60	96	66	40	85	156	104	63	123	231	156	95	159	320	213	130		
	4	NR‡	NR‡	33	19	74	92	63	37	102	152	102	60	146	225	152	90	187	313	208	123		
	6	NR‡	NR‡	31	18	83	89	60	35	114	147	99	58	163	220	148	87	207	307	203	120		
8	0	37	83	50	37	58	164	93	68	83	273	154	112	123	412	234	171	161	580	319	233		
	2	39	56	39	28	59	108	75	55	83	176	119	87	121	261	179	131	155	363	246	180		
	5	NR‡	NR‡	37	26	77	102	69	49	107	168	114	81	151	252	171	121	193	352	235	167		
	8	NR‡	NR‡	33	23	90	95	64	45	122	161	107	76	175	243	163	116	223	342	225	160		
10	0	37	87	53	42	57	174	99	78	82	293	165	130	120	444	254	201	158	628	344	272		
	2	39	61	41	33	59	117	80	64	82	193	128	102	119	287	194	155	153	400	272	218		
	5	52	56	39	31	76	111	76	61	105	185	122	98	148	277	186	149	190	388	261	209		
	10	NR‡	NR‡	34	27	97	100	68	53	132	171	112	87	188	261	171	133	237	369	241	188		
15	0	36	93	57	49	56	190	111	95	80	325	186	160	116	499	283	243	153	713	388	334		
	2	38	69	47	40	57	136	93	80	80	225	149	128	115	337	224	193	148	473	314	270		
	5	51	63	44	37	75	128	86	73	102	216	140	119	144	326	217	184	182	459	298	253		
	10	NR‡	NR‡	39	33	95	116	79	67	128	201	131	111	182	308	203	173	228	438	284	241		
	15	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	72	61	158	186	124	105	220	290	192	163	272	418	269	229		
20	0	35	96	60	54	54	200	118	106	78	346	201	181	114	537	306	275	149	772	428	385		
	2	37	74	50	45	56	148	99	89	78	248	165	149	113	375	248	223	144	528	344	310		

(Continued)

Table C.2 (Continued)

Vent and vent connector diameter, D, in																					
3				4				5				6				7					
Appliance input rating limits, thousands of Btu/h																					
Height, H, ft	Lateral, L, ft	FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*	
		Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†
	5	50	68	47	42	73	140	94	84	100	239	158	141	141	363	239	213	178	514	334	297
	10	NR‡	NR‡	41	36	93	129	86	77	125	223	146	130	177	344	224	199	222	491	316	281
	15	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	80	71	155	208	136	121	216	325	210	187	264	469	301	268
	20	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	186	192	126	112	254	306	196	174	309	448	285	254
25	0	35	98	62	57	54	206	123	112	77	359	210	193	112	561	320	293	147	811	450	412
	2	37	77	53	49	56	156	105	96	77	265	174	160	111	402	264	241	142	569	368	338
	5	50	71	50	45	73	149	100	92	99	255	166	151	139	390	255	233	175	555	358	326
	10	NR	NR	NR	NR	92	137	92	84	124	239	157	143	174	371	241	219	218	531	342	311
	15	NR	NR	NR	NR	NR	NR	NR	NR	153	224	147	134	212	351	226	206	260	508	325	297
	20	NR	NR	NR	NR	NR	NR	NR	NR	184	208	NR	NR	250	332	212	193	304	486	309	282
30	0	34	99	63	59	53	211	127	118	76	372	219	204	110	584	334	311	144	849	472	439
	2	37	80	56	52	55	164	111	103	76	281	183	170	109	429	279	259	139	610	392	365
	5	49	74	52	48	72	157	106	99	98	271	173	161	136	417	271	252	171	595	382	355
	10	NR‡	NR‡	NR‡	NR‡	91	144	98	91	122	255	168	156	171	397	257	239	213	570	367	341
	15	NR‡	NR‡	NR‡	NR‡	115	131	NR‡	NR‡	151	239	157	146	208	377	242	225	255	547	349	325
	20	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	181	223	NR‡	NR‡	246	357	228	212	298	524	333	310
	30	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	389	477	305	284
50	0	33	99	66	63	51	213	133	126	73	394	230	219	105	629	361	343	138	928	515	489
	2	36	84	61	58	53	181	121	115	73	318	205	195	104	495	312	296	133	712	443	421
	5	48	80	NR‡	NR‡	70	174	117	111	94	308	198	188	131	482	305	290	164	696	435	413
	10	NR‡	NR‡	NR‡	NR‡	89	160	NR‡	NR‡	118	292	186	177	162	461	292	277	203	671	420	399

(Continued)

Table C.2 (Continued)

Vent and vent connector diameter, D, in																					
3				4				5				6				7					
Appliance input rating limits, thousands of Btu/h																					
Height, H, ft	Lateral, L, ft	FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*	
		Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†
100	15	NR‡	NR‡	NR‡	NR‡	112	148	NR‡	NR‡	145	275	174	165	199	441	280	266	244	646	405	385
	20	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	176	257	NR‡	NR‡	236	420	267	254	285	622	389	370
	30	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	315	376	NR‡	NR‡	373	573	NR‡	NR‡
	0	NR‡	NR‡	NR‡	NR‡	49	214	NR‡	NR‡	69	403	NR‡	NR‡	100	659	395	375	131	991	555	527
	2	NR‡	NR‡	NR‡	NR‡	51	192	NR‡	NR‡	70	351	NR‡	NR‡	98	563	373	354	125	828	508	483
	5	NR‡	NR‡	NR‡	NR‡	67	186	NR‡	NR‡	90	342	NR‡	NR‡	125	551	366	348	156	813	501	476
	10	NR‡	NR‡	NR‡	NR‡	85	175	NR‡	NR‡	113	324	NR‡	NR‡	153	532	354	336	191	789	486	462
	15	NR‡	NR‡	NR‡	NR‡	132	162	NR‡	NR‡	138	310	NR‡	NR‡	188	511	343	326	230	764	473	449
	20	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	168	295	NR‡	NR‡	224	487	NR‡	NR‡	270	739	458	435
30	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	231	264	NR‡	NR‡	301	448	NR‡	NR‡	355	685	NR‡	NR‡	
50	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	540	584	NR‡	NR‡

Vent and vent connector diameter, D, in																					
8						9						10						12			
Appliance input rating limits, thousands of Btu/h																					
Height, H, ft	Lateral, L, ft	FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*	
		Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†
6	0	211	695	369	229	267	894	469	291	371	1118	569	353	537	1639	849	526				
	2	201	423	284	173	251	541	368	224	347	673	453	276	498	979	648	395				
	4	237	416	277	163	295	533	360	212	409	664	443	261	584	971	638	376				
	6	263	409	271	160	327	526	352	208	449	656	433	255	638	962	627	370				

(Continued)

Table C.2 (Continued)

Height, H, ft		Lateral, L, ft		Vent and vent connector diameter, D, in															
				8				9				10				12			
				Appliance input rating limits, thousands of Btu/h															
FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*					
Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†				
8	0	206	777	414	302	258	1002	536	391	360	1257	658	480	521	1852	967	706		
	2	197	482	321	234	246	617	417	304	339	768	513	374	486	1120	743	542		
	5	245	470	311	221	305	604	404	287	418	754	500	355	598	1104	730	518		
	8	280	458	300	213	344	591	392	278	470	740	486	345	665	1089	715	508		
10	0	202	844	449	355	253	1093	584	461	351	1373	718	567	507	2031	1057	835		
	2	193	531	354	283	242	681	456	365	332	849	559	447	475	1242	848	678		
	5	241	518	344	275	299	667	443	354	409	834	544	435	584	1224	825	660		
	10	296	497	325	254	363	643	423	330	492	808	520	406	688	1194	788	615		
15	0	195	966	523	450	244	1259	681	586	336	1591	838	721	488	2374	1237	1064		
	2	187	631	413	355	232	812	543	467	319	1015	673	579	457	1491	983	845		
	5	231	616	400	340	287	795	526	447	392	997	657	558	562	1469	963	819		
	10	284	592	381	324	349	768	501	426	470	966	628	534	664	1433	928	789		
	15	334	568	367	312	404	742	484	411	540	937	601	511	750	1399	894	760		
20	0	190	1053	573	516	238	1379	750	675	326	1751	927	834	473	2631	1346	1211		
	2	182	708	468	421	227	914	611	550	309	1146	754	679	443	1689	1098	988		
	5	224	692	457	407	279	896	596	530	381	1126	734	653	547	1665	1074	956		
	10	277	666	437	389	339	866	570	507	457	1092	702	625	646	1626	1037	923		
	15	325	640	419	373	393	838	549	489	526	1060	677	603	730	1587	1005	894		
	20	374	616	400	356	448	810	526	468	592	1028	651	579	808	1550	973	866		
25	0	187	1111	610	559	234	1461	801	734	319	1861	992	908	464	2814	1446	1324		
	2	179	766	501	459	223	992	655	600	303	1246	809	741	434	1844	1203	1102		

(Continued)

Table C.2 (Continued)

		Vent and vent connector diameter, D, in															
		8				9				10				12			
		Appliance input rating limits, thousands of Btu/h															
Height, H, ft	Lateral, L, ft	FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*	
		Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†
	5	220	749	489	446	274	973	640	583	374	1225	790	720	536	1818	1179	1075
	10	271	722	469	428	333	942	616	562	449	1190	762	695	633	1777	1140	1040
	15	319	695	450	410	386	912	594	541	517	1156	736	671	716	1736	1105	1008
	20	367	670	431	393	441	883	571	520	581	1122	710	647	794	1696	1070	975
30	0	184	1168	647	602	229	1542	852	792	312	1971	1056	982	454	2996	1545	1437
	2	175	823	533	496	219	1069	698	649	296	1346	863	803	424	1999	1308	1216
	5	215	806	521	485	269	1049	684	636	366	1324	846	787	524	1971	1283	1193
	10	265	777	501	466	327	1017	662	616	440	1287	821	764	620	1927	1243	1156
	15	312	750	481	447	379	985	638	593	507	1251	794	738	702	1884	1205	1121
	20	360	723	461	429	433	955	615	572	570	1216	768	714	780	1841	1166	1084
	30	461	670	426	396	541	895	574	534	704	1147	720	670	937	1759	1101	1024
50	0	176	1292	704	669	220	1724	948	901	295	2223	1189	1130	428	3432	1818	1727
	2	168	971	613	582	209	1273	811	770	280	1615	1007	957	401	2426	1509	1434
	5	204	953	602	572	257	1252	795	755	347	1591	991	941	496	2396	1490	1416
	10	253	923	583	554	313	1217	765	727	418	1551	963	915	589	2347	1455	1382
	15	299	894	562	534	363	1183	736	699	481	1512	934	887	668	2299	1421	1350
	20	345	866	543	516	415	1150	708	673	544	1473	906	861	741	2251	1387	1318
	30	442	809	502	477	521	1086	649	617	674	1399	848	806	892	2159	1318	1252
100	0	166	1404	765	727	207	1900	1033	981	273	2479	1300	1235	395	3912	2042	1940
	2	158	1152	698	663	196	1532	933	886	259	1970	1168	1110	371	3021	1817	1726
	5	194	1134	688	654	240	1511	921	875	322	1945	1153	1095	460	2990	1796	1706

(Continued)

Table C.2 (Continued)

Height, H, ft		Lateral, L, ft		Vent and vent connector diameter, D, in													
				8				9				10				12	
				Appliance input rating limits, thousands of Btu/h													
FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*			
Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†		
	10	238	1104	672	638	293	1477	902	857	389	1905	1133	1076	547	2938	1763	1675
	15	281	1075	656	623	342	1443	884	840	447	1865	1110	1055	618	2888	1730	1644
	20	325	1046	639	607	391	1410	864	821	507	1825	1087	1033	690	2838	1696	1611
	30	418	988	NR‡	NR‡	491	1343	824	783	631	1747	1041	989	834	2739	1627	1546
	50	617	866	NR‡	NR‡	711	1205	NR‡	NR‡	895	1591	NR‡	NR‡	1138	2547	1489	1415

* NAT = natural draft appliance

† DP = depressurization

‡ NR = not recommended

Table C.3
Capacity of Type B double-wall vents with Type B double-wall vent connectors serving two or more Category I appliances

(See Clauses [8.13.2](#), [8.21.5](#), and [C.2.11](#), and Figures [C.2](#), [C.3](#), and [C.11](#).)

Capacity of vent connectors																	
Vent height H, ft		Vent connector diameter, D, in															
		3				4				5				6			
		Appliance input rating limits, thousands of Btu/h															
		FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*	
		Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†
6	1	22	37	26	16	35	66	46	28	46	106	72	44	58	164	104	63
	2	23	41	31	19	37	75	55	34	48	121	86	52	60	183	124	76
	3	24	44	35	21	38	81	62	38	49	132	96	59	62	199	139	85
8	1	22	40	27	19	35	72	48	34	49	114	76	54	64	176	109	77
	2	23	44	32	23	36	80	57	40	51	128	90	64	66	195	129	92
	3	24	47	36	26	37	87	64	45	53	139	101	72	67	210	145	103
10	1	22	43	28	22	34	78	50	39	49	123	78	61	65	189	113	88
	2	23	47	33	26	36	86	59	46	51	136	93	73	67	206	134	105
	3	24	50	37	29	37	92	67	52	52	146	104	81	69	220	150	117
15	1	21	50	30	26	33	89	53	45	47	142	83	71	64	220	120	102
	2	22	53	35	30	35	96	63	54	49	153	99	84	66	235	142	121
	3	24	55	40	34	36	102	71	60	51	163	111	94	68	248	160	136
20	1	21	54	31	28	33	99	56	50	46	157	87	78	62	246	125	113
	2	22	57	37	33	34	105	66	59	48	167	104	94	64	259	149	134
	3	23	60	42	38	35	110	74	67	50	176	116	104	66	271	168	151
25	1	21	58	32	30	32	106	58	53	46	169	90	82	61	267	130	119
	2	22	61	38	35	34	112	68	62	48	179	107	98	63	279	154	141

(Continued)

Table C.3 (Continued)

Capacity of vent connectors																	
		Vent connector diameter, D, in															
		3				4				5				6			
		Appliance input rating limits, thousands of Btu/h															
Vent height H, ft	Vent connector rise, R, ft	FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*	
		Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†
	3	23	63	43	40	35	117	77	70	49	187	120	110	65	290	173	159
30	1	20	62	33	31	31	113	59	55	45	181	93	86	60	288	134	125
	2	21	64	39	36	33	118	70	65	47	190	110	102	62	299	158	147
	3	22	66	44	41	34	123	79	73	48	198	124	115	64	309	178	166
50	1	19	71	35	33	30	133	64	61	43	216	101	96	57	349	145	138
	2	21	73	43	41	32	137	76	72	45	223	119	113	59	358	172	163
	3	22	75	48	46	33	141	86	82	46	229	134	127	61	366	194	184
100	1	18	82	37	36	28	158	66	65	40	262	104	102	53	442	150	147
	2	19	83	44	43	30	161	79	77	42	267	123	121	55	447	178	174
	3	20	84	50	49	31	163	89	87	44	272	138	135	57	452	200	196

Capacity of vent connectors																	
		Vent connector diameter, D, in															
		7				8				9				10			
		Appliance input rating limits, thousands of Btu/h															
Vent height H, ft	Vent con- nector rise, R, ft	FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*	
		Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†
6	1	77	225	142	87	92	296	185	113	109	376	237	145	128	466	289	176
	2	79	253	168	102	95	333	220	134	112	424	282	172	131	526	345	210
	3	82	275	189	115	97	363	248	151	114	463	317	193	134	575	386	235

(Continued)

Table C.3 (Continued)

Capacity of vent connectors																	
Vent height H, ft	Vent connector rise, R, ft	Vent connector diameter, D, in															
		7				8				9				10			
		Appliance input rating limits, thousands of Btu/h															
		FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*	
		Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†
8	1	84	243	148	105	100	320	194	138	118	408	248	176	138	507	303	215
	2	86	269	175	124	103	356	230	163	121	454	294	209	141	564	358	254
	3	88	290	198	141	105	384	258	183	123	492	330	234	143	612	402	285
10	1	89	257	154	120	106	341	200	156	125	436	257	200	146	542	314	245
	2	91	282	182	142	109	374	238	186	128	479	305	238	149	596	372	290
	3	94	303	205	160	111	402	268	209	131	515	342	267	152	642	417	325
15	1	88	298	163	139	110	389	214	182	134	493	273	232	162	609	333	283
	2	91	320	193	164	112	419	253	215	137	532	323	275	165	658	394	335
	3	93	339	218	185	115	445	286	243	140	565	365	310	167	700	444	377
20	1	86	334	171	154	107	436	224	202	131	552	285	257	158	681	347	312
	2	89	354	202	182	110	463	265	239	134	587	339	305	161	725	414	373
	3	91	371	228	205	113	486	300	270	137	618	378	466	164	764	466	419
25	1	85	363	177	162	105	474	231	212	128	601	295	271	155	742	360	329
	2	87	381	209	191	108	499	274	251	132	633	350	320	158	783	427	391
	3	90	397	235	215	111	521	309	283	135	662	392	422	161	819	480	439
30	1	83	391	182	169	103	512	238	221	125	649	305	284	151	802	372	346
	2	85	408	215	200	105	535	282	262	129	679	360	335	155	840	439	408
	3	88	423	242	225	108	555	317	295	132	706	405	377	158	874	494	459
50	1	78	477	197	187	97	627	257	244	120	797	330	314	144	984	403	383
	2	81	490	234	222	100	645	306	291	123	820	392	372	148	1014	478	454

(Continued)

Table C.3 (Continued)

Capacity of vent connectors																	
Vent height H, ft	Vent connector rise, R, ft	Vent connector diameter, D, in															
		7				8				9				10			
		Appliance input rating limits, thousands of Btu/h															
		FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*	
		Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†
	3	83	502	263	250	103	661	343	326	126	842	441	419	151	1043	538	511
100	1	73	611	204	200	91	810	266	261	112	1038	341	334	135	1285	417	409
	2	75	619	242	237	94	822	316	310	115	1054	405	397	139	1306	494	484
	3	78	627	272	267	97	834	355	348	118	1069	455	446	142	1327	555	544
Capacity of vent connectors																	
Vent height H, ft	Vent connector rise, R, ft	Vent connector diameter, D, in															
		12				14				16				18			
		Appliance input rating limits, thousands of Btu/h															
		FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*	
		Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†
6	2	174	764	496	303	223	1046	653	398	281	1371	853	520	346	1772	1080	659
	4	180	897	616	376	230	1231	827	504	287	1617	1081	659	352	2069	1370	836
	6	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§
8	2	186	822	516	366	238	1126	696	494	298	1478	910	646	365	1920	1150	817
	4	192	952	644	457	244	1307	884	628	305	1719	1150	817	372	2211	1460	1037
	6	198	1050	772	548	252	1445	1071	760	313	1902	1390	987	380	2434	1770	1257
10	2	196	870	536	418	249	1195	730	569	311	1570	955	745	379	2049	1205	940
	4	201	997	664	518	256	1371	924	721	318	1804	1205	940	387	2332	1535	1197
	6	207	1095	792	618	263	1509	1118	872	325	1989	1455	1135	395	2556	1865	1455

(Continued)

Table C.3 (Continued)

Capacity of vent connectors																			
Vent height H, ft		Vent connector rise, R, ft		Vent connector diameter, D, in															
				12				14				16				18			
				Appliance input rating limits, thousands of Btu/h															
				FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*	
Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†				
15	2	214	967	568	483	272	1334	790	672	336	1760	1030	876	408	2317	1305	1109		
	4	221	1076	712	605	279	1499	1006	855	344	1978	1320	1122	416	2579	1665	1415		
	6	228	1181	856	728	286	1632	1222	1039	351	2157	1610	1369	424	2796	2025	1721		
20	2	223	1051	596	536	291	1443	840	756	357	1911	1095	986	430	2533	1385	1247		
	4	230	1162	748	673	298	1597	1064	958	365	2116	1395	1256	438	2778	1765	1589		
	6	237	1253	900	810	307	1726	1288	1159	373	2287	1695	1526	450	2984	2145	1931		
25	2	220	1134	614	562	289	1554	875	801	362	2047	1143	1047	446	2712	1463	1340		
	4	227	1239	770	705	296	1700	1112	1019	371	2241	1453	1330	456	2944	1843	1688		
	6	234	1327	926	848	305	1823	1349	1235	379	2406	1763	1614	468	3142	2243	2054		
30	2	216	1217	632	588	286	1664	910	846	367	2183	1190	1107	461	2891	1540	1432		
	4	223	1316	792	737	294	1802	1160	1079	376	2366	1510	1404	474	3110	1920	1786		
	6	231	1400	952	885	303	1920	1410	1311	384	2524	1830	1702	485	3299	2340	2176		
50	2	206	1479	689	655	273	2023	1007	957	350	2659	1315	1249	435	3548	1665	1582		
	4	213	1561	860	817	281	2139	1291	1226	359	2814	1685	1601	447	3730	2135	2028		
	6	221	1631	1031	979	290	2242	1575	1496	369	2951	2055	1952	461	3893	2605	2475		
100	2	192	1923	712	698	254	2644	1050	1029	326	3490	1370	1343	402	4707	1740	1705		
	4	200	1984	888	870	263	2731	1346	1319	336	3606	1760	1725	414	4842	2220	2176		
	6	208	2035	1064	1043	272	2811	1642	1609	346	3714	2150	2107	426	4968	2700	2646		

(Continued)

Table C.3 (Continued)

Capacity of vent connectors													
Vent height H, ft		Vent connector diameter, D, in											
		20				22				24			
		Appliance input rating limits, thousands of Btu/h											
		FAN		NAT*		FAN		NAT*		FAN		NAT*	
		Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†
6	2	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§
	4	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§
	6	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§
8	2	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§
	4	471	2737	1800	1278	560	3319	2180	1548	662	3957	2590	1839
	6	478	3018	2180	1548	568	3665	2640	1874	669	4373	3130	2222
10	2	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§
	4	486	2887	1890	1474	581	3502	2280	1778	686	4175	2710	2114
	6	494	3169	2290	1786	589	3849	2760	2153	694	4593	3270	2551
15	2	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§
	4	523	3197	2060	1751	624	3881	2490	2117	734	4631	2960	2516
	6	533	3470	2510	2134	634	4216	3030	2576	743	5035	3600	3060
20	2	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§
	4	554	3447	2180	1962	661	4190	2630	2367	772	5005	3130	2817
	6	567	3708	2650	2385	671	4511	3190	2871	785	5392	3790	3411
25	2	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§
	4	587	3644	2273	2081	695	4436	2745	2514	810	5306	3270	2994
	6	600	3894	2763	2530	706	4744	3335	3054	823	5677	3970	3636
30	2	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§

(Continued)

Table C.3 (Continued)

Capacity of vent connectors													
Vent height H, ft	Vent connector rise, R, ft	Vent connector diameter, D, in											
		20				22				24			
		Appliance input rating limits, thousands of Btu/h											
		FAN		NAT*		FAN		NAT*		FAN		NAT*	
		Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†
	4	619	3840	2365	2199	728	4681	2860	2660	847	5606	3410	3171
	6	632	4080	2875	2674	741	4976	3480	3236	860	5961	4150	3860
50	2	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§
	4	580	4601	2633	2501	709	5569	3185	3026	851	6633	3790	3601
	6	594	4808	3208	3048	724	5826	3885	3691	867	6943	4620	4389
100	2	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§	N/A§
	4	523	5982	2750	2695	639	7254	3330	3263	769	8650	3950	3871
	6	539	6143	3350	3283	654	7453	4070	3989	786	8892	4810	4714

Capacity of common vent																				
Common vent diameter, D, in																				
4					5					6					7					
Combined appliance input rating, thousands of Btu/h																				
Vent height, H, ft	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†
	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)
6	92	81	49	65	40	140	116	71	103	63	204	161	98	147	90	309	248	151	200	122
8	101	90	64	73	52	155	129	92	114	81	224	178	126	163	116	339	275	195	223	158
10	110	97	76	79	62	169	141	110	124	97	243	194	151	178	139	367	299	233	242	189
15	125	112	95	91	77	195	164	139	144	122	283	228	194	206	175	427	352	299	280	238

(Continued)

Table C.3 (Continued)

Capacity of common vent																				
Common vent diameter, D, in																				
4						5					6					7				
Combined appliance input rating, thousands of Btu/h																				
Vent height, H, ft	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†
	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)
20	136	123	111	102	92	215	183	165	160	144	314	255	230	229	206	475	394	355	310	279
25	144	131	120	110	101	230	197	180	173	158	338	276	253	248	227	511	427	391	335	307
30	152	138	128	118	110	244	210	195	185	172	361	297	276	266	247	547	459	427	360	335
50	167	153	147	134	129	279	244	234	214	205	421	353	339	310	298	641	547	525	423	406
100	175	163	160	NR‡	NR‡	311	277	271	NR‡	NR‡	489	421	413	NR‡	NR‡	751	658	645	479	469

Capacity of common vent																				
Common vent diameter, D, in																				
8						9					10					12				
Combined appliance input rating, thousands of Btu/h																				
Vent height, H, ft	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†
	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)
6	404	314	192	260	159	547	434	265	335	204	672	520	317	410	250	900	696	425	588	359
8	444	348	247	290	206	602	480	341	378	268	740	577	410	465	330	994	773	549	652	463
10	477	377	294	315	246	649	522	407	405	316	800	627	489	495	386	1076	841	656	712	555
15	556	444	377	365	310	753	612	520	465	395	924	733	623	565	480	1247	986	838	825	701
20	621	499	449	405	365	842	688	619	523	471	1035	826	743	640	576	1405	1116	1004	916	824
25	671	542	497	438	401	911	748	685	564	517	1122	901	825	690	632	1532	1222	1119	971	889
30	720	585	544	470	437	979	808	751	605	563	1209	975	907	740	688	1658	1327	1234	1025	953

(Continued)

Table C.3 (Continued)

Capacity of common vent																				
Common vent diameter, D, in																				
8						9					10					12				
Combined appliance input rating, thousands of Btu/h																				
Vent height, H, ft	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†
	+FA-N	+NA-T*	(F+N)	+NA-T*	(N+N)	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)
50	854	706	678	550	528	1164	977	938	705	677	1451	1188	1140	860	826	2024	1640	1574	1280	1229
100	1025	873	856	625	613	1408	1215	1191	800	784	1784	1502	1472	975	956	2569	2131	2088	1670	1637

Capacity of common vent															
Common vent diameter, D, in															
14								16				18			
Combined appliance input rating, thousands of Btu/h															
Vent height, H, ft	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†
	+FAN	+NAT*	(F+N)	+NAT*	(N+N)	+FAN	+NAT*	(F+N)	+NAT*	(N+N)	+FAN	+NAT*	(F+N)	+NAT*	(N+N)
6	1284	990	604	815	497	1735	1336	815	1065	650	2253	1732	1057	1345	820
8	1423	1103	783	912	648	1927	1491	1059	1190	845	2507	1936	1375	1510	1072
10	1542	1200	936	995	776	2093	1625	1268	1300	1014	2727	2113	1648	1645	1283
15	1794	1410	1199	1158	984	2440	1910	1624	1510	1284	3184	2484	2111	1910	1624
20	2006	1588	1429	1290	1161	2722	2147	1932	1690	1521	3561	2798	2518	2140	1926
25	2190	1740	1595	1408	1290	2971	2353	2156	1840	1686	3879	3062	2806	2330	2135
50	2911	2347	2253	1863	1788	3964	3183	3056	2430	2333	5184	4149	3983	3075	2952
100	3732	3076	3014	2450	2401	5125	4202	4118	3200	3136	6749	5509	5399	4050	3969

(Continued)

Table C.3 (Continued)

Capacity of common vent															
Common vent diameter, D, in															
20						22					24				
Combined appliance input rating, thousands of Btu/h															
Vent height, H, ft	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†
	+FAN	+NAT*	(F+N)	+NAT*	(N+N)	+FAN	+NAT*	(F+N)	+NAT*	(N+N)	+FAN	+NAT*	(F+N)	+NAT*	(N+N)
6	2838	2180	1330	1660	1013	3488	2677	1633	1970	1202	4206	3226	1968	2390	1458
8	3162	2439	1732	1860	1321	3890	2998	2129	2200	1562	4695	3616	2567	2680	1903
10	3444	2665	2079	2030	1583	4241	3278	2557	2400	1872	5123	3957	3086	2920	2278
15	4026	3133	2663	2360	2006	4971	3862	3283	2790	2372	6016	4670	3970	3400	2890
20	4548	3552	3197	2640	2376	5573	4352	3917	3120	2808	6749	5261	4735	3800	3420
25	4926	3873	3548	2875	2634	6056	4755	4357	3400	3115	7345	5754	5273	4140	3793
30	5303	4193	3899	3110	2892	6539	5157	4796	3680	3422	7940	6247	5810	4480	4166
50	6567	5240	5030	3800	3648	8116	6458	6200	4500	4320	9837	7813	7500	5475	5256
100	8597	6986	6846	5000	4900	10 681	8648	8475	5920	5802	13 004	10 499	10 289	7200	7056

* NAT = natural draft appliance.

† DP = depressurization.

‡ NR = not recommended.

§ N/A = not applicable.

Table C.4
Capacity of Type B double-wall vents with single-wall vent connectors serving two or more Category I appliances
 (See Clauses [8.13.2](#), [8.21.5](#), and [C.2.11](#) and Figures [C.3](#) and [C.4](#).)

Capacity of vent connectors																			
Vent height, H, ft		Vent connector rise, R, ft		Vent connector diameter, D, in															
				3				4				5				6			
				Appliance input rating limits, thousands of Btu/h															
				FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*	
Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†				
6	1	NR‡	NR‡	26	16	NR‡	NR‡	46	28	NR‡	NR‡	71	43	NR‡	NR‡	102	62		
	2	NR‡	NR‡	31	19	NR‡	NR‡	55	34	NR‡	NR‡	85	52	168	182	123	75		
	3	NR‡	NR‡	34	21	NR‡	NR‡	62	38	121	131	95	58	174	198	138	84		
15	1	NR‡	NR‡	29	25	79	87	52	44	116	138	81	69	177	214	116	99		
	2	NR‡	NR‡	34	29	83	94	62	53	121	150	97	82	185	230	138	117		
	3	NR‡	NR‡	39	33	87	100	70	60	127	160	109	93	193	243	157	133		
20	1	NR‡	NR‡	30	26	78	95	54	47	115	150	84	74	174	235	120	106		
	2	NR‡	NR‡	35	31	82	101	64	56	120	162	100	88	182	250	143	125		
	3	NR‡	NR‡	40	35	86	106	72	64	125	171	113	99	190	262	162	142		
25	1	NR‡	NR‡	30	28	78	102	55	50	114	163	86	78	172	257	125	113		
	2	NR‡	NR‡	36	32	82	108	65	59	118	173	103	93	180	270	147	133		
	3	NR‡	NR‡	41	37	86	113	74	67	124	182	116	106	188	281	167	151		
30	1	47	60	31	29	77	110	57	53	113	175	89	83	169	278	129	120		
	2	50	62	37	34	81	115	67	62	117	185	106	99	177	290	152	141		
	3	54	64	42	39	85	119	76	71	122	193	120	112	185	300	172	160		
50	1	46	69	33	32	75	128	60	58	109	207	96	93	162	336	137	133		
	2	49	71	40	39	79	132	72	70	114	215	113	110	170	345	164	159		
	3	53	72	45	44	83	136	82	80	119	221	128	124	178	353	186	180		

(Continued)

Table C.4 (Continued)

Capacity of vent connectors																	
Vent height, H, ft		Vent connector diameter, D, in															
		7				8				9				10			
		Appliance input rating limits, thousands of Btu/h															
		FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*	
	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	
6	1	207	223	140	85	262	293	183	112	325	373	234	143	447	463	286	174
	2	215	251	167	102	271	331	219	134	334	422	281	171	458	524	344	210
	3	222	273	188	115	279	361	247	151	344	462	316	193	468	574	385	235
15	1	238	291	158	134	312	380	208	177	397	482	266	226	556	596	324	275
	2	246	314	189	161	321	411	248	211	407	522	317	269	568	646	387	329
	3	255	333	215	183	331	438	281	239	418	557	360	306	579	690	437	371
20	1	234	321	164	144	307	419	215	189	391	531	275	242	547	657	335	294
	2	243	342	195	172	316	448	257	226	401	569	328	288	559	704	400	351
	3	251	359	222	195	326	473	290	255	412	601	371	326	571	745	452	397
25	1	230	350	169	153	301	458	223	202	384	581	285	257	537	718	347	314
	2	239	369	202	182	312	484	265	240	395	615	338	306	550	761	412	373
	3	248	386	228	207	321	507	300	271	406	646	383	346	563	800	467	422
30	1	226	380	175	163	296	497	230	214	378	630	294	273	528	779	358	333
	2	236	397	208	193	307	521	274	255	389	662	349	325	541	819	425	395
	3	244	412	235	219	316	542	309	287	400	690	394	366	555	855	482	448
50	1	217	460	188	182	284	604	245	238	364	768	314	305	507	951	384	372
	2	226	473	223	216	294	623	293	284	376	793	375	364	520	983	458	444
	3	235	486	252	244	304	640	331	321	387	816	424	411	535	1013	518	502

(Continued)

Table C.4 (Continued)

Capacity of common vent																				
Common vent diameter, D, in																				
4						5					6					7				
Combined appliance input rating, thousands of Btu/h																				
Vent height, H, ft	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†
	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)
6	89	78	48	64	39	136	113	69	100	61	200	158	96	144	88	304	244	149	196	120
8	98	87	62	71	50	151	126	89	112	80	218	173	123	159	113	331	269	191	218	155
10	106	94	73	76	59	163	137	107	120	94	237	189	147	174	136	357	292	228	236	184
15	121	108	92	88	75	189	159	135	140	119	275	221	188	200	170	416	343	292	274	233
20	131	118	106	98	88	208	177	159	155	140	305	247	222	223	201	463	383	345	302	272
25	138	125	115	106	97	222	190	174	167	153	328	267	244	240	220	498	415	380	326	299
30	145	132	123	113	105	236	202	188	179	166	350	286	266	257	239	533	446	415	349	325
50	159	145	139	128	123	268	233	224	204	196	406	337	324	296	284	622	529	508	410	394

Capacity of common vent																				
Common vent diameter, D, in																				
8									9						10					
Combined appliance input rating, thousands of Btu/h																				
Vent height, H, ft	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†
	+FAN	+NAT*	(F+N)	+NAT*	(N+N)	+FAN	+NAT*	(F+N)	+NAT*	(N+N)	+FAN	+NAT*	(F+N)	+NAT*	(N+N)	+FAN	+NAT*	(F+N)	+NAT*	(N+N)
6	398	310	189	257	157	541	429	262	332	203	665	515	314	407	248					
8	436	342	243	285	202	592	473	336	373	265	730	569	404	460	327					
10	467	369	288	309	241	638	512	399	398	310	787	617	481	487	380					

(Continued)

Table C.4 (Continued)

Capacity of common vent															
Common vent diameter, D, in															
8						9					10				
Combined appliance input rating, thousands of Btu/h															
Vent height, H, ft	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†
	+FAN	+NAT*	(F+N)	+NAT*	(N+N)	+FAN	+NAT*	(F+N)	+NAT*	(N+N)	+FAN	+NAT*	(F+N)	+NAT*	(N+N)
15	544	434	369	357	303	738	599	509	456	388	905	718	610	553	470
20	606	487	438	395	356	824	673	606	512	461	1013	808	727	626	563
25	655	529	484	427	392	891	732	671	553	506	1098	880	806	675	618
30	703	570	530	459	427	958	790	735	593	551	1183	952	885	723	672
50	833	686	659	535	514	1139	954	916	689	661	1418	1157	1111	838	804

* NAT = natural draft appliance.

† DP = depressurization.

‡ NR = not recommended.

Table C.5
Capacity of tile-lined masonry chimneys with Type B double-wall vent connectors serving a single Category I appliance

(See Clauses [8.12.8](#) and [8.13.1](#) and Figures [C.4](#) and [C.5](#).)

Vent connector diameter, D, in To be used with chimney areas within the size limits at bottom of next page																					
3						4				5				6				7			
Appliance input rating limits, thousands of Btu/h																					
Height, H, ft	Lateral, L, ft	FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*	
		Min	Max	Max	DP†	Min	Max	Max	D-P†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†
6	2	N/ A**	N/ A**	28	17	N/ A**	N/A**	52	31	N/ A**	N/ A**	86	51	N/ A**	N/ A**	130	77	N/ A**	N/ A**	180	106
	5	N/ A**	N/ A**	25§	15§	N/ A**	N/A**	49	29	N/ A**	N/ A**	82	48	N/ A**	N/ A**	117	69	N/ A**	N/ A**	165	97
10	2	N/ A**	N/ A**	31	23	N/ A**	N/A**	61	45	N/ A**	N/ A**	103	75	N/ A**	N/ A**	162	118	N/ A**	N/ A**	221	161
	5	N/ A**	N/ A**	28§	20§	N/ A**	N/A**	57	42	N/ A**	N/ A**	96	70	N/ A**	N/ A**	148	108	N/ A**	N/ A**	204	149
	10	N/ A**	N/ A**	25§	18§	N/ A**	N/A**	50§	36§	N/ A**	N/ A**	87	62	N/ A**	N/ A**	139	99	N/ A**	N/ A**	181	129
15	2	N/ A**	N/ A**	35§	30§	N/ A**	N/A**	67	58	N/ A**	N/ A**	114	98	N/ A**	N/ A**	179	154	N/ A**	N/ A**	250	215
	5	N/ A**	N/ A**	33§	28§	N/ A**	N/A**	62	53	N/ A**	N/ A**	107	92	N/ A**	N/ A**	164	141	N/ A**	N/ A**	231	199
	10	N/ A**	N/ A**	28§	24§	N/ A**	N/A**	55§	47§	N/ A**	N/ A**	97	82	N/ A**	N/ A**	153	130	N/ A**	N/ A**	216	184
	15	N/ A**	N/ A**	NR‡	NR‡	N/ A**	N/A**	48§	41§	N/ A**	N/ A**	89§	76§	N/ A**	N/ A**	141	120	N/ A**	N/ A**	201	171
20	2	N/ A**	N/ A**	38§	34§	N/ A**	N/A**	74	67	N/ A**	N/ A**	124	112	N/ A**	N/ A**	201	181	N/ A**	N/ A**	274	247
	5	N/ A**	N/ A**	36§	32§	N/ A**	N/A**	68§	61§	N/ A**	N/ A**	116	104	N/ A**	N/ A**	184	166	N/ A**	N/ A**	254	229

(Continued)

Table C.5 (Continued)

Vent connector diameter, D, in To be used with chimney areas within the size limits at bottom of next page																					
3				4				5				6				7					
Appliance input rating limits, thousands of Btu/h																					
Height, H, ft	Lateral, L, ft	FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*	
		Min	Max	Max	DP†	Min	Max	Max	D- P†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†
	10	N/ A**	N/ A**	NR‡	NR‡	N/ A**	N/A**	60\$	53\$	N/ A**	N/ A**	107\$	95\$	N/ A**	N/ A**	172	153	N/ A**	N/ A**	237	211
	15	N/ A**	N/ A**	NR‡	NR‡	N/ A**	N/A**	NR‡	NR‡	N/ A**	N/ A**	97\$	86\$	N/ A**	N/ A**	159	142	N/ A**	N/ A**	220	196
	20	N/ A**	N/ A**	NR‡	NR‡	N/ A**	N/A**	NR‡	NR‡	N/ A**	N/ A**	83\$	74\$	N/ A**	N/ A**	148\$	132\$	N/ A**	N/ A**	206	183
25	2	N/ A**	N/ A**	40\$	36\$	N/ A**	N/A**	78\$	72\$	N/ A**	N/ A**	131	120	N/ A**	N/ A**	209	191	N/ A**	N/ A**	289	265
	5	N/ A**	N/ A**	NR‡	NR‡	N/ A**	N/A**	72\$	66\$	N/ A**	N/ A**	122\$	112\$	N/ A**	N/ A**	191	175	N/ A**	N/ A**	268	245
	10	N/ A**	N/ A**	NR‡	NR‡	N/ A**	N/A**	NR‡	NR‡	N/ A**	N/ A**	111\$	101\$	N/ A**	N/ A**	178\$	162\$	N/ A**	N/ A**	250	228
	15	N/ A**	N/ A**	NR‡	NR‡	N/ A**	N/A**	NR‡	NR‡	N/ A**	N/ A**	102\$	93\$	N/ A**	N/ A**	165\$	151\$	N/ A**	N/ A**	232\$	21- 1\$
	20	N/ A**	N/ A**	NR‡	NR‡	N/ A**	N/A**	NR‡	NR‡	N/ A**	N/ A**	NR‡	NR‡	N/ A**	N/ A**	NR‡	NR‡	N/ A**	N/ A**	217\$	19- 7\$
30	2	N/ A**	N/ A**	41\$	38\$	N/ A**	N/A**	82\$	76\$	N/ A**	N/ A**	137	127	N/ A**	N/ A**	216	201	N/ A**	N/ A**	303	282
	5	N/ A**	N/ A**	NR‡	NR‡	N/ A**	N/A**	76\$	71\$	N/ A**	N/ A**	128\$	119\$	N/ A**	N/ A**	198	184	N/ A**	N/ A**	281	261
	10	N/ A**	N/ A**	NR‡	NR‡	N/ A**	N/A**	67\$	62\$	N/ A**	N/ A**	115\$	107\$	N/ A**	N/ A**	184\$	171\$	N/ A**	N/ A**	263	245
	15	N/ A**	N/ A**	NR‡	NR‡	N/ A**	N/A**	NR‡	NR‡	N/ A**	N/ A**	107\$	100\$	N/ A**	N/ A**	171\$	159\$	N/ A**	N/ A**	243\$	22- 6\$

(Continued)

Table C.5 (Continued)

Vent connector diameter, D, in To be used with chimney areas within the size limits at bottom of next page																					
3				4				5				6				7					
Appliance input rating limits, thousands of Btu/h																					
Height, H, ft	Lateral, L, ft	FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*	
		Min	Max	Max	DP†	Min	Max	Max	D- P†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†
	20	N/ A**	N/ A**	NR‡	NR‡	N/ A**	N/A**	NR‡	NR‡	N/ A**	N/ A**	91\$	85\$	N/ A**	N/ A**	159\$	148\$	N/ A**	N/ A**	227\$	21- 1\$
	30	N/ A**	N/ A**	NR‡	NR‡	N/ A**	N/A**	NR‡	NR‡	N/ A**	N/ A**	NR‡	NR‡	N/ A**	N/ A**	NR‡	NR‡	N/ A**	N/ A**	188\$	17- 5\$
50	2	N/ A**	N/ A**	NR‡	NR‡	N/ A**	N/A**	92\$	87\$	N/ A**	N/ A**	161\$	153\$	N/ A**	N/ A**	251\$	238\$	N/ A**	N/ A**	351\$	33- 3\$
	5	N/ A**	N/ A**	NR‡	NR‡	N/ A**	N/A**	NR‡	NR‡	N/ A**	N/ A**	151\$	143\$	N/ A**	N/ A**	230\$	219\$	N/ A**	N/ A**	323\$	30- 7\$
	10	N/ A**	N/ A**	NR‡	NR‡	N/ A**	N/A**	NR‡	NR‡	N/ A**	N/ A**	138\$	131\$	N/ A**	N/ A**	215\$	204\$	N/ A**	N/ A**	304\$	28- 9\$
	15	N/ A**	N/ A**	NR‡	NR‡	N/ A**	N/A**	NR‡	NR‡	N/ A**	N/ A**	127\$	121\$	N/ A**	N/ A**	199\$	189\$	N/ A**	N/ A**	282\$	26- 8\$
	20	N/ A**	N/ A**	NR‡	NR‡	N/ A**	N/A**	NR‡	NR‡	N/ A**	N/ A**	NR‡	NR‡	N/ A**	N/ A**	185\$	176\$	N/ A**	N/ A**	264\$	25- 1\$
	30	N/ A**	N/ A**	NR‡	NR‡	N/ A**	N/A**	NR‡	NR‡	N/ A**	N/ A**	NR‡	NR‡	N/ A**	N/ A**	NR‡	NR‡	N/ A**	N/ A**	NR‡	NR‡
Minimum internal area of chimney, in²		12				19				28				38				50			
Maximum internal area of chimney, in²		49				88				137				198				269			

(Continued)

Table C.5 (Continued)

Vent connector diameter, D, in To be used with chimney areas within the size limits at bottom of next page																	
8				9				10				12					
Appliance input rating limits, thousands of Btu/h																	
Heigh- t, H, ft	Later- al, L, ft	FAN		NA*		FAN		NA*		FAN		NA*		FAN		NAT*	
		Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†
6	2	N/A**	N/A**	247	146	N/A**	N/A**	320	189	N/A**	N/A**	401	237	N/A**	N/A**	581	343
	5	N/A**	N/A**	231	136	N/A**	N/A**	298	176	N/A**	N/A**	376	222	N/A**	N/A**	561	331
10	2	N/A**	N/A**	298	218	N/A**	N/A**	388	283	N/A**	N/A**	491	358	N/A**	N/A**	724	529
	5	N/A**	N/A**	277	202	N/A**	N/A**	365	266	N/A**	N/A**	466	340	N/A**	N/A**	712	520
	10	N/A**	N/A**	263	187	N/A**	N/A**	342	243	N/A**	N/A**	444	315	N/A**	N/A**	668	474
15	2	N/A**	N/A**	336	289	N/A**	N/A**	441	379	N/A**	N/A**	562	483	N/A**	N/A**	841	723
	5	N/A**	N/A**	313	269	N/A**	N/A**	416	358	N/A**	N/A**	533	458	N/A**	N/A**	828	712
	10	N/A**	N/A**	296	252	N/A**	N/A**	394	335	N/A**	N/A**	567	482	N/A**	N/A**	777	660
	15	N/A**	N/A**	281	239	N/A**	N/A**	375	319	N/A**	N/A**	485	412	N/A**	N/A**	742	631
20	2	N/A**	N/A**	375	338	N/A**	N/A**	491	442	N/A**	N/A**	627	564	N/A**	N/A**	953	858
	5	N/A**	N/A**	350	315	N/A**	N/A**	463	417	N/A**	N/A**	597	537	N/A**	N/A**	933	840
	10	N/A**	N/A**	332	295	N/A**	N/A**	440	392	N/A**	N/A**	566	504	N/A**	N/A**	879	782
	15	N/A**	N/A**	314	279	N/A**	N/A**	418	372	N/A**	N/A**	541	481	N/A**	N/A**	840	748
	20	N/A**	N/A**	296	263	N/A**	N/A**	397	353	N/A**	N/A**	513	457	N/A**	N/A**	807	718
25	2	N/A**	N/A**	398	365	N/A**	N/A**	525	481	N/A**	N/A**	672	616	N/A**	N/A**	1033	946
	5	N/A**	N/A**	372	340	N/A**	N/A**	495	453	N/A**	N/A**	640	586	N/A**	N/A**	1014	929
	10	N/A**	N/A**	353	321	N/A**	N/A**	470	429	N/A**	N/A**	607	554	N/A**	N/A**	952	868
	15	N/A**	N/A**	334	304	N/A**	N/A**	447	408	N/A**	N/A**	581	530	N/A**	N/A**	911	830

(Continued)

Table C.5 (Continued)

Vent connector diameter, D, in To be used with chimney areas within the size limits at bottom of next page																	
8				9				10				12					
Appliance input rating limits, thousands of Btu/h																	
Heigh- t, H, ft	Later- al, L, ft	FAN		NA*		FAN		NA*		FAN		NA*		FAN		NAT*	
		Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†
	20	N/A**	N/A**	314	286	N/A**	N/A**	424	386	N/A**	N/A**	553	504	N/A**	N/A**	874	796
30	2	N/A**	N/A**	421	392	N/A**	N/A**	558	519	N/A**	N/A**	717	667	N/A**	N/A**	1112	1034
	5	N/A**	N/A**	393	365	N/A**	N/A**	526	489	N/A**	N/A**	683	635	N/A**	N/A**	1094	1017
	10	N/A**	N/A**	373	347	N/A**	N/A**	500	465	N/A**	N/A**	648	603	N/A**	N/A**	1025	953
	15	N/A**	N/A**	353	328	N/A**	N/A**	476	443	N/A**	N/A**	621	578	N/A**	N/A**	981	912
	20	N/A**	N/A**	332	309	N/A**	N/A**	450	419	N/A**	N/A**	592	551	N/A**	N/A**	940	874
	30	N/A**	N/A**	288§	268§	N/A**	N/A**	416	387	N/A**	N/A**	555	516	N/A**	N/A**	877	816
50	2	N/A**	N/A**	477	453	N/A**	N/A**	633	601	N/A**	N/A**	812	771	N/A**	N/A**	1243	1180
	5	N/A**	N/A**	445	423	N/A**	N/A**	596	566	N/A**	N/A**	774	735	N/A**	N/A**	1225	1164
	10	N/A**	N/A**	424§	403§	N/A**	N/A**	567	539	N/A**	N/A**	733	696	N/A**	N/A**	1147	1090
	15	N/A**	N/A**	400§	380§	N/A**	N/A**	539§	512§	N/A**	N/A**	702	667	N/A**	N/A**	1099	1044
	20	N/A**	N/A**	376§	357§	N/A**	N/A**	511§	485§	N/A**	N/A**	669§	636§	N/A**	N/A**	1050	998
	30	N/A**	N/A**	327§	311§	N/A**	N/A**	468§	445§	N/A**	N/A**	623§	592§	N/A**	N/A**	984	935
Minimum internal area of chimney, in²		63				78				95				132			
Maximum internal area of chimney, in²		352				445				550				792			

* NAT = natural draft appliance.

† DP = depressurization.

‡ NR = not recommended.

§ Possibility of continuous condensation.

** N/A = not applicable.

Table C.6
Capacity of tile-lined masonry chimneys with single-wall vent connectors serving a single Category I appliance
 (See Clauses [8.12.8](#) and [8.13.1](#), and Figures [C.5](#) and [C.6](#).)

Vent connector diameter, D, in To be used with chimney areas within the size limits at bottom of next page																					
3				4				5				6				7					
Appliance input rating limits, thousands of Btu/h																					
Height, H, ft	Later- al, L, ft	FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*	
		Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	D- P†
6	2	N/A**	N/A**	28	17	N/ A**	N/ A**	52	31	N/A**	N/A**	86	51	N/ A**	N/ A**	130	77	N/A**	N/ A**	180	106
	5	N/A**	N/A**	25§	15§	N/ A**	N/ A**	48	28	N/A**	N/A**	81	48	N/ A**	N/ A**	116	68	N/A**	N/ A**	164	97
10	2	N/A**	N/A**	31	23	N/ A**	N/ A**	61	45	N/A**	N/A**	102	74	N/ A**	N/ A**	161	118	N/A**	N/ A**	220	161
	5	N/A**	N/A**	28§	20§	N/ A**	N/ A**	56	41	N/A**	N/A**	95	69	N/ A**	N/ A**	147	107	N/A**	N/ A**	203	148
	10	N/A**	N/A**	24§	17§	N/ A**	N/ A**	49§	35§	N/A**	N/A**	86	61	N/ A**	N/ A**	137	97	N/A**	N/ A**	189	134
15	2	N/A**	N/A**	35§	30§	N/ A**	N/ A**	67	58	N/A**	N/A**	113	97	N/ A**	N/ A**	178	153	N/A**	N/ A**	249	214
	5	N/A**	N/A**	32§	28§	N/ A**	N/ A**	61	52	N/A**	N/A**	106	91	N/ A**	N/ A**	163	140	N/A**	N/ A**	230	198
	10	N/A**	N/A**	27§	23§	N/ A**	N/ A**	54§	46§	N/A**	N/A**	96	82	N/ A**	N/ A**	151	128	N/A**	N/ A**	214	182
	15	N/A**	N/A**	NR‡	NR‡	N/ A**	N/ A**	46§	39§	N/A**	N/A**	87§	74§	N/ A**	N/ A**	138	117	N/A**	N/ A**	198	168
20	2	N/A**	N/A**	38§	34§	N/ A**	N/ A**	73	66	N/A**	N/A**	123	111	N/ A**	N/ A**	200	180	N/A**	N/ A**	273	246
	5	N/A**	N/A**	35§	32§	N/ A**	N/ A**	67§	60§	N/A**	N/A**	115	104	N/ A**	N/ A**	183	165	N/A**	N/ A**	252	227

(Continued)

Table C.6 (Continued)

Vent connector diameter, D, in To be used with chimney areas within the size limits at bottom of next page																					
3				4				5				6				7					
Appliance input rating limits, thousands of Btu/h																					
Height, H, ft	Later- al, L, ft	FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*	
		Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	D- P†
	10	N/A**	N/A**	NR‡	NR‡	N/A**	N/A**	59§	53§	N/A**	N/A**	105§	93§	N/A**	N/A**	170	151	N/A**	N/A**	235	209
	15	N/A**	N/A**	NR‡	NR‡	N/A**	N/A**	NR‡	NR‡	N/A**	N/A**	95§	85§	N/A**	N/A**	156	139	N/A**	N/A**	217	193
	20	N/A**	N/A**	NR‡	NR‡	N/A**	N/A**	NR‡	NR‡	N/A**	N/A**	80§	71§	N/A**	N/A**	144§	128	N/A**	N/A**	202	180
25	2	N/A**	N/A**	40§	36§	N/A**	N/A**	77§	71§	N/A**	N/A**	130	119	N/A**	N/A**	208	190	N/A**	N/A**	288	264
	5	N/A**	N/A**	NR‡	NR‡	N/A**	N/A**	71§	65§	N/A**	N/A**	121§	111	N/A**	N/A**	190	174	N/A**	N/A**	266	243
	10	N/A**	N/A**	NR‡	NR‡	N/A**	N/A**	63§	57§	N/A**	N/A**	109§	99§	N/A**	N/A**	176§	16-0§	N/A**	N/A**	248	226
	15	N/A**	N/A**	NR‡	NR‡	N/A**	N/A**	NR‡	NR‡	N/A**	N/A**	100§	92§	N/A**	N/A**	162§	14-8§	N/A**	N/A**	229§	208
	20	N/A**	N/A**	NR‡	NR‡	N/A**	N/A**	NR‡	NR‡	N/A**	N/A**	84§	77§	N/A**	N/A**	150§	13-6§	N/A**	N/A**	213§	194
30	2	N/A**	N/A**	41§	38§	N/A**	N/A**	81§	75§	N/A**	N/A**	136	126	N/A**	N/A**	215	200	N/A**	N/A**	302	281
	5	N/A**	N/A**	NR‡	NR‡	N/A**	N/A**	75§	70§	N/A**	N/A**	127§	11-8§	N/A**	N/A**	196	182	N/A**	N/A**	279	259
	10	N/A**	N/A**	NR‡	NR‡	N/A**	N/A**	66§	61§	N/A**	N/A**	113§	10-5§	N/A**	N/A**	182§	16-9§	N/A**	N/A**	260	242
	15	N/A**	N/A**	NR‡	NR‡	N/A**	N/A**	NR‡	NR‡	N/A**	N/A**	105§	98§	N/A**	N/A**	168§	15-6§	N/A**	N/A**	240§	22-3§

(Continued)

Table C.6 (Continued)

Vent connector diameter, D, in To be used with chimney areas within the size limits at bottom of next page																					
3				4				5				6				7					
Appliance input rating limits, thousands of Btu/h																					
Height, H, ft	Later- al, L, ft	FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*	
		Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	D- P†
	20	N/A**	N/A**	NR‡	NR‡	N/A**	N/A**	NR‡	NR‡	N/A**	N/A**	88\$	82\$	N/A**	N/A**	155\$	14-4\$	N/A**	N/A**	223\$	20-7\$
	30	N/A**	N/A**	NR‡	NR‡	N/A**	N/A**	NR‡	NR‡	N/A*	N/A*	NR‡	NR‡	N/A**	N/A**	NR‡	NR‡	N/A**	N/A**	182\$	16-9\$
50	2	N/A**	N/A**	NR‡	NR‡	N/A**	N/A**	91\$	86\$	N/A**	N/A**	160\$	15-2\$	N/A**	N/A**	250\$	23-8\$	N/A**	N/A**	350\$	33-3\$
	5	N/A**	N/A**	NR‡	NR‡	N/A**	N/A**	NR‡	NR‡	N/A**	N/A**	140\$	13-3\$	N/A**	N/A**	228\$	21-7\$	N/A**	N/A**	321\$	30-5\$
	10	N/A**	N/A**	NR‡	NR‡	N/A**	N/A**	NR‡\$	NR‡	N/A**	N/A**	136\$	12-9\$	N/A**	N/A**	212\$	20-1\$	N/A**	N/A**	301\$	28-6\$
	15	N/A**	N/A**	NR‡	NR‡	N/A**	N/A**	NR‡	NR‡	N/A**	N/A**	124\$	11-8\$	N/A**	N/A**	195\$	18-5\$	N/A**	N/A**	278\$	26-4\$
	20	N/A**	N/A**	NR‡	NR‡	N/A**	N/A**	NR‡	NR‡	N/A**	N/A**	NR‡	NR‡	N/A**	N/A**	180\$	17-1\$	N/A**	N/A**	258\$	24-5\$
	30	N/A**	N/A**	NR‡	NR‡	N/A**	N/A**	NR‡	NR‡	N/A*	N/A*	NR‡	NR‡	N/A**	N/A**	NR‡	NR‡	N/A**	N/A**	NR‡	NR‡
Minimum internal area of chimney, in²		12				19				28				38				50			
Maximum internal area of chimney, in²		49				88				137				198				269			

(Continued)

Table C.6 (Continued)

Vent connector diameter, D, in To be used with chimney areas within the size limits at bottom of next page																	
8				9				10				12					
Appliance input rating limits, thousands of Btu/h																	
Height, H, ft	Later- al, L, ft	FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*	
		Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†
6	2	N/A**	N/A**	247	146	N/A**	N/A**	319	188	N/A**	N/A**	400	236	N/A**	N/A**	580	342
	5	N/A**	N/A**	230	136	N/A**	N/A**	297	175	N/A**	N/A**	375	221	N/A**	N/A**	560	330
10	2	N/A**	N/A**	297	217	N/A**	N/A**	387	283	N/A**	N/A**	490	358	N/A**	N/A**	722	527
	5	N/A**	N/A**	276	201	N/A**	N/A**	364	266	N/A**	N/A**	465	339	N/A**	N/A**	710	518
	10	N/A**	N/A**	261	185	N/A**	N/A**	345	245	N/A**	N/A**	441	313	N/A**	N/A**	665	472
15	2	N/A**	N/A**	335	288	N/A**	N/A**	440	378	N/A**	N/A**	560	482	N/A**	N/A**	840	722
	5	N/A**	N/A**	312	268	N/A**	N/A**	414	356	N/A**	N/A**	531	457	N/A**	N/A**	825	710
	10	N/A**	N/A**	294	250	N/A**	N/A**	392	333	N/A**	N/A**	504	428	N/A**	N/A**	774	658
	15	N/A**	N/A**	278	236	N/A**	N/A**	372	316	N/A**	N/A**	481	409	N/A**	N/A**	738	627
20	2	N/A**	N/A**	374	337	N/A**	N/A**	490	441	N/A**	N/A**	625	563	N/A**	N/A**	950	855
	5	N/A**	N/A**	348	313	N/A**	N/A**	461	415	N/A**	N/A**	594	535	N/A**	N/A**	930	837
	10	N/A**	N/A**	330	294	N/A**	N/A**	437	389	N/A**	N/A**	562	500	N/A**	N/A**	875	779
	15	N/A**	N/A**	311	277	N/A**	N/A**	414	368	N/A**	N/A**	536	477	N/A**	N/A**	835	743
	20	N/A**	N/A**	292	260	N/A**	N/A**	392	349	N/A**	N/A**	510	454	N/A**	N/A**	800	712
25	2	N/A**	N/A**	397	364	N/A**	N/A**	523	479	N/A**	N/A**	670	614	N/A**	N/A**	1030	944
	5	N/A**	N/A**	370	339	N/A**	N/A**	493	451	N/A**	N/A**	637	584	N/A**	N/A**	1010	926
	10	N/A**	N/A**	350	319	N/A**	N/A**	467	425	N/A**	N/A**	603	550	N/A**	N/A**	948	864
	15	N/A**	N/A**	330	301	N/A**	N/A**	443	403	N/A**	N/A**	576	525	N/A**	N/A**	905	825

(Continued)

Table C.6 (Continued)

Vent connector diameter, D, in To be used with chimney areas within the size limits at bottom of next page																	
8				9				10				12					
Appliance input rating limits, thousands of Btu/h																	
Height, H, ft	Later- al, L, ft	FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*	
		Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†
	20	N/A**	N/A**	310	282	N/A**	N/A**	419	382	N/A**	N/A**	548	499	N/A**	N/A**	866	790
30	2	N/A**	N/A**	420	391	N/A**	N/A**	556	517	N/A**	N/A**	715	665	N/A**	N/A**	1110	1032
	5	N/A**	N/A**	391	364	N/A**	N/A**	524	487	N/A**	N/A**	680	632	N/A**	N/A**	1090	1014
	10	N/A**	N/A**	370	344	N/A**	N/A**	496	461	N/A**	N/A**	644	599	N/A**	N/A**	1020	949
	15	N/A**	N/A**	349	325	N/A**	N/A**	471	438	N/A**	N/A*	615	572	N/A**	N/A**	975	907
	20	N/A**	N/A**	327	304	N/A**	N/A**	445	414	N/A**	N/A**	585	544	N/A**	N/A**	932	867
	30	N/A**	N/A**	281§	261§	N/A**	N/A**	408	379	N/A**	N/A**	544	506	N/A**	N/A**	865	804
50	2	N/A**	N/A**	475	451	N/A**	N/A**	631	599	N/A**	N/A**	810	770	N/A**	N/A**	1240	1178
	5	N/A**	N/A**	442	420	N/A**	N/A**	593	563	N/A**	N/A**	770	732	N/A**	N/A**	1220	1159
	10	N/A**	N/A**	420§	399§	N/A**	N/A**	562	534	N/A**	N/A**	728	692	N/A**	N/A**	1140	1083
	15	N/A**	N/A**	395§	375§	N/A**	N/A**	533§	506§	N/A**	N/A**	695	660	N/A**	N/A**	1090	1036
	20	N/A**	N/A**	370§	352§	N/A**	N/A**	504§	479§	N/A**	N/A**	660§	627§	N/A**	N/A**	1040	988
	30	N/A**	N/A**	318§	302§	N/A**	N/A**	458§	435§	N/A**	N/A**	610§	580§	N/A**	N/A**	970	922
Minimum internal area of chimney, in ²		63		78				95				132					
Maximum internal area of chimney, in ²		352		445				550				792					

* NAT = natural draft appliance.

† DP = depressurization.

‡ NR = not recommended.

§ Possibility of continuous condensation.

** N/A = not applicable.

Table C.7
Capacity of tile-lined masonry chimneys with Type B double-wall vent connectors serving two or more Category I appliances

(See Clauses [8.12.8](#), [8.13.2](#), and [C.2.9](#) and Figures [C.6](#) and [C.7](#).)

Capacity of vent connectors																	
Vent height, H, ft		Vent connector rise, R, ft		Vent connector diameter, D, in													
				3				4				5				6	
				Appliance input rating limits, thousands of Btu/h													
		FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*	
		Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†
6	1	24	NR‡	21	13	39	62	40	24	52	106	67	41	65	194	101	62
	2	26	43	28	17	41	79	52	32	53	133	85	52	67	230	124	76
	3	27	49	34	21	42	92	61	37	55	155	97	59	69	262	143	87
15	1	24	48	23	20	38	93	44	37	54	154	74	63	72	277	114	97
	2	25	55	31	26	39	105	55	47	56	174	89	76	74	299	134	114
	3	26	59	35	30	41	115	64	54	57	189	102	87	76	319	153	130
20	1	24	50	24	21	38	99	45	40	53	167	77	67	71	304	118	104
	2	25	57	31	27	39	111	56	49	55	185	91	80	73	325	138	121
	3	26	61	35	31	41	120	65	56	57	200	104	91	75	343	156	137
25	1	24	52	24	22	37	105	47	42	53	179	79	72	70	330	123	111
	2	25	58	32	29	38	116	57	52	55	197	93	84	73	350	141	128
	3	26	62	36	32	40	126	65	59	56	210	105	96	75	368	160	145
30	1	24	54	25	23	37	111	48	45	52	192	82	76	69	357	127	118
	2	25	60	32	30	38	122	58	54	54	208	95	88	72	376	145	135
	3	26	64	36	33	40	131	66	61	56	221	107	100	74	392	163	152
50	1	23	52	26	25	36	116	49	48	51	209	82	80	67	405	133	129
	2	24	59	31	30	37	127	58	56	53	225	96	93	70	421	152	147

(Continued)

Table C.7 (Continued)

Capacity of vent connectors																			
Vent height, H, ft		Vent connector rise, R, ft		Vent connector diameter, D, in															
				3				4				5				6			
				Appliance input rating limits, thousands of Btu/h															
				FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*	
		Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†		
		3		26	64	37	36	39	135	66	64	55	237	108	105	72	435	170	165

Capacity of vent connectors																	
Vent height, H, ft		Vent con- nector rise, R, ft		Vent connector diameter, D, in													
				7				8				9				10	
				Appliance input rating limits, thousands of Btu/h													
				FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN	
		Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†
6	1	87	274	141	86	104	370	201	123	124	479	253	154	145	599	319	195
	2	89	324	173	106	107	436	232	142	127	562	330	201	148	694	378	231
	3	91	369	203	124	109	491	270	165	129	633	349	213	151	795	439	268
15	1	100	384	174	148	125	511	229	195	153	658	297	252	184	824	375	319
	2	103	419	192	163	128	558	260	221	156	718	339	288	187	900	432	367
	3	105	448	215	183	131	597	292	248	159	760	382	325	190	960	486	413
20	1	99	424	178	157	123	567	238	209	150	733	310	272	181	921	394	347
	2	102	456	198	173	126	610	269	236	154	788	352	309	184	990	449	395
	3	104	483	221	194	129	647	300	264	157	829	394	346	187	1047	502	441
25	1	97	464	183	165	121	624	246	223	148	808	324	293	178	1018	413	374
	2	100	494	203	184	124	663	278	252	151	858	365	331	182	1081	467	422
	3	102	519	227	206	127	696	309	279	154	899	406	368	185	1133	519	470

(Continued)

Table C.7 (Continued)

Capacity of vent connectors																				
Vent height, H, ft	Vent connector rise, R, ft	Vent connector diameter, D, in																		
		7				8				9				10						
		Appliance input rating limits, thousands of Btu/h																		
		FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*				
Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†					
30	1	96	504	187	174	119	680	255	237	145	883	337	313	175	1115	432	402			
	2	99	531	209	194	122	715	287	267	149	928	378	352	179	1171	484	450			
	3	101	554	233	217	125	746	317	295	152	968	418	389	182	1220	535	498			
50	1	92	582	198	192	115	798	271	263	140	1049	362	351	168	1334	462	448			
	2	95	604	222	215	118	827	304	295	143	1085	400	388	172	1379	510	495			
	3	98	624	247	240	121	854	334	324	147	1118	439	426	176	1421	558	541			
Capacity of common vent																				
Minimum internal area of chimney, A, in²																				
12				19						28						38				
Vent Height, H, ft	Combined appliance input rating, thousands of Btu/h																			
	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†
	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)	+FA-N	+NA-T*	(F+N)	+NA-T*	(N+N)	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)
6	NR‡	74	45	25	15	NR‡	119	73	46	28	NR‡	178	109	71	43	NR‡	257	157	103	63
8	NR‡	80	57	28	20	NR‡	130	92	53	38	NR‡	193	137	82	58	NR‡	279	198	119	84
10	NR‡	84	66	31	24	NR‡	138	108	56	44	NR‡	207	161	90	70	NR‡	299	233	131	102
15	NR‡	90	77	36	31	NR‡	152	129	67	57	NR‡	233	198	106	90	NR‡	334	284	152	129
20	NR‡	92	83	41	37	NR‡	159	143	75	68	NR‡	250	225	122	110	NR‡	368	331	172	155
25	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	260	238	130	119	NR‡	386	354	185	170

(Continued)

Table C.7 (Continued)

Capacity of common vent																				
Minimum internal area of chimney, A, in ²																				
12						19					28					38				
Vent Height, H, ft	Combined appliance input rating, thousands of Btu/h																			
	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†
	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)
30	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	270	251	137	127	NR‡	404	376	198	184
50	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡
Capacity of common vent																				
Minimum internal area of chimney, A, in ²																				
50						63					78					113				
Vent Height, H, ft	Combined appliance input rating, thousands of Btu/h																			
	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†
	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)
6	NR‡	351	214	143	87	NR‡	458	279	188	115	NR‡	582	355	246	150	NR‡	853	520	NR‡	NR‡
8	NR‡	384	273	163	116	NR‡	501	356	218	155	NR‡	636	452	278	197	NR‡	937	665	408	290
10	NR‡	409	319	177	138	NR‡	538	420	236	184	NR‡	686	535	302	236	NR‡	1010	788	454	354
15	NR‡	467	397	212	180	NR‡	611	519	283	241	NR‡	781	664	365	310	NR‡	1156	983	546	464
20	NR‡	508	457	243	219	NR‡	668	601	325	293	NR‡	858	772	419	377	NR‡	1286	1157	648	583
25	NR‡	536	491	261	239	NR‡	708	648	353	324	NR‡	914	837	458	419	NR‡	1330	1217	699	640
30	NR‡	564	525	278	259	NR‡	747	695	381	354	NR‡	969	901	496	461	NR‡	1373	1277	749	697
50	NR‡	620	595	328	315	NR‡	831	798	461	443	NR‡	1089	1045	606	582	NR‡	1692	1624	922	885

* NAT = natural draft appliance.

† DP = depressurization.

‡ NR = not recommended.

Table C.8

Capacity of tile-lined masonry chimneys with single-wall vent connectors serving two or more Category I appliances

(See Clauses 8.12.8, 8.13.2, and C.2.9, and Figures C.7 and C.8.)

Capacity of vent connectors																	
Total vent height, H, ft	Vent con- nector rise, R, ft	Vent connector diameter, D, in															
		3				4				5				6			
		Appliance input rating limits, thousands of Btu/h															
		FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*	
		Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†
6	1	NR‡	NR‡	21	14	NR‡	NR‡	39	25	NR‡	NR‡	66	43	179	191	100	65
	2	NR‡	NR‡	28	18	NR‡	NR‡	52	34	NR‡	NR‡	84	55	186	227	123	80
	3	NR‡	NR‡	34	22	NR‡	NR‡	61	40	134	153	97	63	193	258	142	92
15	1	NR‡	NR‡	23	20	NR‡	NR‡	43	37	129	151	73	62	199	271	112	95
	2	NR‡	NR‡	30	26	92	103	54	46	135	170	88	75	207	295	132	112
	3	NR‡	NR‡	34	29	96	112	63	54	141	185	101	86	215	315	151	128
20	1	NR‡	NR‡	23	21	NR‡	NR‡	44	40	128	163	75	67	197	296	116	103
	2	NR‡	NR‡	30	27	92	108	55	49	134	181	90	79	205	319	135	120
	3	NR‡	NR‡	34	30	96	117	64	57	140	195	102	91	213	337	154	136
25	1	NR‡	NR‡	24	22	NR‡	NR‡	46	42	127	175	78	71	195	322	120	110
	2	NR‡	NR‡	31	28	91	114	56	51	133	192	91	84	203	342	139	127
	3	NR‡	NR‡	35	32	95	122	64	59	139	206	104	95	211	359	157	144
30	1	NR‡	NR‡	24	23	86	108	47	45	126	187	80	76	193	347	124	118
	2	NR‡	NR‡	31	29	91	119	57	54	132	203	93	88	201	366	142	135
	3	NR‡	NR‡	35	33	95	127	65	62	138	216	105	100	209	381	160	152
50	1	NR‡	NR‡	25	24	85	113	48	46	124	204	80	76	188	392	130	124
	2	NR‡	NR‡	31	29	89	123	57	54	130	218	94	89	196	408	149	142
	3	NR‡	NR‡	35	33	94	131	65	62	136	231	106	101	205	422	167	159

(Continued)

Table C.8 (Continued)

Capacity of vent connectors																	
Total vent height, H, ft	Vent connector rise, R, ft	Vent connector diameter, D, in															
		7				8				9				10			
		Appliance input rating limits, thousands of Btu/h															
		FAN		NAT*		FAN		NAT*		FAN		NAT*		FAN		NAT*	
		Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†	Min	Max	Max	DP†
6	1	231	271	140	91	292	366	200	130	362	474	252	164	499	594	283	184
	2	239	321	172	112	301	432	231	150	373	557	299	194	509	696	331	215
	3	247	365	202	131	309	491	269	175	381	634	348	226	519	793	375	244
15	1	268	376	171	145	349	502	225	191	445	646	291	247	623	808	360	306
	2	277	411	189	161	359	548	256	218	456	706	334	284	634	884	402	342
	3	286	439	213	181	368	586	289	246	466	755	378	321	646	945	437	371
20	1	265	415	175	155	345	556	233	207	440	719	304	269	615	902	392	348
	2	274	447	194	172	355	598	265	235	451	773	347	307	627	971	431	383
	3	283	473	218	193	365	634	297	263	461	819	389	344	639	1028	465	412
25	1	262	453	179	164	342	611	242	222	435	791	317	292	608	995	423	390
	2	272	482	200	184	352	649	273	251	447	841	359	330	620	1058	461	425
	3	280	506	224	206	361	681	304	279	457	882	401	368	633	1110	493	454
30	1	259	492	183	174	338	665	250	238	430	864	330	314	600	1089	455	432
	2	269	518	205	195	348	699	282	268	442	908	372	353	613	1145	490	466
	3	277	540	229	218	358	729	312	296	452	946	412	391	626	1193	521	495
50	1	252	567	194	184	328	778	265	252	417	1022	355	337	582	1302	537	510
	2	262	588	218	207	339	806	298	283	429	1058	393	373	596	1346	567	539
	3	271	607	243	231	349	831	328	312	440	1090	431	409	610	1386	595	565

(Continued)

Table C.8 (Continued)

Capacity of common vent																				
Common vent equivalent chimney diameter, D, in																				
4						5					6					7				
Total vent height, H, ft	Combined appliance input rating, thousands of Btu/h																			
	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†
	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)
6	NR‡	73	44	25	15	NR‡	118	71	45	27	NR‡	176	106	71	43	NR‡	255	153	102	61
8	NR‡	79	58	28	20	NR‡	128	93	52	38	NR‡	190	139	81	59	NR‡	276	201	118	86
10	NR‡	83	66	31	25	NR‡	136	109	56	45	NR‡	205	164	89	71	NR‡	295	236	129	103
15	NR‡	88	75	36	31	NR‡	149	127	66	56	NR‡	230	196	105	89	NR‡	335	285	150	128
20	NR‡	90	80	40	36	NR‡	157	140	74	66	NR‡	247	220	120	107	NR‡	362	322	170	151
25	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	257	234	128	117	NR‡	380	346	183	166
30	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	266	247	135	126	NR‡	398	370	195	181
50	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡	NR‡
Capacity of common vent																				
Common vent equivalent chimney diameter, D, in																				
8						9					10					12				
Total vent height, H ft	Combined appliance input rating, thousands of Btu/h																			
	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†
	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)
6	NR‡	348	209	142	85	NR‡	455	273	187	112	NR‡	579	347	245	147	NR‡	846	508	NR‡	NR‡
8	NR‡	390	285	162	118	NR‡	497	363	217	158	NR‡	633	462	277	202	NR‡	928	677	405	296
10	NR‡	405	324	175	140	NR‡	532	426	234	187	NR‡	680	544	300	240	NR‡	1000	800	450	360

(Continued)

Table C.8 (Continued)

Capacity of common vent																				
Total vent heigh- t, H ft	Common vent equivalent chimney diameter, D, in																			
	8					9					10					12				
	Combined appliance input rating, thousands of Btu/h																			
	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†	FAN	FAN	DP†	NAT*	DP†
	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)	+FAN	+NA-T*	(F+N)	+NA-T*	(N+N)
	15	NR‡	460	391	210	179	NR‡	602	512	280	238	NR‡	772	656	360	306	NR‡	1139	968	540
20	NR‡	503	448	240	214	NR‡	661	588	321	286	NR‡	849	756	415	369	NR‡	1264	1125	640	570
25	NR‡	531	484	258	235	NR‡	700	638	349	319	NR‡	903	823	453	413	NR‡	1356	1236	690	629
30	NR‡	558	519	275	256	NR‡	739	687	377	351	NR‡	957	890	490	456	NR‡	1447	1346	740	688
50	NR‡	612	581	325	309	NR‡	821	780	456	433	NR‡	1076	1022	600	570	NR‡	1672	1588	910	865

* NAT = natural draft appliance.

† DP = depressurization.

‡ NR = not recommended.

Note: See Clause [8.12.8](#).

Table C.9
Single-wall vent connector total length limits for multiple appliance installations
 (See Clause [8.18.7.](#))

Vent connector diameter, in (cm)	3 (7.6)	4 (10)	5 (13)	6 (15)	7 (18)	8 (20)
Maximum vent connector length, ft	5 (1.5)	6 (1.8)	8 (2.4)	10 (3)	11 (3.3)	12 (3.7)

Notes:

- 1) Double-wall vent connectors may exceed these values up to a limit of twice these specified values. However, when these values are exceeded using a Type B vent, the capacity shall be reduced by 10%.
- 2) For configurations in which vent connectors are combined prior to entering the common vent, the vent connector length of a vent connector serving a single appliance shall be measured from the appliance draft-hood outlet or flue collar to the point where the vent gases meet the vent gases from another appliance.

Table C.10
Masonry chimney liner dimensions with circular equivalents
 (See Clause [8.13.3.](#))

Nominal liner size, In (mm)	Inside dimensions of liner, in (mm)	Inside diameter or equivalent diameter, in (mm)	Equivalent area, in² (cm²)
4 × 8 (102) × (203)	2-1/2 × 6-1/2 (64) × (165)	4 (102) 5 (127) 6 (152) 7 (178)	12.2 (81.1) 19.6 (126.7) 28.3 (182.4) 38.3 (248.3)
8 × 8 (203) × (203)	6-3/4 × 6-3/4 (171) × (171)	7.4 (188) 8 (203)	42.7 (277.5) 50.3 (324.3)
8 × 12 (203) × (305)	6-1/2 × 10-1/2 (165) × (267)	9 (229) 10 (254)	63.6 (410.4) 78.5 (506.7)
12 × 12 (305) × (305)	9-3/4 × 9-3/4 (248) × (248)	10.4 (264) 11 (279)	84.9 (548.1) 95 (613.1)
12 × 16 (305) × (406)	9-1/2 × 13-1/2 (241) × (343)	11.8 (300) 12 (305) 14 (356)	109.4 (705.5) 113.1 (729.7) 153.9 (993.1)
16 × 16 (406) × (406)	13-1/4 × 13-1/4 (337) × (337)	14.5 (368) 15 (381)	165.1 (1065.4) 176.7 (1140.1)
16 × 20 (406) × (508)	13 × 17 (330) × (432)	16.2 (411) 18 (457)	206.1 (1329.8) 254.5 (1641.7)
20 × 20 (508) × (508)	16-3/4 × 16-3/4 (425) × (425)	18.2 (462) 20 (508)	260.2 (1678.4) 314.2 (2026.8)
20 × 24 (508) × (610)	16-1/2 × 20-1/2 (419) × (521)	20.1 (511) 22 (559)	317.3 (2047.1) 380.1 (2452.5)

(Continued)

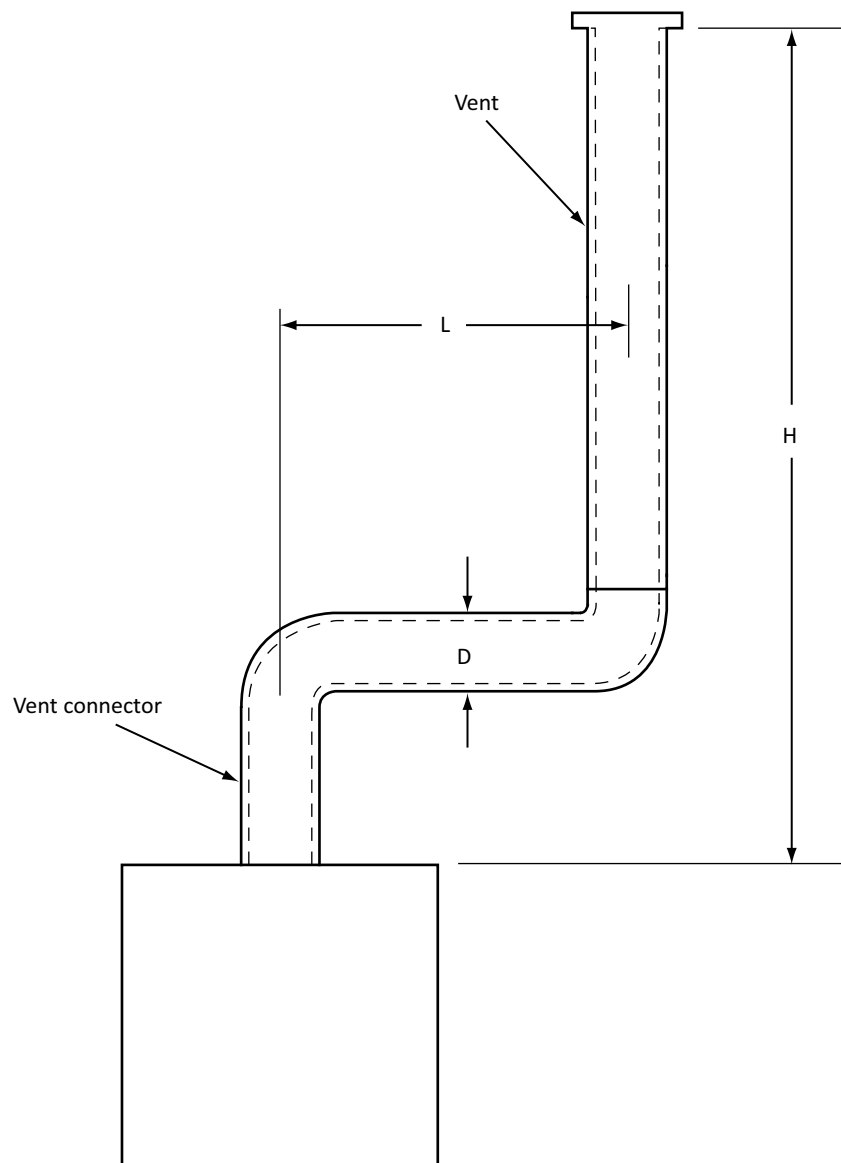
Table C.10 (Continued)

Nominal liner size, In (mm)	Inside dimensions of liner, in (mm)	Inside diameter or equivalent diameter, in (mm)	Equivalent area, in ² (cm ²)
24 × 24 (610) x (610)	20-1/4 × 20-1/4 (514) x (514)	22.1 (561) 24 (610)	383.6 (2474.8) 452.4 (2918.6)
24 × 28 (610) x (711)	20-1/4 × 24-1/4 (514) x (616)	24.1 (612)	456.2 (2943.0)
28 × 28 (711) x (711)	24-1/4 × 24-1/4 (616) x (616)	26.4 (671) 27 (686)	547.4 (3531.5) 572.6 (3693.9)
30 × 30 (762) x (762)	25-1/2 × 25-1/2 (648) x (648)	27.9 (709) 30 (762)	611.4 (3944.3) 706.9 (4560.4)
30 × 36 (762) x (914)	25-1/2 × 31-1/2 (648) x (800)	30.9 (785) 33 (838)	749.9 (4838.1) 855.3 (5518.0)
36 × 36 (914) x (914)	31-1/2 × 31-1/2 (800) x (800)	34.4 (874) 36 (914)	929.4 (5996.2) 1017.9 (6566.9)

Note: When liner sizes differ dimensionally from those shown in this table, equivalent diameters may be determined from published tables for square and rectangular ducts of equivalent carrying capacity or by other engineering methods.

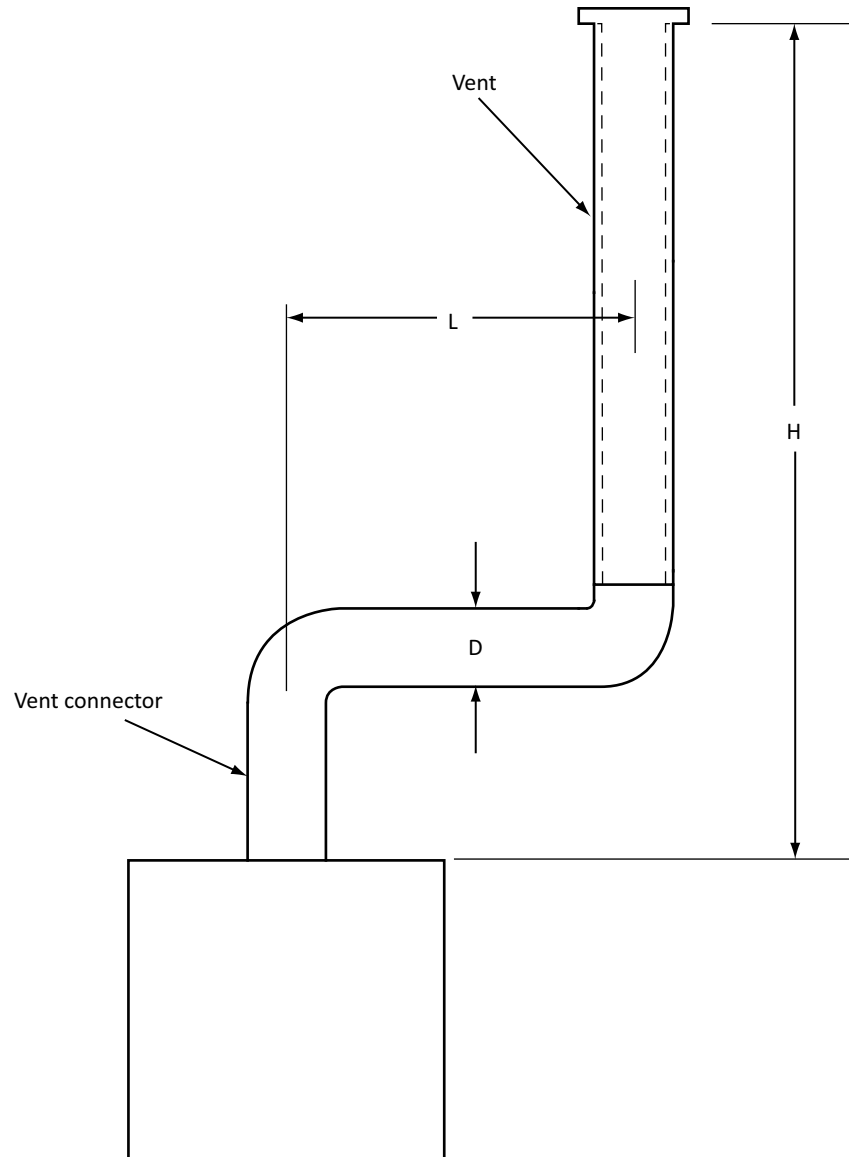
Note: Table [C.1](#) shall be used when a Type B vent is used for both the vent connector and the vertical vent.

Figure C.1
Configuration of Type B double-wall vents with Type B double-wall vent connectors serving a single Category I appliance



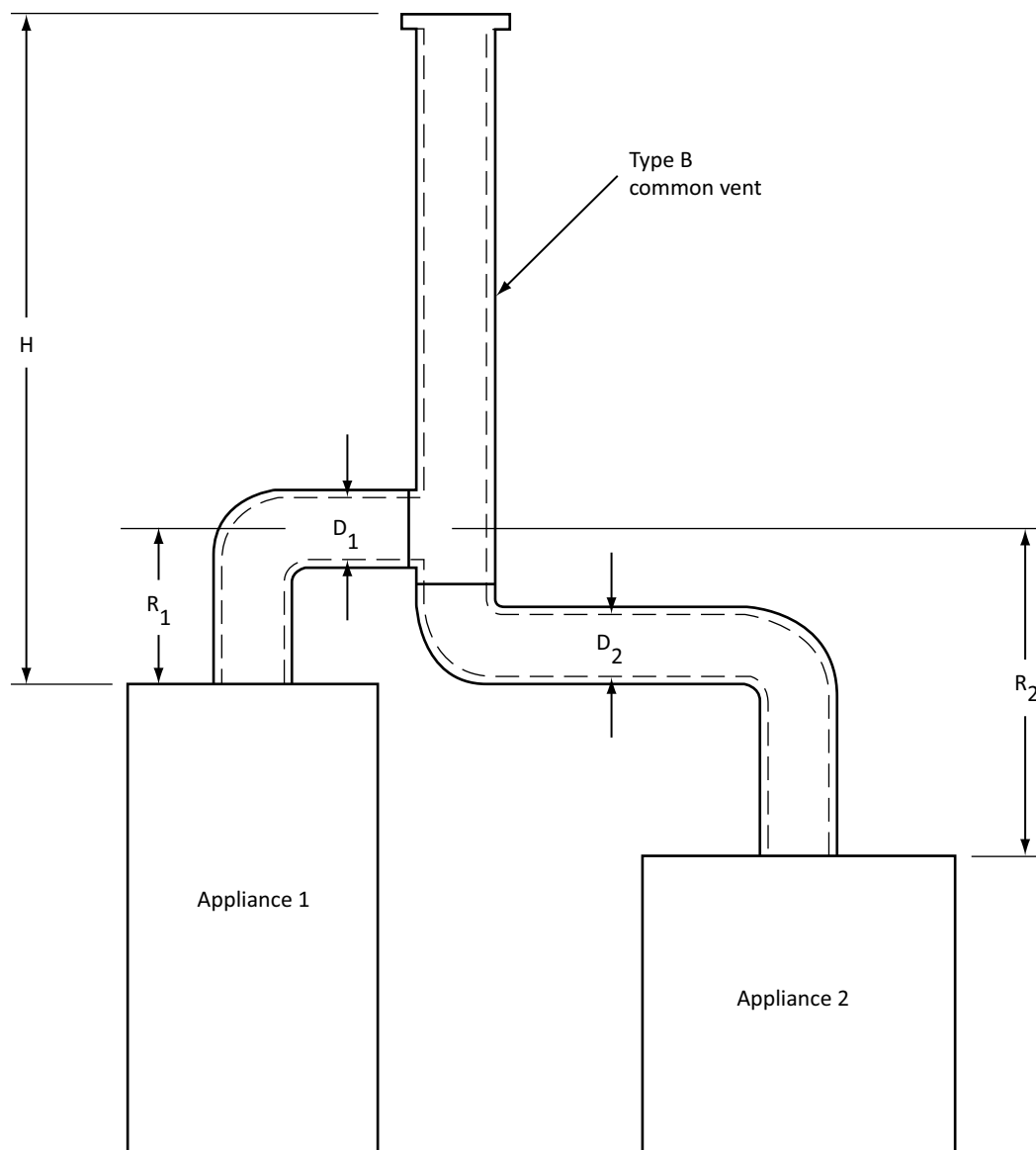
Note: Table [C.2](#) shall be used when a single-wall metal vent connector is attached to a Type B vertical vent.

Figure C.2
Configuration of Type B double-wall vents with single-wall metal vent connectors
serving a single Category I appliance
(See Table [C.2](#).)



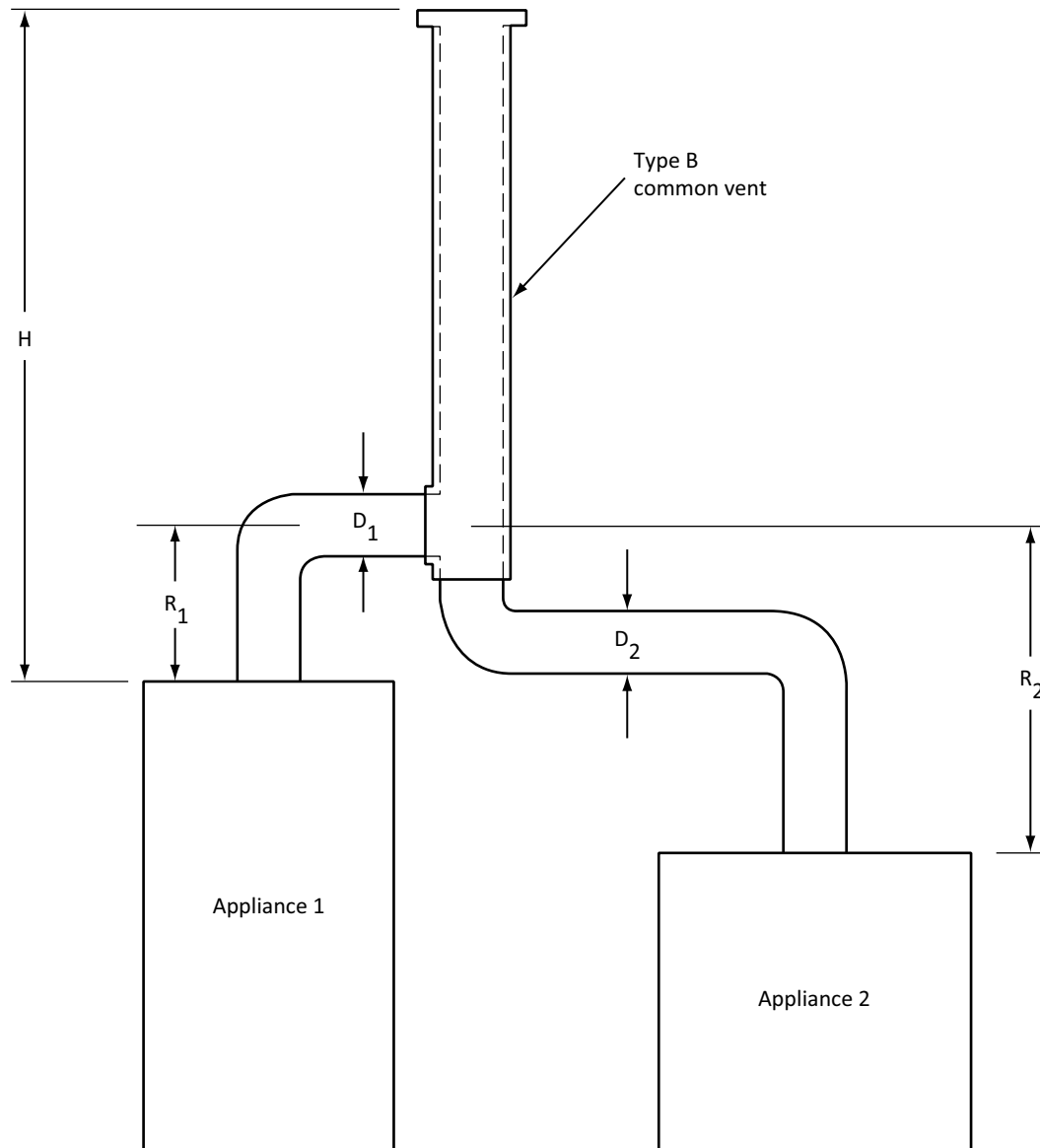
Note: Table [C.3](#) shall be used when Type B vent connectors are attached to a Type B common vent.

Figure C.3
Configuration of Type B double-wall vents with Type B double-wall vent connectors
serving two or more Category I appliances



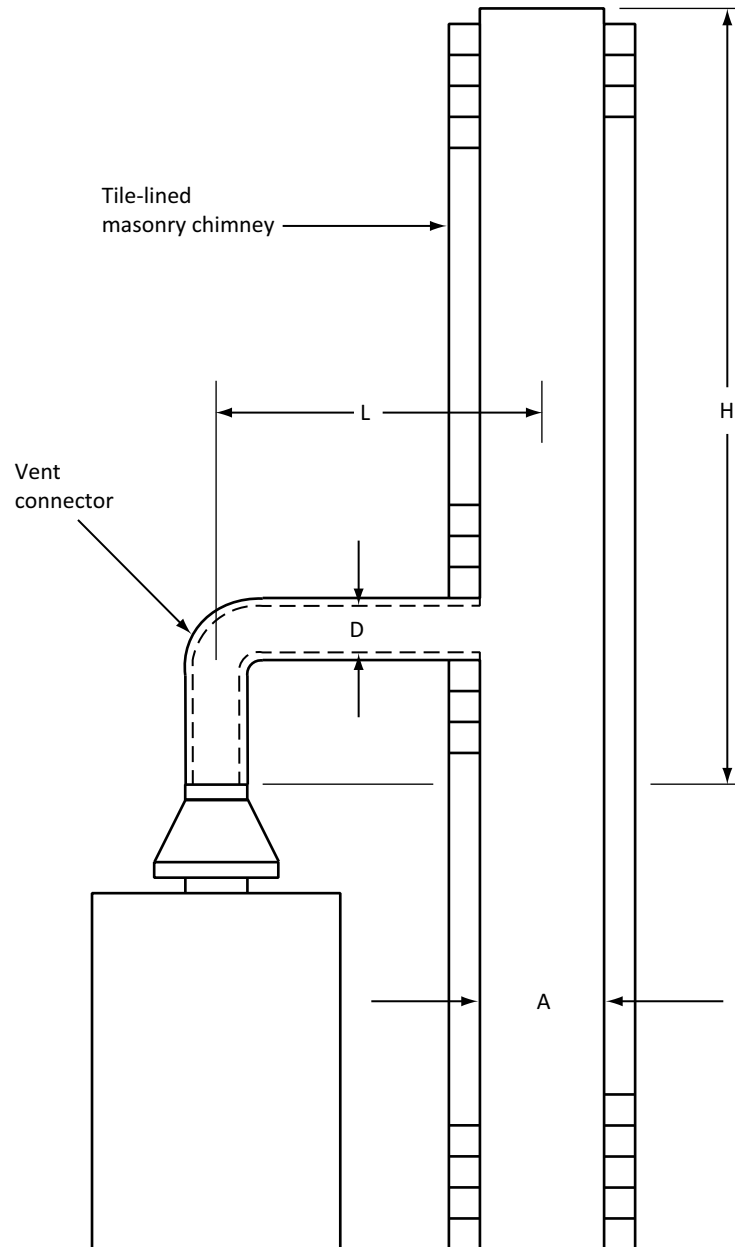
Note: Table [C.4](#) shall be used when single-wall metal vent connectors are attached to a Type B common vent.

Figure C.4
Configuration of Type B double-wall vents with single-wall vent connectors serving
two or more Category I appliances
 (See Table [C.4](#).)



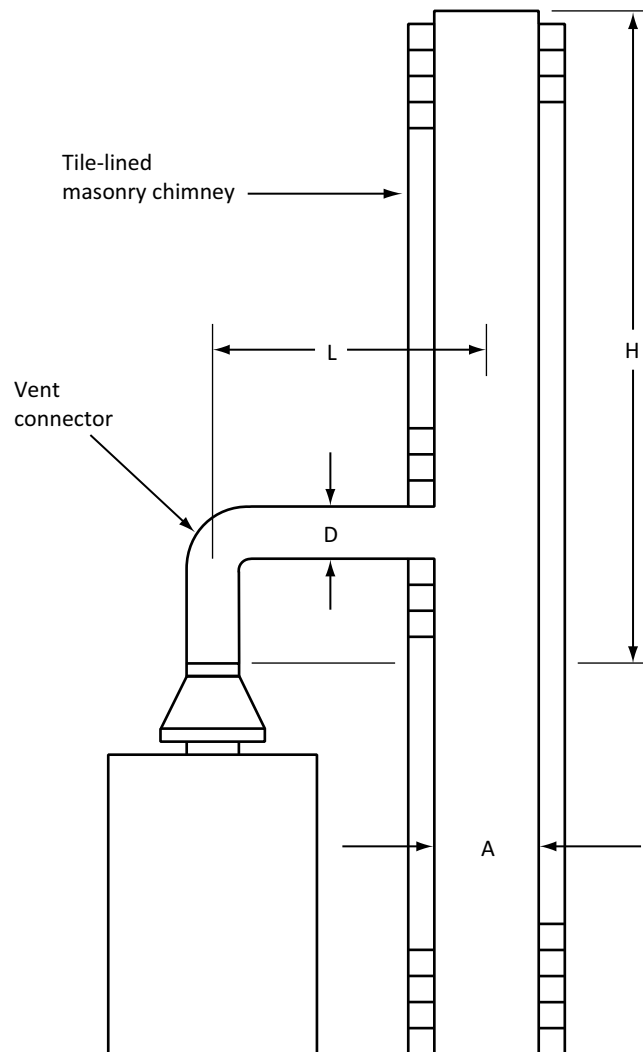
Note: Table [C.5](#) shall be used when a Type B double-wall vent connector is attached to a tile-lined masonry chimney.

Figure C.5
Configuration of tile-lined masonry chimneys with Type B double-wall vent connectors serving a single Category I appliance
 (See Table [C.5](#).)



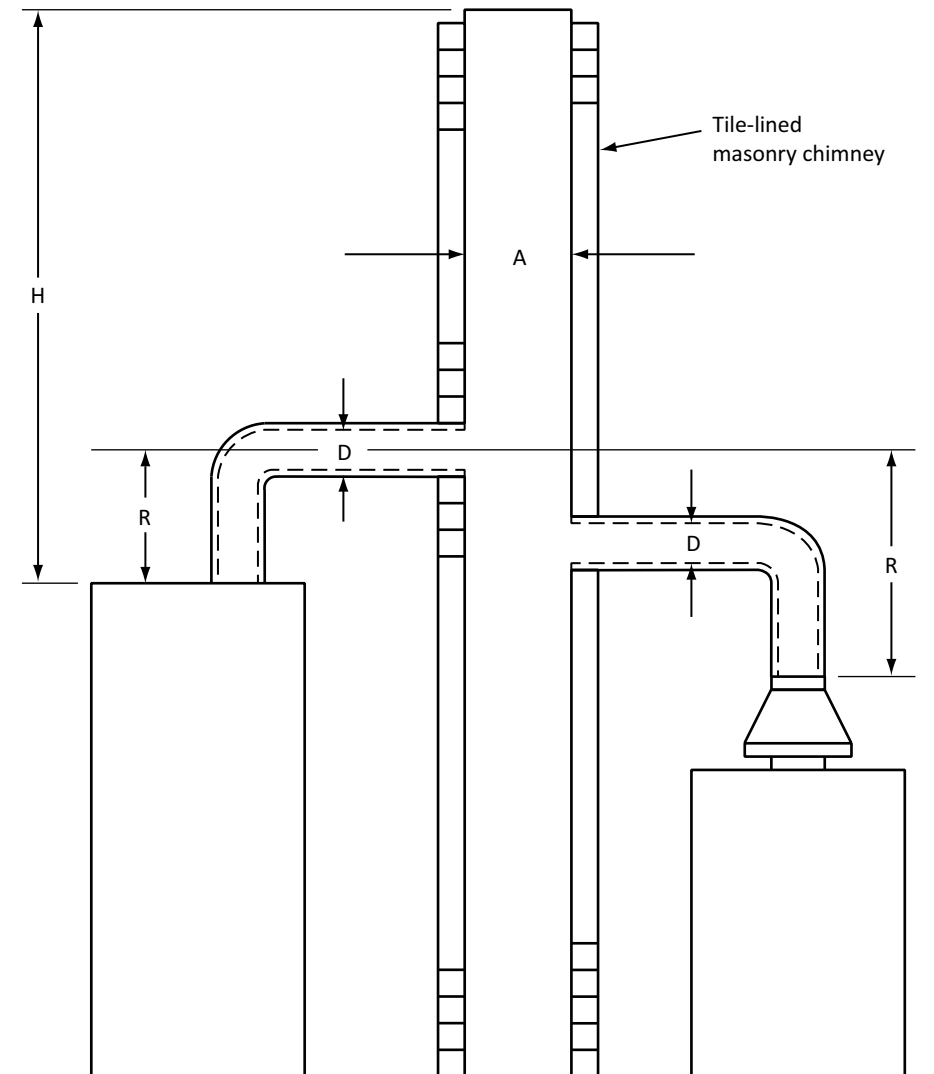
Note: Table [C.6](#) shall be used when a single-wall vent connector is attached to a tile-lined masonry chimney.

Figure C.6
Configuration of tile-lined masonry chimneys with single-wall vent connectors
 serving a single Category I appliance
 (See Table [C.6](#).)



Note: Table [C.7](#) shall be used when Type B double-wall vent connectors are attached to a tile-lined masonry chimney.

Figure C.7
Configuration of tile-lined masonry chimneys with Type B double-wall vent connectors serving two or more Category I appliances
(See Table [C.7](#).)



Note: Table [C.8](#) shall be used when single-wall vent connectors are attached to a tile-lined masonry chimney.

Figure C.8
Configuration of tile-lined masonry chimneys with single-wall vent connectors
serving two or more Category I appliances
(See Table [C.8](#).)

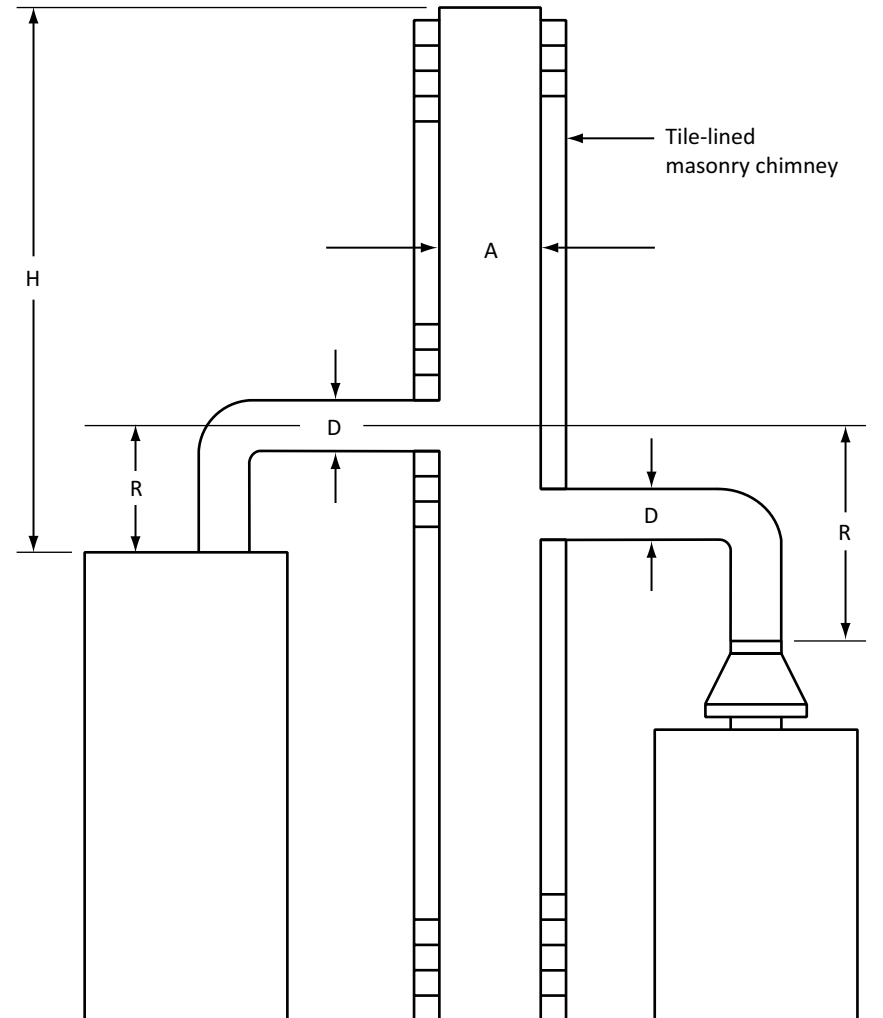


Figure C.9
Schematic diagram showing a typical manifolded common vent section “L” of the vent connector
(See Clauses [C.2.4](#) and [C.2.5](#).)

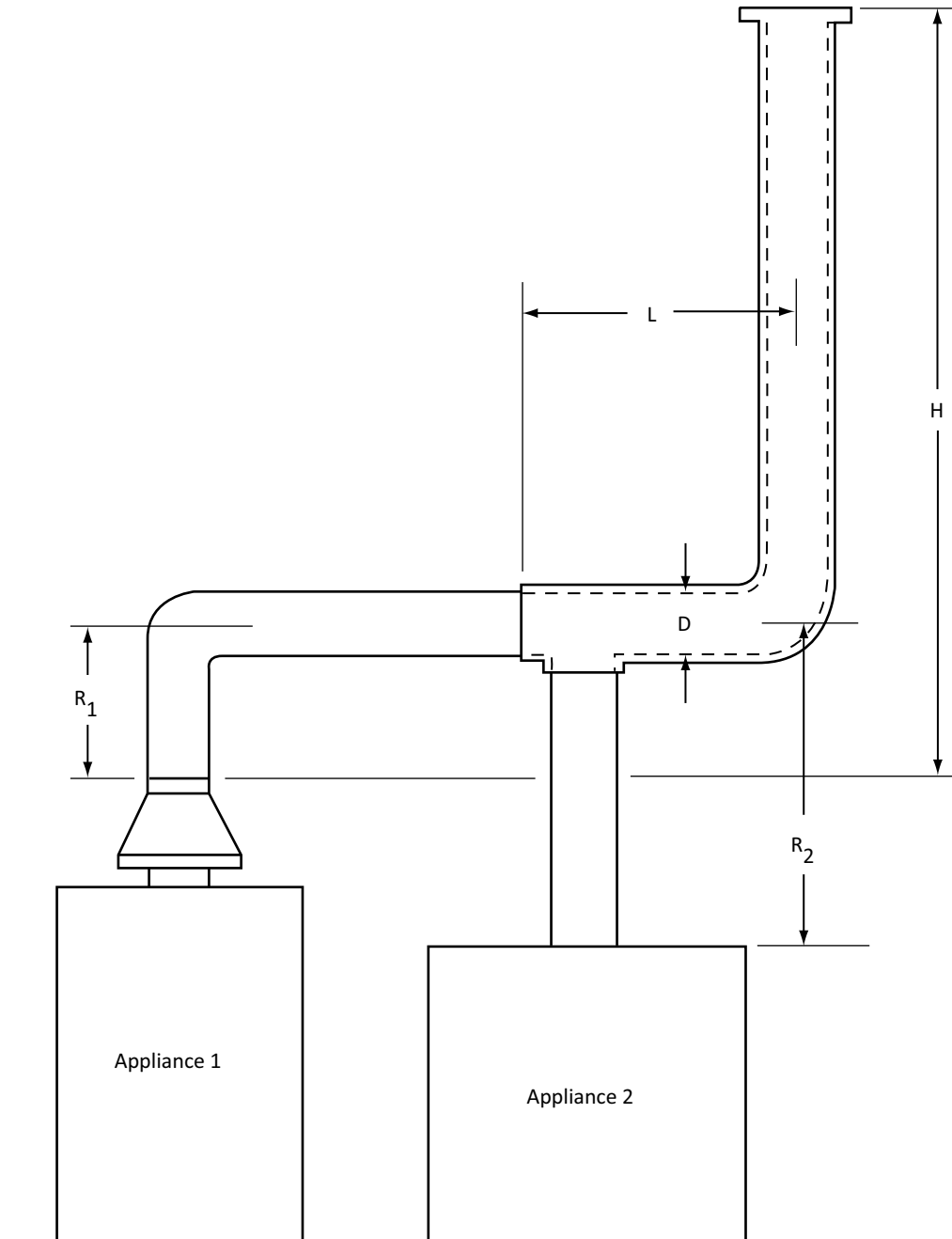


Figure C.10
Schematic diagram showing offset in the common vent section of the vertical vent
(See Clause [C.2.6.](#))

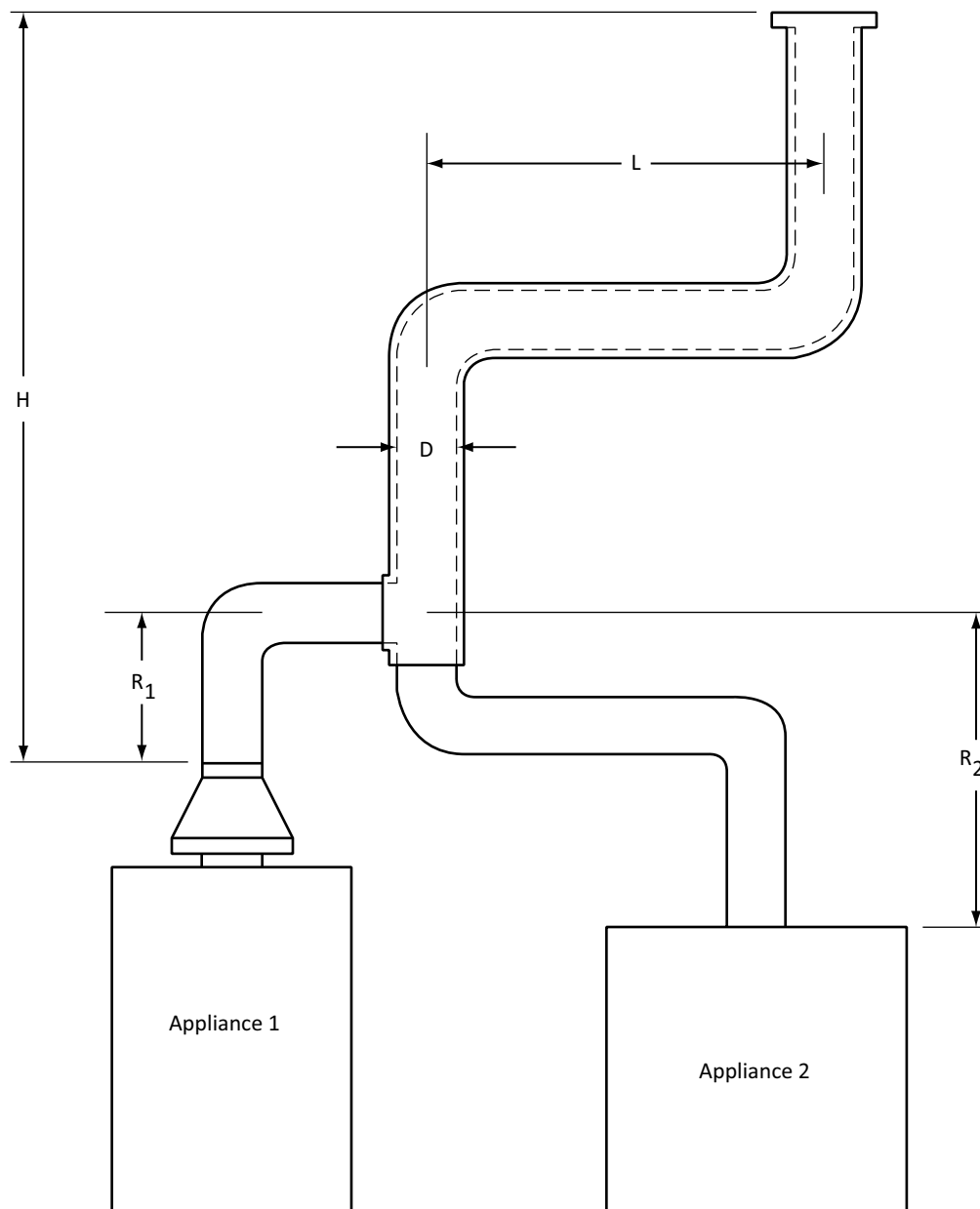
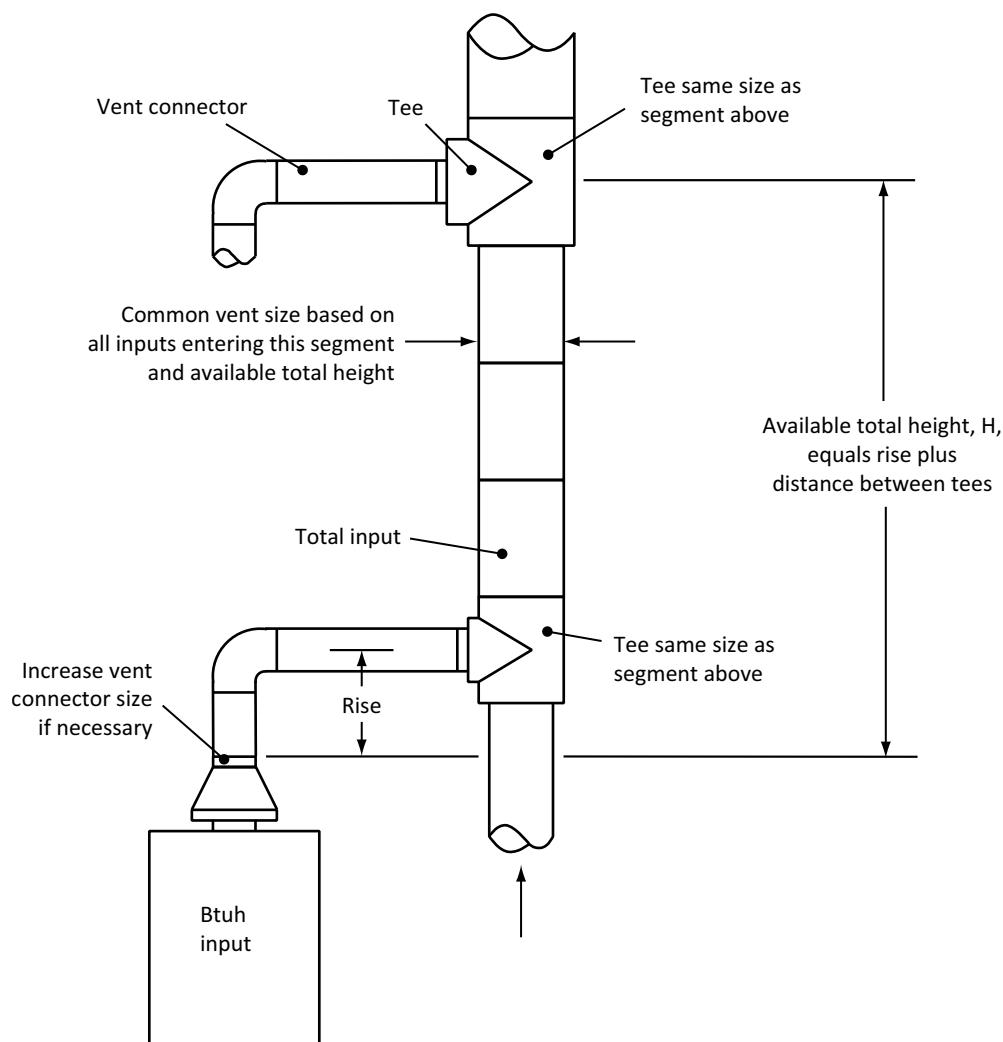


Figure C.11
Multi-storey gas vent design procedure for each segment of system
 (See Clauses [8.21.5](#) and [C.2.13](#).)

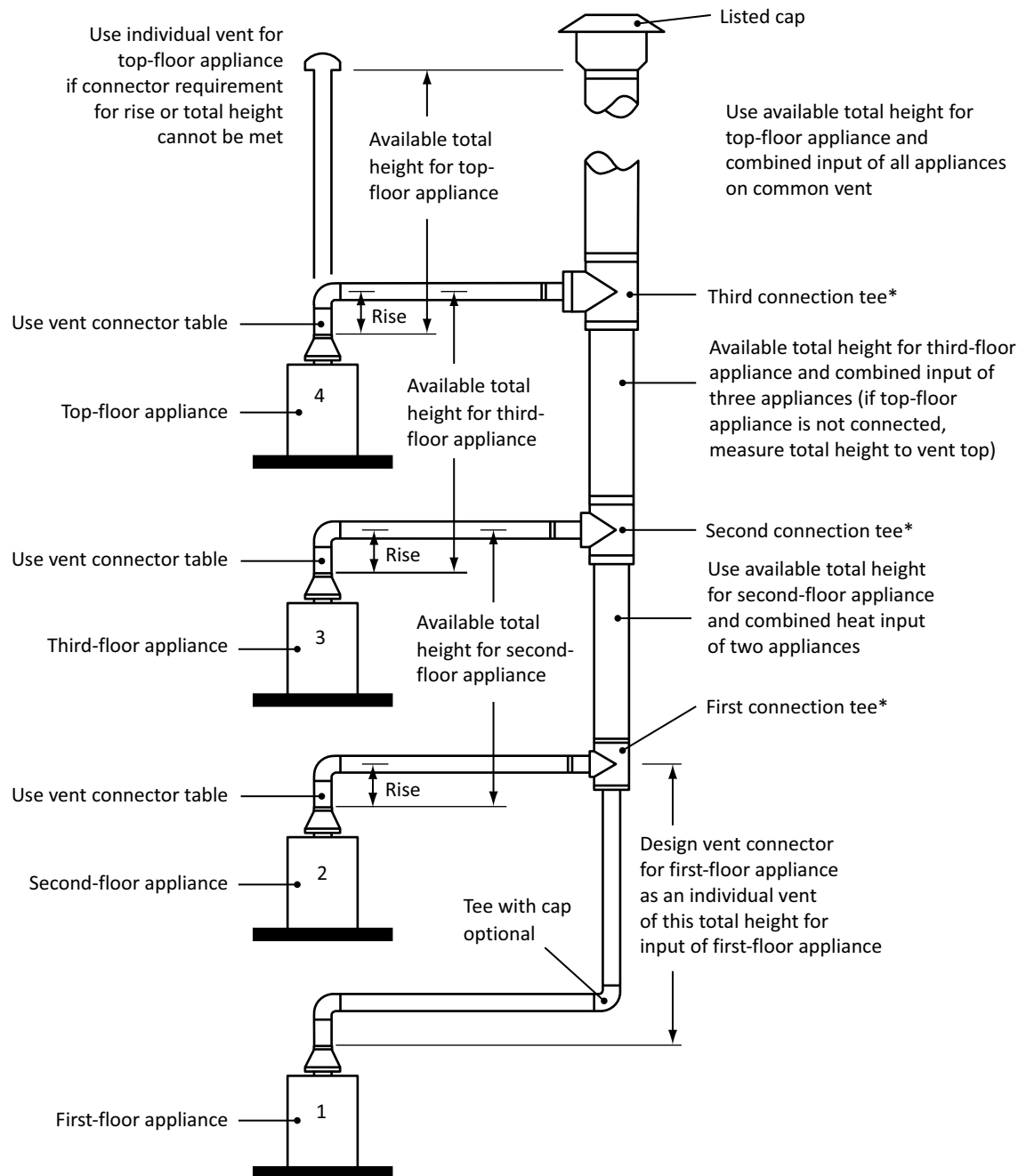


Notes:

- 1) Vent connector size depends on
 - a) input;
 - b) rise;
 - c) available total height, H ; and
 - d) Table [C.3](#) connectors.
- 2) Common vent size depends on
 - a) combined inputs;
 - b) available total height, H ;
 - c) Table [C.3](#) common vent; and
 - d) vertical common vent with no offset.

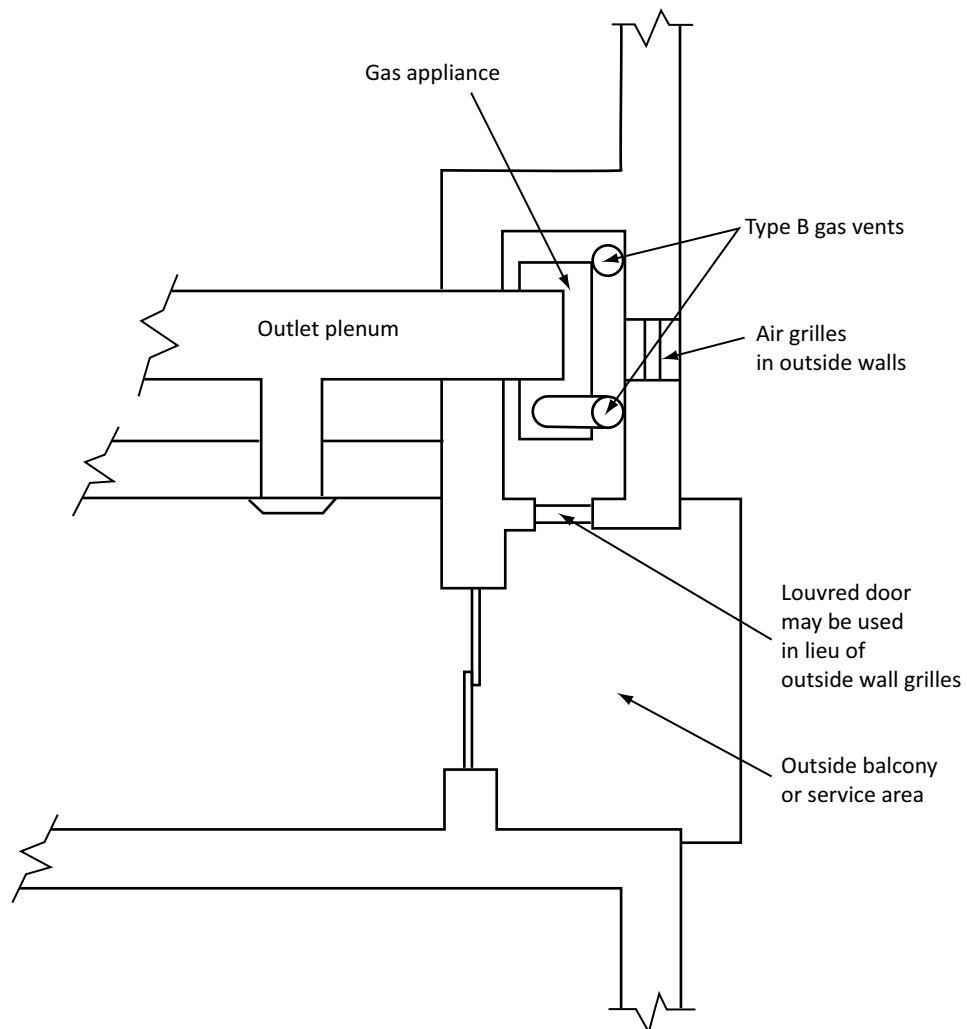
Figure C.12 Principles of design of multi-storey vents using vent connector and common vent design tables

(See Clauses [8.21.5](#) and [C.2.14](#).)



*Each interconnection tee is same size as segment of common vent directly above.

Figure C.13
Plan view of practical separation method for multi-storey venting
(See Clause [8.21.2.](#))



Annex D (informative)

Customer's meter and service regulator installations

Notes:

- 1) *This informative annex has been written in mandatory language to facilitate adoption where users of the Code or regulatory authorities wish to adopt it formally as additional requirements to this Code.*
- 2) *This annex is an extract from Clause 12.4.15 of CSA Z662.*

12.4.15 Customers' meters and service regulators

Note: *Meter location and installation requirements of Clause 12.4.15 apply to meters that are designed and tested as specified in AGA B109.1 or equivalent.*

12.4.15.1

Customers' meters and service regulators may be located either inside or outside buildings, depending on local conditions. If the service regulator is located inside the building

- a) the service line shall be designed to mitigate an uncontrolled release of gas inside the building if a failure occurs; and
- b) the presence of the gas service shall be clearly indicated with permanent marking outside of the building.

Notes:

- 1) *Permanent marking may be accomplished by labelling or yellow paint and, if practical, words indicating the contents of the piping, as follows:*
 - a) *clear indication on the wall of the building using labelling or paint;*
 - b) *clear indication on the service piping by labelling or painting the pipe; or*
 - c) *pavement markers.*
- 2) *Examples of satisfactory mitigation methods include, but are not limited to, the following:*
 - a) *installing an excess flow valve at or close to the service line connection to the main; or*
 - b) *installing an outside above grade shut-off valve prior to the piping entrance.*

12.4.15.2

For service lines requiring series regulation as specified in Clause 12.4.11.1 b), the upstream regulator shall be located outside the building.

12.4.15.3

Where installed within buildings, customers' meters and regulators shall be in readily accessible locations that afford reasonable protection from thermal stresses and sources of heat, mechanical stresses, and chemical deterioration. Service regulators shall be located near the piping entrance and, where practical, the meters shall be installed at the same locations. For service lines supplying large industrial customers and installations where gas is used at higher than standard service pressure, the regulators may be installed at other readily accessible locations.

12.4.15.4

Where located outside buildings, meters and regulators shall be installed in readily accessible locations. Where outside meters and regulators are installed in locations that do not afford reasonable protection from damage, such protection shall be provided. Meter protection configurations shall be included in the operating company's operating and maintenance procedures.

Note: *The operating company can consider other related industry standards when developing meter protection configurations. In addition, consideration should be given to volume of vehicular traffic and proximity of travel,*

severity in the event of impact, space availability for installation, accessibility for future meter set work, reflectivity/retrorreflectivity/visibility and aesthetics.

12.4.15.5

Regulators requiring vents for proper and effective operation, unless manufactured or equipped to limit the escape of gas from their vent opening, even in the event of an operating diaphragm failure, to less than 0.071 m³/h of natural gas or 0.028 m³/h of propane gas, shall be vented to the outside atmosphere and shall terminate in rain- and insect-resistant fittings. A regulator with vent limiting means, up to 0.071 m³/h of natural gas or 0.028 m³/h of propane gas, if installed indoors, shall be installed in a ventilated space only.

Notes:

- 1) *These venting requirements are in accordance with the pressure regulator requirements for propane and natural gas applications in CSA B149.1.*
- 2) *This is a commentary on Clause 12.4.15.5 of CSA Z662:
The term “operating diaphragm” is intended to allow for the use of the T-OPCO (with or without thermal-protection over-pressure cut-off) type of regulators that are designed not to require venting to the outside atmosphere even where an operating diaphragm failure is possible. Since Clause 12.4.15.5 was originally written, there have been ongoing improvements in the technology of natural gas regulators. Between the operating diaphragm and the atmosphere, there is a second diaphragm known as the safety diaphragm. The safety diaphragm limits the escape of gas to the atmosphere to 0.071 m³/h of natural gas or 0.028 m³/h of propane gas in the event of an operating diaphragm failure. When the operating diaphragm fails, the regulator’s OPCO device automatically cuts off supply to the downstream piping, resulting in a service call.*

12.4.15.6

Where regulator failure would result in the release of gas, open ends of the vents shall be located where the gas can escape freely into the atmosphere and away from any openings in the buildings. Clearances from building openings shall be commensurate with local conditions and the volume of gas that can be released, but shall not be less than those specified in Clause 12.4.15.7. Where regulators can be submerged during floods, either a special anti-flood-type breather vent fitting shall be installed or the vent line shall be extended above the height of the expected flood waters.

12.4.15.7

The discharge from overpressure protection devices, relief devices, internal relief valves, and the termination of any other vent not eligible to be discharged in a ventilated space as per Clause 12.4.15.5, shall terminate outdoors with the clearances specified in Table 12.5.

Table 12.5
Clearance from centreline of service regulator relief vent discharge applicable for
natural gas and propane applications
 (See Clause 12.4.15.7.)

Clearance in m (ft) from service regulator vent based on relief capacity				
Clearance from	Natural gas [CSA 6.18 certified OPCO (overpressure cut-off) regulators with limited relief of 1.5 m³/h (50 scf/h) or less]	Natural gas up to an including (55 m³/h) 1900 scf/h	Natural gas over (55 m³/h) 1900 scf/h	Propane
Building opening*	0.3 (1 ft)	0.9 (3 ft)	3.0 (10 ft)	0.9 (3 ft)
Appliance vent outlet	dimension A	dimension B	dimension B	dimension B
Moisture exhaust duct†	0.9 (3 ft)	0.9 (3 ft)	0.9 (3 ft)	0.9 (3 ft)
Mechanical air intake	0.9 (3 ft)	3.0 (10 ft)	3.0 (10 ft)	3.0 (10 ft)
Appliance air intake	0.3 (1 ft)	0.9 (3 ft)	3.0 (10 ft)	3.0 (10 ft)
Source of ignition	0.3 (1 ft)	0.9 (3 ft)	0.9 (3 ft)	3.0 (10 ft)

* Outdoor air intakes that are less than 8 in (200 mm) in diameter or equivalent area shall be considered a building opening in using the Table.

† Applies to gas or electric dryer termination.

Dimension A = 0.3 m (1 ft) in any direction from the vent termination of a gas appliance, and additionally 0.9 m (3 ft) horizontally when within a vertical distance of 4.6 m (15 ft) between the appliance vent termination and the relief vent termination.

Dimension B = 0.9 m (3 ft) in any direction from the vent termination of a gas appliance, and additionally 0.9 m (3 ft) horizontally when within a vertical distance of 4.6 m (15 ft) between the appliance vent termination and the relief vent termination.

Note: The outdoor air intake referred to in Table 12.5 is the ducting that goes from the outside of the structure and terminates into the return air plenum before the appliance, sometimes referred to as a fresh-air intake.

12.4.15.8

Vaults and pits housing meters and regulators shall be designed and constructed to support the loads that can be imposed upon the meters and regulators.

12.4.15.9

Meter support shall be provided if the downstream connecting piping is other than rigid piping.

Note: Inadequate support of the meter can result if the outlet of the meter is connected to flexible metallic tubing.

12.4.15.10

Regulator and relief vent piping shall be

- a) capable of withstanding the maximum pressures as specified in Clauses 12.4.9 and 12.4.10;
- b) included in the determination of the relief system capacity specified in Clause 4.18.3.2;
- c) composed of metallic material where installed within buildings; and
- d) designed, fabricated, and installed to prevent static build-up and mechanical damage where plastic materials are used.

Annex E (informative)

Example of piping design sizing (imperial and metric)

Note: This informative annex has been written in mandatory language to facilitate adoption where users of the Code or regulatory authorities wish to adopt it formally as additional requirements to this Code.

E.1 Piping design sizing (imperial)

E.1.1 General (imperial)

The required minimum size of Schedule 40 iron pipe shall be determined for each section and each outlet of the piping system shown in Figure [E.1](#). The gas has a relative density of 0.60 and a heating value of 1000 Btu/ft³ in the example shown in Figure [E.1](#).

In the example, the gas pressure in Annex [A](#) is 2 psig, and the gas pressure from A to all appliances is less than 14 in w.c.

E.1.2 Size determination from Table [A.2](#)

The pipe size for the sections from Annex [A](#) to all appliance outlets shall be determined.

The distance between the last stage regulator A and the farthest appliance E shall be established (in the example shown in Figure [E.1](#), it is 50 ft) to determine the longest run of the system operating under the pressure of 14 in w.c. This distance shall then be used to size the entire system downstream of the regulator A. The choice of tables for a specific application shall be determined by the pressure supplied to the system and the pressure drop allowed.

Using the 50 ft line of Table [A.2](#) (imperial), the pipe sizes shall be as follows:

- a) D–E for 60 000 Btu/h requires an NPS 1/2;
- b) D–F for 25 000 Btu/h requires an NPS 1/2;
- c) C–G for 150 000 Btu/h requires an NPS 3/4;
- d) B–H for 40 000 Btu/h requires an NPS 1/2;
- e) C–D for 85 000 Btu/h requires an NPS 1/2;
- f) B–C for 235 000 Btu/h requires an NPS 1; and
- g) A–B for 275 000 Btu/h requires an NPS 1.

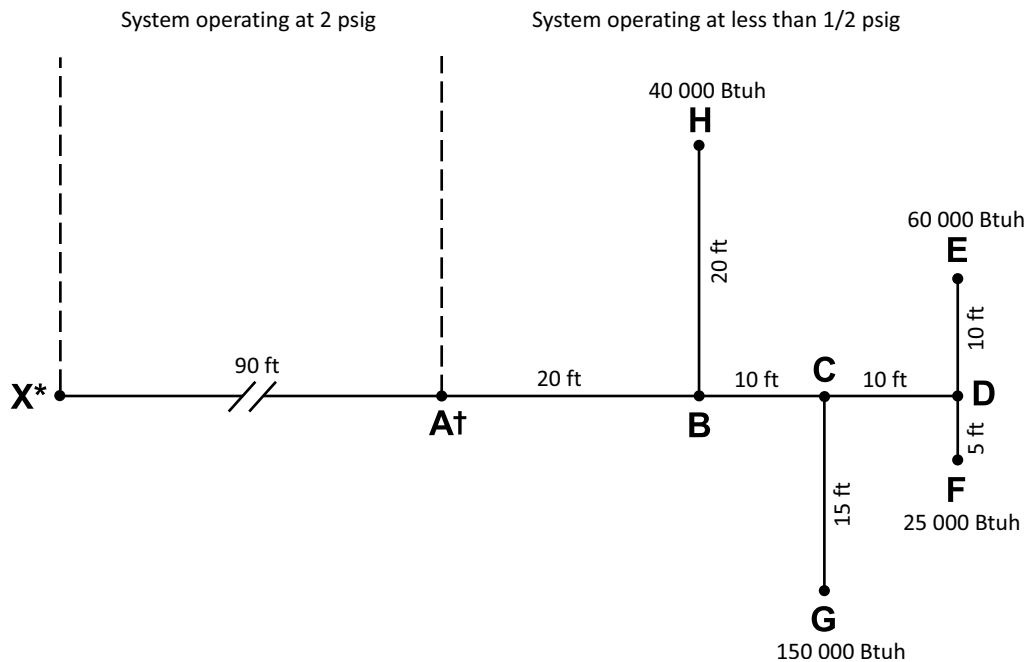
E.1.3 Size determination from Table [A.3](#)

The pipe size for the 90 ft Annex [A](#) operating at 2 psig using Table [A.3](#) (imperial) for 275 000 Btu/h shall be determined. An NPS 1/2 pipe shall be required.

E.1.4 Relative density

Multipliers to be used when the relative density of the gas is other than 0.60 shall be as given in Table [A.15](#).

Figure E.1
Example of pipe design sizing (imperial)
 (See Clauses [E.1.1](#) and [E.1.2](#).)



* Service regulator and meter.

† System regulator.

Note: The example shown is of a general nature and does not necessarily include the allowances that shall be taken into account in each particular installation. In every installation, all pertinent requirements in Clause [6](#) shall be adhered to.

E.2 Piping design sizing (metric)

E.2.1 General (metric)

The required minimum size of Schedule 40 iron pipe shall be determined for each outlet of the piping system shown in Figure [E.2](#). The gas has a relative density of 0.60 and a heating value of 38 MJ/m³ in the example shown in Figure [E.2](#).

In the example, the gas pressure in Annex [A](#) is 14 kPa, and the gas pressure from A to all appliances is less than 3.5 kPa.

E.2.2 Size determination from Table [A.2](#)

The pipe size for the sections from Annex [A](#) to all appliance outlets shall be determined.

The distance between the last stage regulator A and the farthest appliance E shall be established (in the example, it is 15 m) to determine the longest run of the system operating under the pressure of 3.5 kPa. This distance shall then be used to size the entire system downstream of the regulator A. The choice of tables for a specific application shall be determined by the pressure supplied to the system and the pressure drop allowed.

Using the 15 m line of Table [A.2](#) (metric), the pipe sizes shall be as follows:

- a) D–E for 17.5 kW requires an NPS 1/2;

- b) D–F for 7.0 kW requires an NPS 1/2;
- c) C–G for 45.0 kW requires an NPS 3/4;
- d) B–H for 12.0 kW requires an NPS 1/2;
- e) C–D for 24.5 kW requires an NPS 1/2;
- f) B–C for 69.5 kW requires an NPS 1; and
- g) A–B for 81.5 kW requires an NPS 1.

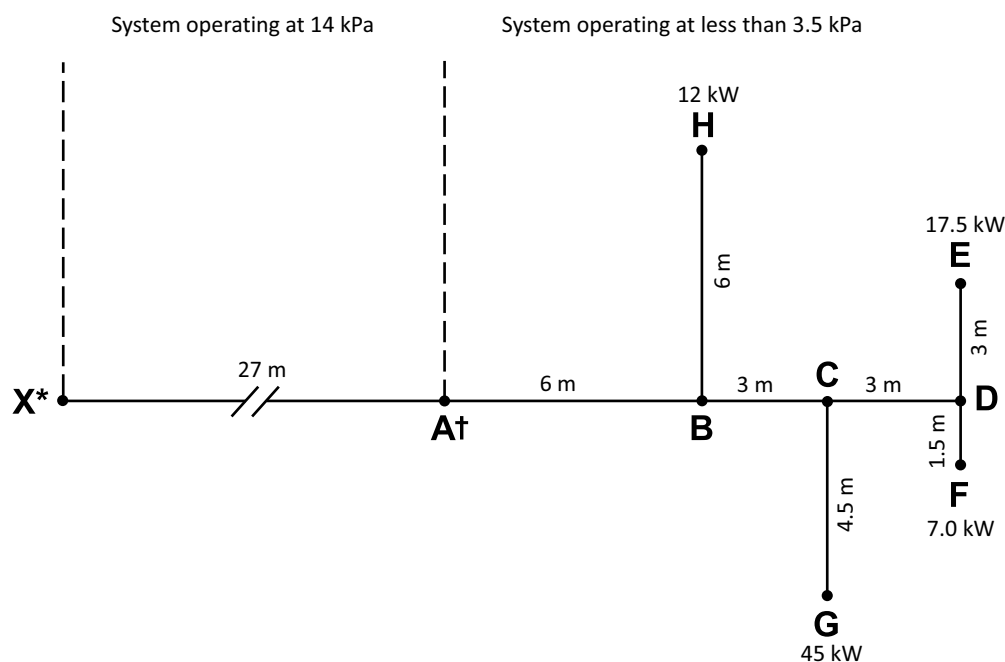
E.2.3 Size determination from Table A.3

The pipe size for the 27 m Annex A operating at 14 kPa using Table A.3 (metric) for 81.5 kW shall be determined. An NPS 1/2 pipe shall be required.

E.2.4 Relative density

Multipliers to be used when the relative density of the gas is other than 0.60 shall be as given in Table A.15.

Figure E.2
Example of pipe design sizing (metric)
 (See Clause E.2.1.)



* Service regulator and meter.

† System regulator.

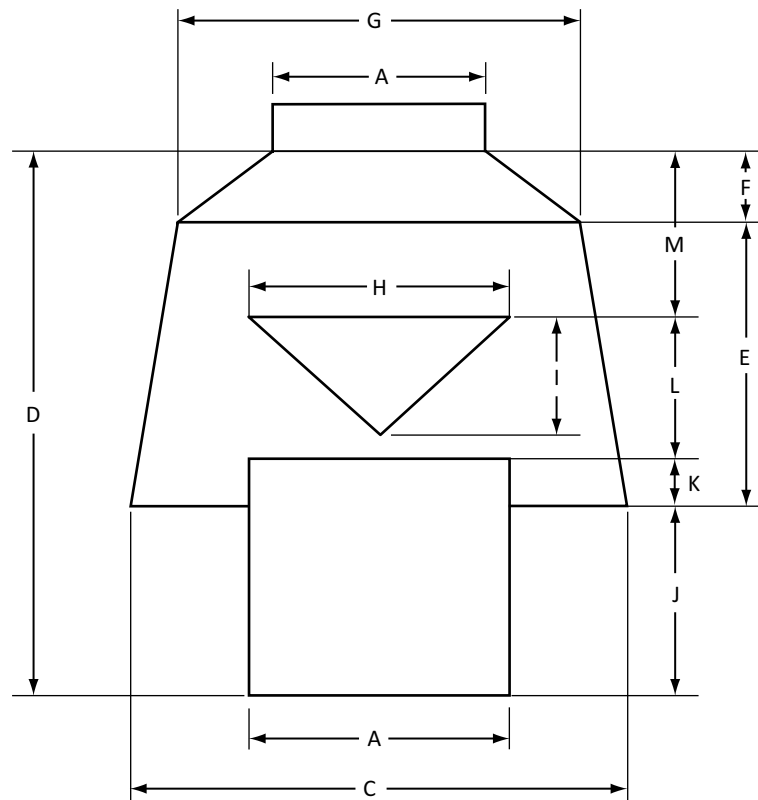
Note: The example shown is of a general nature and does not necessarily include the allowances that shall be taken into account in each particular installation. In every installation, all pertinent requirements in Clause 6 shall be adhered to.

Annex F (informative)

Suggested general dimensions for draft hoods

Note: This annex is not a mandatory part of this Code.

Figure F.1
Suggested general dimensions for a vertical draft hood



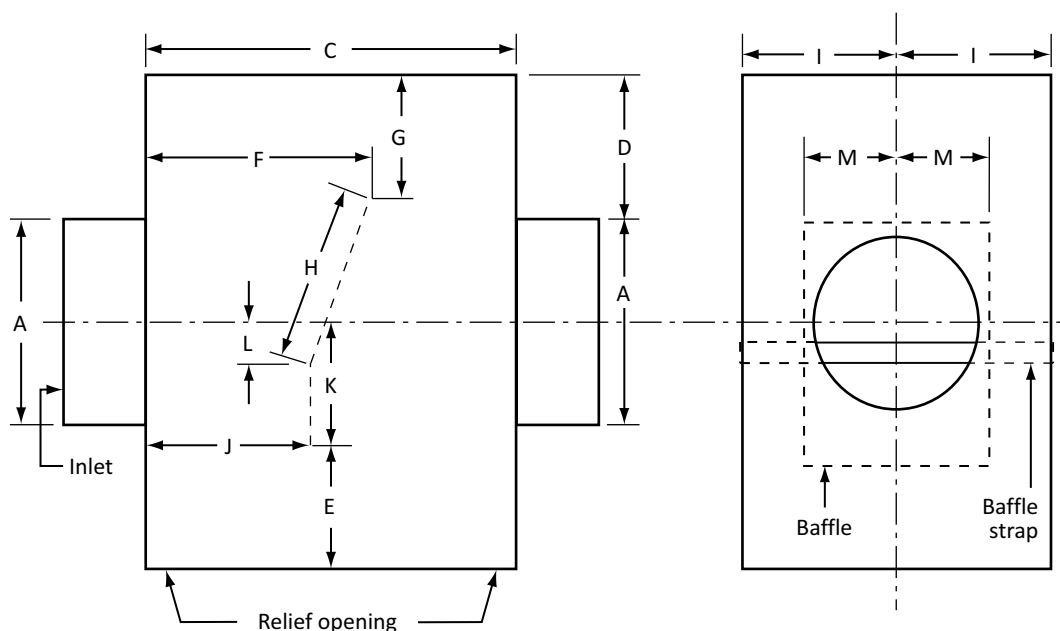
(Continued)

Figure F.1 (Continued)

Dimensions, in (mm)											
A	C	D	E	F	G	H	I	J	K	L	M
3 (75)	5.5 (140)	7.0 (178)	3.8 (97)	0.7 (18)	4.4 (112)	3.0 (76)	1.5 (38)	2.5 (64)	0.7 (18)	1.5 (38)	2.3 (58)
4 (100)	7.2 (183)	9.5 (241)	5.0 (127)	1.0 (25)	6.0 (152)	4.0 (102)	2.0 (51)	3.5 (89)	0.8 (20)	2.0 (51)	3.0 (76)
5 (125)	9.4 (240)	10.8 (274)	5.3 (135)	1.5 (38)	8.0 (203)	5.0 (127)	2.3 (58)	4.0 (102)	0.9 (23)	2.4 (61)	3.5 (89)
6 (150)	11.5 (292)	12.0 (305)	5.6 (142)	1.9 (48)	9.8 (249)	6.0 (152)	2.5 (64)	4.5 (114)	0.8 (20)	2.7 (69)	4.0 (102)
7 (175)	13.5 (343)	13.9 (353)	6.4 (163)	2.3 (58)	11.6 (295)	7.0 (178)	2.9 (74)	5.3 (135)	0.9 (23)	3.1 (79)	4.6 (117)
8 (200)	15.5 (394)	15.8 (401)	7.1 (180)	2.7 (69)	13.4 (340)	8.0 (203)	3.2 (81)	6.0 (152)	1.0 (25)	3.5 (89)	5.3 (135)
9 (225)	17.4 (444)	17.5 (444)	7.7 (196)	3.1 (79)	15.2 (386)	9.0 (229)	3.5 (89)	6.7 (170)	1.0 (25)	4.0 (102)	5.8 (147)
10 (250)	19.7 (500)	18.8 (478)	7.9 (201)	3.6 (91)	17.2 (437)	10.0 (254)	3.8 (96)	7.3 (185)	1.0 (25)	4.3 (109)	6.2 (157)
11 (275)	22.2 (564)	20.7 (526)	8.4 (213)	4.3 (109)	19.6 (498)	11.0 (279)	4.1 (104)	8.0 (203)	1.0 (25)	4.6 (117)	6.6 (168)
12 (300)	24.7 (627)	22.2 (564)	8.7 (221)	5.0 (127)	22.0 (559)	12.0 (305)	4.4 (112)	8.5 (216)	1.5 (38)	5.0 (127)	7.0 (178)
									1.7 (43)		

Note: Figure F.1 illustrates only one design of a vertical draft hood and is not to be construed as the only design that may be used.

Figure F.2
Suggested general dimensions for a horizontal draft hood



(Continued)

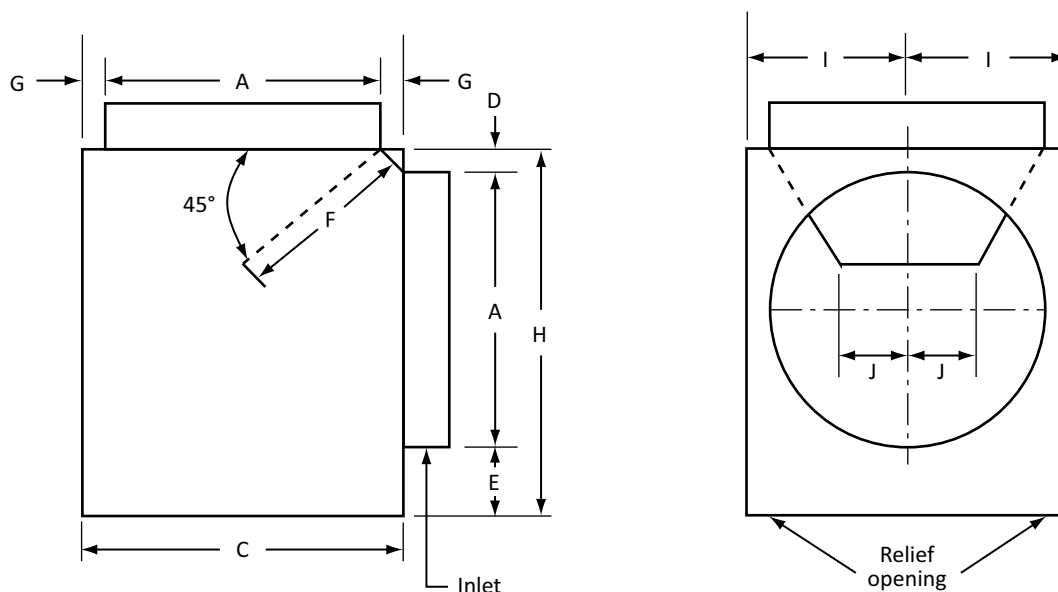
Figure F.2 (Continued)

Dimensions, in (mm)						
A	C	D	E	F	G	H
3 (75)	6 (150)	1-1/2 (38)	4-3/4 (121)	3-3/4 (95)	1-3/8 (35)	2-1/2 (64)
4 (100)	8 (200)	2 (50)	4-3/4 (121)	5 (125)	1-7/8 (48)	3-3/8 (86)
5 (125)	10 (250)	2-1/2 (64)	4-3/4 (121)	6-1/4 (159)	2-3/8 (60)	4-3/16 (106)
6 (150)	12 (300)	3 (75)	4-3/4 (121)	7-1/2 (190)	2-7/8 (73)	5 (125)
7 (175)	14 (350)	3-1/2 (89)	4-3/4 (121)	8-3/4 (222)	3-3/8 (86)	5-7/8 (149)
8 (200)	16 (400)	4 (100)	4-3/4 (121)	10 (250)	3-7/8 (98)	6-11/16 (170)
9 (225)	18 (450)	4-1/2 (114)	4-3/4 (121)	11-1/4 (286)	4-3/8 (111)	7-1/2 (190)
10 (250)	20 (500)	5 (125)	4-3/4 (121)	12-1/2 (318)	4-7/8 (124)	8-3/8 (213)
11 (275)	22 (550)	5-1/2 (140)	4-3/4 (121)	13-3/4 (349)	5-3/8 (136)	9-3/16 (233)
12 (300)	24 (600)	6 (150)	4-3/4 (121)	15 (381)	5-7/8 (149)	10 (250)

Dimensions, in (mm)				
I	J	K	L	M
2-1/2 (64)	2-1/2 (64)	2-1/8 (54)	9/16 (14)	1-3/4 (44)
3-3/8 (86)	3-3/8 (86)	2-7/8 (73)	3/4 (19)	2-5/16 (59)
4-3/16 (106)	4-3/16 (106)	3-1/2 (89)	15/16 (24)	2-15/16 (75)
5 (125)	5 (125)	4-1/4 (108)	1-1/8 (29)	3-1/2 (89)
5-7/8 (149)	5-7/8 (149)	5 (125)	1-5/16 (33)	4-1/16 (102)
6-11/16 (170)	6-11/16 (170)	5-5/8 (143)	1-1/2 (38)	4-11/16 (119)
7-1/2 (190)	7-1/2 (190)	6-3/8 (162)	1-11/16 (43)	5-1/4 (133)
8-3/8 (213)	8-3/8 (213)	7 (175)	1-7/8 (48)	5-13/16 (148)
9-3/16 (233)	9-3/16 (233)	7-3/4 (197)	2-1/16 (57)	6-3/8 (162)
10 (250)	10 (250)	8-1/2 (216)	2-1/4 (57)	7 (175)

Note: Figure F.2 illustrates only one design of a horizontal draft hood and is not to be construed as the only design that may be used.

Figure F.3
Suggested general dimensions for a horizontal to vertical draft hood



Dimensions, in (mm)								
A	C	D	E	F	G	H	I	J
3 (75)	4 (100)	1/2 (13)	3/4 (19)	2 (50)	1/2 (13)	4-1/4 (108)	2 (50)	5/8 (16)
4 (100)	5 (125)	1/2 (13)	1 (25)	2-11/16 (67)	1/2 (13)	5-1/2 (140)	2-1/2 (64)	13/16 (21)
5 (125)	6 (150)	1/2 (13)	1-1/4 (32)	3-5/16 (84)	1/2 (13)	6-3/4 (171)	3 (75)	1 (25)
6 (150)	7 (175)	1/2 (13)	1-1/2 (38)	4 (100)	1/2 (13)	8 (200)	3-1/2 (89)	1-3/16 (30)
7 (175)	8 (200)	1/2 (13)	1-3/4 (44)	4-11/16 (119)	1/2 (13)	9-1/4 (235)	4 (100)	1-3/8 (35)
8 (200)	9 (225)	1/2 (13)	2 (50)	5-3/16 (132)	1/2 (13)	10-1/2 (267)	4-1/2 (114)	1-9/16 (40)
9 (225)	10 (250)	1/2 (13)	2-1/4 (57)	6 (150)	1/2 (13)	11-3/4 (298)	5 (125)	1-3/4 (44)
10 (250)	11 (275)	1/2 (13)	2-1/2 (64)	6-11/16 (170)	1/2 (13)	13 (330)	5-1/2 (140)	1-15/16 (49)
11 (275)	12 (300)	1/2 (13)	2-3/4 (70)	7-5/16 (186)	1/2 (13)	14-1/4 (362)	6 (153)	2-1/8 (54)
12 (300)	13 (325)	1/2 (13)	3 (75)	8 (200)	1/2 (13)	15-1/2 (394)	6-1/2 (165)	2-5/16 (59)

Note: Figure F.3 illustrates only one design of a horizontal to vertical draft hood and is not to be construed as the only design that may be used.

Annex G (informative)

Piping expansion and flexibility

Note: *This informative annex has been written in mandatory language to facilitate adoption where users of the Code or regulatory authorities wish to adopt it formally as additional requirements to this Code.*

G.1 General

Problems are not usually experienced in residential piping systems with thermal expansion because the piping or tubing is relatively short.

However, in commercial and industrial buildings, there can be substantial variations in the lengths of piping or tubing mains as the indoor temperature changes on weekends or between seasons. It is therefore essential that provisions be made in the piping design for flexibility to avoid undesirable bending and strong forces at elbows or joints.

This flexibility is obtained by the use of designed pipe bends, loops, offsets, expansion, hints, or swivel joints. The piping or tubing shall be anchored at appropriate locations to control the direction of expansion and contraction.

G.2 Gas piping to rooftop units

Due to the wide range of temperatures on a roof between summer and winter, piping design shall provide for expansion and contraction. Because of the exposure to sunlight and rain or snow, coated or wrapped gas piping is not recommended. Bare steel piping and fittings shall be cleaned after assembly, painted with a metal primer, and given at least one coat of exterior enamel paint.

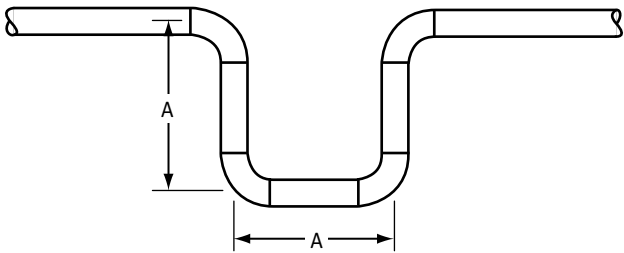
Expansion control loops should be used on pipe sizes of 2 in (50 mm) or less whenever the unrestrained pipe length is at least 100 ft (30 m). Larger diameter piping is generally installed with expansion joints. As shown in Figure [G.1](#), the control loop consists of 90° offsets to the main pipe run so it provides an open-ended box configuration.

For purposes of design, a 90° change of direction or an offset may be considered the equivalent to a control loop when the offset length is at least three times Dimension A in Figure [G.1](#).

Some form of thermal expansion control shall be provided for each 100 ft (30 m). For example, a 200 ft (61 m) length shall be anchored in the centre between each loop. The anchor of an offset or 90° bend should be not closer than 10 ft (3 m) from the bend to permit expansion.

Suitable pipe supports are commercially available to permit the free movement of piping when it expands and contracts.

Figure G.1
Expansion control loop
(See Clause [G.2.](#))



Iron pipe size, in	Wall thickness (Schedule 40), in (mm)	Dimension A, ft (m)
3/4	0.113 (2.87)	4.5 (1.4)
1	0.133 (3.38)	5.0 (1.5)
1-1/4	0.140 (3.56)	5.5 (1.6)
1-1/2	0.145 (3.68)	6.0 (1.8)
2	0.154 (3.91)	6.5 (2.0)

Annex H (normative)

Purging of piping and tubing systems where a readily accessible burner is not available or where an appliance is not equipped with a continuous pilot

Note: *This annex is a mandatory part of this Code.*

H.1 General

Purging of piping and tubing systems where a readily accessible burner is not available or where an appliance is not equipped with a continuous pilot shall be undertaken as outlined in Clauses [H.2](#) to [H.7](#).

H.2 Applicability

The procedure shall be applicable to the purging of air or inert gas using natural gas from a piping/tubing system downstream of the meter.

H.3 No smoking

No smoking shall be permitted when purging.

H.4 Sources of ignition

Prior to commencing the purge, all sources, or potential sources, of ignition shall be removed or shut off.

H.5 Restrictions

The purging procedure shall be used only under the following conditions:

- a) maximum piping system size is NPS 1;
- b) maximum tubing size is NTS 3/4;
- c) maximum piping or tubing system gas pressure is 2 psig (14 kPa);
- d) maximum gas pressure at purging point is 11 in w.c. (2.74 kPa) or less; and
- e) the longest run of pipe or tubing in the system is 100 ft (30 m) or less.

H.6 Exposure

Piping and tubing systems not meeting the conditions listed in Clause [H.5](#) shall be purged to the outdoors. Those that do meet these conditions shall be purged in accordance with Clause [H.7](#).

H.7 Procedure

The purging procedure shall be as follows:

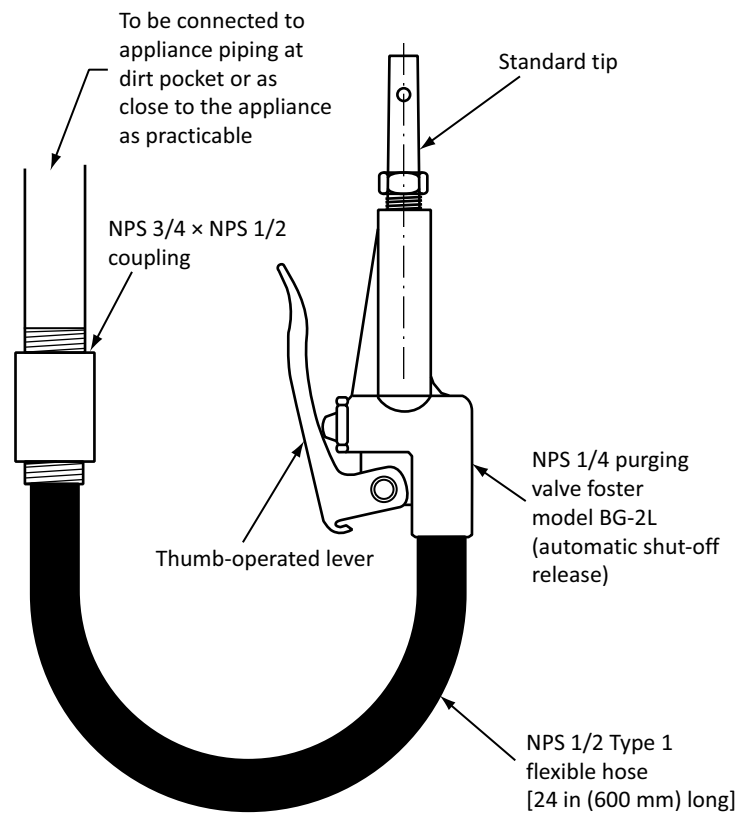
- a) Use only the approved purging assembly. See Figure [H.1](#).
- b) Inspect the purging assembly prior to use. Replace worn or damaged hose, and ensure that all joints are tight.
- c) Ensure that the manual shut-off valve for the appliance is closed.
- d) Connect the purging assembly to the appliance dirt pocket, where so equipped, or to the appliance piping, as close to the appliance as is practicable.
- e) Ensure that no sources of ignition are present within the purge area and that the purge area is well ventilated.
- f) Determine the approximate purging time from Table [H.1](#). Ensure that the purge does not exceed 100 s.

- g) Open the manual shut-off valve for the appliance.
- h) Begin purging by fully opening the automatic shut-off release purging valve on the purging assembly. Purge in a continuous and uninterrupted manner. Caution: Under no circumstances shall the automatic shut-off release purging valve be fixed in the open position.
- i) Terminate purging when the smell of gas is detected at the purging valve outlet or when the approximate purging time has been reached. Do not exceed the purging time by more than 10 s.
- j) When purging is terminated, close the manual shut-off valve and disconnect the purging assembly from the piping system.
- k) Reassemble the appliance piping, open the manual shut-off valve, and soap test the affected joints to ensure that there are no leaks.

Table H.1
Approximate purging times at 7 in w.c. (1.74 kPa)
 (See Clause [H.7.](#))

Pipe/tubing size, in	Length of pipe/tubing, ft (m)	Time to purge, s
NPS 1/2	10 (3)	3
	20 (6)	5
	60 (18)	15
	100 (30)	26
NPS 1	10 (3)	7
	20 (6)	14
	60 (18)	43
	100 (30)	84
NTS 1/2	10 (3)	1
	20 (6)	2
	60 (18)	6
	100 (30)	10
NTS 3/4	10 (3)	3
	20 (6)	5
	60 (18)	15
	100 (30)	26

Figure H.1
Approved purging assembly
(See Clause [H.7.](#))



Annex I (informative)

General information

Note: This annex is not a mandatory part of this Code.

I.1 Abbreviations

I.1.1 Organizations

The following abbreviations of names of organizations apply in this Code:

ANSI	— American National Standards Institute
ASME	— The American Society of Mechanical Engineers
ASTM	— American Society for Testing and Materials
CGA	— Canadian Gas Association (CGA Standards, Recommended Practices, and Codes are now published by CSA Group.)
CGSB	— Canadian General Standards Board
CSA	— CSA Group
DOT	— Department of Transportation
NRC	— National Research Council Canada NRC Corporate Communications
TC	— Transport Canada
UL	— Underwriters Laboratories Inc.
ULC	— Underwriters' Laboratories of Canada

I.1.2 Words and phrases

The following abbreviations of words and phrases apply in this Code:

B&SG	— Brown and Sharpe gauge
Btu/h	— British thermal unit per hour
°C	— degree Celsius
cf	— cubic foot
cfm	— cubic foot per minute
CGI	— combustible gas indicator
CSST	— corrugated stainless steel tubing
DFIAH	— direct gas-fired industrial air heater
DFMAH	— direct-fired make-up air heater
DFPAH	— direct gas-fired process air heater
dm ³	— cubic decimeter
°F	— degree Fahrenheit
ft	— foot
FVIR	— flammable vapours ignition resistant
g	— gram
gal	— gallon

GSG	—	galvanized sheet gauge
GPH	—	gallon per hour
h	—	hour
kg	—	kilogram
kJ	—	kilojoule
kPa	—	kilopascal
kW	—	kilowatt
lb	—	pound
L	—	litre
LEL	—	lower explosive level
m	—	metre
m ³	—	cubic metre
mA	—	milliampere
min	—	minute
mm	—	millimetre
MSG	—	manufacturers standard gauge
N	—	newton
NGV	—	natural gas vehicle
No.	—	number
NPS	—	nominal pipe size
NTS	—	nominal tube size
oz	—	ounce
Pa	—	pascal
ppm	—	parts per million
psia	—	pound per square inch absolute
psig	—	pound per square inch gauge
pt	—	pint
RFA	—	residential fuelling appliance
s	—	second
scf	—	standard cubic feet
SI	—	International System of Units
SWG	—	steel wire gauge
USWG	—	United States water gallon
V	—	volt
VAV	—	variable air volume
VFA	—	vehicle fuelling appliance
VRA	—	vehicle refuelling appliance
W	—	watt
w.c.	—	water column
WC	—	water capacity

wt — weight

I.2 Properties of fuels for an average commercial fuel grade data

The following data have been taken from various sources, and values shown related to the properties of fuels are for an average commercial fuel grade. It should be kept in mind that external variables for much of the data provided in this annex are significant, and the user must take care using this information. Where the information is to be used for a significant purpose, it is recommended that the user search out other sources of reference and investigate and verify the relevancy of the specific variables used to determine the constants provided.

Table I.1
Properties of commercial propane, butane, and natural gas

a) Imperial			
Properties	Propane	Butane	Natural gas
Chemical formula	C ₃ H ₈	C ₄ H ₁₀	CH ₄
Molecular weight	44.06	58.1	16
Freezing point, °F, at atmospheric pressure	−310	−260	−296
Boiling point, °F	−44	32	−260
Normal state, at 60°F, atmospheric pressure	Gas	Gas	Gas
Relative density of gas (air = 1)	1.52	2	0.6
Relative density of liquid (water = 1)	0.51	0.58	0.3
Mass per gal of liquid at 60 °F, lb	5.1	5.8	59.8
Calorific value, Btu/ft³ (vapourized)	2 520	3 260	1 000
Btu/lb (vapourized)*	21 622	21 190	23 875
Btu/Imperial gal (vapourized)	109 956	122 900	101 600
ft³ of vapour from 1 lb of liquid at 60 °F	8.5	6.5	23
ft³ of vapour from 1 gal of liquid at 60 °F	44	38	59
Latent heat of vapourization at boiling point, Btu/lb	185	166	—
Latent heat of vapourization at boiling point, Btu/gal	944	968	—
Specific heat of vapour, Btu/lb/°F at 60 °F	0.405	0.385	0.53
Specific heat of liquid, Btu/lb/°F at 60 °F	0.59	0.55	—
Mass density of vapour, lb/ft³	0.116	0.153	0.042

(Continued)

Table I.1 (Continued)

a) Imperial			
Properties	Propane	Butane	Natural gas
Mass density of liquid, lb/ft³	31.7	36	—
<i>* Based on the gross heating value.</i>			
b) Metric			
Properties	Propane	Butane	Natural gas
Chemical formula	C ₃ H ₈	C ₄ H ₁₀	CH ₄
Molecular weight	44.06	58.1	16
Freezing point, °C, at atmospheric pressure	−190	−162	−182
Boiling point, °C	−42	0	−162
Normal state, at 15 °C, atmospheric pressure	Gas	Gas	Gas
Relative density of vapour (air = 1)	1.52	2	0.6
Relative density of liquid (water = 1)	0.51	0.58	0.3
Mass per L of liquid at 15.5 °C, kg	2.3	2.6	27.2
Calorific value, kJ/m³ (vapourized)	93 800	121 500	37 240
kJ/kg (vapourized)	49 700	49 400	55 600
kJ/L (vapourized)	24 700	27 600	25 400
m³ of vapour from 1 kg of liquid at 15 °C	0.53	0.41	—
m³ of vapour from 1 L of liquid at 15 °C	0.265	0.22	—
Specific heat of liquid, J/kg/°C at 15 °C	2 470	2 300	—
Mass density of vapour, kg/m³	1.86	2.45	0.676
Mass density of liquid, kg/L	0.51	0.58	—

Table I.2
Combustion data

a) Imperial			
Properties	Propane	Butane	Natural gas
ft³ of air required to burn 1 ft³ of gas	23.5	30	9.5
ft³ of oxygen required to burn 1 ft³ of gas	5	6.25	2
ft³ of air required to burn 1 lb of gas	200	195	226
Approximate ignition temperature, °F	920	900	1300
Approximate maximum flame temperature, °F	3600	3600	3600
Limits of flammability, percentage of gas in air mixture:			
Percentage of gas at lower limit	2.1	1.9	4
Percentage of gas at upper limit	9.5	8.5	15
Approximate theoretical percentage of gas at maximum rate of flame propagation	5	4	0
Products of complete combustion:			
ft³ of carbon dioxide (CO₂)/ft³ of gas	3	3.9	1
ft³ of water vapour/ft³ of gas	4	5	2
ft³ of nitrogen (N₂)/ft³ of gas	18.5	23.6	8
lb of water (H₂O)/lb of gas	1.6	1.5	2.25
lb of nitrogen (N₂)/lb of gas	12	11.8	13.2
Ultimate carbon dioxide (CO₂) percentage by volume	13.9	14.1	11.7
b) Metric			
Properties	Propane	Butane	Natural gas
m³ of air required to burn 1 m³ of gas	23.5	30	9.5
m³ of oxygen required to burn 1 m³ of gas	5	6.25	2
m³ of air required to burn 1 kg of gas	12.3	11.7	14.1
Approximate ignition temperature, °C	490	480	700
Approximate maximum flame temperature, °C	1980	1980	1980
Limits of flammability, percentage of gas in air mixture:			
Percentage of gas at lower limit	2.1	1.9	4
Percentage of gas at upper limit	9.5	8.5	15
Approximate theoretical percentage of gas at	4.8	3.8	0

(Continued)

Table I.2 (Continued)

b) Metric			
Properties	Propane	Butane	Natural gas
maximum rate of flame propagation			
Products of complete combustion:			
m³ of carbon dioxide (CO₂)/m³ of gas	3	3.9	1
m³ of water vapour/m³ of gas	4	5	2
m³ of nitrogen (N₂)/m³ of gas	18.5	23.6	8
kg of water (H₂O)/kg of gas	1.6	1.5	2.25
kg of nitrogen (N₂)/kg of gas	12	11.8	13.2
Ultimate carbon dioxide (CO₂) percentage by volume	13.9	14.1	11.7

Table I.3
Orifice capacity — Heat input calculation

For standard conditions, the heat input calculation is as follows:

$$Q = HHV \cdot (1658.5 \cdot K \cdot A) \cdot (\sqrt{H/P})$$

where

Q = heat input, Btu/h

HHV = gross heating value, Btu/ft³

K = coefficient of discharge

A = area of orifice, in²

H = manifold pressure, in w.c.

ρ = specific gravity of gas

Type of gas	Gross heating value, Btu/ft³	Orifice factor	Manifold pressure, in w.c.	Specific gravity
Propane	2520	0.8	11.00	1.52
Butane	3260	0.8	11.00	2.00
Natural	1000	0.779	3.50	0.6

(Continued)

Table I.3 (Continued)

Drill size	Diameter, in	Diameter, mm	Orifice capacity, Btu/h (kW)					
			11 in w.c. (2.75 kPa)				3.5 in w.c. (0.88 kPa)	
			Propane		Butane		Natural gas	
			Btu/h	kW	Btu/h	kW	Btu/h	kW
80	0.0135	0.343	1 288	0.378	1 453	0.426	467	0.137
79	0.0141	0.358	1 403	0.411	1 583	0.464	509	0.149
78	0.0160	0.406	1 805	0.529	2 036	0.597	654	0.192
77	0.0180	0.457	2 287	0.670	2 579	0.756	829	0.243
76	0.0200	0.508	2 826	0.828	3 187	0.934	1 024	0.300
75	0.0210	0.533	3 111	0.912	3 508	1.028	1 127	0.330
74	0.0225	0.572	3 583	1.050	4 040	1.184	1 298	0.380
73	0.0240	0.61	4 074	1.194	4 595	1.347	1 476	0.433
72	0.0250	0.635	4 415	1.294	4 979	1.459	1 600	0.469
71	0.0260	0.66	4 770	1.398	5 379	1.577	1 728	0.507
70	0.0280	0.711	5 535	1.622	6 243	1.830	2 006	0.588
69	0.0293	0.743	6 045	1.772	6 817	1.998	2 190	0.642
68	0.0310	0.787	6 782	1.988	7 649	2.242	2 457	0.720
67	0.0320	0.813	7 237	2.121	8 162	2.392	2 622	0.769
66	0.0330	0.838	7 689	2.254	8 672	2.542	2 786	0.817
65	0.0350	0.889	8 654	2.536	9 760	2.860	3 136	0.919
64	0.0360	0.914	9 147	2.681	10 316	3.024	3 315	0.971
63	0.0370	0.94	9 675	2.836	10 911	3.198	3 506	1.027
62	0.0380	0.965	10 197	2.988	11 500	3.370	3 695	1.083
61	0.0390	0.991	10 754	3.152	12 128	3.554	3 897	1.142
60	0.0402	1.02	11 392	3.339	12 848	3.765	4 128	1.210
59	0.0409	1.04	11 843	3.471	13 357	3.915	4 291	1.258
58	0.0421	1.07	12 536	3.674	14 138	4.144	4 543	1.331
57	0.0429	1.09	13 009	3.813	14 672	4.300	4 714	1.382
56	0.0465	1.18	15 246	4.468	17 195	5.039	5 524	1.619
55	0.0520	1.32	19 079	5.592	21 517	6.306	6 913	2.026
54	0.0551	1.4	21 461	6.290	24 204	7.094	7 776	2.279
53	0.0594	1.51	24 966	7.317	28 157	8.252	9 047	2.651

(Continued)

Table I.3 (Continued)

Drill size	Diameter, in	Diameter, mm	Orifice capacity, Btu/h (kW)					
			11 in w.c. (2.75 kPa)				3.5 in w.c. (0.88 kPa)	
			Propane		Butane		Natural gas	
			Btu/h	kW	Btu/h	kW	Btu/h	kW
52	0.0634	1.61	28 383	8.319	32 009	9.381	10 284	3.014
51	0.0669	1.7	31 645	9.275	35 688	10.460	11 466	3.361
50	0.0701	1.78	34 693	10.168	39 126	11.467	12 571	3.684
49	0.0728	1.85	37 475	10.983	42 264	12.387	13 579	3.980
48	0.0760	1.93	40 787	11.954	45 998	13.481	14 779	4.331
47	0.0783	1.99	43 362	12.709	48 903	14.333	15 712	4.605
46	0.0811	2.06	46 466	13.618	52 404	15.359	16 837	4.935
45	0.0819	2.08	47 373	13.884	53 426	15.658	17 165	5.031
44	0.0858	2.18	52 037	15.251	58 687	17.200	18 856	5.526
43	0.0890	2.26	55 927	16.391	63 073	18.486	20 265	5.939
42	0.0933	2.37	61 503	18.026	69 362	20.329	22 286	6.532
41	0.0961	2.44	65 190	19.106	73 520	21.548	23 622	6.923
40	0.0980	2.49	67 889	19.897	76 564	22.440	24 600	7.210
39	0.0996	2.53	70 088	20.542	79 044	23.166	25 396	7.443
38	0.1016	2.58	72 886	21.362	82 199	24.091	26 410	7.740
37	0.1039	2.64	76 315	22.367	86 067	25.225	27 653	8.105
36	0.1067	2.71	80 416	23.569	90 691	26.580	29 138	8.540
35	0.1098	2.79	85 234	24.981	96 125	28.173	30 884	9.052
34	0.1110	2.82	87 077	25.521	98 203	28.782	31 552	9.247
33	0.1130	2.87	90 192	26.434	101 716	29.811	32 681	9.578
32	0.1161	2.95	95 290	27.928	107 466	31.496	34 528	10.120
31	0.1201	3.05	101 860	29.853	114 875	33.668	36 909	10.817
1/64	0.0156	0.397	1 726	0.506	1 946	0.570	625	0.183
1/32	0.0313	0.704	5 427	1.591	6 120	1.794	1 966	0.576
3/64	0.0469	1.19	15 506	4.545	17 487	5.125	5 619	1.647
1/16	0.0626	1.59	27 682	8.113	31 219	9.150	10 030	2.940
5/64	0.0780	1.98	42 927	12.581	48 412	14.189	15 555	4.559
3/32	0.0937	2.38	62 024	18.178	69 949	20.501	22 474	6.587
7/64	0.1094	2.78	84 624	24.802	95 437	27.971	30 663	8.987
1/8	0.1252	3.18	110 728	32.453	124 877	36.599	40 122	11.759

Table I.4
Conversion of volumes of one kind of gas to volumes of a different kind of gas
passing through an orifice or regulator

If you have a volume of natural gas and want to know equivalent volume of	Propane:	Multiply by 0.63
	Butane:	Multiply by 0.55
	Air:	Multiply by 0.77
If you have a volume of butane and want to know equivalent volume of	Propane:	Multiply by 1.15
	Natural gas:	Multiply by 1.83
	Air:	Multiply by 1.42
If you have a volume of air and want to know equivalent volume of	Propane:	Multiply by 0.81
	Butane:	Multiply by 0.71
	Natural gas:	Multiply by 1.29
If you have a volume of propane and want to know equivalent volume of	Butane:	Multiply by 0.87
	Natural gas:	Multiply by 1.58
	Air:	Multiply by 1.23

Table I.5
Conversion table

Multiply	By	To obtain
Btu/h	0.00029295	kilowatts
cubic feet	28.31685	cubic decimetres
cubic feet	0.028317	cubic metres
degrees Fahrenheit	$(\text{temp in } ^\circ F - 32) \times 5 / 9$	degrees Celsius
feet	0.3048	metres
inches	25.4	millimetres
inches water column	249	pascals
inches water column	0.249	kilopascals
pounds force	4.44822	newtons
pounds per square inch	6.894757	kilopascals
square inches	645.16	square millimetres

Annex J (normative)

Acceptance criteria for visual inspection of welds

Note: This annex is a mandatory part of this Code

J.1 General

J.1.1 Application

This annex applies where referenced by the Code.

J.1.2 Procedure

The completed welds on the outside surface of the piping shall be visually inspected for 100% of the weld length for any imperfections in accordance with the requirements of Clause [J.2](#). The inspection shall be performed in accordance with documented procedures approved by the installation contractor. Such procedures shall include requirements for extent and frequency of visual examination, personnel qualifications and visual acuity, maximum viewing distance and angle, lighting conditions, evaluation tools, and reporting.

Note: The requirements for qualification of welding inspectors and procedures can be subject to other provincial or territorial regulations.

J.1.3 Results

Results of the visual inspection of completed welds shall be reported in a format approved by the installation contractor. Reports of defective welds shall include

- a) weld identification;
- b) description, position, and length of defects;
- c) date; and
- d) signature of qualified inspector.

J.2 Acceptance criteria

J.2.1 Weld crown

At no point shall the outside surface of welds be below the surface of the adjacent base metal or above it by more than the amount shown in Table [J.1](#) except that an additional 1.0 mm shall be allowed for localized deviation.

Table J.1
Outside crown height
(See Clause [J.2.1](#).)

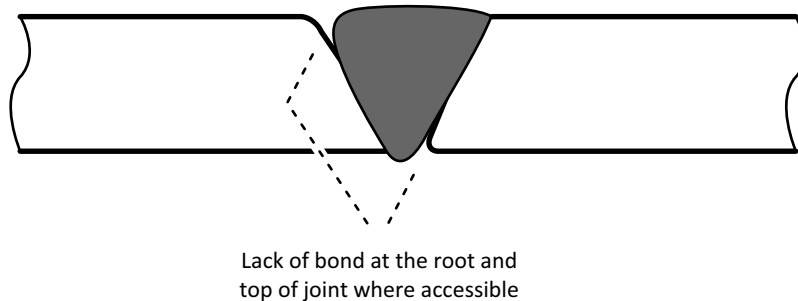
Nominal wall thickness, in (mm)	Outside crown height, maximum, mm
3/8 (10.0) or less	1/10 (2.5)
Greater than 3/8 (10.0)	1/8 (3.5)

J.2.2 Incomplete fusion

Individual indications of incomplete fusion conditions at the crown of the weld, and at the root of the weld where accessible, shall not exceed 1/2 in (12 mm) in length. The cumulative length of such

indications in any 12 in (300 mm) length of weld shall not exceed 1 in (25 mm), except that for welds less than 300 mm long, the cumulative length of such indications shall not exceed 8% of the weld length (refer to Figure J.1).

Figure J.1
Incomplete fusion
(See Clause J.2.2.)

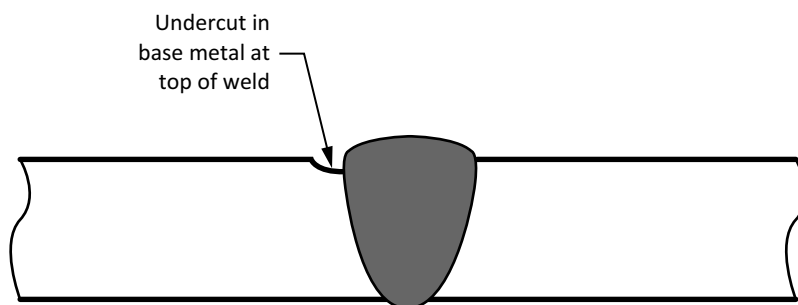


J.2.3 Undercut

The following shall apply for acceptance of undercut in the top of the weld:

- Except as permitted in Clause J.2.3 c), individual lengths of indications of undercut shall not exceed 2 in (50 mm) nor exceed a depth of 1/32 in (1.0 mm).
- Except as permitted in Clause J.2.3 c), the cumulative length of such indications in any 12 in (300 mm) length of weld shall not exceed 2 in (50 mm), except that for welds less than 12 in (300 mm long), the cumulative length of such indications shall not exceed 16% of the weld length.
- Undercut depths less than 1/64 in (0.5 mm) or 6% of the nominal wall thickness, whichever is the lesser, shall be acceptable regardless of length, provided that a mechanical method of assessing the depth is used (see Figure J.2).

Figure J.2
Undercut
(See Clause J.2.3.)



J.2.4 Cracks and arc burns

Indications of cracks shall be unacceptable regardless of location (weld metal or HAZ). Indications of arc burns shall be unacceptable regardless of location.

J.2.5 Unequal leg length — Fillet welds

Except where required by design, there shall be not more than 1/8 in (3 mm) difference between the leg lengths of each fillet weld.

Annex K (informative)

Allowable leakage of gas appliance valve train components

Note: *This annex is not a mandatory part of this Code.*

This annex describes how gas appliance valve train components should be assessed for leakage at time of initial putting into use and reactivating after temporarily discontinuing the fuel supply to the appliance. This applies only to

- automatic gas valves certified to CSA/ANSI Z21.21/CSA 6.5;
- gas appliance regulators certified to CSA: ANSI Z21.18/CSA 6.3; and
- combination gas control valves certified to ANSI Z21.78/CSA 6.20.

These devices are used on mass market gas appliances for the residential and small commercial/ industrial markets, and these certification standards allow them to leak as much as 200 cubic centimetres per hour (200 cm³/h and equivalent to 0.00706 ft³/h) through the valve body. This leakage is extremely small and not considered to be a safety hazard.

When using an electronic combustible gas indicator (CGI), this leakage can sometimes be detected when using the most sensitive detection scale on the CGI. To confirm if the condition is potentially hazardous, the technician should switch the detector's scale to the "% LEL scale" (or similar name; the less sensitive scale) and position the sensing probe approximately 2 in (50 mm) from the source. If the detector indicates the presence of gas, then it should be treated as potentially hazardous; if not, then it should be treated as non-hazardous.

This leakage rate is small in relation to the minimum ventilation rates specified in provincial and territorial building codes, so any leakage from these devices will almost certainly dissipate, consistently keeping indoor concentrations much lower than one that could ignite, usually without being detected by smell.

Annex L (informative)

Recommended requirements for automatic safety shut-off valves and automatic vent valves installed on gas turbines having capacities greater than 12.5 MMBtu/h (3.66 MW) and inlet pressures greater than 150 PSI (1034.3 kPa)

Note: This annex is not a mandatory part of this Code.

L.1 Recommended requirements

When using automatic valves handling inlet gas pressures over 150 psig (1034.3 kPa) and capacities greater than 12.5 MMBtu/h (3.66 MW), the valves should be approved for a given application. When approving a valve for this kind of application, the following performance requirements should be taken into consideration:

- a) The automatic valve should not incorporate a bypass or an external means that prevents it from closing.
- b) The pressure rating, temperature rating, and corrosion resistance should be taken into account.
- c) There should be UV protection if installed outside.
- d) The automatic valve should not utilize fuel gas pressure/flow through the valve or an external power source for closure.
- e) The automatic valve should not have a closing time so slow that it would adversely affect the safety of downstream equipment.
- f) The automatic valve should use elastomers resistant to the effects of the gas.
- g) The automatic valve should comply with the following clauses in CSA/ANSI Z21.21/CSA 6.5:
 - i) Clause 4.1, General requirements;
 - ii) Clause 4.2, Equipment and data to be furnished by the manufacturer;
 - iii) Clause 4.3, Assembly;
 - iv) Clause 4.8, Materials;
 - v) Clause 4.9, Instructions; and
 - vi) Clause 5.4, Leakage (number of cycles, as declared by the automatic valve manufacturer, with which the automatic valve can comply; the recommended capability is 20 000 cycles).
- h) Closing times should be not more than 5 s.

Annex M (normative)

Requirements for the operation of appliances at shows, exhibitions, or other similar events

Note: *This annex is a mandatory part of this Code.*

M.1 Application

This annex applies to appliances that

- a) are on display at shows, exhibitions, or other similar events; and
- b) are designed to be used outdoors or vented to the outdoors.

M.2 Exposure

An appliance may be operated and vented indoors if it meets the requirements of this annex.

M.3 Purpose

An appliance shall only be used for the purpose of demonstrating its operation and shall not be used for heating space, water, or any other thing or for any other purpose.

M.4 Marking

An appliance certified or approved for outdoor use being operated indoors for the purpose of demonstration shall be clearly marked with a warning that this appliance is for outdoor use only and the sign shall read:

WARNING

THE USE OF THIS TYPE OF APPLIANCE IS PROHIBITED FOR INDOOR USE. FOR YOUR SAFETY THE UNIT IN THIS DISPLAY IS CONSTANTLY MONITORED FOR THE PRESENCE OF CARBON MONOXIDE.

TO PROTECT YOU AND YOUR FAMILY NEVER USE A (name of the appliance i.e. BBQ, Patio Heater, Fire Pit, etc.) INDOORS, INCLUDING A GARAGE.

ATTENTION

L'UTILISATION DE CE TYPE D'APPAREIL EST INTERDITE POUR L'UTILISATION À L'INTÉRIEUR. POUR VOTRE SÉCURITÉ L'APPAREIL DANS CETTE EXPOSITION EST SURVEILLÉ CONSTANTMENT POUR LA PRÉSENCE DE MONOXYDE DE CARBONE.

POUR PROTÉGER VOUS ET VOTRE FAMILLE, N'UTILISER JAMAIS (nom de l'appareil c.-à-d. BBQ, appareil de chauffage de patio, foyer, etc.) À L'INTÉRIEUR, Y COMPRIS UN GARAGE.

The sign shall be located immediately adjacent to the appliance and in clear view of the public, and the letters shall be a minimum 1 in (25 mm) high.

M.5 Installation

An appliance shall be installed and activated initially by a person holding an appropriate valid certificate under the authority having jurisdiction.

M.6 Operation

A person who has knowledge of the manufacturer's operating instructions for the appliance shall be in constant and immediate control of the operation of the appliance. A copy of the manufacturer's instructions shall be left with the appliance.

M.7 Certification

An appliance shall be certified or approved.

M.8 Carbon monoxide

The level of carbon monoxide in the vicinity of an appliance shall

- a) be measured at intervals not exceeding 3 h;
- b) be measured 4 ft (1.2 m) above the floor and 4 ft (1.2 m) horizontally from the appliance; and
- c) be recorded with the date and time the measurements were made.

The record of levels of carbon monoxide made shall be kept where the appliance is displayed and for the entire period of its display.

M.9 Maximum carbon monoxide

An appliance shall be shut down if the carbon monoxide level determined under Clause [M.8](#) exceeds 25 ppm.

M.10 Protection

A means shall be provided to physically protect any person from contact with hot surfaces, hot gases, or flames resulting from operation of an appliance.

M.11 Fire extinguisher

A certified portable fire extinguisher classified in accordance with ULC CAN/ULC-S508 of not less than 10-B:C rating shall be located at each booth or stall displaying appliances.

Annex N (informative)

Generators, compressors/pressure boosters, engines, and turbines

Note: This informative annex has been written in mandatory language to facilitate adoption where users of the Code or regulatory authorities wish to adopt it formally as additional requirements to this Code.

N.1 Compressors/pressure boosters

N.1.1 Low pressure cut off

A compressor/pressure booster or similar equipment capable of reducing pressure in the service piping to a point lower than the required service pressure shall be provided with a low pressure cut-off device of the manual reset type.

N.1.2 Vibration isolation

A compressor shall be isolated from vibration at the inlet or outlet by a hose compliant with Clause 10.10 of CSA B149.3.

N.1.3 High gas pressure limit control

A pressure booster capable of creating an outlet pressure higher than the normal operating pressure shall be equipped with a mechanical bypass around the booster and a high gas pressure safety device installed in the booster outlet piping set to prevent the system pressure from exceeding the normal operating pressure by 20%.

N.1.4 Application

Hermetically sealed compressors used for pressure boosting for the purpose of supplying burners, torches, or cylinder-filling applications shall comply with the requirements of Clause 17 of CSA B149.3 and shall not have a capacity greater than 500 scf/h (14.15 m³/h).

N.2 Engines and turbines

N.2.1 Emergency generators

If the generator is used as an emergency electrical supply system as defined by CSA C282, the gas train safety interlocks, if installed, shall be permitted to be wired and operated in accordance with Table 1 of CSA C282.

N.2.2 Requirements

A turbine not falling under Clause [N.2.5](#), or an engine shall be equipped with

- a) a safety shut-off valve or valves as required in [N.2.3](#) that are certified in accordance with the requirements of CSA/ANSI Z21.21/CSA 6.5. It shall be controlled by a vacuum switch, oil pressure switch, or an equivalent device to prevent the flow of gas to the fuel system on the engine or turbine when it is not running;
- b) an automatic speed governor;
- c) a vacuum switch or low-oil-pressure switch;
- d) a zero-governor-type regulator or gas control valve; and
- e) a hose compliant with Clause 10.10 of CSA B149.3, not exceeding 6 ft (2 m) in length, where the gas hose is installed downstream of the safety shut-off valve or valves required under Item a). The

valve train upstream of the gas hose shall be mounted, anchored, and supported in such a manner as to minimize damage to the valve train from the engine or turbine vibration.

N.2.3 Safety shut-off valves for input up to 12.5 MMBtu/h (3.66 MW) and inlet pressure up to 150 psig (1034 kPa)

A turbine not falling under Clause [N.2.5](#), or an engine shall comply with the following. Where the input to an engine or turbine is

- a) up to and including 2.5 MMBtu/h (732 kW), one safety shut-off valve marked C/I or two safety shut-off valves. However, if propane-fueled, a minimum of two safety shut-off valves shall be provided;
- b) over 2.5 MMBtu/h (732 kW), and up to and including 5 MMBtu/h (1464 kW), at least two safety shut-off valves shall be provided, each marked C/I, and shall be piped in series and wired in parallel;
- c) over 5 MMBtu/h (1464 kW), and up to and including 12.5 MMBtu/h (3660 kW), two safety shut-off valves shall be in series, each marked C/I. At least one safety shut-off valve shall be equipped with a proof of closure switch that is integrated with the start-up circuit; and
- d) over 12.5 MMBtu/h (3660 kW), at least two safety shut-off valves in series, each marked C/I. Each safety shut-off valve shall be equipped with a proof of closure switch that is integrated with the start-up circuit. The two safety shut-off valves shall be supervised by an approved valve proving system (VPS), which is integrated into the start-up circuit and prevent safety shut-off valves from opening when a leak is detected, or be equipped with an automatic vent valve installed in a vent line that is connected into the valve train immediately downstream of the first safety shut-off valve.

N.2.4 Pressure safety devices

Where the input to an engine or turbine is in excess of 2.5 MMBtu/h (732 kW), a high gas pressure safety device and a low gas pressure safety device shall be installed and set to detect incorrect outlet pressure ranges of the pressure regulator.

N.2.5 For input greater than 12.5 MMBtu/h (3.66 MW) and inlet pressure greater than 150 psig (1034 kPa)

A turbine having capacities greater than 12.5 MMBtu/h (3.66 MW) and inlet pressures greater than 150 psig shall be equipped with

- a) two safety shut-off valves in series, each with proof of closure, and an automatic vent valve installed downstream of the first safety shut-off valves. Each safety shutoff valve and the automatic vent valve shall be approved for use in the application;
Note: See Annex K of CSA B149.3 for recommended requirements automatic valves to be approved for use.
- b) a control valve and a turbine controller to maintain proper turbine speed; and
- c) a gas hose, having a length recommended by the hose manufacturer for the application, shall be installed between the turbine and the control valve or between the turbine and the externally installed safety shutoff valve.

N.2.6 Overcrank protection

Where an engine or turbine is of an automatic-start type, it shall be equipped with overcrank protection in addition to the requirements of Clauses 17.3.2 to 17.3.4 of CSA B149.3.

N.2.7 Inlet pressure regulator

Where the inlet pressure is in excess of 0.5 psig (3.5 kPa), a pressure regulator of the lock-up type shall be provided to the entrance to the valve train.

N.2.8 Overpressure protection

The valve train shall meet the over-pressure protection requirements of Clause 10.4 of CSA B149.3.

N.2.9 Venting of pressure control devices

A regulator or relief valve shall be vented to a safe location outdoors and shall meet the requirements for venting of pressure control devices per Clause [5.5](#).

N.3 Additional requirements for engines and turbines in buildings

N.3.1 Compliance

An engine or turbine installation in a building shall be in compliance with Clause [7.2.5](#).

N.3.2 Gas detection

The equipment for indoor installation or in an enclosure shall be equipped with a gas detector that

- a) is installed in accordance with the detector manufacturer's instructions for the type of gas;
- b) is set to activate at gas detection levels at and above one-fifth of the lower limit of flammability;
- c) upon activation, produces an audible and visual alarm;
- d) is interlocked with the mechanical ventilation system; and
- e) is interlocked to shut off the equipment.

N.3.3 Rating plate

Each engine or turbine shall be provided with a clearly legible, permanent rating plate indicating

- a) manufacturer's or vendor's name;
- b) type of fuel(s);
- c) electrical rating;
- d) maximum input rating in Btu/h (kW); and
- e) minimum and protected inlet pressure at the point of connection.

Annex O (informative)

Special considerations for industrial occupancy

Note: This informative annex has been written in mandatory language to facilitate adoption where users of the Code or regulatory authorities wish to adopt it formally as additional requirements to this Code.

1.1 Inclusions [Substitution]

This annex applies to the installation of appliances, equipment, components, accessories, and piping and tubing systems extending from the termination of the utility installation where gas is to be used for fuel purposes in locations classified as industrial occupancy.

1.7 Application [New]

The requirements of the main body of this Standard apply in addition to this annex. Where a conflict or inconsistency exists between the main body and this annex, the requirements of this annex shall take precedence.

Note: The clauses in this annex are one of the following with respect to clauses in the main body of this Standard:

- a) “[Addition]” — supplemental requirements have been added to an existing clause;
- b) “[Deleted]” — clause deleted;
- c) “[Modification]” — clarification or rewording has been made to the corresponding clause, but not a substitution invalidating the clause. New words are displayed with underlined text and struck words are shown with strikethrough text;
- d) “[New]” — a new clause is added; and
- e) “[Substitution]” — clause supersedes the clause in the main body in its entirety.

2 Reference publications [Modification]

ASME International

B31.1-2022

Power Piping [New]

4.3.1 Initial installation [Modification]

Before leaving installations, installers, including the professional of record if applicable, shall ensure that the appliance, accessory, component, equipment, or piping and tubing they installed complies with the requirements of this Code, and the person initially activating the appliance shall ensure that the appliance is in safe working order.

4.3.2 User instructions [Modification]

Installers, including the professional of record if applicable, shall instruct users in the safe and correct operation of all appliances or equipment that they install.

4.3.3 Manufacturer instructions [Modification]

The installer, including the professional of record if applicable, shall ensure that the manufacturer’s instructions supplied with the appliance are left with the user.

4.3.4 Replacement parts [Modification]

Before installing any replacement part of an appliance, the installer, including the professional of record if applicable, shall ensure that the replacement part provides operational characteristics at least equivalent to those of the original part.

4.3.5 Appliance conversion [Modification]

When the installation or conversion of an appliance constitutes a conversion from another form of energy, the installer, including the professional of record if applicable, shall advise the user of the appliance, at the time of installation or conversion, to have the former form of energy either removed or left safe and secure from accidental activation. For example, the user shall be advised

- a) in the case of a fuel oil supply tank
 - i) to remove the fill pipe, and cap or plug the exposed fill pipe opening to an inside tank;
 - ii) to shut off the tank outlet valve, remove the filter, and plug or cap the valve outlet; and
 - iii) where the tank is located outdoors, to disconnect all exposed piping or tubing, and cap or plug the piping or tubing as close as practicable to the tank;
- b) in the case of a fuel oil central distributing system
 - i) to shut off the fuel oil supply line valve located within the building; and
 - ii) to disconnect the fuel oil supply line immediately downstream of the meter, and cap or plug the outlet of the meter;
- c) in the case of a propane system
 - i) to shut off the cylinder or tank valve; and
 - ii) to disconnect and cap or plug the propane gas piping system outdoors; and
- d) in the case of an electrical appliance
 - i) to shut off the power supply to the electrical appliance at the switch; and
 - ii) to ensure that the overcurrent protection, fuse, or circuit breaker has been removed or put in the off position.

6.1.3 Welding [New]

Welded steel piping shall conform with the requirements of CSA Z662, ASME B31.1, or ASME B31.3. Piping systems with design pressure of 100 psig (700 kPa) or less may conform to the requirements of Clause 6.

6.2.10 Hoses and hose fittings [Modification]

Except as required in Clauses [6.2.11](#) and ~~6.2.12~~ [6.20.5](#), every non-metallic hose and hose fitting shall have a minimum working pressure of 350 psig (2400 kPa) and shall comply with CSA 8.1 or CSA 8.3.

Note: Refer to Clause [6.20.5](#) for metallic hoses.

6.4 Volume of gas to be used for sizing gas piping systems [New]

The volumetric flow of gas to be used for sizing gas piping systems shall equal the maximum expected system flow for all appliances, except appliances deemed as a spare appliance, plus a 20% margin.

Note: The maximum expected system flow may be determined from historical gas flow records for existing facilities or, for new facilities, from the mass and energy balance calculations used to design the facility. [New]

6.4.1 Total volume [Deleted]

6.4.2 Volume determination [Deleted]

6.4.3 Diversity load [Deleted]

6.8.3 Piping support [Modification]

Piping or tubing shall not be supported by other piping or tubing, and shall be installed with individual

supports of sufficient strength and quality. Supports shall be spaced in accordance with one of the following requirements:

- a) Table [6.2](#);
- b) [ASME B31.1](#); or
- c) [ASME B31.3](#).

6.8.6 Filter requirements [Modification]

When a filter is used in a piping system, a ~~low pressure indicator~~ indication device shall be installed ~~directly downstream of the filter to detect excessive pressure drop across the filter.~~ This pressure indication device shall not be used to satisfy the requirements of Clause [6.8.10](#).

6.8.10 Test ports [Modification]

A test port or pressure indication device shall be installed immediately downstream of a line pressure regulator or an industrial pressure regulator to allow adjustment of the set point of the regulator except where the pressure regulator can be adjusted while measuring and observing the supply pressure at any appliance being served by the pressure regulator. Where opening the test port could create an uncontrolled release of gas, the test port shall be equipped with a manual shut-off valve that is either capped or plugged.

6.8.11 Air and oxygen [New]

Air or oxygen shall not be introduced to the gas supply upstream of the manual shutoff valve specified in Clause [6.18.2](#).

6.9.2 Type [Modification]

Piping of NPS 2-1/2 to 4 shall be either press-connect fittings certified to CSA/ANSI LC 4/CSA 6.32 or welded or welded flange pipe joints.

Piping greater than NPS 4 shall have ~~be~~ welded or welded flanged pipe joints.

6.9.4 Welding acceptance criteria [Substitution]

The weld acceptance criteria for any steel piping with design pressure of 100 psig (700 kPa) or less may be by visual inspection of the external weld surface, as well as the internal weld surface where accessible without the use of special tools. Acceptance criteria of the weld shall be in accordance with the requirements of Annex J.

Note: *Hand mirrors and flashlights are not considered to be special tools.*

6.13.1 Dirt pocket installation [Modification]

A dirt pocket shall be installed at the bottom of any piping or tubing on the final drop serving an appliance other than

- a) an illuminating appliance;
- b) a range;
- c) a clothes dryer;
- d) an outdoor grill;
- e) a portable appliance or equipment;
- f) a decorative appliance;
- g) a gas log;
- h) a room heater; ~~and~~
- i) an appliance incorporating a sediment trap; and

- j) an appliance where equipment used for sediment removal is installed upstream.

6.13.5 Drip pocket installation [Modification]

A drip pocket shall be provided at all points in a piping system where condensation can collect, such as points where the piping is exposed to either wide ranges or sudden changes in temperature, except where other equipment is installed upstream of the appliance for moisture removal.

6.18.1 Certification or approval [Addition]

A manual shut off valve shall be certified to CSA 3.11, CSA 3.16, or CSA/ANSI Z21.15/CSA 9.1, or approved for use with gas, and it shall not be subjected to either a temperature or a pressure outside of its certified rating range. For manual shut-off valves having a diameter larger than NPS 8, a manual shut-off valve certified by a nationally recognized certification organization acceptable to the authority having jurisdiction may be used.

6.18.4 Type and size [Modification]

A readily accessible manual shut-off valve shall be of either the ball, butterfly, eccentric, or lubricated-plug-type where

- a) the piping is larger than NPS 1;
- b) the tubing is 1 in (25.4 mm) OD or larger; or
- c) the pressure exceeds 0.5 psig (3.5 kPa).

6.18.5 Several piping systems [Modification]

When a shut off valve controls several piping systems, it shall be

- a) readily accessible for operation at all times;
- b) provided with an installed handle or hand wheel;
- c) installed to provide protection from damage; and
- d) clearly marked with an enamelled metal, substantial fibre, or other permanent tag, so that the piping system it controls can be readily identified.

6.20.6 Interconnected equipment [Modification]

When tanks or pieces of equipment are interconnected, provisions shall be made to compensate for vibration and differential settling of the tanks, equipment, and interconnecting piping. Where a gas hose is used for this purpose, it shall be a metallic gas hose complying with ULC/ORD 20.6, CGA CR96-001, ANSI/CAN/UL 536, or a Type II or Type III gas hose complying with CSA 8.1.

6.22.1 Requirements [Modification]

The source of test pressure shall be isolated while the piping or tubing system is under test, and the system shall retain the test pressure for the minimum duration required in Table [6.3](#) and Clause [6.22.4](#) without showing any drop in pressure.

6.22.2 Before connection requirements [Substitution]

Before an appliance is connected, a piping and tubing system that contains fittings or joints shall be pressure tested either pneumatically or hydrostatically.

Table 6.3 [Modification]

Pressure test requirements

(See Clauses [6.22.1](#) and ~~6.22.2~~ [6.22.7](#))

6.22.7 Pneumatic testing [New]

If the system is pressure tested pneumatically either by using air, inert gas or carbon dioxide,

- a) appliance shut-off valves not rated for the test pressure being used and meters and pressure regulators shall not be connected to the piping or tubing system under test;
- b) one or more pressure-indicating gauges shall be connected to the piping or tubing system under test in a way in which each gauge shall experience the full test pressure;
- c) pressure-indicating gauges shall be
 - i) graduated dial indicating pressure gauges with a minimum diameter of 3 in (75 mm) ranged no less than 15% nor more than 300% of the test pressure or test gauges with an accuracy of $\pm \frac{1}{4}\%$;
 - ii) each gauge shall be calibrated against a standard dead weight tester or a calibrated master gauge;
 - iii) the gauge shall have been calibrated within 12 months prior to each test, or any time there is reason to believe the gauge is in error;
- d) dial increments shall be the lesser of 2 psig (14 kPa) or 2% of the maximum dial reading of the pressure gauge;
- e) a pressure recorder when used for this test shall be calibrated to the requirements of Item c);
- f) the possibility of brittle fracture shall be considered when conducting pressure tests at low temperatures;
- g) all wrapped and/or factory-coated piping and tubing systems (except for CSST and copper) of all sizes and lengths shall be tested at a minimum pressure of 100 psig (700 kPa); and
- h) the pressure and duration of the test shall be in accordance with Table 6.3.

Note: These test pressures and test durations are minimum requirements. Circumstances can require test pressures and test durations in excess of those shown in the Table 6.3.

6.22.8 Hydrostatic testing [New]

If the system is pressure tested hydrostatically,

- a) appliance shut-off valves not rated for the test pressure being used and meters and pressure regulators shall not be connected to the piping or tubing system under test;
- b) one or more pressure-indicating gauges shall be connected to the piping or tubing system under test in a way in which each gauge shall experience the full test pressure;
- c) pressure-indicating gauges shall be
 - i) graduated dial indicating pressure gauges with a minimum diameter of 3 in (75 mm) ranged no less than $1\frac{1}{2}$ nor more than 4 times the test pressure or test gauges with an accuracy of $\pm \frac{1}{4}\%$;
 - ii) each gauge shall be calibrated against a standard dead weight tester or a calibrated master gauge;
 - iii) the gauge shall have been calibrated within 12 months prior to each test, or any time there is reason to believe the gauge is in error;
- d) dial increments shall be the lesser of 2 psig (14 kPa) or 2% of the maximum dial reading of the pressure gauge;
- e) a pressure recorder when used for this test shall be calibrated to the requirements of Item c);
- f) the possibility of brittle fracture shall be considered when conducting pressure tests at low temperatures;
- g) the fluid shall be water unless there is the possibility of damage due to freezing in which case a mixture of water and methanol (or equivalent alternative) shall be used as the medium;
- h) the hydrostatic test pressure at every point in a piping system shall be not less than 1.5 times the design pressure;

- i) all wrapped and/or factory-coated piping and tubing systems (except for CSST and copper) of all sizes and lengths shall be tested at a minimum pressure of 100 psig (700 kPa);
- j) the hydrostatic test shall be maintained for at least 30 minutes or whatever time is required to complete the inspection of joints for leakage; and
- k) once the hydrostatic test is successfully completed, the piping and tubing system shall be fully drained and then dried.

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